BCSP CERTIFIED SAFETY PROFESSIONAL · CSP11

CSP11 Study Book

Topic-wise notes for all 7 domains and 47 blueprint competencies: definition, simple explanation, real work scenario, exam tips, every blueprint point, and 10 CSP-style questions per topic.

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47 competencies · 434 sub-topic notes · 470 practice questions with explanations · 13-week study plan · formula sheet
Sources: the CSP11 blueprint, your study guides and workbooks ("My research"), John's question bank and Yates (4th ed.). Personal study copy, not for distribution.

How to use this book

Each competency follows the same seven sections. Work through them in order: read sections 1 to 6, close the book, answer the ten questions on paper, then mark them with the answer key and write every miss in the error log at the back.

Daily routine (about 2 hours on weekdays)

Time	Activity
45 min	Read one competency (sections 1 to 6). Underline numbers, standards and formulas.
20 min	Answer its 10 practice questions closed-book. Circle the key words: BEST, FIRST, MOST, EXCEPT, NOT.
15 min	Mark answers, read every explanation (including the ones you got right) and log your misses.
40 min	In the app: "Timed drill on this topic" (25 questions) for the same competency.

Weekends: review the week's topics, redo all logged misses, and take a mixed timed quiz in the app (50 to 100 questions). Exam pace is about 1.65 minutes per question (200 questions in 5.5 hours).

Targets: 8/10 or better on every topic set; 80%+ on domain tests; 85%+ on Domain 6 calculations. A topic is not done until you hit the target twice on different days.

13-week study plan

Week	Focus	Competencies	What to do
1	Diagnostic + Domain 2 (part 1)	2.1 - 2.7	Take a 50-question mixed quiz in the app first to get a baseline. Then study benchmarking, standards, culture, incident investigation, MOC, system safety (FTA/FMEA) and indicators.
2	Domain 2 (part 2)	2.8 - 2.14	ISO 45001/14001/19011 and Z10, policies, records retention, finance (ROI, payback, NPV), leadership, project management, statistics. Do every calculation by hand.
3	Domain 1 (part 1)	1.1 - 1.4	Prevention through Design, process safety/PSM, common workplace hazards (largest topic), life safety and occupant load.
4	Domain 1 (part 2) + D1 test	1.5 - 1.7	Fleet safety, materials handling and rigging (sling-angle math), tools and machines. Weekend: 50-question Domain 1 + 2 timed test; target 80%.
5	Domain 3	3.1 - 3.4	Risk evaluation process, JHA/PHA, hierarchy of controls, financial risk (avoid, retain, transfer, share), risk ranking matrices.
6	Domain 6 (part 1)	6.1 - 6.3	Exposure measurement, noise, radiation, heat/cold, ventilation, toxicology, epidemiology. Calculation-heavy: drill the formula sheet daily.
7	Domain 6 (part 2) + formula week	6.4 - 6.6	Ergonomics (NIOSH lifting), chemistry (containment volume), physics. Redo every calculation you missed in weeks 2, 4 and 6.
8	Domain 4	4.1 - 4.5	ERPs, ICS/business continuity, fire prevention and suppression, hazmat transportation, workplace violence.
9	Domain 7	7.1 - 7.6	Needs assessment, ADDIE, Bloom, Kirkpatrick, methods, adult learning. Weekend: 100-question mixed mock (2 h 45 min).
10	Domain 5 + Mock exam 1	5.1 - 5.5	Spill containment, GHS, waste, sustainability, ESG. Weekend: full 200-question timed mock (5.5 h). Log every miss.
11	Weak-area retraining	Your weakest 6-8 topics	Re-read the notes for the topics where you scored under 80%, redo their 10 MCQs, then 25-question topic drills in the app.

12	Mock exam 2 + calculations	All domains	Second full timed mock. Target 75-80%+. Re-drill calculations and every question in your error log.
13	Final review	Formula sheet + error log	Light mixed quizzes only. Re-read the CSP tips and exam traps for every topic. Rest well the day before the exam.

Exam weighting (CSP11 blueprint)

Domain	Weight	About how many of 200 questions*
D1 Advanced Application of Safety Principles	25%	~50
D2 Program Management	25%	~50
D3 Risk Management	15%	~30
D4 Emergency Management	9%	~18
D5 Environmental Management	6%	~12
D6 Occupational Health and Applied Science	10%	~20
D7 Training	10%	~20

*Estimated from the weights. BCSP sets the exact number of scored questions; check bcsp.org for current exam details.

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DOMAIN 1

Advanced Application of Safety Principles

25% of the exam · 7 competencies

1.1	Prevention through Design (PtD) Describe the principles of minimizing hazards using Prevention-Through-Design (e.g., avoidance, elimination, substitution, safety design criteria for workplace facilities, machines, and practices)
1.2	Process Safety Apply the principles of process safety (e.g., pressure relief systems, chemical compatibility, management of change, materials of construction, process flow diagrams)
1.3	Common Workplace Hazards Evaluate common workplace hazards (e.g., electrical, falls, confined spaces, lockout/tagout, working around water, caught in, struck by, excavation)
1.4	Facility Life Safety Evaluate facility life safety features (e.g., public space safety, floor loading, occupancy loads)
1.5	Fleet Safety Describe fleet safety principles (e.g., driver and equipment safety, maintenance, surveillance equipment, GPS monitoring, telematics, hybrid vehicles, fuel systems, driving under the influence, fatigue)
1.6	Materials Handling Evaluate materials handling methods and controls (e.g., forklifts, aerial lifts, and other powered industrial trucks; cranes, hand trucks, hoists, rigging, manual handling, drones)
1.7	Tools, Machines and Equipment Evaluate the use of tools, machines, and equipment (e.g., hand tools, power tools, ladders, grinders, hydraulics, robotics)

DOMAIN 1 · COMPETENCY 1.1

Prevention through Design (PtD)

CSP11 blueprint: Describe the principles of minimizing hazards using Prevention-Through-Design (e.g., avoidance, elimination, substitution, safety design criteria for workplace facilities, machines, and practices)

1. Definition

Prevention through Design is the practice of anticipating hazards and designing them out of facilities, equipment, tools, processes and work methods at the earliest stage of the life cycle (concept and design), so that risk is reduced before workers are ever exposed. It is described in ANSI/ASSP Z590.3 and NIOSH's PtD initiative.

2. Explanation in simple words

Fix the hazard on the drawing board, not on the shop floor. Changing a drawing costs a few dollars; changing a built plant costs thousands, and changing nothing can cost a life. The earlier you act, the more options you have (eliminate or substitute) and the cheaper they are.

3. Key points to remember

- Life-cycle thinking: concept → design → construction/installation → operation → maintenance → decommissioning/demolition. Consider every phase, including maintenance and demolition.
- Hierarchy applied at design: avoidance (do not introduce the hazard), elimination, substitution, engineering controls, warnings, administrative controls, PPE.
- Avoidance vs elimination: avoidance means the hazard is never introduced into a new design; elimination removes a hazard that already exists or was proposed.
- Safety design criteria come from consensus and code bodies: ANSI/ASSP (Z590.3, Z10, Z244.1), NFPA codes, UL listings, FM Global data sheets, ISO 12100 (machinery risk assessment), NIOSH guidance, OSHA standards (minimum).
- Design reviews: preliminary hazard analysis at concept, HAZOP/What-If at detailed design, pre-startup safety review (PSSR) before operation.
- Typical PtD solutions: grade-level access for maintenance, parapets 42 in. high on roofs, tie-off anchors built in, remote lube/fill lines, interlocked guarding, ergonomic workstation heights, wider aisles to separate forklifts and pedestrians.
- Business case: lower retrofit cost, lower injury costs, better productivity; cost of change rises steeply as the project moves from concept to operation.

4. Real work scenario

> A new distribution center is being designed. The safety professional reviews the drawings and notices that HVAC units on the roof will need monthly filter changes. Instead of relying on harnesses later, she asks the architect to add a 42-inch parapet and a fixed stair to the roof. Mechanics now work without fall-arrest equipment, and the change cost less than 1% of the roof budget.

5. CSP exam tips and traps

- When a question says 'design phase', 'blueprint review' or 'new facility', look for the answer that removes the hazard by design (relocate, eliminate, substitute) rather than training, signs or PPE.
- If two answers are both engineering controls, pick the one applied earlier in the life cycle.
- Remember the standard: ANSI/ASSP Z590.3 = Prevention through Design.

- Trap: a 'good' engineering control (platform with guardrails) is weaker than eliminating the task at height (remote access, grade-level equipment).

6. Every blueprint point (6 blueprint items + 2 more)

Avoidance BLUEPRINT

Avoidance means a hazard is never introduced in the first place - the safest design choice is made before anything exists, e.g., choosing a site, layout or process that has no fall, chemical or traffic exposure.

- Applied at concept stage (cheapest point to change).
- Example: placing rooftop equipment at grade so no roof access is ever needed.
- Difference from elimination: elimination removes a hazard that exists or was proposed; avoidance never creates it.

> **Exam tip:** Keywords 'concept stage', 'new facility', 'layout planning' point to avoidance/elimination.

Elimination BLUEPRINT

Remove the hazard entirely at the design stage so it no longer exists (Z590.3, NIOSH PtD). Top of the hierarchy of controls.

- Example: pre-assembled components remove on-site cutting and welding.
- Example: remote lube/fill points remove work at height.
- PtD addresses the top two levels of the hierarchy: elimination and substitution.

> **Exam tip:** If elimination or substitution is feasible, it is usually the correct answer over engineering, admin or PPE.

Substitution BLUEPRINT

Replace a hazardous material, process or equipment with a less hazardous one.

- Water-based paints instead of solvent-based (removes VOC exposure).
- Non-silica blasting media instead of sand.
- Make sure the substitute does not introduce a new, worse hazard.

Minimization, moderation and simplification (inherently safer design)

Inherently safer design principles in the guide: minimize quantity/concentration, moderate (lower) operating conditions, and simplify systems to reduce human error.

- Minimization: smaller containers, lower inventory, less concentrated solutions.
- Moderation: run reactions at ambient temperature/pressure, lower voltage or speed.
- Simplification: standardize control layouts across lines to prevent operator confusion.
- Integration of engineering controls: design fixed guards, interlocks and fail-safe shutdowns into machines from the start.

Safety design criteria for workplace facilities BLUEPRINT

Facilities are designed against codes and consensus standards so that egress, fall protection, ventilation, fire protection and traffic separation are built in.

- Sources: OSHA (minimum), NFPA (101, 13, 30), IBC/IFC, ANSI/ASSP, FM Global data sheets, UL listings, NIOSH guidance.
- Examples: 42-in. parapets/guardrails, separated pedestrian and forklift routes, adequate exits, ventilation designed to ASHRAE/ACGIH.
- ANSI/ASSP Z590.3 = PtD guideline; ISO 12100 = machinery design risk assessment.

Safety design criteria for machines BLUEPRINT

Machine design should remove hazards (inherently safe design), then add safeguarding, then information for use - the ISO 12100 three-step method.

- Fixed guards, interlocks, two-hand controls and safety-rated control circuits designed in, not added later.
- Lockable energy-isolating devices on new equipment (1910.147 requires new or overhauled equipment to accept a lock).
- Ergonomic heights and reach distances; noise specifications in purchase orders.

Safety design criteria for practices BLUEPRINT

Work methods and procedures are designed so the safe way is the easy way: sequencing tasks, eliminating non-routine hazardous steps and building checks into the process.

- Design procedures with the people who do the work.
- Plan maintenance access during design (grade-level access, isolation points, lifting points).
- Review procedures under MOC when equipment or materials change.

Life-cycle consideration and benefits of PtD

Hazards are evaluated across the full life cycle: concept, design, construction, operation, maintenance, decommissioning and demolition.

- Example: modular equipment that can be dismantled and recycled safely at end of life.
- Benefits: lower PPE/training/admin costs, injury prevention before exposure, regulatory compliance before commissioning, long-term reduction in catastrophic failures.
- If the hazard already exists and redesign is impractical, PtD is not the answer - pick the best engineering or administrative control.

Read more: Yates, Safety Professional's Reference and Study Guide (4th ed.): Ch. 35 Prevention through Design (PtD), p. 780

7. Ten CSP-style practice questions

Answer all ten on paper first, then check the key. Score: _____ / 10 Date: _____ Retest: _____ / 10

1.1-Q1. Which Prevention through Design (PtD) strategy aims to completely remove a hazard during the design phase?

A. Substitution
B. Minimization
C. Elimination
D. Moderation

1.1-Q2. During design review for a new packaging line, maintenance will need to grease bearings on a drive located 14 ft above the floor every week. Which design decision BEST applies Prevention through Design?

A. Relocate the grease points with remote lube lines so the task is done from floor level
B. Specify a permanent platform with guardrails at the drive
C. Require a 100% tie-off procedure and provide harnesses
D. Add the task to the weekly JHA and train the mechanics

1.1-Q3. A new conveyor system is being installed in a distribution center. During design review, engineers propose adding emergency-stop pull cords but leaving several exposed pinch points near the rollers because "guards can be added later if injuries occur." Which Prevention through Design action is MOST appropriate?

A. Install signs warning employees to keep hands clear of the rollers
B. Provide gloves to protect workers if their hands get too close
C. Train workers to avoid pinch points during startup
D. Eliminate or enclose the pinch points with fixed guards before installation

1.1-Q4. A safety manager is reviewing drawings for a new CNC machining area. The design places electrical panels directly behind the operators, requiring them to reach across the work zone to access them. What is the BEST Prevention through Design recommendation?

A. Require operators to wear additional PPE when accessing the panels
B. Relocate the panels to a safe, accessible location during design
C. Train workers to de-energize machines before accessing the panels
D. Post warning signs on the panels

1.1-Q5. Under inherently safer design, redesigning a reaction to run at near-ambient temperature and pressure instead of high temperature and pressure demonstrates which principle?

A. Moderation (attenuation)
B. Simplification
C. Minimization (intensification)
D. Substitution

1.1-Q6. A company is automating a welding workstation. The project manager skips the hazard review during initial design to "save time" and plans to do it at commissioning. Engineers later find that the robot's reach envelope intersects the operator walkway. Which step was missed?

A. Vendor warranty review
B. Purchasing specification verification
C. Pre-startup safety review
D. Early-stage hazard identification and risk assessment

1.1-Q7. Designing the control panel of a new unit with fewer, clearly labeled controls to reduce operator confusion during startup is an example of which inherently safer design principle?

A. Minimization
B. Simplification
C. Substitution
D. Moderation

1.1-Q8. During design of a new paint finishing line, the engineer specifies a water-based coating in place of the flammable solvent-based coating used at the old plant. This is BEST described as:

A. Elimination
B. An engineering control
C. Substitution
D. An administrative control

1.1-Q9. Which consensus standard provides guidance specifically on Prevention through Design - addressing occupational hazards and risks in design and redesign processes?

A. ANSI/ASSP Z590.3
B. ANSI/ASSP Z359.1
C. ANSI Z16.1
D. ANSI/ASSP Z10

1.1-Q10. ANSI/ASSP Z590.3 places one control ABOVE elimination in its hierarchy. It refers to preventing a hazard from ever entering the workplace, such as siting a new building outside a flood plain. What is it called?

A. Attenuation
B. Isolation
C. Substitution
D. Avoidance

Answer key and explanations

Q	Ans	Why
1	**C**	Elimination designs the hazard out so it no longer exists; it is the most effective control. Substitution and minimization only reduce the hazard.
2	**A**	Remote lube lines eliminate the need to work at height. A guarded platform is a good engineering control but still leaves exposure; tie-off and training are lower-order controls.

3	**D**	PtD addresses hazards at the design stage rather than after injuries. Fixed guarding designed in now beats signs, training or PPE, which are lower in the hierarchy.
4	**B**	Changing the layout on paper removes the exposure at almost no cost. PPE, training and signs accept the hazard instead of designing it out.
5	**A**	Moderation uses less hazardous conditions (lower temperature, pressure or concentration) to reduce severity. Minimization reduces the quantity; substitution swaps the material.
6	**D**	PtD requires hazard identification and risk assessment at the conceptual/design stage, when changes are cheapest. A PSSR at commissioning is too late to redesign the layout cheaply.
7	**B**	Simplification removes unnecessary complexity so errors are less likely and easier to detect. The trap is minimization, which refers to reducing hazardous inventory.
8	**C**	Replacing a hazardous material with a less hazardous one is substitution. The hazard is reduced, not removed, so it is not elimination.
9	**A**	ANSI/ASSP Z590.3 is the PtD standard. Z10 is the OHS management system standard, Z359 covers fall protection, and Z16.1 covers safety metrics/recordkeeping.
10	**D**	Avoidance (prevention at concept/siting) sits at the top of the Z590.3 hierarchy, followed by elimination, substitution, engineering controls, warnings, administrative controls and PPE.

DOMAIN 1 · COMPETENCY 1.2

Process Safety

CSP11 blueprint: Apply the principles of process safety (e.g., pressure relief systems, chemical compatibility, management of change, materials of construction, process flow diagrams)

1. Definition

Process safety is the management of hazards in processes that handle highly hazardous chemicals to prevent fires, explosions and toxic releases. In the U.S. it is governed by OSHA 29 CFR 1910.119 (Process Safety Management, PSM) and EPA 40 CFR 68 (Risk Management Program, RMP).

2. Explanation in simple words

Personal safety stops slips and cuts; process safety stops the big events that kill many people at once (Bhopal, Texas City). It works by knowing your chemicals and equipment, analyzing what can go wrong, keeping equipment in good condition, and controlling every change.

3. Key points to remember

- PSM has 14 elements: employee participation, process safety information (PSI), process hazard analysis (PHA), operating procedures, training, contractors, pre-startup safety review (PSSR), mechanical integrity, hot work permit, management of change (MOC), incident investigation, emergency planning and response, compliance audits, trade secrets.
- PHA must be revalidated at least every 5 years; compliance audits every 3 years; operator refresher training at least every 3 years; incident investigation must start within 48 hours and reports kept 5 years.
- PHA methods: What-If, Checklist, What-If/Checklist, HAZOP (guide words: no, more, less, reverse, other than), FMEA, Fault Tree Analysis.
- Pressure relief systems: relief valves (reclose), safety valves (gas, pop action), rupture discs (non-reclosing). A rupture disc upstream of a relief valve protects the valve from corrosion; the space between must be monitored. Never put a block valve in a relief path without a car-seal/lock-open program.
- Chemical compatibility: segregate incompatibles (acids/bases, oxidizers/flammables, hypochlorite/acids → chlorine, water-reactives). Use a compatibility chart, SDS Section 10 and separate secondary containment.
- Materials of construction: choose metals, gaskets and linings compatible with the chemical, temperature and pressure (e.g., chloride stress-corrosion cracking of 304/316 stainless; PTFE gaskets creep). A material change is not 'replacement in kind' and needs MOC.
- PFD vs P&ID: the process flow diagram shows major equipment, flow paths and operating conditions; the piping and instrumentation diagram adds every pipe, valve, instrument, interlock and relief device. Both are part of process safety information.
- MOC covers changes to chemicals, technology, equipment, procedures and facilities (and organization/staffing in good practice), including temporary changes. Replacement in kind is excluded.

4. Real work scenario

> A refinery wants to raise the throughput of a distillation column by 10%. The MOC review checks whether the existing relief valve is still sized for the new maximum flow, updates the P&ID and operating procedures, trains operators, and completes a PSSR before startup. The review finds the relief valve is undersized, and it is replaced before the change goes live.

5. CSP exam tips and traps

- Questions often hide an MOC trigger inside 'it's basically the same part' or 'it's only temporary'. If the specification changes, MOC applies.
- Know which PSM element matches a scenario: missing drawings = PSI; corroded pipe left unrepaired = mechanical integrity; no PHA follow-up = PHA recommendations must be resolved.
- HAZOP uses guide words; FMEA lists component failure modes; FTA works top-down from an event.
- Numbers to memorize: 5-year PHA revalidation, 3-year audit, 3-year refresher training, 48-hour investigation start.

6. Every blueprint point (5 blueprint items + 3 more)

PSM scope and the 14 elements (29 CFR 1910.119)

PSM applies to processes with highly hazardous chemicals (Appendix A) at or above threshold quantities, or 10,000 lb or more of flammable liquids/gases.

- 14 elements: employee participation, PSI, PHA, operating procedures, training, contractors, PSSR, mechanical integrity, hot work permit, MOC, incident investigation, emergency planning and response, compliance audits, trade secrets.
- Exempt: retail facilities, oil and gas well drilling/servicing, hydrocarbon fuels used only for workplace heating/cooking, atmospheric tanks not connected to a process, farms (fertilizer use), flammables below 10,000 lb, non-production labs, normally unoccupied remote facilities.
- EPA's Risk Management Program (40 CFR 68) parallels PSM for community protection.

> **Exam tip:** 'Operator not included in PHA team' = employee participation violation.

Process safety information (PSI)

Complete, accurate, up-to-date written information compiled BEFORE the PHA: chemical hazards, process technology and process equipment.

- Chemicals: SDSs, toxicity/exposure limits, physical, reactivity, corrosivity, thermal stability, mixing (incompatibility) data.
- Technology: block/process flow diagrams, safe operating limits, process chemistry, maximum intended inventory, consequences of deviation.
- Equipment: P&IDs, relief system design basis, materials of construction, electrical classification, ventilation, design codes (ASME, API, NFPA), material/energy balances, safety systems.

> **Exam tip:** Outdated P&IDs, missing reactivity data or unknown maximum inventory = PSI failure.

Process hazard analysis and other PSM numbers

PHA is required before startup, by a team with process and methodology expertise, and must be revalidated at least every 5 years.

- PHA must address: process hazards, previous incidents with catastrophic potential, controls and detection, consequences of control failure, facility siting, human factors, qualitative evaluation of effects.
- Methods: What-If, Checklist, What-If/Checklist, HAZOP, FMEA, FTA (plus LOPA in practice).
- Timeframes: operating procedures certified annually; refresher training at least every 3 years; incident investigation starts within 48 hours, reports kept 5 years; compliance audits every 3 years (keep the two most recent reports).
- PSSR is required before hazardous chemicals are introduced to new or modified facilities.
- Trade secrets can never be used to withhold PSI needed for safety (confidentiality agreements allowed).

> **Exam tip:** Facility siting is often overlooked in exam scenarios involving community impact or weather.

Pressure relief systems BLUEPRINT

Relief devices protect equipment from overpressure by releasing to a safe location.

- Design and sizing per ASME BPVC Section VIII and API 520/521; discharge away from people and ignition sources.
- Inspection: visual annually (API 576); bench test/recertification typically every 5 years, sooner in corrosive or fouling service.
- Relief valve recloses; rupture disc is non-reclosing; disc upstream of a valve protects it from corrosion (monitor the space between).
- A rupture disc alone is not a substitute for a relief valve unless designed as part of a dual system.

Chemical compatibility BLUEPRINT

Prevent dangerous reactions by keeping incompatible chemicals, materials and equipment apart.

- Check SDS Section 10 (stability and reactivity) and compatibility charts for containers, piping and gaskets.
- Segregate per NFPA 400 (e.g., oxidizers away from flammables; acids away from cyanides - produces HCN; hypochlorite away from acids - chlorine).
- Storing incompatibles in the same containment area without separation is a violation even if containers are sealed.

Management of change in process safety BLUEPRINT

All modifications to HHC processes are evaluated, authorized and documented before implementation.

- Applies to equipment changes, raw material type/quality/supplier, operating conditions, control systems/software, procedures.
- Required considerations: technical basis, impact on health and safety, modification of SOPs, time period (temporary/permanent), authorization.
- Communicate and train affected employees and contractors before startup.
- 'Replacement in kind' excluded - but PSSR may still apply.

> **Exam tip:** Trigger words: new material supplier, modifying reactor temperature, upgrading control software.

Materials of construction BLUEPRINT

Select materials that withstand the pressures, temperatures and chemicals for the life of the process.

- Select per ASME, API 650/653 (tanks) and ASTM material standards; consider corrosion allowance and coatings.
- Document specifications in PSI before startup.
- A material change is not replacement in kind - it requires MOC (e.g., stainless steel in hot chloride service can stress-crack).

> **Exam tip:** Choosing material based only on cost, without corrosion resistance, can invalidate code compliance.

Process flow diagrams and P&IDs BLUEPRINT

PFDs show major equipment, streams and operating conditions; P&IDs add every pipe, valve, instrument, interlock and relief device.

- Both are part of PSI and must match field conditions.
- Update P&IDs after every approved MOC.
- Block flow diagram < PFD < P&ID in detail.

> **Exam tip:** Which diagram shows safety interlocks and relief valves? The P&ID, not the PFD.

Read more: Yates, Safety Professional's Reference and Study Guide (4th ed.): Ch. 4 Regulations (PSM), Ch. 38 Management of Change, p. 859; Ch. 27 Hazardous Materials Management

7. Ten CSP-style practice questions

Answer all ten on paper first, then check the key. Score: _____ / 10 Date: _____ Retest: _____ / 10

1.2-Q1. Which process safety drawing shows process piping together with instrumentation, control loops, interlocks and safety devices such as relief valves?

A. Process Flow Diagram (PFD)
B. Piping and Instrumentation Diagram (P&ID)
C. Block Flow Diagram (BFD)
D. Plot plan

1.2-Q2. A rupture disc is installed directly upstream of a spring-loaded relief valve in corrosive service. What is the PRIMARY reason for this arrangement, and what must be provided between them?

A. To isolate the relief valve from the corrosive process fluid; a pressure gauge or telltale indicator to detect disc leakage
B. To increase relieving capacity; a check valve
C. To allow the valve to reseat faster; a block valve
D. To reduce the set pressure; a second rupture disc

1.2-Q3. A chemical plant substitutes a new raw material in a batch recipe. The engineer assumes it is compatible because it is "chemically similar" and skips the hazard review. At startup, excessive heat causes the reactor relief valve to lift. Which MOC requirement should have prevented this?

A. Training operators after the change
B. Inspecting the relief valve after installation
C. Reviewing safety data and performing a hazard evaluation of the change before implementation
D. Performing a post-incident investigation

1.2-Q4. A warehouse stores drums of oxidizers next to flammable liquids with no separation. Which process safety concept is MOST directly violated?

A. Pressure relief design
B. Process flow design
C. Materials of construction
D. Chemical compatibility

1.2-Q5. Piping for a new hydrochloric acid line is selected only on lowest cost, without considering corrosion resistance. Which process safety element is being ignored?

A. Process flow diagram review
B. Contractor safety
C. Emergency planning
D. Materials of construction

1.2-Q6. Operators at a blending unit rely on tribal knowledge because the written procedures were never updated after a recipe change. During startup an operator adds an ingredient too early, causing a runaway reaction that lifts the relief system. Which PSM element was MOST directly violated?

A. Operating procedures
B. Mechanical integrity
C. Contractors
D. Incident investigation

1.2-Q7. During a turnaround, operations wants to bypass a high-level interlock on a tank for 6 hours while a transmitter is replaced. Under a PSM-compliant program, this temporary bypass should be:

A. Allowed without paperwork because it is temporary
B. Managed through MOC with a defined time limit, compensating measures and authorization
C. Approved verbally by the shift supervisor
D. Documented only after the work is complete

1.2-Q8. While reviewing a PFD for a high-pressure steam loop, you notice steam passes directly from a boiler to a reactor with no symbol for a pressure relief device. What is the BEST action?

A. Approve the design if the piping is sized correctly
B. Assume relief is shown on the P&ID and not relevant
C. Flag the drawing and confirm whether overpressure protection was omitted or is provided elsewhere
D. Confirm the pipe routing and allow commissioning

1.2-Q9. Which of the following changes would require a Management of Change (MOC) review under OSHA PSM?

A. Replacing a valve with an identical spare part
B. Reviewing a closed incident report
C. Performing routine scheduled lubrication
D. Changing a reactor's operating temperature range

1.2-Q10. A single pressure relief valve protects a pressure vessel. Under ASME pressure vessel practice, the relief valve set pressure should be no higher than the vessel's:

A. Maximum allowable working pressure (MAWP)
B. Hydrostatic test pressure
C. Normal operating pressure
D. Burst pressure

Answer key and explanations

Q	Ans	Why
1	B	P&IDs show valves, instruments, interlocks and relief devices. PFDs show only major equipment, main flows and operating conditions.
2	A	The disc protects the valve internals from corrosion and fouling. The space between them must be monitored, because a pinhole leak can build back-pressure and keep the disc from bursting at its set point.
3	C	MOC requires the technical basis and safety impact of a change, including chemical reactivity and compatibility, to be evaluated BEFORE the change is made. Everything else in the list happens after the hazard has been introduced.
4	D	Oxidizers greatly intensify the burning of flammables. Incompatible materials must be separated by distance, barriers or cabinets.
5	D	Materials of construction must be compatible with the process fluid, temperature and pressure. Under-specified materials corrode and lead to loss of containment.
6	A	1910.119(f) requires written, current operating procedures, certified annually. An MOC failure contributed, but the most direct gap is outdated procedures.
7	B	Temporary changes, including interlock bypasses, fall under MOC. They need a technical review, a time limit, authorization and compensating measures such as a dedicated operator watching the level.
8	C	Never assume overpressure protection exists. Verify it against the P&ID and relief design basis before approving; a missing relief device is a potentially catastrophic gap.
9	D	MOC applies to changes in chemicals, technology, equipment, procedures and operating limits. A "replacement in kind" (identical part) is specifically excluded.
10	A	The set pressure of a single relief device must not exceed MAWP; allowable accumulation above that (for example 10%) occurs only while relieving. Setting it at operating pressure would cause nuisance lifting.

DOMAIN 1 · COMPETENCY 1.3

Common Workplace Hazards

CSP11 blueprint: Evaluate common workplace hazards (e.g., electrical, falls, confined spaces, lockout/tagout, working around water, caught in, struck by, excavation)

1. Definition

Common workplace hazards are the recurring energy sources and conditions that cause most serious injuries and fatalities: electricity, falls from height, hazardous atmospheres and engulfment in confined spaces, unexpected release of energy during servicing, drowning, caught-in/between and struck-by events, and trench cave-ins. OSHA's 'Focus Four' in construction are falls, struck-by, caught-in/between and electrocution.

2. Explanation in simple words

Every one of these hazards is an energy that can hurt you: electrical, gravity, mechanical, chemical or soil weight. The safety professional finds the energy, keeps people away from it or controls it, and checks that the control is working before anyone is exposed.

3. Key points to remember

- Electrical (NFPA 70E, OSHA 1910 Subpart S, 1926 Subpart K): de-energize first (electrically safe work condition); energized work only if de-energizing creates greater hazard or is infeasible, with an energized work permit, shock and arc-flash risk assessment and PPE. GFCIs trip at about 4-6 mA imbalance; let-go threshold ~10 mA; ventricular fibrillation ~50-100+ mA.
- Falls: general industry protection at 4 ft (1910.28), construction at 6 ft (1926.501), scaffolds at 10 ft, steel erection 15 ft. Guardrail top rail 42 in. ± 3 in., withstand 200 lb. PFAS: anchor 5,000 lb per worker, free fall ≤ 6 ft, maximum arresting force 1,800 lb with harness, deceleration distance ≤ 3.5 ft. Body belts are prohibited for fall arrest.
- Confined spaces (1910.146 / 1926 Subpart AA): large enough to enter, limited entry/exit, not designed for continuous occupancy. A permit space also has a hazardous atmosphere, engulfment, inwardly converging walls or another serious hazard. Acceptable oxygen 19.5-23.5%; flammables < 10% LFL. Test in order oxygen → flammables → toxics. Attendant never enters to rescue. Cancelled permits kept 1 year.
- Lockout/tagout (1910.147): written energy control program, machine-specific procedures, authorized vs affected employees, periodic inspection at least annually, verify zero energy, release stored energy (springs, hydraulics, capacitors, gravity). Group lockout uses a lockbox; shift change requires continuity (the incoming lock goes on before the outgoing lock comes off).
- Working around water (1926.106): USCG-approved PFDs, ring buoys with at least 90 ft of line no more than 200 ft apart, and a lifesaving skiff immediately available.
- Caught-in/between: unguarded rotating parts, in-running nip points, shifting loads, trench collapse, equipment rollover. Struck-by: falling/flying objects, vehicles and mobile equipment, swinging loads. Controls: guarding, tool tethers, toe boards, exclusion zones, high-visibility clothing, spotters, backup alarms.
- Excavation (1926 Subpart P): protective system required at 5 ft or deeper (unless stable rock); designed by a registered professional engineer over 20 ft. Ladder/ramp within 25 ft of lateral travel at 4 ft or deeper; test atmosphere at > 4 ft where hazardous atmosphere is possible; spoils at least 2 ft back. Soil: Stable rock (vertical), Type A ¾:1, Type B 1:1, Type C 1½:1. A competent person inspects daily and after rain.

4. Real work scenario

> A utility crew is replacing a water main in a 7-ft trench in Type B soil next to a busy road. The safety professional requires a trench box rated for the depth, spoils 2 ft back from the edge, a ladder within 25 ft, atmospheric testing because a sewer line runs nearby, a daily competent person inspection, and a flagger and barriers for traffic. When it rains overnight, the competent person re-inspects before anyone enters.

5. CSP exam tips and traps

- Numbers are heavily tested: 4 ft / 6 ft / 10 ft fall thresholds, 5 ft and 20 ft trench thresholds, 19.5-23.5% oxygen, 10% LFL, 25 ft to a ladder.
- For electrical scenarios, the BEST answer is almost always 'de-energize and lock out' before any PPE answer.
- For confined space rescue questions, the attendant summons rescue and never enters.
- LOTO shift change: watch for answers that remove the old lock first; continuity of protection is required.
- Fall clearance = free fall + deceleration (3.5 ft) + D-ring height/worker height + safety factor (often 2-3 ft).

6. Every blueprint point (8 blueprint items + 3 more)

Electrical hazards BLUEPRINT

Shock, arc flash, arc blast, burns and fires from contact with energized parts or faults. Governed by OSHA 1910 Subpart S, 1926 Subpart K and NFPA 70E.

- Establish an electrically safe work condition first; energized work only when de-energizing creates greater hazard or is infeasible (energized work permit, shock and arc-flash risk assessment, PPE).
- Current, not voltage, injures: ~1 mA perception, ~10 mA let-go, 50-100+ mA ventricular fibrillation.
- GFCI trips on a hot/neutral imbalance of about 4-6 mA within ~1/40 s; required for temporary wiring/construction (or assured equipment grounding program).
- Approach boundaries (70E): limited (unqualified may enter only with qualified escort), restricted (qualified only, with shock PPE). Arc flash boundary = 1.2 cal/cm² onset of second-degree burn.
- Insulating gloves: Class 00 500 V, 0 1,000 V, 1 7,500 V, 2 17,000 V, 3 26,500 V, 4 36,000 V AC; dielectric test every 6 months, inspect before each use, wear leather protectors.

> **Exam tip:** Electrical scenario: 'de-energize and lock out' beats any PPE answer.

Falls (working at heights, slips and trips) BLUEPRINT

Falls from height and same-level slips/trips are leading causes of serious injury.

- Thresholds: general industry 4 ft (1910.28), construction 6 ft (1926.501), scaffolds 10 ft, fixed ladders > 24 ft need ladder safety system or PFAS (cages being phased out by 2036).
- Hierarchy of fall protection: elimination > passive (guardrails, covers, nets) > fall restraint > fall arrest > administrative (warning lines, safety monitor).
- Guardrail: top rail 42 in. ± 3 in., withstand 200 lb; mid-rail ~21 in.; toe board ≥ 3.5 in.; floor holes > 2 in. covered.
- PFAS (ABC: Anchorage 5,000 lb per worker, Body harness, Connector): free fall ≤ 6 ft, deceleration ≤ 3.5 ft, arresting force ≤ 1,800 lb. Positioning devices: free fall ≤ 2 ft. Safety nets ≤ 30 ft below, drop-tested with 400-lb bag.
- Warning lines ≥ 6 ft from roof edge; safety monitor on same level with no other duties.
- Suspension trauma: serious effects in 5-15 min; rescue promptly, ideally < 5 min (ANSI Z359.2); trauma relief straps.
- Stair rail required with ≥ 4 risers or > 30 in. rise; handrail 30-38 in. (construction).

> **Exam tip:** 'Works automatically without worker input' = passive protection; 'prevents reaching the edge' = restraint.

Confined spaces BLUEPRINT

A confined space is large enough to enter, has limited entry/exit and is not designed for continuous occupancy. A permit-required space also has a serious hazard.

- PRCS hazards: hazardous atmosphere (O_2 < 19.5% or > 23.5%, flammables ≥ 10% LFL, toxics above PEL), engulfment, inwardly converging walls/sloping floor, other serious hazards.
- Test order: oxygen → flammables → toxics.
- Roles: entrant; attendant (stays outside, monitors, summons rescue, never enters); entry supervisor (authorizes/cancels entry, signs permit).
- Rescue: in-house or third party, trained and equipped; non-entry retrieval (tripod, lifeline, full-body harness) for vertical spaces > 5 ft deep where feasible.
- Cancelled permits kept 1 year; program reviewed annually. Construction: 1926 Subpart AA.

> **Exam tip:** 'Who authorizes entry?' = entry supervisor.

Lockout/tagout (hazardous energy control) BLUEPRINT

1910.147 protects workers from unexpected start-up or release of stored energy during servicing and maintenance.

- Energy types: electrical, hydraulic, pneumatic, mechanical (springs, flywheels, gravity), thermal, chemical, magnetic.
- Written program + machine-specific energy control procedures; lockout devices durable, standardized, substantial, identifiable.
- Lock if the device can be locked; tagout only if it cannot, with additional measures (remove handle, block switch).
- 8 steps: prepare, notify affected employees, shut down, isolate, apply locks/tags, release stored energy, verify isolation (try start, return controls to off), return to service (only the person who applied the lock removes it).
- Authorized vs affected vs other employees; retraining when assignments, machines or procedures change.
- Periodic inspection at least annually per procedure, by an authorized employee other than the one using it; certify machine, date, employees, inspector.
- Shift change: documented transfer so protection is continuous.

> **Exam tip:** Stored energy is the most-missed LOTO concept: bleed hydraulics, discharge capacitors, block raised parts.

Working around water BLUEPRINT

Drowning hazards on bridges, docks, barges and work over or near water (1926.106).

- USCG-approved PFDs (life jackets) where there is a danger of drowning.
- Ring buoys with at least 90 ft of line, no more than 200 ft apart.
- At least one lifesaving skiff immediately available.
- Also consider cold-water shock, currents and electrical hazards near water.

Caught-in/between hazards BLUEPRINT

Being squeezed, crushed or pinched between objects - machinery, collapsing materials, trench walls, vehicles.

- Sources: unguarded rotating parts, in-running nip points, shear points, trench collapse, equipment rollover, materials shifting.
- Controls: guarding (point of operation, power transmission < 7 ft above floor), LOTO, trench protective systems, ROPS + seat belts, staying out of swing radius.

Struck-by hazards BLUEPRINT

Being hit by falling, flying, swinging or rolling objects or vehicles.

- Sources: falling tools/materials, flying particles from grinding, swinging crane loads, backing vehicles, mobile equipment.
- Controls: toe boards, tool tethers, debris nets, exclusion zones, barricades below overhead work, high-visibility apparel, backup alarms, spotters, eye/face protection (Z87+), hard hats (Type II for lateral impact).

Excavation and trenching BLUEPRINT

1926 Subpart P. Cave-ins are the most common and fatal hazard; also falling loads, hazardous atmospheres, water accumulation, mobile equipment.

- Trench: narrow excavation deeper than wide, bottom width ≤ 15 ft.
- 4-5-20 rule: ≥ 4 ft access/egress (within 25 ft lateral travel) and atmosphere testing; ≥ 5 ft protective system unless stable rock; > 20 ft protective system designed by a registered PE.
- Soil types (by competent person; visual + manual tests): Stable rock 90°; Type A > 1.5 tsf, ¾:1 (53°); Type B 0.5-1.5 tsf, 1:1 (45°); Type C ≤ 0.5 tsf, 1½:1 (34°). Fissured, previously disturbed or vibrated soil cannot be Type A.
- Protective systems: sloping/benching, shoring, shielding (trench boxes).
- Spoil piles ≥ 2 ft from edge; no work in accumulated water without precautions; daily inspection by a competent person and after rain.

> **Exam tip:** Even < 5 ft may need protection if the competent person finds unstable soil.

Machine guarding (caught-in/struck-by control)

Guards protect against point-of-operation, power transmission and other moving parts; covered in detail under 1.7.

- Guard types: fixed, interlocked, adjustable, self-adjusting; devices: light curtains, pressure mats, pullback, restraint, two-hand controls.
- Any moving power transmission part < 7 ft above floor/platform must be guarded.

Personal protective equipment (supporting control)

PPE is the last line of defense, used when engineering and administrative controls are not enough (1910 Subpart I; hazard assessment and written certification required).

- Eye/face: ANSI Z87.1; Z87+ = high impact; goggles for chemical splash; face shield only with glasses/goggles underneath; laser eyewear by wavelength and optical density.
- Head: ANSI Z89.1 - Class G 2,200 V, Class E 20,000 V, Class C no electrical; Type I top impact, Type II top + lateral; replace suspension ~12 months, shell ~5 years.
- Foot: ASTM F2413 markings I/75 C/75 (toe), Mt (metatarsal), EH (electrical hazard), PR (puncture), Cd (conductive/static).
- HAZWOPER PPE levels: A (vapor-tight + SCBA), B (splash + SCBA - highest respiratory, lower skin), C (APR, known contaminant, O_2 ≥ 19.5%), D (work clothes).
- Employer pays for required PPE (with limited exceptions).

Compressed gases and pressure vessels

Cylinders (DOT) and stationary pressure vessels (ASME BPVC) store large amounts of energy.

- Cylinders: hydrostatic retest typically every 5 years; valve caps on when not in use; upright and secured; oxygen ≥ 20 ft from fuel gases or a 5-ft-high, ½-hour fire-rated barrier; identify by label/stencil, not color alone.
- Acetylene: never used above 15 psig (unstable).
- Pressure vessel: internal pressure > 15 psig; MAWP marked and not exceeded; ASME Section VIII (U stamp), Section I power boilers (S stamp).

> **Exam tip:** DOT governs cylinders (transport); ASME governs pressure vessels (stationary).

Read more: Yates, Safety Professional's Reference and Study Guide (4th ed.): Ch. 16 Electrical Safety, p. 427; Ch. 25 Construction Safety, p. 567; Ch. 29 Walking and Working Surfaces, p. 657

7. Ten CSP-style practice questions

Answer all ten on paper first, then check the key. Score: _____ / 10 Date: _____ Retest: _____ / 10

1.3-Q1. A worker uses a personal fall arrest system with a 6-ft shock-absorbing lanyard. Because of the work position, the worker's dorsal D-ring is 2 ft ABOVE the anchorage connection. The shock absorber can extend up to 3.5 ft. What is the free fall distance before the system begins to arrest the fall?

A. 8 ft
B. 9.5 ft
C. 11.5 ft
D. 13.5 ft

1.3-Q2. A worker uses a 6-ft shock-absorbing lanyard with the anchor at the same height as the D-ring. Deceleration distance is 3.5 ft, the D-ring is 5 ft above the worker's feet, and a 2-ft safety margin is required. What is the minimum clearance needed below the anchor (ignore harness stretch)?

A. 11.5 ft
B. 14.5 ft
C. 16.5 ft
D. 18.5 ft

1.3-Q3. A crew uses a 150 gpm pump to dewater a flooded trench 16 ft long and 3 ft wide, holding 3 ft of water. Using 7.48 gal per cubic foot and assuming no inflow, about how long will pumping take?

A. 5.2 minutes
B. 6.2 minutes
C. 6.5 minutes
D. 7.2 minutes

1.3-Q4. Workers are installing pipe in a 9-ft-deep trench in Type B soil with no protective system in place. The competent person has stepped away. What corrective action is required before work can continue?

A. Provide hard hats and high-visibility vests
B. Get workers out and install a protective system (sloping, shoring or shielding) before re-entry
C. Backfill the trench at the end of the shift
D. Place a ladder at one end of the trench

1.3-Q5. In a 6-ft-deep excavation, spoil piles have been placed right at the trench edge and rain has made the soil wet and heavy. What is the MOST appropriate corrective action?

A. Place plywood over the trench as temporary protection
B. Have the spoil moved at least 2 ft back from the edge (or retained by a device)
C. Allow work to continue because the trench is less than 7 ft deep
D. Build a berm of spoil at the edge to hold the soil in place

1.3-Q6. Under OSHA 29 CFR 1910.147, when may a tagout program be used instead of lockout?

A. When an OSHA inspector authorizes it
B. When two or more employees are involved
C. When the work is considered low risk
D. When the energy-isolating device is not capable of being locked out

1.3-Q7. An attendant outside a permit-required confined space sees that the entrant has stopped moving and is not responding. What should the attendant do FIRST?

A. Summon rescue and emergency services and begin non-entry retrieval if equipped
B. Enter the space to help the entrant
C. Leave to find a supervisor with rescue equipment
D. Shut down the ventilation system

1.3-Q8. An electrician is told to replace a breaker in an energized 480-V panel to avoid a production shutdown. Under NFPA 70E, what should be the FIRST consideration?

A. Select arc-rated PPE from the PPE category tables
B. Use insulated tools rated for 1,000 V
C. Establish an electrically safe work condition (de-energize, lock out, test for absence of voltage)
D. Have the electrician sign an energized work permit

1.3-Q9. Construction workers are placing forms on a bridge deck over a river, and fall protection does not fully eliminate the drowning hazard. Under OSHA 1926.106, which combination is required?

A. USCG-approved life jackets, ring buoys with at least 90 ft of line, and a lifesaving skiff immediately available
B. Hard hats and a high-visibility vest only
C. A rescue plan that calls the local fire department
D. Swim training for all employees on the crew

1.3-Q10. At a busy construction site, laborers on foot are repeatedly found in the travel paths of backing dump trucks. Which control is MOST effective at reducing the struck-by risk?

A. Require high-visibility Class 3 vests for all laborers
B. Implement an internal traffic control plan that physically separates pedestrian and vehicle routes and minimizes backing
C. Remind drivers to sound the horn before backing
D. Hold a toolbox talk on struck-by hazards

Answer key and explanations

Q	Ans	Why
1	**A**	When the D-ring is above the anchor, free fall = lanyard length + D-ring height above the anchor = 6 + 2 = 8 ft. The deceleration distance (3.5 ft) is added only for total fall/clearance, not free fall. Note this exceeds the 6-ft free fall limit.
2	**C**	Clearance = free fall (6) + deceleration (3.5) + D-ring height (5) + safety margin (2) = 16.5 ft. Leaving out the D-ring height gives the 11.5-ft trap.
3	**D**	Volume = 16 x 3 x 3 = 144 ft3; 144 x 7.48 = 1,077 gal; 1,077 / 150 = 7.18 min.
4	**B**	Trenches 5 ft or deeper require a protective system unless they are in stable rock (1926.652). PPE and a ladder do not protect against a cave-in.
5	**B**	1926.651(j)(2) requires spoil and equipment to be kept at least 2 ft from the edge, or held back by retaining devices, to reduce surcharge loading and falling material.
6	**D**	Tagout alone is allowed only when the isolating device cannot accept a lock. If it can, lockout is required unless tagout is shown to give full employee protection.
7	**A**	Attendants must not enter (1910.146(i)). They call rescue, attempt non-entry retrieval and stay at their post. Untrained would-be rescuers account for many confined space deaths.
8	**C**	NFPA 70E requires de-energizing as the primary protection method. Energized work is justified only if de-energizing creates a greater hazard or is infeasible, and production convenience is not a valid reason.
9	**A**	1926.106 requires approved life jackets or buoyant work vests, ring buoys with 90 ft of line no more than 200 ft apart, and at least one lifesaving skiff immediately available.
10	**B**	Separating people from moving equipment, and designing routes that minimize backing, removes the exposure. Vests, horns and training are lower-order controls that still leave workers in the path.

DOMAIN 1 · COMPETENCY 1.4

Facility Life Safety

CSP11 blueprint: Evaluate facility life safety features (e.g., public space safety, floor loading, occupancy loads)

1. Definition

Facility life safety is the design, construction and operation of buildings so occupants can survive a fire or other emergency long enough to escape or be protected in place. NFPA 101 Life Safety Code and the International Building/Fire Codes set requirements for means of egress, occupant load, compartmentation, alarms, sprinklers and emergency lighting. Floor loading ensures structures are not overloaded.

2. Explanation in simple words

A safe building lets everyone get out quickly and keeps the floor from collapsing. You count how many people can be in a space, make sure there are enough wide, unlocked, well-lit exits, keep smoke out of escape routes, and never put more weight on a floor than it was designed to carry.

3. Key points to remember

- Means of egress has three parts: exit access (path to an exit), exit (protected route, e.g., enclosed stair) and exit discharge (to the public way).
- Occupant load = area ÷ occupant load factor (ft² per person) from the code table. Examples (NFPA 101/IBC): business ~150 ft² gross; assembly with tables and chairs ~15 ft² net; standing assembly ~5-7 ft² net; storage ~300-500 ft² gross. Always use the factor given in the question and round up.
- Egress capacity: width = occupant load × capacity factor (typically 0.2 in./person for level components and doors, 0.3 in./person for stairs). Minimum door width about 32 in. clear; exit doors must open without keys or special knowledge.
- Number of exits: generally 2 minimum; 3 when occupant load is 501-1,000; 4 when over 1,000. Travel distance and dead-end limits depend on occupancy and sprinklers.
- OSHA 1910.36/.37: exit routes kept free of obstructions, exits marked with illuminated signs, doors side-hinged and swinging in the direction of travel when the room holds more than 50 people or the area is high hazard.
- Public space safety: crowd management (NFPA 101 requires trained crowd managers for large assembly occupancies, 1 per 250 occupants), emergency lighting (90 minutes), slip resistance, CPTED (crime prevention through environmental design).
- Floor loading: rated live load in pounds per square foot (psf). Load intensity = weight ÷ contact area. Post the rated load where required (1910.22(b)); check mezzanines and racking. Point loads (rack legs, forklift wheels) can exceed slab capacity even when average psf is acceptable.

4. Real work scenario

> A company converts a 6,000 ft² warehouse bay into a training hall with tables and chairs. Using 15 ft² per person, the occupant load becomes 400. The safety professional confirms the room now needs at least 80 in. of egress width (400 × 0.2), two separate exits remotely located, doors that swing outward with panic hardware, emergency lighting and a posted occupant load sign.

5. CSP exam tips and traps

- Occupant load and floor load questions are straight calculations; read the units (gross vs net area, psf).
- An increased occupant load (change of use) triggers new egress requirements - a favorite scenario.

- Propped-open fire or smoke doors and locked exits are always serious life-safety violations.
- Floor load trap: divide by AREA (ft²), not by one side length.

6. Every blueprint point (3 blueprint items + 0 more)

Public space safety BLUEPRINT

Places open to the public (assembly, retail, healthcare) must protect occupants who are unfamiliar with the building.

- Clear, marked and lit means of egress; exit signs; doors with panic hardware in assembly occupancies.
- Emergency lighting: ≥ 1 foot-candle (10 lux) average at floor along exit paths, 90-minute duration; test monthly ≥ 30 seconds and annually for 90 minutes.
- Crowd managers for large assembly occupancies (NFPA 101: 1 per 250 occupants).
- Slip resistance, lighting, CPTED (crime prevention through environmental design), smoke compartments kept closed (never prop smoke/fire doors).

> **Exam tip:** Remember emergency lighting: monthly 30 s + annual 90 min.

Floor loading BLUEPRINT

Floors, mezzanines and platforms must not be loaded beyond their design (maximum intended) load (1910.22(b)).

- Load intensity (psf) = weight ÷ contact area (ft²).
- Post and enforce the maximum load limit where required (warehouses, mezzanines, storage areas).
- Point loads (rack legs, forklift wheels) can exceed slab capacity even when average psf is acceptable.

> **Exam tip:** Heavy-storage warehouse scenario: post and enforce floor load limits.

Occupancy loads BLUEPRINT

Occupant load = floor area ÷ occupant load factor (ft² per person) from NFPA 101/IBC tables; it drives the number and width of exits.

- Factors in the guide (NFPA 101 table): assembly standing ~7 ft²/person net (5 in concentrated areas), assembly with chairs ~15 ft² net, business 100 ft² (current editions use 150 ft² gross), industrial 100 ft², storage 300 ft².
- Egress width = occupant load × 0.2 in. (level/doors) or 0.3 in. (stairs).
- Exits: 2 minimum; 3 for 501-1,000; 4 for > 1,000 occupants.
- Use the factor given in the question and round up.

> **Exam tip:** Change of use (storage → training room) increases occupant load and egress needs.

Read more: Yates, Safety Professional's Reference and Study Guide (4th ed.): Ch. 12 Fire Prevention and Protection, p. 350; Ch. 29 Walking and Working Surfaces, p. 657

7. Ten CSP-style practice questions

Answer all ten on paper first, then check the key. Score: _____ / 10 Date: _____ Retest: _____ / 10

1.4-Q1. During an inspection of a business occupancy, you find a storage room converted to an office. The occupant load rose from 4 to 12, but the only egress door from the room is 24 in. wide and no exits were added. Which NFPA 101 life-safety issue does this MOST clearly create?

A. Failure to install emergency lighting in rooms with more than 10 occupants
B. Excessive travel distance for a business occupancy
C. Lack of a posted occupant-load sign in the office
D. Insufficient means-of-egress capacity/width for the increased occupant load

1.4-Q2. An assembly room has an occupant load of 300. The egress capacity factor for level components (doors) is 0.2 in. per person. What is the minimum total egress door width required?

A. 45 in.
B. 60 in.
C. 75 in.
D. 90 in.

1.4-Q3. A mezzanine floor is rated for a uniform live load of 125 psf. A pallet weighing 2,400 lb rests on a 4 ft x 4 ft footprint. Which statement is correct?

A. The pallet imposes 100 psf and is acceptable
B. The pallet imposes 125 psf, exactly at the rating
C. The pallet imposes 150 psf and exceeds the rating
D. The pallet imposes 600 psf and exceeds the rating

1.4-Q4. A new office (business occupancy) has 9,000 ft2 of gross floor area. Using an occupant load factor of 150 ft2 gross per person, what is the occupant load?

A. 60
B. 90
C. 100
D. 900

1.4-Q5. A banquet hall used for tables and chairs (less concentrated assembly) has 6,000 ft2 of net floor area. Using 15 ft2 net per person, what is the occupant load?

A. 300
B. 400
C. 600
D. 857

1.4-Q6. Under NFPA 101, emergency lighting along the path of egress must initially provide an average illumination at floor level of not less than:

A. 1 foot-candle
B. 5 foot-candles
C. 10 foot-candles
D. 20 foot-candles

1.4-Q7. Under NFPA 101, battery-powered emergency lighting must be tested:

A. Weekly for 5 minutes
B. Quarterly for 10 minutes
C. Monthly for at least 30 seconds and annually for 90 minutes
D. Every 6 months for 60 minutes

1.4-Q8. In a new healthcare wing, contractors prop open smoke-barrier doors for long periods while moving equipment between patient sleeping units. Under NFPA 101, what is the GREATEST concern?

A. Door hardware will wear out prematurely
B. The facility may be cited for fire drill documentation
C. Employees may trip on staged equipment
D. The propped doors compromise smoke-compartment integrity, which defend-in-place protection depends on

1.4-Q9. A warehouse has 150,000 ft2 total area, of which 30,000 ft2 is not usable (offices, equipment, aisles). If the maximum allowable stack height is 30 ft, what is the usable storage volume?

A. 900,000 ft3
B. 3,600,000 ft3
C. 4,500,000 ft3
D. 5,400,000 ft3

1.4-Q10. To deter shoplifting, a retail store manager chains one of the two required exit doors during business hours. What is the BEST response from the safety professional?

A. Allow it if an employee carries a key to the chain
B. Post an "Emergency Exit - Use Other Door" sign on the chained door
C. Have the chain removed immediately; required exits must be openable from the egress side without keys or special knowledge (consider approved delayed-egress hardware)
D. Allow it as long as the other exit is unobstructed

Answer key and explanations

Q	Ans	Why
1	D	A change of use that raises occupant load requires a re-check of egress capacity and door width. A 24-in. door is below the minimum clear width for a means-of-egress door.
2	B	Required width = 300 x 0.2 in. = 60 in. That total is then divided among the required exits, each of which must also meet the minimum door width. Using 0.3 in./person (the stair factor) gives 90 in.
3	C	Load = weight / area = 2,400 lb / 16 ft2 = 150 psf, which exceeds 125 psf. The 600 psf distractor divides by one side (4 ft) instead of the area.
4	A	Occupant load = area / factor = 9,000 / 150 = 60 persons. Use the factor given in the stem; dividing by 100 (an older business factor) gives 90.
5	B	6,000 / 15 = 400 persons. The 857 distractor uses 7 ft2 per person, which applies to concentrated assembly with chairs only.
6	A	NFPA 101 requires an initial average of 1 ft-candle (10.8 lux) and a minimum of 0.1 ft-candle at any point, for at least 90 minutes.
7	C	NFPA 101 requires a 30-second functional test every month and a full 1.5-hour test every year, with written records kept.
8	D	Healthcare occupancies use a defend-in-place strategy that relies on smoke compartments. Doors must self-close, or be held open only by devices that release on alarm.
9	B	Usable area = 150,000 - 30,000 = 120,000 ft2; x 30 ft = 3,600,000 ft3. Using the gross area gives 4,500,000 ft3, which is the trap. Actual loading must still be checked against floor-load ratings and sprinkler clearance.
10	C	NFPA 101 and OSHA 1910.36/.37 prohibit locking required exits against egress while the building is occupied. Approved delayed-egress or alarmed hardware addresses theft without trapping occupants.

DOMAIN 1 · COMPETENCY 1.5

Fleet Safety

CSP11 blueprint: Describe fleet safety principles (e.g., driver and equipment safety, maintenance, surveillance equipment, GPS monitoring, telematics, hybrid vehicles, fuel systems, driving under the influence, fatigue)

1. Definition

Fleet safety is the management system that controls risk from company-owned or operated vehicles: selecting and training drivers, maintaining vehicles, monitoring driving behavior, and controlling impairment and fatigue. Commercial motor vehicles in the U.S. are regulated by FMCSA (49 CFR Parts 382-396), and ANSI/ASSP Z15.1 gives a fleet safety management framework.

2. Explanation in simple words

Motor vehicle crashes are a leading cause of work deaths. A good fleet program hires good drivers, keeps vehicles in good repair, watches how people drive with technology, and makes sure nobody drives tired, drunk or distracted.

3. Key points to remember

- ANSI/ASSP Z15.1 elements: management leadership, operational environment, driver selection and qualification (MVR checks), driver behavior/training, vehicle selection and maintenance, incident reporting and analysis.
- FMCSA hours of service (property carriers): 11 hours driving after 10 hours off duty, 14-hour on-duty window, 30-minute break after 8 hours driving, 60/70-hour weekly limits; electronic logging devices (ELDs) record duty status.
- Drug and alcohol testing (49 CFR 382 / Part 40): pre-employment, random, reasonable suspicion, post-accident, return-to-duty and follow-up. Alcohol 0.02-0.039 BAC = removed for 24 hours; 0.04+ = violation requiring SAP evaluation before return.
- Maintenance: driver pre-trip/post-trip inspection (DVIR), preventive maintenance schedule, annual inspection for CMVs, records retained.
- Telematics/GPS: data on speeding, harsh braking/acceleration, cornering, seat belt use, idling, location; inward/outward cameras. Use data for coaching and root cause (routes, schedules, fatigue), not only discipline.
- Hybrid/electric vehicles: high-voltage systems (orange cables), follow OEM disable procedure and capacitor discharge time, HV-rated gloves for service, lithium battery thermal runaway risk.
- Fuel systems: gasoline (flammable, vapor heavier than air), diesel (combustible), CNG/LNG/propane (pressure, leaks, ventilation in garages), hydrogen. Static bonding during fueling, no smoking, engine off.
- Fatigue: circadian low 2-6 a.m., sleep debt, shift length; controls include scheduling limits, rest breaks, fatigue risk management systems. Distracted driving: hands-free policies, no texting (FMCSA bans hand-held phones for CMV drivers).
- Rates: crashes per million miles = crashes × 1,000,000 ÷ miles driven.

4. Real work scenario

> Telematics shows that three drivers on overnight routes generate most harsh-braking events. Instead of firing them, the safety manager reviews the routes and schedules with the drivers and finds they are working 13-hour shifts with tight delivery windows. The company adds a second driver to the longest route and moves delivery windows, and harsh-braking events drop by 60%.

5. CSP exam tips and traps

- When telematics data appear in a question, the BEST answer usually analyzes and coaches before punishing.
- Know the BAC thresholds (0.02 and 0.04) and the 11/14-hour rules.
- For hybrids and EVs, the answer involves isolating the high-voltage system, not only the 12-V battery.
- Fleet rate calculations use per 1,000,000 miles (sometimes 100,000 miles) - read the stem.

6. Every blueprint point (7 blueprint items + 0 more)

Driver safety, licensing and training BLUEPRINT

Driver selection and qualification (MVR checks, CDL where required) and training in defensive driving, hazard recognition and company policy.

- ANSI/ASSP Z15.1 = safe practices for motor vehicle operations.
- Defensive driving systems: Smith System (5 keys: aim high in steering, get the big picture, keep your eyes moving, leave yourself an out, make sure they see you), IPDE (identify, predict, decide, execute), SEE (search, evaluate, execute).
- Following distance 3-4 s cars, 6-8 s trucks; reduce speed by 1/3 in rain, by 1/2 on snow/ice; seat belts required (FMCSA 392.16).
- DOT physical every 2 years (vision 20/40 corrected, forced whisper at 5 ft, blood pressure); untreated sleep apnea can disqualify.

Equipment safety and maintenance BLUEPRINT

Vehicles are selected for safety features and kept in safe condition.

- Safety features: ABS, electronic stability control, backup cameras, collision mitigation, lane departure warning.
- Driver pre-trip/post-trip inspection (DVIR); preventive maintenance schedule; defects corrected before dispatch; annual inspection for CMVs; records kept.

Surveillance equipment, GPS monitoring and telematics BLUEPRINT

Technology to monitor and coach driver behavior.

- GPS monitoring: routes, speeds, idle time, location.
- Telematics: GPS + vehicle diagnostics - engine faults, seat belt use, harsh braking/acceleration, real-time data, predictive maintenance.
- Cameras/event recorders: dash and inward-facing; used for incident investigation and coaching with recorded footage; balance with privacy.
- Use data to find root causes (schedules, routes, fatigue) and coach, not only to discipline.

> **Exam tip:** Engine faults/seat belt/real-time data = telematics; routes/speed/idle = GPS; training with footage = event recorder.

Hybrid and alternative-fuel vehicles BLUEPRINT

High-voltage battery packs and alternative fuels add shock, arc flash, fire and explosion hazards.

- Orange cables = high voltage; follow OEM disable procedure and capacitor discharge time; HV-rated gloves and arc-rated gear for service (NFPA 70E).
- Lithium battery fires: thermal runaway, re-ignition; water alone may not be effective quickly - special responder training.
- Emergency responders need EV/hybrid-specific procedures.

Fuel systems BLUEPRINT

Each fuel has specific fire, pressure and static hazards.

- Gasoline/diesel: flammable/combustible; bond and ground to control static.
- CNG/LNG/propane: high-pressure or cryogenic storage per DOT/ASME; upright, ventilated; garages need ventilation/gas detection.
- Hydrogen: extremely flammable, governed by NFPA 2.
- Fueling: engine off, no smoking, no cell phone use near pumps.

> **Exam tip:** 'Which fuel system requires NFPA 2?' = hydrogen.

Driving under the influence (DUI) and drug/alcohol testing BLUEPRINT

FMCSA (49 CFR 382, Part 40) testing for CDL drivers.

- Tests: pre-employment, random, reasonable suspicion, post-accident, return-to-duty, follow-up.
- BAC 0.04% or higher = violation for CDL drivers (stricter than public 0.08%); 0.02-0.039 = off duty for 24 hours.
- Zero tolerance for illegal drugs; violation = removal from safety-sensitive work, SAP evaluation before return.
- Signs of impairment: slurred speech, erratic driving, odor of alcohol, pinpoint pupils (opioids).

> **Exam tip:** Impaired operator observed: remove from safety-sensitive work immediately and follow the reasonable-suspicion process.

Fatigue management (hours of service) BLUEPRINT

Fatigue is as dangerous as alcohol impairment; microsleeps at highway speed cover long distances blind.

- Property carriers: 11 hours driving after 10 consecutive hours off; 14-hour on-duty window; 30-minute break after 8 hours driving; 60 h/7 days or 70 h/8 days.
- Electronic logging devices record hours.
- Controls: scheduling, route planning, rest breaks, fatigue risk management, sleep apnea screening.

> **Exam tip:** Numbers 11/14/30/60/70 = hours-of-service question.

Read more: Yates, Safety Professional's Reference and Study Guide (4th ed.): Ch. 25 Construction Safety (vehicles); Ch. 30 Materials Handling and Storage (powered vehicles)

7. Ten CSP-style practice questions

Answer all ten on paper first, then check the key. Score: _____ / 10 Date: _____ Retest: _____ / 10

1.5-Q1. A CDL driver's random alcohol test shows 0.03 BAC. Under FMCSA rules (49 CFR Part 382), the employer must:

A. Remove the driver from safety-sensitive functions for at least 24 hours
B. Allow the driver to continue because the result is below 0.04
C. Terminate the driver immediately
D. Retest after 8 hours and then return the driver to duty

1.5-Q2. The Smith System defensive-driving principle "Leave Yourself an Out" emphasizes:

A. Avoiding distractions inside the cab
B. Keeping space cushions around the vehicle so there is an escape path
C. Obeying posted traffic laws
D. Wearing seatbelts at all times

1.5-Q3. A logistics company's telematics show that a driver's lane-departure warnings and harsh-braking events rise sharply after 9 hours on shift. What is the MOST effective immediate action?

A. Disable the lane-departure warnings to reduce distraction
B. Move the driver to night routes when traffic is lighter
C. Assign an online fatigue-awareness module after the shift
D. Require an immediate rest break and review shift scheduling practices

1.5-Q4. During scheduled maintenance on a hybrid delivery van, a technician smells fuel vapor near the rear tank. The van also has a high-voltage battery system. What is the safest procedure?

A. Continue working and finish the battery inspection first
B. Apply sealant to the suspected leak before shutting down power
C. Disconnect only the 12-V battery and proceed
D. Ventilate the area, isolate both the fuel system and the high-voltage system, and follow the manufacturer's hybrid lockout procedure

1.5-Q5. A fleet drove 4,500,000 miles last year and had 18 preventable crashes. What is the preventable crash rate per million miles?

A. 0.25
B. 2.5
C. 4.0
D. 40

1.5-Q6. Your company drove 57,553,441 miles last year and had 42 collisions. Using Fleet incident rate = (incidents x 1,000,000) / miles driven, what is the rate?

A. 0.42
B. 0.57
C. 0.73
D. 1.37

1.5-Q7. Under FMCSA Hours-of-Service rules for property-carrying drivers, what is the maximum driving time allowed after 10 consecutive hours off duty?

A. 8 hours
B. 10 hours
C. 11 hours
D. 14 hours

1.5-Q8. A skid steer is stolen overnight from a construction site. The machine's telematics show it was moved off-site at 2:15 a.m., and the security cameras on the south fence were offline. What should be the FIRST preventive improvement?

A. Provide reliable/redundant power to the cameras and set geofence alerts that trigger when equipment moves after hours
B. Have operators take equipment home after each shift
C. Leave all keys in the cab at night
D. Remove GPS units so thieves cannot tamper with them

1.5-Q9. What is the PRIMARY safety purpose of event recorders (inward- and outward-facing cameras) in a fleet program?

A. Monitoring vehicle emissions
B. Capturing driving events for investigation and targeted driver coaching
C. Replacing GPS tracking
D. Automatically enforcing seatbelt use

1.5-Q10. A truck driver's post-trip Driver Vehicle Inspection Report (DVIR) lists defective brakes. Under 49 CFR 396.11, before the vehicle is dispatched again the motor carrier must:

A. Let the next driver decide whether the vehicle is safe
B. Reduce the vehicle's load by 25% until repaired
C. Record the defect in the maintenance log for the next scheduled PM
D. Repair the defect and certify on the DVIR that it was repaired (or that repair was unnecessary)

Answer key and explanations

Q	Ans	Why
1	A	A result of 0.02-0.039 requires removal from safety-sensitive functions for at least 24 hours. A result of 0.04 or more is a violation that triggers removal and the SAP return-to-duty process.
2	B	"Leave yourself an out" means managing the space around the vehicle so you can always steer away from a hazard. The other Smith keys are: aim high, get the big picture, keep your eyes moving, and make sure they see you.
3	D	The data show fatigue in progress. Stopping the exposure now and fixing the scheduling (the root cause) is best. Night routes make circadian fatigue worse.
4	D	Hybrids combine a flammable fuel hazard with a high-voltage electrical ignition source. Both energy sources must be controlled per the OEM procedure; the 12-V battery alone does not de-energize the HV system.
5	C	Rate = crashes x 1,000,000 / miles = 18 x 1,000,000 / 4,500,000 = 4.0. Always check the normalizing factor (per million miles versus per 100,000 miles).
6	C	42 x 1,000,000 / 57,553,441 = 0.73 collisions per million miles. 1.37 is the inverse (miles/incidents scaled); 0.42 and 0.57 simply misread the inputs.
7	C	Property carriers may drive up to 11 hours within a 14-hour on-duty window after 10 consecutive hours off. The 14-hour figure is the duty window, a classic distractor.
8	A	Combining surveillance with telematics geofencing gives real-time detection. The other options remove protection or increase the risk of theft.
9	B	Event recorders capture video around harsh events, which supports accurate investigations and individualized coaching. Coaching is what produces the safety benefit.
10	D	Defects that affect safe operation must be repaired, and the repair certified on the DVIR, before the vehicle is operated again. The next driver must review that certification.

DOMAIN 1 · COMPETENCY 1.6

Materials Handling

CSP11 blueprint: Evaluate materials handling methods and controls (e.g., forklifts, aerial lifts, and other powered industrial trucks; cranes, hand trucks, hoists, rigging, manual handling, drones)

1. Definition

Materials handling covers the lifting, moving, storing and placing of materials by hand or by machine. Hazards include tip-over, struck-by, crush, overload, dropped loads, musculoskeletal injury and collisions with pedestrians. Key standards: OSHA 1910.178 (powered industrial trucks), 1910.179/1926 Subpart CC (cranes), 1910.184/1926.251 (slings and rigging), ANSI A92 (aerial work platforms), NIOSH lifting equation, FAA Part 107 (drones).

2. Explanation in simple words

Anything that lifts or moves a load can drop it, tip over or hit someone. You choose the right equipment, train and evaluate the operator, inspect the gear, know the weight and center of gravity, and keep people out of the path of the load.

3. Key points to remember

- Forklifts/PITs: operator training and evaluation every 3 years (and after incidents, near misses, unsafe operation or new truck type); daily pre-use inspection; stability triangle - lateral tip-over from turning with a raised load, longitudinal tip-over from overload; travel with load low and tilted back; loaded travel up a ramp with load upgrade; no modifications affecting capacity without manufacturer's written approval.
- Aerial lifts (boom and scissor, ANSI A92.22/.24): boom lifts require fall restraint/arrest attached to the basket anchor; operators must not climb the rails; check ground conditions, overhead power lines and wind limits.
- Cranes: qualified/certified operators (1926.1427), load charts, critical lift plans (often > 75% of capacity, multiple cranes, personnel lifting, near power lines), anti-two-block devices, outrigger setup on firm ground, 20-ft clearance from power lines up to 350 kV (1926.1408), inspections each shift/monthly/annually. Personnel platforms: total load ≤ 50% of rated capacity, trial lift and proof test at 125%.
- Rigging: know load weight and center of gravity. Sling tension per leg = (W ÷ number of legs) ÷ sin(angle from horizontal). Angle factors: 60° ×1.155, 45° ×1.414, 30° ×2.0. Avoid angles below 30°. Inspect slings before each use; wire rope removal criteria (e.g., broken wires, kinks, birdcaging, heat damage). Three wire rope clips minimum, U-bolts on the dead end ('never saddle a dead horse').
- Hoists and hand trucks: rated capacity marking, hook latches, push rather than pull hand trucks, keep the load center over the wheels.
- Manual handling: NIOSH Revised Lifting Equation RWL = LC × HM × VM × DM × AM × FM × CM (LC = 51 lb); Lifting Index = load ÷ RWL; LI > 1 increases risk, LI > 3 high risk. Controls: lift assists, adjust heights (waist level), reduce reach, team lifts as a last resort.
- Drones (FAA Part 107): remote pilot certificate, visual line of sight, ≤ 400 ft AGL, daylight/twilight rules, restrictions over people; include flight paths and no-fly zones in the JHA.

4. Real work scenario

> A plant must lift a 6,000-lb motor with a two-leg bridle. The rigger's first plan gives a 30° sling angle because the headroom is low, which would put 6,000 lb on each leg. The safety professional asks for a spreader beam to make the legs vertical, which drops the tension to 3,000 lb per leg, and confirms the crane load chart includes the weight of the hook block and rigging.

5. CSP exam tips and traps

- Sling-angle calculations appear often: divide by the number of legs FIRST, then by sin θ (θ from horizontal).
- In a three- or four-leg bridle, assume only two legs carry the load unless told otherwise.
- Total crane load = load + hook block + rigging; compare to the chart at the actual radius.
- Forklift questions: lateral tip-over is about center of gravity leaving the stability triangle; the answer to tip-over survival is 'stay in the seat, hold on, lean away'.

6. Every blueprint point (8 blueprint items + 1 more)

Forklifts and other powered industrial trucks BLUEPRINT

1910.178 covers forklifts, pallet jacks, reach trucks, tow tractors; hazards include tip-over, struck-by, falling loads, pedestrian collisions, fuel/battery hazards.

- Training: formal + practical + evaluation; evaluation at least every 3 years; refresher after unsafe operation, accident/near miss, poor evaluation, new truck type or workplace change.
- Daily pre-use inspection; defective trucks removed from service; design per ANSI/ITSDF B56.1; no capacity-affecting modifications without manufacturer's written approval.
- Safe operation: seat belt (tip-over protection), load low and tilted back, no riders, avoid turning on grades, travel loaded up a ramp with load upgrade, horn at intersections.
- Classes: I-III electric; IV-V internal combustion (cushion/pneumatic tires); VI tractors; VII rough terrain. IC trucks indoors need CO monitoring/ventilation.
- Stability triangle: lateral tip-over when combined CG leaves the triangle (turning with raised load).

> **Exam tip:** Tip-over scenario: seat belt; stay in seat, hold on, lean away.

Aerial lifts BLUEPRINT

Boom lifts and scissor lifts (ANSI A92.20/.22/.24); OSHA treats scissor lifts as mobile scaffolds.

- Boom lifts: body harness with lanyard attached to the basket anchor; stay tied off when repositioning.
- Do not climb or stand on rails; check ground conditions, slopes, overhead power lines and wind limits.
- Operator training and familiarization; pre-use inspection and function test.

Cranes BLUEPRINT

Mobile/tower cranes (1926 Subpart CC) and overhead/gantry cranes (1910.179).

- Operators trained and certified/qualified (1926.1427); qualified signal persons; qualified riggers.
- Load chart for configuration and radius; total load includes hook block and rigging; critical lift plans (often > 75% capacity, tandem, personnel, near power lines).
- Power lines up to 350 kV: maintain 20 ft (1926.1408); if contact occurs, operator stays in the cab.
- Inspections: each shift, monthly, annual. Anti-two-block devices required.
- Personnel platforms (1926.1431): total load ≤ 50% of rated capacity, trial lift, proof test at 125%, crane level within 1%.

Hand trucks BLUEPRINT

Two-wheel and four-wheel hand trucks reduce manual carrying but still cause strains and crush injuries.

- Keep load center of gravity low and over the axle; tilt so weight is on wheels.
- Push rather than pull where possible; go down ramps with the truck ahead (or backwards with load uphill as manufacturer directs).
- Rated capacity respected; secure loads; keep hands inside handles.

Hoists BLUEPRINT

Chain hoists, wire rope hoists and monorails used for lifting within plants.

- Rated load marked; hook safety latches; inspect hooks, chains and wire rope before use.
- Never lift people with material hoists; keep clear of suspended loads.

Rigging BLUEPRINT

Slings and hardware connecting the load to the hook.

- Know load weight and center of gravity before selecting slings.
- Sling tension per leg = (W ÷ legs) ÷ sin θ (θ from horizontal); 60° ×1.155, 45° ×1.414, 30° ×2.0 (capacity reduced ~50%); avoid < 30°.
- Wire rope slings typically 5:1 design factor; chain slings must be alloy (Grade 80+) with tags; synthetic slings protected from chemicals, heat and sharp edges.
- Inspect slings before each shift/use; remove damaged slings from service.
- Taglines control swinging loads; never stand under a suspended load.

> **Exam tip:** As sling angle decreases, tension increases.

Manual handling BLUEPRINT

Lifting, lowering, carrying, pushing and pulling by hand - a major source of MSDs.

- NIOSH RWL = LC × HM × VM × DM × AM × FM × CM; LC = 51 lb; HM = 10/H; VM = 1 − 0.0075|V − 30|; DM = 0.82 + 1.8/D; AM = 1 − 0.0032A; FM and CM from tables (coupling good 1.0, fair 0.95, poor 0.90).
- LI = load ÷ RWL; LI ≤ 1.0 acceptable for most workers; > 1.0 increased risk - redesign.
- Ideal lift height ~30 in. (knuckle height); keep loads close; avoid floor and overhead lifts; use mechanical aids.

> **Exam tip:** Most tested NIOSH fact: LC = 51 lb; twisting (AM) and frequency (FM) lower the safe weight the most.

Drones BLUEPRINT

Unmanned aircraft systems for inventory, inspection, delivery and remote monitoring.

- FAA Part 107: Remote Pilot certificate, aircraft < 55 lb, ≤ 400 ft AGL, visual line of sight, restrictions over people.
- Benefits: inspect high racks, tanks and roofs without ladders or scaffolds.
- Controls: planned flight paths away from people, no-fly zones, barriers, pre-flight checks.

> **Exam tip:** Drone compliance keywords: FAA Part 107, < 55 lb, < 400 ft.

Scaffolds (supported and suspended)

Temporary elevated platforms (1926 Subpart L).

- Support own weight + 4× intended load; suspension ropes 6× load.
- Fully planked, gaps < 1 in.; guardrails at ≥ 10 ft (construction), top rail 38-45 in.
- Ties every 26 ft vertically and 30 ft horizontally (when height-to-base ratio > 4:1).
- Competent person supervises erection and inspects before each shift and after events; never climb cross-braces.
- Platform ≤ 14 in. from the face of work unless guarded.

> **Exam tip:** Frequently tested: 4× load, 26 ft vertical / 30 ft horizontal ties, 10 ft fall protection.

Read more: Yates, Safety Professional's Reference and Study Guide (4th ed.): Ch. 30 Materials Handling and Storage, p. 687; Ch. 24 Ergonomics (lifting), p. 546

7. Ten CSP-style practice questions

Answer all ten on paper first, then check the key. Score: _____ / 10 Date: _____ Retest: _____ / 10

1.6-Q1. A 500-lb pipe is lifted by a two-leg bridle sling with the load centered. Each leg makes a 60-degree angle with the horizontal. What is the tension in each leg?

A. 250 lb
B. 289 lb
C. 354 lb
D. 500 lb

1.6-Q2. A 2,000-lb load is supported by a two-leg sling, and each leg makes a 45-degree angle with the horizontal. What force is exerted in each leg?

A. 1,000 lb
B. 1,155 lb
C. 1,414 lb
D. 2,000 lb

1.6-Q3. A forklift travels at 6 mph (about 8.8 ft/s). Operator perception-reaction time is 1.5 s and braking deceleration is 4 ft/s2. What is the total stopping distance (reaction distance + braking distance, where braking = v2/2a)?

A. 9.7 ft
B. 13.2 ft
C. 22.9 ft
D. 25.0 ft

1.6-Q4. A mobile crane load chart shows 10,000 lb capacity at the planned radius. The load weighs 8,200 lb, the hook block 450 lb and the rigging 150 lb. What percentage of rated capacity is the lift?

A. 82%
B. 85%
C. 88%
D. 92%

1.6-Q5. Using the NIOSH lifting equation: LC = 51 lb and horizontal distance H = 20 in. (HM = 10/H). All other multipliers equal 1.0. The load weighs 30 lb. What is the Lifting Index?

A. 0.59
B. 0.85
C. 1.18
D. 1.70

1.6-Q6. A crane will work near a power line whose voltage has not been confirmed by the utility. Under OSHA 1926.1408(a)(2), with the line assumed to be up to 350 kV, no part of the crane, load line or load may come closer to the line than:

A. 5 ft
B. 10 ft
C. 15 ft
D. 20 ft

1.6-Q7. A team must lift a 3,000-lb component with a 5-ton bridge crane. The only available wire rope sling has a visible cut in its outer strands. What is the safest course of action?

A. Remove the damaged sling from service and defer the lift until an inspected, compliant sling is available
B. Proceed but reduce crane speed to compensate
C. Have a qualified rigger certify the damaged sling for one-time use
D. Use the sling only if the load is under 50% of its rated capacity

1.6-Q8. A warehouse is piloting drones (UAS) to move small parts between workstations. During testing, a drone flies over occupied work areas and briefly loses stability in an HVAC air current. What is the BEST control?

A. Limit drone use to break and lunch periods
B. Redesign flight paths to avoid flying over people, using barriers or no-fly zones
C. Require hard hats while drones operate
D. Continue testing and add signage

1.6-Q9. A worker is operating a boom-supported aerial lift (articulating boom). Under OSHA 1926.453, what fall protection practice is required?

A. Wear a body harness or restraint with the lanyard attached to the boom or basket, never to an adjacent pole or structure
B. No fall protection is needed if the guardrails are intact
C. Tie off to a nearby structure for maximum anchorage strength
D. Stand on the midrail to reach higher work

1.6-Q10. A sit-down counterbalanced forklift begins to tip over sideways. What should the operator do?

A. Jump clear on the downhill side
B. Lower the forks quickly to the ground
C. Jump clear on the uphill side
D. Stay in the seat with the seatbelt on, grip the wheel, brace the feet and lean away from the direction of the tip

Answer key and explanations

Q	Ans	Why
1	B	Tension per leg = (W/2) / sin(angle) = 250 / 0.866 = 289 lb. 250 lb ignores the angle; 354 lb uses 45 degrees.
2	C	Each leg = (2,000/2) / sin 45 = 1,000 / 0.707 = 1,414 lb. 1,155 lb would be the answer at 60 degrees. As the angle from horizontal decreases, leg tension increases.
3	C	Reaction = 8.8 x 1.5 = 13.2 ft; braking = 8.8^2 / (2 x 4) = 77.4 / 8 = 9.7 ft; total = 22.9 ft. Picking only one of the two parts is the trap.
4	C	Total = 8,200 + 450 + 150 = 8,800 lb; 8,800 / 10,000 = 88%. Hook block and rigging count against chart capacity, and many companies classify a lift above 75-85% of capacity as critical.
5	C	HM = 10/20 = 0.5; RWL = 51 x 0.5 = 25.5 lb; LI = 30 / 25.5 = 1.18. An LI above 1.0 means increased risk, so redesign the task, for example by reducing H.
6	D	1926.1408 sets a 20-ft minimum approach distance for lines up to 350 kV unless the utility confirms the voltage, in which case Table A applies (for example 10 ft up to 50 kV). Over 350 kV the distance is 50 ft.
7	A	1910.184 requires slings with damage such as cut strands to be removed from service immediately. Derating or "one-time use" is not permitted.
8	B	Keeping the flight path away from people removes exposure to falling drones and payloads. Hard hats and signs are lower-order controls.
9	A	1926.453(b)(2)(v) requires a body belt/harness with the lanyard attached to the boom or basket. Tying off to an adjacent structure can pull the worker out if the lift moves. Standing on rails is prohibited.
10	D	Most forklift tipover deaths occur when operators try to jump and are crushed by the overhead guard. Staying belted in the operator compartment is the trained response.

DOMAIN 1 · COMPETENCY 1.7

Tools, Machines and Equipment

CSP11 blueprint: Evaluate the use of tools, machines, and equipment (e.g., hand tools, power tools, ladders, grinders, hydraulics, robotics)

1. Definition

This competency covers the selection, guarding, inspection and safe use of hand tools, power tools, ladders, abrasive grinders, hydraulic systems, machines and industrial/collaborative robots. Key references: OSHA 1910 Subpart O (machine guarding, 1910.212), Subpart P (hand and portable powered tools, 1910.242-.244), 1910.23 and 1926 Subpart X (ladders), ANSI B11 series, ANSI/RIA R15.06 and ISO 10218 (robots).

2. Explanation in simple words

Tools and machines hurt people through the moving parts, flying pieces, electricity and stored pressure. Guard the moving parts, inspect tools before use, use the right tool for the job, and lock out energy before maintenance.

3. Key points to remember

- Machine hazards: point of operation, power transmission (belts, gears, shafts) and other moving parts; motions (rotating, reciprocating, transverse) and actions (cutting, punching, shearing, bending). Guard types: fixed (preferred - simple and permanent), interlocked, adjustable, self-adjusting. Devices: presence-sensing (light curtains), pullback, restraint, two-hand controls/trips, gates.
- Guard requirements: prevent contact, secure (not easily removed), protect from falling objects, create no new hazards, allow safe lubrication.
- Hand tools: most injuries come from misuse and poor maintenance (mushroomed chisel heads, cracked handles, wrong tool). Employer is responsible for tool condition, including employee-owned tools.
- Power tools: double-insulated or grounded; GFCI on construction sites; never carry by the cord; disconnect before changing blades; guards in place on circular saws; powder-actuated tools require training and licensed operators; compressed air for cleaning limited to < 30 psi with chip guarding (never on the body).
- Grinders (1910.215): work rest ≤ 1/8 in. from the wheel; tongue guard ≤ 1/4 in.; ring test wheels before mounting; never exceed rated RPM; stand aside during first minute of startup; face shield plus safety glasses.
- Ladders: 4:1 rule (base 1 ft out for every 4 ft of height, about 75.5°); side rails extend 3 ft above the landing; Type IA = 300 lb, Type I = 250 lb, Type II = 225 lb, Type IAA = 375 lb; top step/cap of stepladder not used; inspect before use; three points of contact. New fixed ladders over 24 ft need a ladder safety system or PFAS (1910.28).
- Hydraulics: high-pressure fluid injection injuries are surgical emergencies; never check for leaks with hands; relieve pressure and block raised parts (gravity energy) before work.
- Robotics: perimeter guarding with interlocked gates, teach mode at reduced speed (≤ 250 mm/s) with enabling device, LOTO for maintenance; collaborative robots use power and force limiting, speed and separation monitoring, hand guiding or safety-rated monitored stop - all based on a risk assessment (ISO/TS 15066).

4. Real work scenario

> Operators on a packaging line keep opening a guard to clear jams, and one suffers a finger injury. The safety professional finds that the jams happen at a fixed spot. Engineering adds an interlocked access door with a jam-clearing tool slot and adjusts the infeed guide so that jams drop by 80%. Training alone had not stopped the behavior because the guard was blocking the job.

5. CSP exam tips and traps

- When workers bypass guards, the BEST answer fixes the reason (design the task so the guard is not in the way) rather than only disciplining.
- Fixed guards are generally the preferred guard type; light curtains are devices, not guards.
- Grinder numbers: 1/8 in. work rest, 1/4 in. tongue guard - learn them together.
- Ladder angle calculations use the 4:1 rule and the Pythagorean theorem (ladder length = $\sqrt{(\text{height}^2 + \text{base}^2)}$ + 3 ft extension).

6. Every blueprint point (6 blueprint items + 1 more)

Hand tools BLUEPRINT

Most hand-tool injuries come from misuse and poor maintenance.

- Employer is responsible for safe condition of all tools, including employee-owned.
- Hazards: mushroomed chisel heads, cracked handles, wrong tool, dull blades; inspect, dress or remove from service.
- Use spark-resistant tools near flammables; tether tools at height.

Power tools BLUEPRINT

Electric, pneumatic, hydraulic, powder-actuated and fuel-powered tools.

- Electrical: double-insulated or grounded; GFCI on construction sites; never carry by the cord or yank to disconnect; disconnect before changing blades/bits.
- Guards stay on (circular saw lower guard); anti-kickback pawls and riving knives on saws.
- Pneumatic: secure hoses, safety clips on tools; compressed air for cleaning < 30 psi with chip guarding, never on the body.
- Powder-actuated tools: trained operators, never point at anyone, wait 30 s after misfire.
- Locking triggers/continuous-run features need dead-man or automatic shutoff; sequential-trip nailers reduce injuries.

Ladders BLUEPRINT

Portable and fixed ladders (1910.23, 1926 Subpart X).

- 4:1 rule (~75°); side rails extend 3 ft above the landing; secure top and bottom; three points of contact.
- Duty ratings: IAA 375 lb, IA 300 lb, I 250 lb, II 225 lb, III 200 lb.
- Do not use the top step/cap of a stepladder; non-conductive side rails near electricity; inspect by a competent person; remove defective ladders.
- Rungs 10-14 in. apart; a ladder or stairway is required at a break in elevation of 19 in. or more (construction).

Grinders (abrasive wheels) BLUEPRINT

Bench, pedestal and portable grinders can burst and throw fragments.

- Work rest ≤ 1/8 in. from wheel; tongue guard ≤ 1/4 in.
- Ring-test wheels before mounting; never exceed the rated RPM; stand aside during start-up.
- Safety glasses plus face shield; side guards in place.

> **Exam tip:** Grinder clearances 1/8 in. and 1/4 in. appear often.

Hydraulics BLUEPRINT

Hydraulic presses, jacks, lifts and lines store high pressure and gravity energy.

- Relieve pressure (bleed lines, accumulators) and block raised parts before maintenance - part of LOTO.
- Never search for leaks with hands; use cardboard. Fluid injection injuries are surgical emergencies.
- Inspect hoses and fittings; rated components only.

Robotics and emerging technology BLUEPRINT

Industrial robots, cobots and other new technologies (AI, AR, digital twins, data mining).

- Robots: perimeter guarding with interlocked gates, light curtains/pressure mats, accessible e-stops; ANSI/RIA R15.06 and ISO 10218 require risk assessment; LOTO for maintenance; teach mode at reduced speed.
- Cobots share workspace with humans (ISO/TS 15066): power and force limiting, speed and separation monitoring, hand guiding, safety-rated monitored stop.
- AI: computer vision for PPE compliance, NLP of safety reports, wearables for fatigue; concerns - privacy, bias, over-reliance; AI supports, does not replace compliance.
- Digital twin = virtual replica to simulate hazards; AR = real-time hazard overlays; predictive analytics/data mining find leading indicators (ANSI/ASSP Z16.1).

> **Exam tip:** Shared workspace with humans = cobot; virtual replica = digital twin; real-time visual overlay = AR.

Machine guarding requirements

OSHA 1910.212 general requirements and Subpart O.

- Guards must prevent contact, be secured, protect from falling objects, create no new hazard, allow safe lubrication and not obstruct visibility (e.g., polycarbonate).
- Point-of-operation guarding on presses, shears, saws; power-transmission parts < 7 ft enclosed.
- Guard types: fixed (preferred), interlocked (stops machine when opened), adjustable, self-adjusting; devices: light curtains, mats, pullback, restraint, two-hand controls.
- Woodworking: circular saw guard covers blade above table; radial arm saw returns automatically.

> **Exam tip:** CSP prefers engineering controls (guards, interlocks) over PPE.

Read more: Yates, Safety Professional's Reference and Study Guide (4th ed.): Ch. 25 Construction Safety (tools, ladders), p. 567; Ch. 29 Walking and Working Surfaces, p. 657

7. Ten CSP-style practice questions

Answer all ten on paper first, then check the key. Score: _____ / 10 Date: _____ Retest: _____ / 10

1.7-Q1. James must change a hydraulic hose on a press used earlier that day. The press is powered off, but he does not bleed residual pressure before loosening the fitting, and a burst of fluid causes an injection injury to his forearm. What should he have done before loosening any fittings?

A. Removed the drip pan from under the couplings, then powered off the press
B. Opened the reservoir vent cap and loosened the fittings immediately
C. Raised the hose 6 in. so gravity could drain the fluid
D. Shut off and locked out the press, then relieved and verified dissipation of stored hydraulic pressure

1.7-Q2. On a bench grinder, what is the maximum distance allowed between the wheel periphery and the adjustable tongue guard (upper peripheral guard)?

A. 1/16 in.
B. 1/8 in.
C. 1/4 in.
D. 1/2 in.

1.7-Q3. A worker mounts a new grinding wheel that fits the arbor correctly but has not been ring tested. When the grinder is started, the wheel shatters. Which action would have MOST effectively prevented this?

A. Inspecting and ring testing the wheel before mounting
B. Wearing a face shield and gloves during startup
C. Running the grinder at full speed for 10 minutes before use
D. Reducing the tool rest clearance

1.7-Q4. A robotic welding cell is running in automatic mode. The operator powers off the teach pendant, assumes the robot cannot move, and reaches between the arm and fixture to clear a jam. The robot cycles and pinches his hand. Which procedure would MOST effectively have prevented the injury?

A. Keep the pendant off and wear cut-resistant gloves when reaching in
B. Lock out the robot's main power (and other energy sources) and verify zero energy before entering
C. Use single-block mode to jog the arm away without powering off the controller
D. Press the pendant E-stop and rely on it to de-energize the robot

1.7-Q5. A robotic cell has interlocked gates, but workers prop a gate open with a tool so they can make minor adjustments without stopping the robot. What is the BEST corrective measure?

A. Install cameras to monitor worker behavior
B. Install tamper-resistant interlocks and investigate why workers feel the need to bypass, retraining as needed
C. Post signs telling workers not to enter
D. Reduce the robot's operating speed

1.7-Q6. A roofer shoots his thumb with a pneumatic nailer set to contact (bump) actuation. He was holding the trigger down and tapping the nose on the shingles to drive multiple nails. The tool was in good working order. What is the BEST recommendation?

A. Issue verbal discipline for the injury
B. Require puncture-resistant gloves
C. Tell the worker to go slower
D. Switch to nailers with a full sequential trigger

1.7-Q7. A technician leans a straight ladder against a wall to reach a platform 15 ft high. The base is only 2 ft from the wall, and the ladder shifts during use. What is the proper correction?

A. Reposition the ladder using the 4-to-1 rule (base about 3.75 ft out), on firm footing, and secure it
B. Have a coworker hold the base
C. Use a shorter ladder
D. Tie off the top but leave the base unsecured

1.7-Q8. A ladder is used to access a roof 20 ft high. It will be set at a 75-degree angle and must extend 3 ft above the roof edge. What ladder length is needed?

A. 20 ft
B. 22 ft
C. 24 ft
D. 26 ft

1.7-Q9. Operators on a conveyor repeatedly remove an interlocked guard to clear jams. What is the BEST long-term solution?

A. Train operators and enforce discipline for violations
B. Install a camera to monitor compliance
C. Post warning signs about removing guards
D. Modify the design to provide a jam-clearing tool or access point that does not require removing the guard (or eliminate the jam cause)

1.7-Q10. During a tool crib inspection, several cold chisels and drift pins are found with mushroomed heads. What is the BEST action?

A. Issue them only to experienced workers
B. Require safety glasses whenever they are used
C. Remove them from service until the heads are dressed or the tools replaced
D. Tape the heads to contain fragments

Answer key and explanations

Q	Ans	Why
1	D	LOTO covers stored energy: residual hydraulic pressure must be relieved and verified as zero. High-pressure fluid injection injuries are surgical emergencies.
2	C	OSHA 1910.215 sets the tongue guard at 1/4 in. or less and the work rest at 1/8 in. or less. Mixing up the two is a classic trap; remember 'rest 1/8, tongue 1/4'.
3	A	1910.215(d)(1) requires wheels to be inspected and sound (ring) tested before mounting to detect cracks. PPE only reduces the injury after failure. Standing aside during the 1-minute run-up is also good practice.
4	B	Entering the robot envelope for jam clearing requires hazardous energy control. E-stops and pendants are control-circuit devices, not energy-isolating devices.
5	B	Defeat-resistant interlocks restore the engineering control. Addressing the production pressure behind the bypass prevents it from recurring.
6	D	NIOSH found full sequential triggers cut nail gun injury risk roughly in half compared with contact triggers. This is an engineering fix; gloves, pace and discipline are weaker.
7	A	The 4:1 rule (1 ft out for every 4 ft of working length, about 75 degrees) gives stability. For 15 ft that is about 3.75 ft; the side rails should also extend 3 ft above the landing.
8	C	Length to the eave = 20 / sin 75 = 20 / 0.966 = 20.7 ft; adding 3 ft gives 23.7 ft, so a 24-ft ladder is needed. Forgetting the 3-ft extension gives about 21 ft.
9	D	Fixing the design removes the reason to bypass the guard. Discipline, cameras and signs treat the symptom and are weaker controls.
10	C	OSHA 1926.301 and 1910.242 require unsafe hand tools to be removed from service. Mushroomed heads can shatter and throw fragments when struck.

DOMAIN 2

Program Management

25% of the exam · 14 competencies

2.1	Benchmarking and Gap Analysis Compare performance against established benchmarks (e.g., gap analysis)
2.2	Performance Standards and Action Planning Analyze performance standards to determine plan of action
2.3	EHS Culture: Measure, Analyze and Improve Determine how to measure, analyze, and improve EHS culture
2.4	Incident Investigation and Root Cause Analysis Determine appropriate incident investigation techniques (root causes) and apply corrective actions
2.5	Management of Change (MOC) Describe the Management of Change process (prior, during, after)
2.6	System Safety Analysis Describe system safety analysis techniques (e.g., fault tree analysis, failure modes and effects analysis [FMEA], Safety Case approach, risk summation)
2.7	Leading and Lagging Indicators Evaluate leading and lagging indicators
2.8	EHS Management and Audit Systems Recognize safety, health, and environmental management and audit systems (e.g., ISO 14000 series, 45001, 19011, ANSI Z10)
2.9	Plans, Systems and Policies Describe required components for plans, systems, and policies (e.g., safety, health, and environmental regulations and standards)
2.10	Document Retention and Management Utilize document retention or management principles (e.g., incident investigation, training records, exposure records, maintenance records, environmental management system, audit results, privacy, trade secrets, personal information)
2.11	Budgeting, Finance and Economic Analysis Apply budgeting, finance, and economic analysis techniques and principles (e.g., timelines, budget development, resourcing, return on investment, cost/benefit analysis, role in procurement process)

2.12	Management and Leadership Differentiate management leadership techniques (e.g., management theories, leadership theories, motivation, discipline, authority, responsibility, accountability, communication styles)
2.13	Project Management Apply project management principles and techniques (e.g., RACI charts, project timelines)
2.14	Data Analysis Analyze and/or interpret data (e.g., exposure, release concentrations, sampling data, mean, median, mode, confidence intervals, probabilities, Pareto analysis)

DOMAIN 2 · COMPETENCY 2.1

Benchmarking and Gap Analysis

CSP11 blueprint: Compare performance against established benchmarks (e.g., gap analysis)

1. Definition

Benchmarking is the systematic comparison of an organization's safety performance, processes and practices with a reference point - industry averages, best-in-class peers, internal sites or a standard. Gap analysis identifies the difference between the current state and the desired state and the reasons for that difference.

2. Explanation in simple words

Benchmarking tells you where you stand compared with others. Gap analysis tells you how far you are from where you want to be and why. Together they show what to fix first.

3. Key points to remember

- Types: internal (site vs site), competitive (same industry peers, e.g., BLS rates by NAICS code), functional/generic (best practice from any industry), performance (metrics such as TRIR, DART) and process (how work is done).
- Common benchmarks: BLS injury and illness rates by NAICS code, EMR (1.0 = industry average), industry association data, ISO 45001/ANSI Z10 requirements, OSHA VPP criteria.
- Gap analysis steps: define the desired state (standard, benchmark, objective) → assess the current state (audit, data, interviews) → identify and quantify the gaps → analyze causes → prioritize → action plan with owners, resources and dates → monitor.
- Compare like with like: same NAICS code, same rate formula (200,000 hours), same time period, similar size.
- A gap is a starting point for root-cause thinking, not the conclusion. Do not 'fix' a gap by changing how cases are recorded.

4. Real work scenario

> A plant's DART rate is 2.1 and the BLS rate for its NAICS code is 1.4. The gap analysis shows that 70% of DART cases are hand injuries at two press lines, where guarding is older than the rest of the plant. The action plan targets guarding upgrades at those lines first.

5. CSP exam tips and traps

- The next step after finding a gap is almost always 'analyze the cause and create an action plan', not setting a slogan goal.
- Recognize that benchmarking against BLS data must use the correct NAICS code and the 200,000-hour base.
- An EMR above 1.0 means worse than the industry average and higher premiums.

6. Every blueprint point (2 blueprint items + 3 more)

Benchmarking BLUEPRINT

Benchmarking compares your safety performance against a reference point so you know where you stand. The reference can be regulatory, industry, internal or best-in-class, and it must fit your risks, size, processes and safety maturity.

- Regulatory/national: BLS injury and illness rates by NAICS code; OSHA recordkeeping formulas (TRIR, DART).
- Industry: NSC annual averages and best-in-class statistics, trade association data.
- Internal: your best site, best year or an aspirational company target.
- Best-in-class/world-class: leading organizations in or outside your sector.
- Process (functional) benchmarking compares how a task is done (e.g., how a peer runs MOC), not just the outcome number.
- The guide states regulatory benchmarks are 'legally required' (e.g., keep TRIR at or below the BLS average) - in fact OSHA sets no required TRIR; BLS averages are comparison data only.

> **Exam tip:** Pick the benchmark that is relevant (same NAICS, size, hazards). A small specialty shop compared to a national all-industry average is a weak benchmark.

Gap analysis BLUEPRINT

Gap analysis is a systematic comparison of current performance (or current program) against a benchmark or standard, followed by finding why the gap exists and closing it with an action plan.

- Steps: select benchmark -> collect and validate data -> compare and quantify gap -> analyze root causes -> action plan -> engage management and workers -> communicate -> monitor and adjust.
- Quantify clearly: 'Our TRIR is 2.5, industry average 1.6, best-in-class 1.0' -> gap to average = 0.9 (56% above).
- Program gap analysis: compare your SMS element by element against ISO 45001 or ANSI Z10 clauses before certification.
- Use both lagging (DART, lost-time rate) and leading (near-miss reports, observations) data.
- Raise the benchmark as performance improves (continuous improvement).

> **Exam tip:** A gap analysis is done BEFORE implementing a new management system - it tells you what is missing. The first step is choosing an appropriate benchmark/criteria.

Data collection and validation

Benchmarking is only as good as the data. Records must be accurate, complete and comparable (same definitions, same time period, same hours basis).

- Sources: OSHA 300/300A logs, audit results, training records, workers' comp claims, observation data.
- Under-reporting makes lagging data look better than reality - validate against first-aid logs, clinic visits and comp claims.
- Normalize to exposure: rates per 200,000 hours, not raw counts.
- Compare like with like: same NAICS code, similar size and work type.

> **Exam tip:** A low injury rate with very few near-miss reports suggests under-reporting, not excellent performance.

Root cause of performance gaps

Numbers show a gap exists; analysis shows why. Look for system causes such as training, leadership, equipment, procedures or reporting culture.

- Tools: interviews, 5 Whys, fishbone diagrams, audit findings.
- ANSI/ASSP Z16.1 and Z10 support analyzing causes, not just measuring outcomes.
- Segment the data (by site, shift, department, task) to find hot spots - Pareto helps here.

Communicating and sustaining benchmark results

Share results and the improvement plan widely and track progress over time with dashboards; technology makes benchmarking visible at all levels.

- Communicate through dashboards, reports, toolbox talks and digital platforms - transparency builds buy-in.
- Review gaps in management meetings and safety committees.
- Document results; they are evidence of improvement during audits.
- EHS software automates data collection and visualization (NSC best practice).

Read more: Yates, Safety Professional's Reference and Study Guide (4th ed.): Ch. 36 Measuring Safety and Health Performance, p. 797; Ch. 37 Safety Program Auditing Techniques, p. 818

7. Ten CSP-style practice questions

Answer all ten on paper first, then check the key. Score: _____ / 10 Date: _____ Retest: _____ / 10

2.1-Q1. Your plant's DART rate is 2.1. The industry benchmark (same NAICS) is 1.4. Management asks what to do next. What is the BEST next step?

A. Set a goal of matching the benchmark next year and announce it
B. Analyze the causes of the 0.7 gap (injury types, tasks, controls) and build an action plan with owners and dates
C. Change the recordkeeping method to reduce DART cases
D. Start an incentive program for zero injuries

2.1-Q2. You manage safety operations at a construction company aiming to reduce workplace incidents by comparing your safety performance WITH industry leaders. What benchmarking procedure would you initially follow?

A. Internal Benchmarking - Analyze safety performance within your own teams first.
B. Functional Benchmarking - Examine safety practices in different industries.
C. Competitive Benchmarking - Directly compare safety incident rates against similar companies.
D. Generic Benchmarking - Benchmark against unrelated industries to find innovative safety practices.

2.1-Q3. As a safety coordinator in a logistics company, you're tasked with improving vehicle incident rates. Which benchmarking approach would provide the MOST practical insights?

A. Performance Benchmarking - Measuring specific safety processes against industry best practices.
B. Strategic Benchmarking - Evaluating long-term safety strategies of competitors.
C. Product Benchmarking - Comparing safety equipment with competitors' offerings.
D. Financial Benchmarking - Focusing strictly on financial investment in safety.

2.1-Q4. The main purpose of benchmarking in safety program management is to:

A. Compare safety budgets with competitors
B. Eliminate need for management review
C. Replace internal audits
D. Identify best practices and performance gaps for improvement

2.1-Q5. Benchmarking is a/an

A. Form of injury/illness measurement used by large corporations
B. Process to develop an annual safety and health budget
C. Process to compare company safety program to industry peers
D. Systematic approach to organizational change

2.1-Q6. Plant A recorded 6 recordable cases in 300,000 hours worked. Plant B recorded 4 recordable cases in 160,000 hours. The industry (BLS) TRIR benchmark for the same NAICS code is 3.0. Which statement is correct?

A. Plant A performs worse because it had more recordable cases
B. Plant B has the highest rate (5.0); both plants exceed the 3.0 benchmark
C. Both plants are below the industry benchmark
D. Rates cannot be compared between plants of different sizes

2.1-Q7. In an EHS management system context, a gap analysis compares:

A. The organization's current practices and performance with a desired state such as a standard, regulation or benchmark
B. The safety budget with the operations budget
C. Last year's injury count with this year's injury count
D. Worker perception survey results with supervisor survey results only

2.1-Q8. Senior management decides to pursue ISO 45001 certification for a site that already has an OSHA-compliant safety program. What should the safety professional do FIRST?

A. Train all employees on ISO 45001 clauses
B. Schedule the certification audit with a registrar
C. Rewrite all safety procedures in ISO format
D. Perform a gap analysis of the existing program against ISO 45001 requirements

2.1-Q9. A safety manager wants to compare her plant's DART rate with national data. Which approach gives the MOST valid benchmark comparison?

A. Compare the plant's total number of DART cases with the national number of DART cases
B. Compare the plant's DART rate with BLS data for the same NAICS industry code, establishment size and year
C. Compare the plant's DART rate with the all-industry national average
D. Compare the plant's DART rate with a competitor's TRIR

2.1-Q10. A company with 12 plants compares lockout/tagout audit scores and practices among its own sites to identify the best-performing site's methods and spread them. This is an example of:

A. Competitive benchmarking
B. Generic benchmarking
C. Internal benchmarking
D. Strategic benchmarking

Answer key and explanations

Q	Ans	Why
1	B	Benchmarking shows where you stand; gap analysis must then find why the gap exists and drive an action plan with owners, dates and measures. Goals without causes, recordkeeping changes and incentives do not close the gap.
2	C	Comparing incident rates directly against similar companies/industry leaders is competitive benchmarking. Internal compares your own units; functional/generic look at other industries.
3	A	Performance benchmarking compares specific measurable processes (e.g., vehicle incident rates, driver programs) against best practice, giving the most practical, actionable insight.
4	D	The purpose of benchmarking is to identify best practices and performance gaps that drive improvement; it does not replace audits or management review.
5	C	Benchmarking is the process of comparing your safety program and performance to industry peers or recognized leaders.

6	**B**	TRIR = cases x 200,000 / hours: A = 6 x 200,000 / 300,000 = 4.0; B = 4 x 200,000 / 160,000 = 5.0. Both exceed 3.0. Trap: comparing raw counts instead of normalized rates.
7	**A**	Gap analysis = current state vs. desired/required state (e.g., ISO 45001 clauses, OSHA requirements, best-in-class benchmark); the gaps become the basis of the action plan.
8	**D**	A gap analysis first shows which requirements are already met and which are missing, so effort and resources go to real gaps. Scheduling, rewriting or training before knowing the gaps is premature.
9	**B**	Valid benchmarking compares like with like: the same metric, normalized to hours (rate), for the same industry, size class and period. Raw counts, all-industry averages or different metrics distort the comparison.
10	**C**	Comparing performance and practices among units within the same organization is internal benchmarking; data are easy to obtain and comparable, but the view is limited to company practice.

DOMAIN 2 · COMPETENCY 2.2

Performance Standards and Action Planning

CSP11 blueprint: Analyze performance standards to determine plan of action

1. Definition

Performance standards are the defined expectations for safety performance - regulatory requirements, consensus standards, company policies and objectives. Action planning turns gaps against those standards into specific, resourced, time-bound actions with accountable owners.

2. Explanation in simple words

First decide what 'good' looks like and how you will measure it. Then write a plan that says who will do what, by when, with what money and how you will know it worked.

3. Key points to remember

- SMART objectives: Specific, Measurable, Achievable, Relevant, Time-bound.
- Action plan contents: problem/gap statement, objective, actions, owner (one accountable person), resources/budget, due dates, success measures, status tracking and verification of effectiveness.
- Prioritize by risk (severity × likelihood), legal requirements, cost and feasibility; quick wins plus long-term fixes.
- Performance standards hierarchy: law/regulation (minimum) → consensus standards (ANSI, NFPA, ISO) → company standards (often stricter) → site procedures.
- PDCA ties it together: Plan (objectives, plans), Do (implement), Check (measure, audit), Act (correct and improve).

4. Real work scenario

> After an audit shows that 30% of forklift operators have expired evaluations, the safety manager writes an objective: 'All 120 operators will have a current evaluation by March 31; monthly status report to the plant manager.' Supervisors are named as owners for their crews, and trainer time is budgeted.

5. CSP exam tips and traps

- Choose the answer that contains measurable criteria and a deadline.
- 'Improve safety' or 'zero injuries' without measures are weak objectives on the exam.
- Action plans need verification of effectiveness - closing an action is not the same as solving the problem.

6. Every blueprint point (2 blueprint items + 3 more)

Performance standards BLUEPRINT

A performance standard defines the expected level of performance (an outcome, rate or activity level) that people and programs are measured against. Analyzing performance against the standard tells you what action plan is needed.

- Examples: 100% of corrective actions closed within 30 days; 95% training completion; TRIR below 1.5; 4 safety observations per supervisor per month.
- Regulatory 'performance' standards state the required outcome but let the employer choose the method (e.g., PSM, General Duty); 'specification' standards prescribe exact means (e.g., 42-in. guardrail).
- Standards should be measurable, achievable and tied to accountability (performance reviews).
- ANSI/ASSP Z16.1 provides consistent metric definitions for comparing performance.

> **Exam tip:** If a question asks why a facility met all OSHA specification standards but still had a fatality, the answer usually involves missing performance/management system elements.

Action planning BLUEPRINT

An action plan converts a gap or objective into specific tasks with owners, resources, deadlines and measures, and is managed through PDCA.

- Each action: what, who (single named owner), by when, resources, how success is verified.
- Objectives must be SMART: Specific, Measurable, Achievable, Relevant, Time-bound.
- Prioritize actions by risk (severity x likelihood) and regulatory urgency.
- Track in an action register; review status in management meetings; escalate overdue items.

> **Exam tip:** 'Reduce injuries' is not SMART; 'Reduce forklift-pedestrian near misses at Plant 2 by 50% by Dec 31 through separated walkways' is.

SMART objectives

SMART is the standard test for whether an objective is usable for planning and accountability.

- Specific - clear scope; Measurable - a metric; Achievable - realistic with resources; Relevant - tied to key risks; Time-bound - a deadline.
- ISO 45001 clause 6.2 requires OH&S objectives that are measurable (if practicable), monitored, communicated and updated.
- Mix leading and lagging objectives.

Plan-Do-Check-Act (PDCA) cycle

PDCA (Deming/Shewhart cycle) is the engine of continual improvement in every modern management system standard.

- Plan: policy, objectives, hazard identification, risk assessment, actions.
- Do: implement, train, communicate, provide resources.
- Check: monitor, measure, audit, investigate incidents.
- Act: management review, corrective action, improve - then repeat.
- ISO 45001, ISO 14001 and ANSI Z10 are all structured on PDCA.

> **Exam tip:** Audits and incident investigations are 'Check'; management review and corrective action are 'Act'.

Accountability and performance review

Performance standards only work if people are held accountable and results are reviewed regularly.

- Tie supervisor and manager appraisals to safety activities (leading) not only injury counts.
- Avoid incentive programs that reward 'zero injuries' - they suppress reporting (OSHA 1904.35 prohibits retaliation for reporting).
- Management review at least annually or after serious incidents.

Read more: Yates, Safety Professional's Reference and Study Guide (4th ed.): Ch. 31 Safety Management System, p. 717; Ch. 36 Measuring Safety and Health Performance

7. Ten CSP-style practice questions

Answer all ten on paper first, then check the key. Score: _____ / 10 Date: _____ Retest: _____ / 10

2.2-Q1. Which safety objective is written in the BEST SMART format?

A. Improve safety culture this year
B. Reduce hand injuries
C. Complete machine-guarding risk assessments on all 24 presses by June 30 and close 90% of high-risk actions by September 30
D. Everyone will work safely

2.2-Q2. A cross-functional team (engineering, operations, maintenance and safety) meets before a new press is installed to systematically identify its hazards, assess the severity and likelihood of each, and select risk controls (guarding, interlocks, PPE). Which phase of the Plan-Do-Check-Act (PDCA) cycle does this activity represent?

A. Plan
B. Do
C. Check
D. Act

2.2-Q3. Safety data are reviewed but the checklist and training are never changed. Which PDCA phase is failing?

A. Plan
B. Do
C. Check
D. Act

2.2-Q4. A program evaluation determines that training, audit, and inspection goals were met but incident rates remain unchanged. What should be the next step?

A. Reassess whether indicators measure true system effectiveness
B. Close the program review
C. Reduce training frequency
D. Discontinue leading indicators

2.2-Q5. A program effectiveness review finds that objectives were achieved, but employee engagement scores remain low. What should management prioritize?

A. Develop new technical controls only
B. Enhance leadership visibility and communication initiatives
C. Reduce reporting transparency
D. Focus solely on lagging indicators

2.2-Q6. How does a Safety Manager's input in the development of a corporate business plan influence safety management and performance goals?

A. It ensures the allocation of resources to achieve a zero-injury target.
B. It identifies areas for improvement and sets specific, measurable safety objectives.
C. It integrates safety goals into the overall business strategy and resource allocation.
D. It determines the company's financial projections and resource allocation, with no direct impact on safety goals.

2.2-Q7. Safety goals for any organization should possess the following characteristics:

A. They should be sufficiently broad to ensure their attainment.
B. They should be established by the executive management of the organization.
C. They should be based on injury rates.
D. Achievable and measurable.

2.2-Q8. In PDCA, which step focuses on implementing processes and operational controls?

A. Plan
B. Do
C. Check
D. Act

2.2-Q9. Following an audit, a safety manager writes an action plan to close 15 identified gaps. Which set of elements MUST the action plan contain to be effective?

A. A list of the gaps and the auditor's name
B. A description of each gap and a general statement that it will be fixed
C. Specific actions, a responsible person for each, target completion dates, required resources and a method to verify completion
D. The audit score and a comparison with last year's score

2.2-Q10. A gap analysis identifies these deficiencies, but the budget will fund only one this quarter. Which should be addressed FIRST?

A. Outdated safety bulletin board postings
B. Missing fall protection at an unguarded 4.5 m (15 ft) mezzanine edge used daily
C. Training records stored in two different databases
D. Safety committee minutes not posted on time

Answer key and explanations

Q	Ans	Why
1	C	SMART = Specific, Measurable, Achievable, Relevant, Time-bound. Only C states a scope (24 presses), measures (assessments, 90% closure) and deadlines; the others are vague slogans.
2	A	Identifying hazards, assessing risk and selecting controls before implementation is planning (ISO 45001 clause 6). Do = implementing the controls; Check = monitoring; Act = improving.
3	D	Reviewing data is Check; changing the system (checklists, training) based on findings is Act. If nothing ever changes, the Act phase is failing.
4	A	When activity targets are met but outcomes do not improve, the indicators may not measure what actually drives risk. Reassess and realign indicators rather than closing the review or cutting activity.
5	B	Meeting numeric objectives with low engagement signals a culture gap; visible leadership and two-way communication address engagement. Technical controls alone or lagging-only focus do not.
6	C	Safety input to the business plan integrates safety goals into the overall strategy and resource allocation, so safety is resourced like any other business objective.
7	D	Effective safety goals must be achievable and measurable (SMART). Broad goals cannot be tracked, and injury rates alone are lagging measures.
8	B	Do = implementing the plan: operational controls, training, communication and procedures.
9	C	An action plan turns findings into accountable work: what will be done, by whom, by when, with what resources, and how completion/effectiveness will be verified. Lists without owners and dates rarely get closed.
10	B	Action plans are prioritized by risk (severity x likelihood). A daily exposure to a potentially fatal fall outranks administrative and documentation gaps.

DOMAIN 2 · COMPETENCY 2.3

EHS Culture: Measure, Analyze and Improve

CSP11 blueprint: Determine how to measure, analyze, and improve EHS culture

1. Definition

Safety culture is the shared values, beliefs, attitudes and behaviors that determine how safety is actually managed in an organization - 'how we do things here when no one is watching'. Safety climate is a snapshot of employee perceptions of that culture at a point in time.

2. Explanation in simple words

Culture is what people really do and believe, not what the policy says. You measure it by asking and watching, look for gaps between what managers think and what workers experience, and improve it through visible leadership, involvement and trust.

3. Key points to remember

- Measurement tools: perception surveys (anonymous, repeated over time), focus groups, interviews, observation, behavior-based safety data, near-miss reporting rates, participation in safety activities, corrective action closure.
- Look for perception gaps between management and front-line responses; follow up surveys with focus groups to understand why.
- Maturity models: Hudson ladder (pathological → reactive → calculative → proactive → generative); DuPont Bradley curve (reactive → dependent → independent → interdependent).
- Characteristics of a strong culture: visible management commitment, employee involvement, just culture (fair accountability, no blame for honest errors), open reporting without retaliation, learning from incidents, stop-work authority.
- Behavior-based safety (BBS): identify critical behaviors, observe and measure, give feedback (positive reinforcement), improve barriers. ABC model: Antecedent → Behavior → Consequence; consequences that are soon, certain and positive drive behavior most.
- Culture killers: incentives tied to 'zero injuries' (suppress reporting), punishing reporting, production pressure, inconsistent discipline.

4. Real work scenario

> A survey shows 92% of managers agree that 'production never comes before safety', but only 41% of operators agree. Focus groups reveal that supervisors praise crews for meeting quotas but rarely mention safety. Leadership starts daily safety conversations, adds safety metrics to supervisors' reviews, and repeats the survey a year later; the operator score rises to 68%.

5. CSP exam tips and traps

- Culture questions usually reward involvement, communication and leadership by example over discipline or posters.
- A rise in near-miss reports after a campaign is usually a positive sign of reporting culture.
- BBS feedback should be immediate, specific and positive; public correction or write-ups are distractors.

6. Every blueprint point (3 blueprint items + 4 more)

Measuring EHS culture BLUEPRINT

Culture is the shared values, beliefs, attitudes and behaviors about safety. It is measured with several methods together, because no single tool captures it.

- Safety climate/perception surveys (validated, anonymous) - climate is a snapshot of culture.
- NSC 8 safety climate factors: management commitment, leadership communication, employee involvement, peer support, hazard recognition and reporting, safety training, supervisor support, accountability.
- Behavioral observations (PPE use, procedure compliance).
- Leading indicator review: near-miss reports, meeting participation, corrective-action closure.
- Focus groups and interviews; system review of committees, communications and leadership visibility.

> **Exam tip:** Best answer for 'how to assess culture' combines surveys + observations + leading indicators + interviews, not one tool.

Analyzing EHS culture BLUEPRINT

Analysis turns survey and indicator results into understanding of where and why culture is weak.

- Gap analysis against ANSI/ASSP best practice and NSC benchmarks.
- Segment data by department, shift, role - management often rates culture higher than front-line workers (perception gap).
- Root cause tools (5 Whys, fishbone) for under-reporting or low engagement.
- Benchmark against NSC national/sector data.

> **Exam tip:** A large gap between management and worker survey scores signals a leadership communication/credibility problem.

Improving EHS culture BLUEPRINT

Culture improves through visible leadership, real worker engagement, open blame-free communication, empowerment and accountability, repeated through PDCA.

- Leaders set expectations, lead by example, are visible on the floor.
- Involve workers in committees, hazard ID, investigations and solutions; recognize positive behavior.
- Share survey results and action plans; remove barriers to near-miss reporting.
- Empower stop-work authority; train on the 'why'.
- Tie manager performance reviews to culture goals; re-measure to track progress.

> **Exam tip:** Model answer: measure (surveys, observations, indicators) -> analyze (gap/root cause) -> leadership-driven participative improvement plan.

Safety culture vs. safety climate

Culture is deep and slow to change (values, assumptions); climate is the measurable surface perception at a point in time.

- Climate surveys measure perceptions now; culture change takes years.
- Schein: artifacts, espoused values, basic assumptions.
- Repeat surveys (e.g., every 1-2 years) to trend climate.

Safety culture maturity models

Maturity models describe stages organizations move through toward a generative culture.

- Hudson ladder: pathological -> reactive -> calculative -> proactive -> generative.
- DuPont Bradley Curve: reactive -> dependent -> independent -> interdependent.
- Higher maturity = workers own safety, reporting is valued, leading indicators drive decisions.

> **Exam tip:** 'We do safety because we were caught' = reactive; 'safety is how we do business' = generative.

Just culture and reporting

A just culture separates honest error from reckless behavior so people report errors and near misses without fear, while accountability for reckless acts remains.

- Human error -> console/fix system; at-risk behavior -> coach; reckless behavior -> discipline.
- Blame-and-retrain cycles suppress reporting and hide system causes.
- OSHA 1904.35: employees must have a reasonable reporting procedure and be protected from retaliation.

Behavior-based safety (BBS)

BBS uses peer observation and feedback on critical safe behaviors to reinforce safe practices and generate leading data.

- Elements: identify critical behaviors, observe, give feedback, analyze data, remove barriers.
- ABC model: Antecedent -> Behavior -> Consequence; consequences that are soon, certain and positive drive behavior most.
- BBS must not replace hazard control - fix conditions first (hierarchy of controls).

> **Exam tip:** BBS that blames workers while ignoring system hazards is a weak answer.

Read more: Yates, Safety Professional's Reference and Study Guide (4th ed.): Ch. 21 Management Theories, p. 505; Ch. 36 Measuring Safety and Health Performance

7. Ten CSP-style practice questions

Answer all ten on paper first, then check the key. Score: _____ / 10 Date: _____ Retest: _____ / 10

2.3-Q1. Organizational culture greatly affects implementation of a safety program, and each organization has its own culture. What is one way to measure safety culture improvement?

A. Regular safety perception surveys of all workers
B. Implementation of OSHA safety and health guidelines
C. Number of OSHA violations
D. Implement a formal Pareto Perception Analysis For All

2.3-Q2. A perception survey shows 92% of managers agree that 'safety is never compromised for production,' but only 41% of front-line workers agree. What is the BEST interpretation?

A. The workers misunderstood the question
B. A perception gap exists; investigate through focus groups and look at how production pressure is handled in practice
C. The culture is strong because management is committed
D. Repeat the survey until results match

2.3-Q3. Which of the following is NOT one of the five maturity levels in Hudson's safety culture maturity ladder (based on Reason and Westrum)?

A. Pathological
B. Reactive
C. Predictive
D. Generative

2.3-Q4. A "Just Culture" differs from a "Blame Culture" by:

A. Ignoring unsafe acts
B. Making safety optional
C. Allowing repeat violations without consequence
D. Focusing on learning and system improvements rather than punishment

2.3-Q5. Which of the following is the best indicator of a strong safety culture?

A. Compliance with documentation
B. High rate of near-miss reporting and visible management response
C. Low number of hazard reports
D. Strict enforcement without feedback

2.3-Q6. A safety team notices that workers consistently bypass a machine guard to clear jams, even after reminders. Which BBS-based strategy should be prioritized to improve safe behavior?

A. Increase the number of disciplinary actions for non-compliance
B. Replace supervisors who fail to enforce the rule
C. Investigate workflow and equipment design to remove barriers to safe behavior
D. Post more signs and distribute reminder emails weekly

2.3-Q7. A construction site has gone 60 days without a recordable injury, but unsafe shortcuts continue. Which BBS strategy best shifts focus to positive behaviors?

A. Track and reward safe behaviors like pre-task planning, safety suggestions or PPE compliance
B. Give gift cards only when there are no OSHA recordables
C. Focus future training on incident response
D. Publicly warn workers about disciplinary actions for violations

2.3-Q8. As an organization's culture transitions towards a safety-centric approach, which indicators should stakeholders expect to observe?

A. Increased safety audits, revised disciplinary program, and mandatory performance goals.
B. Reduction in injury reports, fewer regulatory citations, and higher profits.
C. Additional safety professionals, more formal training, and monetary incentives
D. Management support, open communication, and company-wide participation.

2.3-Q9. During a routine observation, you notice a worker using a grinder without safety glasses. When asked, he says, "I've done this for 10 years and never had a problem." What is the best BBS-based response?

A. Write him up to show accountability
B. Engage in a one-on-one conversation about risk perception and eye injury potential
C. Ignore the behavior since he's experienced
D. Add signage near the grinder as a reminder

2.3-Q10. An organization that openly shares lessons learned from incidents across departments is demonstrating:

A. Generative culture focused on learning and prevention
B. Reactive safety culture
C. Calculative culture based on data alone
D. Pathological culture that hides problems

Answer key and explanations

Q	Ans	Why
1	**A**	Perception surveys of all workers, repeated over time, directly measure shared beliefs and attitudes, i.e., culture. Citation counts and guideline adoption are compliance measures, not culture measures.
2	**B**	A large manager-worker perception gap is a key culture finding. Follow up with focus groups/interviews to learn how production pressure is handled in practice; dismissing or repeating the survey ignores the signal.
3	**C**	Hudson's ladder levels are Pathological, Reactive, Calculative, Proactive and Generative. 'Predictive' is not a level.

4	**D**	A just culture separates honest error from reckless behavior, emphasizing learning and system fixes with fair accountability; it neither ignores unsafe acts nor blames by default.
5	**B**	High near-miss reporting combined with visible management response shows trust and engagement. Low hazard reports usually mean underreporting, not safety.
6	**C**	Repeated guard bypassing despite reminders points to a system barrier (jam frequency, design, workflow). Remove the reason for the at-risk behavior; discipline and signs do not address the cause.
7	**A**	Rewarding injury-free outcomes encourages underreporting. Recognizing measurable safe behaviors (pre-task planning, suggestions, PPE use) reinforces the behaviors that prevent incidents.
8	**D**	A culture shift shows up as management support, open communication and company-wide participation. Fewer injuries and citations are lagging results, and more audits or incentives are inputs.
9	**B**	BBS uses respectful one-on-one feedback that addresses risk perception ('never had a problem' is optimism bias). Write-ups or ignoring the behavior do not change beliefs; signs are weak controls.
10	**A**	Openly sharing lessons learned across the organization is a hallmark of a generative (learning) culture; pathological cultures hide problems and reactive cultures act only after events.

DOMAIN 2 · COMPETENCY 2.4

Incident Investigation and Root Cause Analysis

CSP11 blueprint: Determine appropriate incident investigation techniques (root causes) and apply corrective actions

1. Definition

Incident investigation is the systematic process of collecting facts about an incident, determining direct, contributing and root causes, and implementing corrective actions that prevent recurrence. Root causes are the underlying management-system failures that allowed the immediate causes to exist.

2. Explanation in simple words

Ask what happened, how it happened and why - then keep asking why until you reach something the organization can fix in its systems. Blame finds a person; investigation finds the fix.

3. Key points to remember

- Steps: secure the scene and care for the injured → preserve evidence → gather information (people, positions, parts, paper - the 4 Ps) → interview witnesses individually and promptly → build a timeline → analyze causes → develop corrective actions → report → track to closure → verify effectiveness → share lessons.
- Interview tips: private setting, explain the purpose is prevention not blame, open-ended questions ('Tell me what you saw'), let the witness talk, review notes with the witness.
- Causes: direct/immediate (unsafe act or condition, energy release), contributing, root (system: training, procedures, design, supervision, management commitment).
- Techniques: 5 Whys, fishbone/Ishikawa (people, methods, machines, materials, environment, management), fault tree analysis, events and causal factors charting, change analysis, barrier analysis, MORT, multilinear events sequencing (MES), TapRooT, SCAT (systematic cause analysis technique), Swiss cheese model, bow-tie.
- Accident theories: Heinrich domino (unsafe acts), Bird/Loss Causation (lack of control → basic causes → immediate causes → incident → loss), Reason's Swiss cheese (aligned holes in defenses), energy transfer (Haddon).
- Corrective actions should follow the hierarchy of controls; 'retrain' and 'discipline' alone are weak. Assign owners and dates and verify that the actions work.

4. Real work scenario

> An operator's hand is caught in a conveyor while clearing a jam. A 5-Whys analysis shows the guard was removed because jams were frequent, jams were frequent because a new box size was introduced without changing the guides, and the box change was never reviewed under MOC. Corrective actions: adjust the guides, install an interlocked access door and add packaging changes to the MOC procedure.

5. CSP exam tips and traps

- 'The worker didn't follow the procedure' is a symptom - the BEST answer keeps asking why.
- First action at an incident scene: care for the injured and make the scene safe.
- Interview witnesses separately and as soon as possible.
- Fishbone = categorize causes; FTA = top-down logic; FMEA = bottom-up component failures.

6. Every blueprint point (3 blueprint items + 6 more)

Incident investigation process BLUEPRINT

A systematic, standards-based process (OSHA Incident Investigation guide, ANSI Z10, NSC) to learn why an incident happened and prevent recurrence - learning, not blaming.

- Step 1: secure the scene and provide aid - protect people first, stop hazardous operations, LOTO, control spills; then preserve the scene.
- Step 2: report and notify - internally at once; OSHA: fatality within 8 h, in-patient hospitalization/amputation/eye loss within 24 h.
- Step 3: assemble a trained multidisciplinary team led by a neutral leader not involved in the incident.
- Step 4: collect evidence - private witness interviews ASAP with open-ended questions, photos/sketches, physical evidence with chain of custody, records (SOPs, training, maintenance), video/logs.
- Step 5: reconstruct the timeline (sequence diagrams, flow charts), then root cause analysis, corrective actions, report, follow-up.
- Investigate near misses too - they are free lessons.

> **Exam tip:** FIRST action at an incident scene is always care for the injured and make the scene safe - not interviewing or photographing.

Root cause analysis BLUEPRINT

Root causes are the underlying, systemic reasons (management, design, procedures, culture) that allowed the immediate causes. Fixing them prevents recurrence; fixing only immediate causes leads to 'blame and retrain'.

- Immediate causes = unsafe acts (bypassed guard) and unsafe conditions (wet unmarked floor).
- Root causes = why the system allowed them (no procurement policy requiring operator input, production pressure, poor design).
- Tools: 5 Whys, fishbone, change analysis, barrier analysis, FTA, MORT.
- ANSI Z10: RCA expected for significant incidents and near misses.
- Never accept 'human error' or 'worker carelessness' as the final cause.

> **Exam tip:** If an answer choice stops at 'employee failed to follow procedure', look for the one that asks why (training, design, supervision, pressure).

Corrective actions BLUEPRINT

Corrective and preventive actions must target root causes, follow the hierarchy of controls, be owned, tracked, verified and embedded into the system.

- Prioritize by risk (risk matrix, bow-tie); life-threatening and regulatory items first.
- Single named owner and due date for every action; track in an action register (NSC).
- Verify completion AND effectiveness (re-assess risk) before closing; escalate overdue actions to management.
- Update SOPs, training, permits, drawings; share lessons learned organization-wide.
- Include preventive actions: 'how do we stop it happening anywhere else?'

> **Exam tip:** 'Retrain the employee' or 'discipline the worker' is rarely the best corrective action; engineering/system fixes are preferred.

5 Whys

Ask 'why?' repeatedly, using each answer as the basis for the next question, until you reach a system, policy or management cause (often about five whys).

- Best for routine incidents, near misses and simple recurring problems; first step before more detailed tools.
- Example chain: bypassed guard -> slowed production -> guard hard to access -> chosen on cost -> purchasing need not consult operators -> no policy for operator input (root cause).
- Document each why; avoid blaming questions; go beyond five if needed.

> **Exam tip:** If you reach a policy, culture or design issue, you are near the true root cause.

Fishbone (Ishikawa) diagram

A cause-and-effect diagram with the problem at the head and main cause categories as bones; a team brainstorms causes in each category.

- Common categories: People, Equipment (Machines), Process (Methods), Materials, Environment, Management (6 Ms variant: Man, Machine, Method, Material, Measurement, Mother nature).
- Best for complex incidents with many possible contributors and group brainstorming.
- Combine with 5 Whys on each branch; attach to the investigation record.

Change analysis

Compare the incident situation with the normal, incident-free situation and list every difference; then test how each change contributed.

- Use when 'this has never happened before' or a system that normally works failed.
- Typical changes: substitute operator, new material or pallet type, wet surface, time pressure, temporary workaround.
- Ask: who approved the change, was risk assessed, were controls adjusted? Link findings to the MOC process.

> **Exam tip:** Key question for change analysis: 'What was different today?'

Barrier analysis

List every barrier (engineering, administrative, PPE, alarms, physical) that should have stopped the incident and determine whether each was missing, failed, bypassed, ineffective or not updated.

- Best in high-hazard settings with layers of defense (chemical, oil and gas, confined spaces).
- Example: double-wall piping missing; permit bypassed; relief valve stuck; alarm ignored (alarm fatigue); face shield not worn; SOP not updated.
- Leads to system-level fixes rather than behavior fixes; supports defense-in-depth.

Accident causation theories

Theories explain why incidents occur and shape where investigators look.

- Heinrich domino theory: ancestry/social environment -> fault of person -> unsafe act/condition -> accident -> injury; remove the middle domino.
- Heinrich ratio 1 major : 29 minor : 300 no-injury; Bird ratio 1 serious : 10 minor : 30 property damage : 600 near misses.
- Reason's Swiss cheese model: latent conditions (organizational) line up with active failures to let a hazard through barriers.
- Multiple causation: incidents have many contributing factors - system, management, culture.
- Energy release theory (Haddon): incidents are unwanted energy transfer; control the energy or the barrier.

> **Exam tip:** Swiss cheese = latent organizational failures + active errors; questions about 'layers of defense' point to Reason.

Interviews and evidence handling

Credible investigations depend on timely, unbiased, well-documented evidence.

- Interview witnesses separately and privately, soon after the event, with open-ended non-judgmental questions.
- Photograph before moving anything; tag and log physical evidence (chain of custody).
- Compare what should have happened (procedures) with what did happen.
- Gather environmental data (air, noise, lighting), control-system logs and video.

> **Exam tip:** Group interviews create groupthink - interview individually.

Read more: Yates, Safety Professional's Reference and Study Guide (4th ed.): Ch. 22 Accident Causation and Investigation Techniques, p. 515

7. Ten CSP-style practice questions

Answer all ten on paper first, then check the key. Score: _____ / 10 Date: _____ Retest: _____ / 10

2.4-Q1. Which of the following should be the first step in an incident investigation?

A. Interview witnesses
B. Secure and preserve the scene
C. Collect physical evidence
D. Develop recommendations

2.4-Q2. Which investigation tool focuses on "Why" questions to trace a causal chain until a systemic cause is found?

A. Fault Tree Analysis
B. Fishbone Diagram
C. 5 Whys
D. MORT

2.4-Q3. Following a chemical spill, the safety manager assembles a multidisciplinary team to analyze deviations in valve operation, human error, and procedural gaps. The team uses a structured worksheet to identify causes and corrective actions for People, Procedures, Equipment, Environment, and Management Systems. Which investigation technique is being applied?

A. Cause-and-effect (fishbone/Ishikawa) analysis
B. Barrier analysis
C. Event and causal factor charting
D. Energy trace and barrier analysis

2.4-Q4. After a worker falls from a ladder, the investigation reveals the ladder had worn, uneven steps. What type of cause does this represent?

A. Root cause
B. Personal failure
C. Contributing cause
D. Direct cause

2.4-Q5. Verification of corrective actions should occur:

A. Before the root cause is identified
B. Immediately after implementation
C. After a reasonable period to confirm effectiveness
D. Only during management review

2.4-Q6. The most critical output of an incident investigation report is:

A. Recommendations and verified corrective actions
B. Assignment of blame
C. A record of injuries
D. Witness statements

2.4-Q7. When should the investigation be conducted following an injury?

A. Based on the evidence only because witness statements are usually biased or hearsay.
B. Immediately, with all witnesses simultaneously to get the most accurate statements.
C. The next working day after emotions have settled down.
D. Individually, with all direct and indirect witnesses as soon as possible.

2.4-Q8. A supervisor is being interviewed about a forklift near-miss. The investigator asks, "Did you see the forklift operator speeding?" The supervisor answers "No." What would have been a better question to encourage useful information?

A. "Why didn't you stop the operator if they were speeding?"
B. "Tell me what you observed when the forklift entered the aisle."
C. "Did the forklift almost hit the pedestrian?"
D. "Do you think the operator is reckless?"

2.4-Q9. The Bowtie method in incident investigation is best described as:

A. A brainstorming technique to list every possible hazard
B. A timeline that shows human errors in sequence
C. A method to calculate injury severity after incidents
D. A model that links threats, preventive barriers, the top event, and consequences

2.4-Q10. An investigation finds that an operator's hand was caught while clearing jams on a conveyor because the task requires reaching into the nip point while the belt runs, and jams occur about 20 times per shift. Which corrective action is MOST effective?

A. Retrain all operators on the hazards of nip points
B. Discipline the operator for reaching into the conveyor
C. Redesign the infeed to reduce jams and install an interlocked guard that stops the belt when opened
D. Post a warning sign at the jam location

Answer key and explanations

Q	Ans	Why
1	B	First secure the scene to prevent further harm and preserve evidence; interviews and evidence collection follow.
2	C	The 5 Whys repeatedly asks 'why' to move from the immediate cause to the underlying system cause.
3	A	Categorizing causes into branches such as People, Procedures, Equipment, Environment and Management Systems is the cause-and-effect (fishbone/Ishikawa) method.
4	D	Worn, uneven ladder steps are the unsafe condition that directly produced the fall (direct/immediate cause). The root cause would be why the ladder was not inspected or removed from service.
5	C	Corrective actions should be verified after a reasonable period to confirm they are effective and sustained and did not create new hazards; checking on the day of installation only confirms completion.
6	A	An investigation's value is in actionable, verified corrective actions that prevent recurrence, not in blame or data collection alone.
7	D	Interview witnesses individually and as soon as possible, while memories are fresh and before accounts are influenced by others. Group interviews contaminate testimony.
8	B	Open-ended, non-leading questions ('Tell me what you observed...') produce more complete, unbiased information than yes/no or accusatory questions.
9	D	A bowtie links threats, preventive barriers, the top event, mitigative barriers and consequences in one diagram.
10	C	Corrective actions should follow the hierarchy of controls and address the root cause: reducing jams (elimination of the task) plus interlocked guarding (engineering) beats training, signs and discipline.

DOMAIN 2 · COMPETENCY 2.5

Management of Change (MOC)

CSP11 blueprint: Describe the Management of Change process (prior, during, after)

1. Definition

Management of Change is a formal process to identify, evaluate and control the hazards and risks introduced by changes to processes, equipment, chemicals, procedures, facilities, organizations or personnel before the change is implemented. It is required by OSHA PSM (1910.119(l)) and is a key element of ISO 45001 (8.1.3) and ANSI Z10.

2. Explanation in simple words

Every change can bring new hazards. Before you change something, stop and check what could go wrong, fix it, tell and train people, and check afterwards that it worked.

3. Key points to remember

- Before (prior): request and describe the change, technical basis, hazard identification and risk assessment, impact on procedures, training, PSI and permits, approvals by authorized people.
- During: implement controls, update drawings (P&IDs) and procedures, train and inform affected employees and contractors, pre-startup safety review (PSSR) for PSM changes.
- After: verify effectiveness, monitor for unintended consequences, close out and document, retain records.
- Types of change: permanent, temporary (with an expiry date), emergency (expedited but still reviewed); organizational changes (staffing, reorganizations, mergers) and personnel changes are often missed.
- Replacement in kind (identical specification) does not need MOC; anything that changes specifications, materials, settings or conditions does.
- Change management (people side): communicate the reason, involve those affected, address resistance, provide resources (e.g., Kotter's 8 steps, Lewin's unfreeze-change-refreeze).

4. Real work scenario

> Maintenance wants to replace a pump with a higher-pressure model. The MOC review finds that the downstream piping, flange gaskets and relief valve are rated below the new pump's shutoff pressure. The relief valve is resized, gaskets are upgraded, procedures are updated and operators are trained before the pump is started.

5. CSP exam tips and traps

- The FIRST step of MOC is to evaluate the hazards and risks of the proposed change before implementation.
- Temporary changes (bypassing an interlock) and organizational changes are MOC triggers.
- Watch for 'it's the same thing' traps - different material, rating or supplier formulation is not replacement in kind.

6. Every blueprint point (1 blueprint items + 5 more)

MOC process BLUEPRINT

MOC ensures any change to equipment, processes, materials, personnel or organization is identified, assessed and controlled before, during and after implementation so it does not create new hazards or weaken controls.

- Required for PSM-covered processes by 29 CFR 1910.119(l) and expected by ANSI Z10 and ISO 45001 (clause 8.1.3).
- PSM MOC must address: technical basis, impact on safety and health, modifications to operating procedures, time period (temporary changes), authorization requirements.
- Affected employees and contractors must be informed and trained BEFORE startup of the changed process.
- Update PSI (P&IDs, procedures) after the change; PSSR for new or significantly modified facilities.
- 'Replacement in kind' (same specification) is not an MOC change.

> **Exam tip:** Most-missed step: the risk assessment of a 'small' or 'temporary' change. Every change - even temporary - goes through MOC.

Before the change (pre-implementation)

Identify and log the change, define scope and impact, assess risk, check regulations, involve stakeholders and obtain documented authorization.

- Triggers: new equipment, revised procedures, raw materials or supplier, software/controls, layout, staffing levels, organizational changes, temporary modifications.
- Consider direct and indirect impacts (new chemical -> waste handling hazards).
- Risk tools: JHA, What-If, HAZOP, FMEA.
- Check permits, notifications, fire codes, EPA requirements.
- Formal change request + risk assessment record + management/EHS approval before work starts.

> **Exam tip:** If the scenario says the change was implemented and then reviewed, the MOC failed at the 'before' stage.

During the change (implementation)

Communicate, train, install and verify controls, supervise closely and document everything.

- Tell affected personnel what, why and when; give clear new instructions.
- No change is 'safe' until everyone affected is competent.
- Engineering, administrative and PPE controls in place and verified before the system is activated.
- Have contingency plans to pause or reverse the change.
- Use an MOC checklist (ANSI Z10).

After the change (post-implementation)

Verify and validate that controls work, track deviations, update documents, review outcomes and share lessons.

- Inspect/audit the changed system; get feedback from front-line users.
- Track incidents and near misses linked to the change; update the risk assessment.
- Revise SOPs, training, P&IDs, permits, emergency plans.
- Review MOC outcomes in management review; close the MOC only when documentation is complete.

Organizational and personnel change

Changes in people and structure (downsizing, reassignment, new contractors, loss of experienced staff, shift changes) can degrade safety as much as hardware changes.

- Assess loss of knowledge, reduced staffing at critical tasks, changed responsibilities.
- ISO 45001 8.1.3 explicitly includes changes in work organization, staffing and knowledge.
- Temporary changes need an end date and must be reversed or re-approved.

> **Exam tip:** A reorganization that removes an experienced operator from a critical task should trigger MOC.

Common MOC pitfalls

Most change-related incidents come from skipped steps rather than missing procedures.

- Skipping risk assessment for 'minor' changes.
- Failing to update procedures, drawings and training afterward.
- Not involving the people who will work with the change daily.
- Allowing 'temporary' changes to become permanent without review.

Read more: Yates, Safety Professional's Reference and Study Guide (4th ed.): Ch. 38 Management of Change, p. 859

7. Ten CSP-style practice questions

Answer all ten on paper first, then check the key. Score: _____ / 10 Date: _____ Retest: _____ / 10

2.5-Q1. Which step must occur first in an effective Management of Change (MoC) process?

A. Approve implementation
B. Identify the proposed change and screen risk
C. Train employees on new procedures
D. Close the change file

2.5-Q2. A facility plans to replace a key piece of production equipment with a newer model that operates at higher temperatures. What should a safety professional do FIRST as part of the Management of Change process?

A. Approve the purchase order to avoid project delays.
B. Update the safety signage to reflect the higher operating temperature.
C. Conduct a risk assessment to evaluate potential safety and health impacts.
D. Notify employees of the upcoming change via email.

2.5-Q3. Which of the following would NOT normally require a formal Management of Change review?

A. Replacing a pump seal with an identical part number from the same manufacturer
B. Increasing a reactor's operating temperature by 15 deg C
C. Changing a solvent supplier to a different chemical formulation
D. Reducing the number of operators on night shift

2.5-Q4. A company merges two production departments to reduce costs. After the merger, safety performance declines, and employees report confusion about responsibilities and supervision. The MOC review focused only on equipment relocation, not personnel or procedural impacts. Which aspect of MOC was overlooked?

A. Engineering control verification
B. Shift handoff documentation
C. Vendor qualification for new suppliers
D. Organizational change impact on roles and communication

2.5-Q5. Which step in the Management of Change (MoC) process verifies that modified equipment operates safely before returning to service?

A. Identification
B. Authorization
C. Pre-startup safety review
D. Documentation closure

2.5-Q6. The final step in an MoC process is to:

A. Verify implementation effectiveness and close documentation
B. Train affected employees
C. Conduct a risk assessment
D. Approve engineering design

2.5-Q7. A written Management of Change Program is required by OSHA for:

A. Confined Space
B. Process Safety Management
C. Lockout
D. Emergency Response Program

2.5-Q8. Which of the following is the best verification method for determining MoC effectiveness?

A. Using only documentation reviews without observation
B. Management's verbal assurance
C. Delay of verification until next year's audit
D. Employee feedback and audit of implemented controls

2.5-Q9. During MoC, a new control room software is installed. Which hazard is most likely if communication training is skipped?

A. Physical hazard
B. Ergonomic hazard
C. Human factors / operational error
D. Environmental hazard

2.5-Q10. The management of change process covers four primary concepts, regardless of the type of change. Three of these are: Identification of hazards associated with the change Assessment of the risks associated with the change Consideration of the hazards and risks prior to the change Which is the fourth primary concept?

A. Implementation of controls needed to address hazards and risks associated with the change
B. Performance of a "what if" analysis to identify expected risks
C. Hiring of additional management personnel to oversee and direct organizational change
D. Compliance with all applicable regulatory guidelines regarding changes in an occupational health and safety management program

Answer key and explanations

Q	Ans	Why
1	B	MOC begins by identifying the proposed change and screening it for hazards/risk; authorization, training and closure come later.
2	C	Before approval, the change must be risk assessed for safety and health impacts (higher temperature, burns, materials, procedures). Purchasing, signs or emails before assessment skip the 'prior' stage of MOC.
3	A	Replacement-in-kind (identical specifications) is excluded from MOC. Changes to chemicals, process conditions and staffing are all changes that need review.
4	D	MOC covers organizational changes too (staffing, roles, supervision, communication), not just equipment. The review missed the people/organizational impacts.
5	C	The pre-startup safety review (PSSR) verifies that construction, procedures, training and controls are in place before the changed equipment is returned to service.
6	A	MOC closes by verifying the change was implemented as approved and is effective, then updating and closing documentation (P&IDs, SOPs, training records).

7	B	OSHA's PSM standard (29 CFR 1910.119(l)) explicitly requires written MOC procedures. The other standards do not contain a formal MOC element.
8	D	Effectiveness after a change is best verified by observing/auditing the implemented controls and getting feedback from affected users, not by verbal assurance or paperwork alone.
9	C	Changing an operator interface without training increases the likelihood of human error (human factors). MOC must address training and communication during the change.
10	A	The fourth element is implementing controls to address the hazards and risks identified for the change; identification and assessment are useless without control.

DOMAIN 2 · COMPETENCY 2.6

System Safety Analysis

CSP11 blueprint: Describe system safety analysis techniques (e.g., fault tree analysis, failure modes and effects analysis [FMEA], Safety Case approach, risk summation)

1. Definition

System safety applies engineering and management principles to identify hazards and control risk across the entire life cycle of a system. Analysis techniques can be inductive (bottom-up: what happens if this part fails?) or deductive (top-down: what could cause this undesired event?).

2. Explanation in simple words

Some tools start from a single part and ask what happens if it fails (FMEA). Others start from the disaster and work backwards to find every way it could happen (fault tree). A safety case puts all the evidence together to prove the system is safe enough.

3. Key points to remember

- Preliminary Hazard Analysis (PHA): early, concept stage, broad hazard list and rough risk ranking.
- FMEA: inductive, bottom-up; for each component list failure modes, effects and causes; rate Severity, Occurrence and Detection (1-10); RPN = S × O × D. FMECA adds criticality. Limitation: single failures, does not handle combinations or human error well.
- Fault Tree Analysis (FTA): deductive, top-down from a top (undesired) event using Boolean logic. AND gate: all inputs must occur (multiply probabilities). OR gate: any input ($P = 1 - \Pi(1 - Pi)$, or approximately add for rare events). Minimal cut sets identify the smallest combinations causing the top event.
- Event Tree Analysis (ETA): inductive, forward from an initiating event through success/failure of safeguards to outcomes.
- HAZOP: team-based, guide words (no, more, less, as well as, part of, reverse, other than) applied to process parameters at nodes.
- Safety Case: a structured, evidence-based argument that a system is acceptably safe (often to ALARP) for a defined use; required in UK offshore, nuclear, rail; goal-based rather than prescriptive.
- Risk summation: total risk = Σ (frequency × consequence) for all scenarios, used to understand overall risk exposure and compare with tolerability criteria.
- Other: Energy Trace and Barrier Analysis (ETBA), MORT (management oversight and risk tree), Job Safety Analysis, What-If, Checklist. Design terms: fail-safe, fail-passive (shuts down safely), fail-active (alarms/defaults to safe state), fail-operational (keeps working safely).

4. Real work scenario

> A chemical company models a reactor overpressure (top event) with a fault tree. Overpressure occurs if cooling fails (OR gate of pump failure 0.01 and power loss 0.02) AND the relief valve fails (0.001). Cooling failure ≈ 0.0298; top event ≈ $0.0298 \times 0.001 = 3.0 \times 10^{-5}$ per year, which the safety case compares with the company's tolerability criterion.

5. CSP exam tips and traps

- Top-down/deductive = FTA. Bottom-up/inductive = FMEA (and ETA).
- AND gate multiplies; OR gate adds (approximately). Series systems fail if any part fails; parallel/redundant systems fail only if all fail.

- A high RPN ranks first, but many companies act on any severity 9-10 regardless of RPN.
- Safety Case = an argument plus evidence that risk is acceptable.

6. Every blueprint point (4 blueprint items + 6 more)

Fault Tree Analysis (FTA) BLUEPRINT

FTA is a deductive, top-down (backward) technique that starts with an undesired Top Event and uses Boolean logic gates to map all combinations of equipment failures, human errors and external events that could cause it. It can be qualitative or quantitative.

- AND gate: output occurs only if ALL inputs occur - P = P1 x P2 (independent events).
- OR gate: output occurs if ANY input occurs - P = 1 - (1-P1)(1-P2); approx. P1 + P2 for small probabilities.
- Example: AND of 0.01 and 0.02 = 0.0002; OR of 0.01 and 0.02 = 1 - (0.99)(0.98) = 0.0298.
- Minimal cut set = smallest combination of basic events that causes the top event; a single-event cut set = single point of failure.
- Best for complex, high-risk technical systems (process, aviation, nuclear, utilities); output = fault tree diagram and cut sets.

> **Exam tip:** FTA = deductive, top event, logic gates. Adding redundancy changes an OR into an AND and sharply lowers top-event probability.

FMEA / FMECA BLUEPRINT

FMEA is an inductive, bottom-up, proactive technique that lists each component, every way it can fail (failure mode) and the effect on the system. FMECA adds a criticality ranking of severity and likelihood.

- RPN = Severity x Occurrence (probability) x Detection, each usually rated 1-10 (RPN 1-1000).
- Example: S = 8, O = 5, D = 4 -> RPN = 160; act on highest RPN and on any high-severity mode.
- Detection is rated inversely: 10 = almost impossible to detect.
- Good for design, manufacturing and equipment reliability; poor at human error and combined failures.
- Not a root cause/investigation tool - it anticipates what could fail, not why something did fail. PEM chart summarizes ratings.

> **Exam tip:** FMEA = component failure modes, bottom-up, single failures; FTA = top event, top-down, combinations.

Safety Case BLUEPRINT

A structured, evidence-based argument showing that a system or facility achieves and maintains an acceptable level of safety, prepared for regulators and stakeholders.

- Three elements: Claim (the system is safe under stated conditions), Argument (reasoning), Evidence (tests, analyses, records, audits).
- Living document - updated after changes, incidents, audits or new regulations.
- Must be documented, traceable, auditable and defensible.
- Used in high-hazard regulated sectors: offshore oil and gas (UK), nuclear, rail, aviation, defense; often required for licensing.

> **Exam tip:** If the question says 'demonstrate to a regulator that risks are reduced to ALARP with supporting evidence', the answer is a Safety Case.

Risk summation (risk aggregation) BLUEPRINT

Risk summation combines many individual risk values into a total risk profile so cumulative risk is not underestimated.

- Risk = Probability x Consequence for each hazard; total = sum of individual risks.
- Example: 0.1/yr x $50,000 = $5,000; 0.05/yr x $200,000 = $10,000; 0.5/yr x $4,000 = $2,000 -> total expected loss $17,000/yr.
- Compare total to risk tolerance/ALARP; the largest contributor ($10,000) gets resources first.
- Use consistent scales (all 1-5 or all real probabilities and dollars).

> **Exam tip:** Several 'low' risks can sum to an unacceptable total - that is the point of risk summation.

Event Tree Analysis (ETA)

ETA is inductive and forward-looking: from one initiating event, it branches at each safety function (success/failure) to show all possible outcomes. No logic gates.

- Outcome frequency = initiating frequency x product of branch probabilities along the path.
- Example: fire starts 0.1/yr; alarm fails 0.05; sprinkler fails 0.02 -> worst outcome = 0.1 x 0.05 x 0.02 = 0.0001/yr.
- Used to evaluate barrier effectiveness and redundancy and model consequences.

> **Exam tip:** ETA = initiating event -> consequences (forward); FTA = top event -> causes (backward).

Cause-consequence analysis and bow-tie

CCA combines FTA (causes) and ETA (consequences). The bow-tie diagram shows threats and preventive barriers on the left, the hazardous top event (knot) in the middle, and consequences with mitigating barriers on the right.

- Left side = prevention barriers; right side = mitigation/recovery barriers.
- Fishbone often used first to brainstorm causes.
- Excellent for communicating risk to non-technical audiences.

HAZOP

Team-based, node-by-node study of deviations from design intent using guide words combined with process parameters.

- Guide words: No/None, More, Less, As well as, Part of, Reverse, Other than (plus High/Low).
- Parameters: flow, pressure, temperature, level, time, composition.
- Deviation = guide word + parameter (e.g., 'No flow', 'High pressure').
- Led by a trained facilitator with a multidisciplinary team; output = report with recommendations; aims for ALARP. A PSM-accepted PHA method.

> **Exam tip:** Guide words + nodes = HAZOP.

THERP (human reliability)

Technique for Human Error Rate Prediction breaks a task into steps and assigns human error probabilities to each to estimate overall task failure probability.

- Uses task analysis and standard error probability tables, adjusted by performance shaping factors (stress, training, design).
- Overall success = product of step success probabilities; e.g., 3 steps each 0.99 -> 0.97 success, 0.03 failure.
- Feeds FTA/PRA; used in nuclear and process industries for routine procedural tasks.

MORT (Management Oversight and Risk Tree)

MORT is a pre-drawn logic tree used to find management and organizational causes of incidents ('oversights and omissions' and 'assumed risks').

- Developed for the U.S. Department of Energy; very detailed checklist-style tree.
- Focus is on management system failures, not just technical causes.
- Used reactively (investigations) and proactively (program evaluation).

> **Exam tip:** Management/organizational root causes + tree = MORT.

MIL-STD-882E and other hazard analyses

MIL-STD-882E is the DoD system safety standard defining risk assessment matrices and hazard analysis tasks across the system life cycle.

- Severity: 1 Catastrophic, 2 Critical, 3 Marginal, 4 Negligible.
- Probability: A Frequent, B Probable, C Occasional, D Remote, E Improbable, F Eliminated.
- Preliminary Hazard List/Analysis (PHL/PHA) is done early in design; What-If and checklist analyses are simpler qualitative methods.

Read more: Yates, Safety Professional's Reference and Study Guide (4th ed.): Ch. 22 Accident Causation and Investigation Techniques (FTA), p. 515; Ch. 26 Risk Assessment and Management, p. 604; Ch. 15 Statistics (probability)

7. Ten CSP-style practice questions

Answer all ten on paper first, then check the key. Score: _____ / 10 Date: _____ Retest: _____ / 10

2.6-Q1. Two failure modes are analyzed in an FMEA. Mode X: Severity 10, Occurrence 2, Detection 3. Mode Y: Severity 5, Occurrence 6, Detection 5. Using RPN alone, which mode ranks higher and what is its RPN?

A. Mode X, RPN 60
B. Mode Y, RPN 150
C. Mode X, RPN 150
D. They are equal

2.6-Q2. In a fault tree, the top event occurs through an OR gate with two inputs: Event A (an AND gate of B = 0.1 and C = 0.2) and Event D = 0.01. What is the approximate probability of the top event?

A. 0.002
B. 0.03
C. 0.12
D. 0.31

2.6-Q3. A facility has three loss scenarios: (1) 0.01/yr x $500,000; (2) 0.1/yr x $20,000; (3) 0.5/yr x $2,000. Using risk summation, what is the total annualized risk?

A. $5,000
B. $7,000
C. $8,000
D. $522,000

2.6-Q4. Which statement BEST describes a Safety Case?

A. A structured argument, supported by evidence, that a system is acceptably safe for a given application and environment
B. A legal defense file prepared after an incident
C. A list of OSHA citations and abatement dates
D. A bottom-up analysis of component failure modes

2.6-Q5. Which system safety tool begins with an initiating event and explores possible success or failure outcomes?

A. Fault Tree Analysis
B. FMEA
C. MORT
D. Event Tree Analysis

2.6-Q6. Which analysis technique uses logic gates ("AND", "OR") to identify combinations of failures that lead to a top event?

A. Fault Tree Analysis (FTA)
B. Event Tree Analysis (ETA)
C. HAZOP
D. THERP

2.6-Q7. Which of the following hazard analysis techniques uses an inductive approach, starting with specific failures to determine their potential impacts on system safety?

A. Fault Tree Analysis
B. Job Hazard Analysis
C. Failure Modes and Effects Analysis
D. Common Cause Failure Analysis

2.6-Q8. A safety engineer is part of a design team evaluating a new chemical batching process. They break down the process into individual steps and use guide words like "more," "less," and "none" to explore deviations in flow, pressure, and temperature. Which method are they applying?

A. Preliminary Hazard Screening
B. Root Cause Analysis
C. Bowtie Analysis
D. Hazard and Operability Study (HAZOP)

2.6-Q9. Which of the following best describes a "fail-passive" system?

A. When a system fails, it activates a backup system to maintain its operation.
B. Upon failure, the system might cause a new hazard that wasn't previously present.
C. If a system encounters a failure, it ceases its operation without creating any adverse effects.
D. The system continuously monitors and corrects its own failures without human intervention

2.6-Q10. What is the primary purpose of Energy Trace and Barrier Analysis (ETBA)?

A. To design new energy-efficient systems
B. To measure the energy consumption of a facility
C. To determine the most cost-effective energy sources for a project
D. To identify energy sources and evaluate the barriers protecting against hazardous events

Answer key and explanations

Q	Ans	Why
1	B	RPN = S x O x D. X = 10 x 2 x 3 = 60; Y = 5 x 6 x 5 = 150, so Y ranks higher. Trap: many organizations also act on any severity 10 regardless of RPN.
2	B	AND gate multiplies: 0.1 x 0.2 = 0.02. OR gate: 1 - (1 - 0.02)(1 - 0.01) = 0.0298, about 0.03 (rare-event approximation: 0.02 + 0.01).
3	C	Risk summation adds each scenario's frequency x consequence: 5,000 + 2,000 + 1,000 = $8,000 per year. $522,000 adds consequences without frequency.
4	A	A Safety Case is a structured, evidence-supported argument that a system is acceptably safe for a given application and environment (used in offshore, nuclear, rail, defense).
5	D	Event tree analysis is inductive: it starts with an initiating event and maps success/failure paths of safeguards to outcomes. FTA works top-down from an undesired event.
6	A	Fault tree analysis uses Boolean AND/OR gates to show combinations of basic events that produce a top event; it is deductive (top-down).
7	C	FMEA is inductive (bottom-up): it starts with individual component failure modes and determines their effects on the system. FTA is the deductive opposite.
8	D	Breaking a process into nodes and applying guide words (more, less, none, reverse) to parameters like flow, pressure and temperature is HAZOP.
9	C	Fail-passive design: on failure the system goes to its lowest-energy/inoperative state without creating an adverse effect (e.g., a circuit breaker opening). Fail-active/operational designs keep functioning.
10	D	ETBA identifies energy sources in a system and evaluates the adequacy of barriers between those energies and people or property.

DOMAIN 2 · COMPETENCY 2.7

Leading and Lagging Indicators

CSP11 blueprint: Evaluate leading and lagging indicators

1. Definition

Lagging indicators measure outcomes after harm or loss has occurred (injuries, illnesses, lost days, costs). Leading indicators measure the activities, conditions and behaviors that predict and prevent future harm (inspections completed, hazards corrected, training, near-miss reports, observations).

2. Explanation in simple words

Lagging indicators look in the rear-view mirror - they tell you how many people already got hurt. Leading indicators look through the windshield - they tell you whether you are doing the things that prevent people from getting hurt.

3. Key points to remember

- Lagging examples: TRIR, DART rate, LTIR, severity rate, fatalities, workers' compensation costs, EMR, property damage.
- Leading examples: % inspections completed on time, % corrective actions closed by due date, near-miss reports, safety observations, training completion, PTW/JHA quality audits, management walk-arounds, perception survey scores.
- Formulas: Incidence rate = (number of cases × 200,000) ÷ hours worked; 200,000 = 100 full-time workers × 40 h × 50 weeks. Severity rate = (days lost × 200,000) ÷ hours.
- Good leading indicators are linked to specific risks, actionable, measurable, timely and validated against outcomes; avoid metrics that can be 'gamed'.
- Use both: leading to drive prevention, lagging to verify results and benchmark. TRIR alone hides low-frequency, high-severity (SIF) risks.
- Interpreting changes: rising near-miss reports after a reporting campaign usually indicates better reporting; a sudden drop may indicate underreporting.

4. Real work scenario

> A construction firm with a TRIR of 1.2 had a fatality from a trench collapse. Its TRIR had looked good for years because minor injuries were rare. It adds leading indicators focused on serious-injury potential: daily trench inspections completed, % of excavations with protective systems verified, and critical-lift plans reviewed.

5. CSP exam tips and traps

- Indicator classification questions are common - ask 'did harm already happen?' If yes, lagging.
- Calculation: 14 recordables in 350,000 hours → (14 × 200,000) ÷ 350,000 = 8.0.
- Lagging indicators are useful for benchmarking; leading indicators guide proactive action.

6. Every blueprint point (2 blueprint items + 3 more)

Leading indicators BLUEPRINT

Proactive, predictive measures of activities, behaviors or conditions that prevent incidents and show whether safety systems are working.

- Examples: training sessions completed, % corrective actions closed on time, near-miss and hazard reports, safety observations/BBS checks, toolbox talk completion, inspections done, PM completed on time.
- Strengths: early warning, encourage prevention and participation.
- Limitations: harder to define and standardize; need a reporting culture.
- Sources: ANSI/ASSP Z16.1, NSC leading indicator guidance, OSHA.

> **Exam tip:** Number of near-miss reports is LEADING; number of injuries is LAGGING. Rising near-miss reporting is usually good news.

Lagging indicators BLUEPRINT

Reactive, outcome-based measures of harm or loss that has already occurred.

- Examples: TRIR, DART rate, lost-time injury rate, severity rate, workers' comp claims and costs, citations, property damage.
- TRIR = recordable cases x 200,000 / hours worked. Example: 12 cases, 480,000 h -> 5.0.
- DART rate = DART cases x 200,000 / hours. Example: 6 cases, 480,000 h -> 2.5.
- Severity rate = lost workdays x 200,000 / hours. Example: 90 days, 480,000 h -> 37.5.
- 200,000 = 100 full-time workers x 40 h x 50 weeks.
- Limitations: show only past failures, don't say what to fix, can be masked by under-reporting.

> **Exam tip:** 'Managing by the accident book' (lagging only) is a weak program. Use both types.

Selecting and evaluating indicators

Indicators should align with key risks and be balanced between leading and lagging.

- Choose metrics tied to your significant risks; set baselines and targets (OSHA/NSC data).
- Track trends - do increases in leading activity precede decreases in lagging outcomes?
- Communicate and act on results; re-assess indicators as performance improves.
- Good indicator: valid, reliable, actionable, understood, not easily manipulated.

Experience Modification Rate (EMR)

EMR is a workers' compensation rating comparing an employer's actual losses to expected losses for its industry and payroll.

- EMR = actual losses / expected losses (simplified); 1.0 = industry average.
- EMR 0.80 -> premium about 20% below manual; EMR 1.25 -> 25% above.
- Based on about 3 prior policy years (excluding the most recent year); often used to prequalify contractors.
- It is a lagging indicator.

> **Exam tip:** EMR < 1.0 is better than average.

Pitfalls of indicators

Poorly designed metrics drive the wrong behavior.

- Injury-free incentive programs can suppress reporting.
- Counting activities (e.g., number of audits) without measuring quality.
- Using lagging data alone for small sites - too few events to be statistically meaningful.

Read more: Yates, Safety Professional's Reference and Study Guide (4th ed.): Ch. 3 Recording and Reporting Occupational Injuries, p. 44; Ch. 36 Measuring Safety and Health Performance, p. 797

7. Ten CSP-style practice questions

Answer all ten on paper first, then check the key. Score: _____ / 10 Date: _____ Retest: _____ / 10

2.7-Q1. During a quarterly safety review, a site had 5 recordable injuries with 250,000 work hours. What is the Total Recordable Incident Rate (TRIR)?

A. 3.2
B. 4.0
C. 2.5
D. 1.6

2.7-Q2. A company records 11 DART cases in 452,000 hours worked. DART rate?

A. 3.4
B. 4.9
C. 7.8
D. 11.0

2.7-Q3. A site recorded 8 OSHA recordable cases in 400,000 hours: 3 involved days away from work, 2 involved restricted duty/job transfer, and 3 were medical treatment only. What is the DART rate?

A. 1.5
B. 2.5
C. 4.0
D. 1.0

2.7-Q4. During a monthly safety review, a supervisor presents data showing a drop in near-miss reports compared to the previous quarter. What type of indicator is this, and how should it be interpreted?

A. Leading indicator; it shows workers are becoming less engaged in hazard reporting
B. Lagging indicator; it confirms fewer injuries occurred
C. Leading indicator; it may reflect underreporting or lack of hazard awareness
D. Lagging indicator; it demonstrates completed corrective actions

2.7-Q5. A construction company uses its Total Recordable Incident Rate (TRIR) as the sole measure of safety program success. What is the main limitation of relying only on this lagging indicator?

A. It tracks leading behaviors, not outcomes
B. It requires extensive job hazard analysis
C. It provides real-time predictive insight
D. It only reflects past injuries and may ignore hidden risks

2.7-Q6. A general contractor is evaluating the safety performance of subcontractors. Which of the following is the BEST combination of leading and lagging indicators?

A. Pre-task hazard analyses completed and number of recordable cases
B. Lost-time injuries and workers' compensation costs
C. Number of OSHA citations and experience modification rate
D. Days without injury and safety posters displayed

2.7-Q7. Which of the following is the most effective leading performance indicator?

A. Percentage of near-miss reports closed with action verification
B. Annual OSHA recordable rate
C. Workers' compensation claim cost
D. Lost-time injury frequency rate

2.7-Q8. Which key question should management ask to verify KPI effectiveness?

A. Can the KPI be compared to competitors?
B. Is the KPI simple to calculate?
C. Does the KPI align with organizational goals and drive desired behavior?
D. Does the KPI reduce workload for data teams?

2.7-Q9. A construction company has experienced a rise in near-miss incidents over the past quarter, though no recordable injuries have occurred. The safety manager decides to start tracking the number of completed pre-task hazard analyses (PTHAs), the frequency of supervisor safety observations, and attendance at toolbox talks each week. Which Of the following best describes the type Of safety performance measurement being implemented?

A. Lagging indicators that evaluate past incidents and injuries
B. Predictive analytics that estimate total recordable incident rates (TRIR) based on prior data
C. Financial performance metrics that track the cost of safety incidents
D. Leading indicators that measure proactive actions taken to prevent incidents

2.7-Q10. Your company wants to get to an annual incident rate of 0.5 recordable injuries. This indicator is a:

A. Lagging indicator
B. Occupational indicator
C. Leading indicator
D. Cumulative indicator

Answer key and explanations

Q	Ans	Why
1	B	TRIR = 5 x 200,000 / 250,000 = 4.0. The 200,000 represents 100 full-time workers x 2,000 hours.
2	B	DART = 11 x 200,000 / 452,000 = 4.87, about 4.9 (OSHA 1904 rate formula).
3	B	DART counts only days-away plus restricted/transfer cases: (3 + 2) x 200,000 / 400,000 = 2.5. Trap: using all 8 cases gives the TRIR (4.0), and days-away only gives 1.5.
4	C	Near-miss reporting is a leading indicator; a drop may mean underreporting or lower hazard awareness rather than a safer workplace, so investigate before celebrating.
5	D	TRIR is lagging: it reflects past injuries, is statistically unstable at small hours, and can hide high-potential hazards and underreporting.
6	A	Pre-task hazard analyses completed (leading) paired with recordable cases (lagging) gives a meaningful balanced view. The other pairs are weak, mostly one type, or not meaningful measures.
7	A	Percentage of near-miss reports closed with verified action measures proactive hazard control and follow-through. The other options are lagging outcomes.
8	C	A KPI is effective only if it aligns with organizational goals and drives the desired behavior; ease of calculation or comparability are secondary.
9	D	PTHAs completed, supervisor observations and toolbox talk attendance measure proactive activities before incidents: leading indicators.
10	A	A target incident rate is still an outcome measure based on injuries that already occurred: a lagging indicator.

DOMAIN 2 · COMPETENCY 2.8

EHS Management and Audit Systems

CSP11 blueprint: Recognize safety, health, and environmental management and audit systems (e.g., ISO 14000 series, 45001, 19011, ANSI Z10)

1. Definition

Management systems are structured frameworks that integrate EHS into business processes using the Plan-Do-Check-Act cycle. ISO 45001 (OH&S), ISO 14001 (environmental), ANSI/ASSP Z10 (OHSMS, U.S.) and ISO 19011 (guidelines for auditing management systems) are the principal standards.

2. Explanation in simple words

A management system is the company's 'operating system' for safety and environment: set policy, plan, do the work, check results, and keep improving. Audits check whether the system is in place and working.

3. Key points to remember

- ISO 45001 (2018): Annex SL high-level structure - context, leadership and worker participation, planning (hazards, risks, opportunities, legal requirements, objectives), support (resources, competence, communication, documented information), operation (hierarchy of controls, MOC, procurement, contractors, emergency), performance evaluation (monitoring, internal audit, management review), improvement (incidents, nonconformity, corrective action, continual improvement). Requires consultation and participation of non-managerial workers.
- ISO 14001: environmental aspects (causes, e.g., solvent use) and impacts (effects, e.g., air pollution), compliance obligations, life-cycle perspective. ISO 14000 family includes 14031 (performance evaluation), 14040/44 (life-cycle assessment), 14064 (greenhouse gases).
- ANSI/ASSP Z10: U.S. OHSMS standard; hierarchy of controls includes 'warnings' between engineering and administrative controls; emphasizes management leadership and employee participation.
- ISO 19011: guidelines for auditing management systems. Principles: integrity, fair presentation, due professional care, confidentiality, independence, evidence-based approach, risk-based approach. Audit program → plan → opening meeting → collect and verify evidence → findings → closing meeting → report → follow-up.
- Other: ISO 9001 (quality), ISO 31000 (risk management), ISO 45003 (psychological health and safety at work), ISO 50001 (energy), ISO 27001 (information security), OSHA VPP, OHSAS 18001 (withdrawn, replaced by 45001).
- Audit types: first-party (internal), second-party (customer/supplier), third-party (certification). Findings: major/minor nonconformity, observation, opportunity for improvement.

4. Real work scenario

> A manufacturer pursues ISO 45001 certification. The gap assessment shows strong procedures but no formal process for non-managerial worker participation in hazard identification. The company forms worker-led JHA teams and a safety committee with elected members, and documents their input before the certification audit.

5. CSP exam tips and traps

- Aspect = cause; impact = effect (ISO 14001).
- ISO 19011 is about how to audit; it is not a certifiable management system standard.

- Auditors should not audit their own work (independence).
- Management review is part of 'Check/Act'; internal audit is 'Check'.

6. Every blueprint point (4 blueprint items + 4 more)

ISO 14000 series BLUEPRINT

ISO 14001:2015 is the certifiable environmental management system (EMS) standard; the wider 14000 family supports it with guidance on performance, life-cycle and greenhouse gases.

- ISO 14001 focuses on environmental aspects and impacts (emissions, discharges, waste, resource use, biodiversity), compliance obligations, and pollution prevention.
- Same high-level (harmonized) structure as ISO 45001 and ISO 9001: clauses 4 Context, 5 Leadership, 6 Planning, 7 Support, 8 Operation, 9 Performance evaluation, 10 Improvement.
- ISO 14004 = EMS implementation guidance; ISO 14031 = environmental performance evaluation; ISO 14040/14044 = life-cycle assessment; ISO 14064 = GHG accounting.
- Certification is voluntary, by third-party registrars.

> **Exam tip:** Aspect = the activity element (solvent use); impact = the change to the environment (air pollution).

ISO 45001 BLUEPRINT

ISO 45001:2018 is the first international certifiable occupational health and safety management system standard; it replaced OHSAS 18001 (withdrawn 2021).

- Clauses: 1 Scope, 2 Normative references, 3 Terms, 4 Context of the organization, 5 Leadership and worker participation, 6 Planning, 7 Support, 8 Operation, 9 Performance evaluation, 10 Improvement.
- Core principles: leadership commitment, worker consultation and participation (including non-managerial workers), risk-based thinking, context, PDCA continual improvement.
- Clause 8 covers hierarchy of controls, MOC, procurement/contractors and emergency preparedness.
- Certification is voluntary but often required for contracts and supply chains.

> **Exam tip:** Choose ISO 45001 when international recognition or certification is required.

ISO 19011 BLUEPRINT

ISO 19011:2018 gives guidelines for auditing any management system (45001, 14001, 9001); it is guidance, not certifiable.

- Audit principles: integrity, fair presentation, due professional care, confidentiality, independence, evidence-based approach, risk-based approach.
- Covers managing an audit program, conducting audits and auditor competence (knowledge, skills, behavior).
- Audit steps: initiate (objectives, scope, criteria) -> prepare (review documents, audit plan) -> conduct (opening meeting, interviews, observation, sampling, evidence) -> report (closing meeting, major/minor nonconformities, opportunities) -> follow-up (verify corrective action effectiveness).
- Close findings only when evidence of real improvement is confirmed.

> **Exam tip:** Auditor independence: auditors should not audit their own work.

ANSI Z10 BLUEPRINT

ANSI/ASSP Z10.0-2019 is the U.S. consensus OHSMS standard developed by ANSI and ASSP (first published 2005). It is a guidance standard, not a certification scheme.

- Structure (PDCA): management leadership and employee participation; planning; implementation and operation; evaluation and corrective action; management review.
- Strong emphasis on prevention through design, employee participation and integration with business processes.
- U.S. legal context (references OSHA and NIOSH); flexible language for any size organization.
- Compared with ISO 45001: both PDCA and require worker participation; 45001 is certifiable and global, Z10 is not certifiable and U.S.-focused.

> **Exam tip:** U.S. company wanting a practical, flexible framework without certification -> Z10; international contracts -> ISO 45001.

Safety management system (SMS) basics

An SMS is a formal, organization-wide framework of interrelated elements to identify, assess, control and review OH&S risks - an integrated, dynamic approach, not a manual on a shelf.

- Key functions: hazard identification, risk assessment and control, regulatory compliance, promotion of safety culture.
- Effective SMS: visible leadership, active worker participation, defined accountability, integration with business operations, measurement and continual improvement.
- Reasons: legal compliance, risk reduction, continual improvement, reputation/competitive advantage, moral responsibility.
- Hazard = source of potential harm; risk = likelihood x severity of harm.

> **Exam tip:** An SMS is most effective when supported by top management and integrated into business processes; otherwise it becomes a 'paper system'.

OSHA Recommended Practices (SHMS)

OSHA's Recommended Practices for Safety and Health Programs (2016) are voluntary best-practice guidance with seven core elements.

- 7 elements: management leadership; worker participation; hazard identification and assessment; hazard prevention and control; education and training; program evaluation and improvement; communication and coordination for host employers, contractors and staffing agencies.
- OSHA cannot cite an employer simply for lacking a comprehensive program.
- Specific OSHA standards remain mandatory.

> **Exam tip:** Voluntary guidance vs mandatory standard is a common distractor.

Integrated management systems (IMS)

Because ISO 9001, 14001 and 45001 share a common high-level structure, organizations often combine them into one IMS.

- Shared policy, unified procedures (incident reporting, audits, corrective action), combined audits, joint management reviews.
- Benefits: saves time and resources, less duplication, simpler documentation, consistent compliance, aligned safety and environmental goals.
- Example: one solvent-reduction objective cuts emissions and worker exposure.

Types of audits

Audits are classified by who performs them.

- First-party (internal): organization audits itself - self-assessment, preparation.
- Second-party: customer (or agent) audits a supplier or contractor.
- Third-party: independent certification body (registrar) - leads to certification.
- Compliance audit (against regulations) vs management system audit (against a standard) vs program audit.
- Nonconformities: major (system failure/absence of requirement) vs minor (isolated lapse).

> **Exam tip:** Customer auditing its supplier = second-party.

Read more: Yates, Safety Professional's Reference and Study Guide (4th ed.): Ch. 31 Safety Management System, p. 717; Ch. 37 Safety Program Auditing Techniques, p. 818; Ch. 34 Environmental Management

7. Ten CSP-style practice questions

Answer all ten on paper first, then check the key. Score: _____ / 10 Date: _____ Retest: _____ / 10

2.8-Q1. Following ISO 19011 guidelines. During the preparation the week before, you notice that the client has not clearly defined the audit objectives, scope, and criteria, and the audit team is unsure about which standard requirements apply to the facility being audited. According to ISO 19011, what should you do next?

A. Contact the auditee immediately to obtain and confirm the missing information before finalizing the audit plan.
B. Proceed with drafting the audit schedule using assumptions based on past audits and update it during the opening meeting.
C. Wait until the on-site audit begins, then clarify scope and objectives during the opening meeting with the client.
D. Cancel the audit and submit a report stating that the organization is not audit-ready per ISO 19011 clause 6.3.2.

2.8-Q2. An internal OHSMS audit team is being formed. The maintenance supervisor volunteers to audit the maintenance department's LOTO program, which he wrote. Under ISO 19011, what is the concern?

A. He lacks technical knowledge
B. Independence/objectivity - auditors should not audit their own work
C. Audits must be done only by external auditors
D. There is no concern

2.8-Q3. According to ISO 19011, which principle underpins all auditing?

A. Completeness
B. Confidentiality
C. Evidence-based approach
D. Leniency toward management

2.8-Q4. Which of the following is a required input for management review under ISO 45001?

A. Employee vacation schedules
B. Budget for entertainment
C. Departmental profit margins
D. Results of audits, performance evaluations, and incidents

2.8-Q5. An internal audit follow-up finds the same nonconformity from a prior cycle. This indicates:

A. Improved performance
B. Ineffective corrective action or lack of verification
C. Strong audit planning
D. Expanded system coverage

2.8-Q6. Under ISO 14001, an 'environmental aspect' is BEST described as:

A. An element of the organization's activities, products or services that can interact with the environment
B. Any change to the environment resulting from the organization
C. A legal requirement related to emissions
D. An environmental audit finding

2.8-Q7. An Integrated Management System (IMS) typically combines which ISO standards for unified auditing and documentation?

A. ISO 22000, ISO 22301, and ISO 26000
B. ISO 31000, ISO 27001, and ISO 50001
C. ISO 9001, ISO 14001, and ISO 45001
D. ISO 10002, ISO 21500, and ISO 37001

2.8-Q8. Aside from OSHA's safety and health program guidelines, which consensus standards BEST establish occupational safety and health management system approaches?

A. Lean manufacturing and ISO 14001
B. ISO 9001 and ISO 19011
C. ISO 9001 and Lean manufacturing
D. ISO 45001 and ANSI/ASSP Z10.0

2.8-Q9. Which of the following is a requirement for an organization implementing ISO 45001?

A. The organization must use a specific set of tools for risk management as prescribed by ISO.
B. The organization must ensure that workers from external providers are trained in the internal safety practices.
C. The organization must commit to eliminating hazards and minimizing OH&S risks, including the continual improvement of its OH&S management system.
D. All members of the organization must be certified individually in OH&S practices.

2.8-Q10. In a management system audit, nonconformity refers to:

A. An unsafe act by an employee
B. Failure to meet a specified requirement
C. An incomplete checklist
D. A pending corrective action

Answer key and explanations

Q	Ans	Why
1	A	ISO 19011 requires audit objectives, scope and criteria to be determined before the audit plan is finalized; contact the auditee/client to confirm them. Assuming or waiting compromises the audit.
2	B	ISO 19011 principles include independence: auditors should not audit their own work, to remain free from bias and conflict of interest.
3	C	The evidence-based approach is the rational method for reaching reliable, reproducible audit conclusions from verifiable evidence.
4	D	ISO 45001 clause 9.3 management review inputs include audit results, incidents, objectives, performance trends and opportunities for improvement.
5	B	A repeat nonconformity indicates that the earlier corrective action did not address the root cause or was not verified for effectiveness.
6	A	Aspect = an element of activities, products or services that can interact with the environment (the cause); impact = the resulting change to the environment.
7	C	Integrated management systems typically combine ISO 9001 (quality), ISO 14001 (environment) and ISO 45001 (OH&S), which share the same high-level structure and PDCA basis.
8	D	ISO 45001 and ANSI/ASSP Z10.0 are the principal consensus standards for occupational health and safety management systems. ISO 9001 is quality, ISO 14001 environmental, ISO 19011 audit guidance, and Lean a production method.
9	C	ISO 45001 policy must include commitments to eliminate hazards and reduce OH&S risks and to continual improvement; it does not prescribe specific tools or individual certification.
10	B	A nonconformity is non-fulfillment of a requirement (standard, procedure, legal or organizational requirement).

DOMAIN 2 · COMPETENCY 2.9

Plans, Systems and Policies

CSP11 blueprint: Describe required components for plans, systems, and policies (e.g., safety, health, and environmental regulations and standards)

1. Definition

Written plans, programs and policies document how an organization meets regulatory and consensus requirements and controls its risks. Many OSHA and EPA standards require specific written programs with defined elements.

2. Explanation in simple words

A policy says what the company commits to; a program or plan says how it will do it, who is responsible and how it is checked. Regulators ask, 'Show me it is written, trained and working.'

3. Key points to remember

- Safety and health policy: signed top-management commitment, scope, responsibilities, compliance, continual improvement, worker participation.
- Core program elements (OSHA Recommended Practices for Safety and Health Programs, 2016): management leadership, worker participation, hazard identification and assessment, hazard prevention and control, education and training, program evaluation and improvement, communication and coordination for host employers, contractors and staffing agencies.
- Examples of required written programs: Hazard Communication (1910.1200), Emergency Action Plan (1910.38 - may be oral for 10 or fewer employees), Fire Prevention Plan (1910.39), Respiratory Protection (1910.134), Energy Control/LOTO (1910.147), Permit-Required Confined Space (1910.146), Bloodborne Pathogens Exposure Control Plan (1910.1030, reviewed annually), Hearing Conservation (1910.95), PSM (1910.119); EPA SPCC (40 CFR 112), RMP, stormwater SWPPP.
- Typical program structure: purpose, scope, definitions, responsibilities, procedures, training, inspections/audits, recordkeeping, program review.
- Multi-employer worksites: OSHA citation policy - creating, exposing, correcting and controlling employers; host and contractor coordination.
- Legal hierarchy: OSH Act (General Duty Clause 5(a)(1)) → OSHA standards (vertical/specific vs horizontal/general) → consensus standards incorporated by reference → company rules.

4. Real work scenario

> A host employer brings in a contractor to repair a tank. The plans required are the contractor's confined-space program, the host's permit system and hazard information exchange, a site-specific safety plan defining responsibilities, and the emergency/rescue plan. Both employers sign the entry permit and coordinate rescue.

5. CSP exam tips and traps

- If a scenario asks what is missing from a program, look for the required element (e.g., employee accountability in an EAP, worker participation in an SMS).
- A policy without signed management commitment is weak on legal defensibility.
- Know who is the controlling employer on multi-employer sites (the general contractor with authority over the site).

6. Every blueprint point (1 blueprint items + 5 more)

Plans, systems and policies BLUEPRINT

A policy states commitment and direction; plans translate the policy into specific objectives, actions and resources; systems (procedures, programs, controls) carry the plan out day to day.

- Hierarchy: policy (why, signed by top management) -> plans/programs (what, who, when) -> procedures/SOPs (how) -> records (proof).
- Policy = concise statement of intent; procedure = specified step-by-step way to do an activity.
- Plans and policies must be based on applicable regulations and standards (OSHA, EPA, NFPA, ANSI, ISO).
- Review regularly and after significant change or incidents.

> **Exam tip:** A strong plan is SMART, reviewed and ties policy to action; a weak plan is vague with unclear roles and no measurement.

Essential elements of an SH&E policy

A credible policy is short, signed by top management, communicated to all and reviewed.

- Statement of intent/commitment to SH&E.
- Assignment of roles and responsibilities (accountability).
- Risk management approach (hazard ID, hierarchy of controls).
- Commitment to comply with legal requirements and to continual improvement.
- Commitment to communication and worker consultation.
- ISO 45001 5.2 also requires commitments to eliminate hazards/reduce risks and to worker participation.

> **Exam tip:** Policy signed by the CEO/top management shows commitment - the safety manager signing alone is weaker.

Components of an SH&E plan

The plan provides the roadmap from policy to results.

- SMART goals and objectives.
- Legal compliance list (OSHA, EPA, consensus standards).
- Hazard identification methods (walkthroughs, JHA, investigations, worker reports, audits).
- Training schedules; emergency plans; audit and inspection checklists.
- Record retention schedule (what, how long, where).

Core SMS elements in practice

Leadership, planning, implementation, evaluation and continual improvement must all be present.

- Leadership: signs policy, allocates budget and staff, chairs safety committee, responds to audit findings.
- Planning: hazard ID, risk matrices, SMART objectives, legal requirements.
- Implementation: roles, training and competence, operational controls (SOPs, contractors, safe purchasing), communication.
- Evaluation: leading/lagging measures, incident investigation, audits, management review at least annually.
- If incidents continue despite an 'SMS', look for the missing element: leadership, participation, risk assessment or effective corrective action.

> **Exam tip:** Strong SMS: proactive, resourced, data-driven, management-led, employee-involved. Weak: reactive, underfunded, compliance-only.

OSHA mandatory written programs

Although overall SHMS is voluntary, many OSHA standards require written programs and training.

- Written programs: Hazard Communication (1910.1200), energy control/LOTO (1910.147), respiratory protection (1910.134), permit-required confined spaces (1910.146), PSM (1910.119), emergency action and fire prevention plans (1910.38/.39; may be oral if 10 or fewer employees), bloodborne pathogens exposure control plan (reviewed annually).
- PPE: written certification of hazard assessment (1910.132(d)); PPE provided at no cost.
- Training examples: forklift operator evaluation at least every 3 years; BBP annual; LOTO for authorized, affected and other employees.
- General Duty Clause 5(a)(1): workplace free from recognized hazards likely to cause death or serious harm (used for heat, workplace violence, etc.).

OSHA recordkeeping and reporting (29 CFR 1904)

Employers record work-related injuries and illnesses and report severe events.

- Forms: 300 log, 300A annual summary, 301 incident report (within 7 calendar days of learning of the case).
- Report fatality within 8 hours; in-patient hospitalization, amputation or loss of an eye within 24 hours.
- Post 300A from February 1 to April 30; signed by a company executive.
- Retain 5 years following the end of the calendar year covered.
- Partial exemption: 10 or fewer employees, and certain low-hazard industries.

> **Exam tip:** Fatality = 8 hours; hospitalization/amputation/eye = 24 hours.

Read more: Yates, Safety Professional's Reference and Study Guide (4th ed.): Ch. 4 Regulations, p. 55; Ch. 31 Safety Management System, p. 717

7. Ten CSP-style practice questions

Answer all ten on paper first, then check the key. Score: _____ / 10 Date: _____ Retest: _____ / 10

2.9-Q1. While reviewing a corporate safety policy, the regional director identifies a missing component that could compromise legal defensibility and leadership accountability. Which of the following should be included in the policy?

A. A written commitment from top management affirming the company's dedication to safety and health
B. A rotating schedule for team-building exercises among departments
C. A training calendar with optional safety workshops for interested employees
D. A suggestion to include safety data sheets in the employee handbook

2.9-Q2. Which option lists the key elements of an effective health and safety performance management system?

A. Measuring performance, reviewing performance, and auditing near-misses
B. Auditing and review, policy, organizing, planning and implementation, and measuring performance
C. Policy, organizing, planning and implementation, and measuring performance
D. Planning and organization, policy, and review performance

2.9-Q3. During a roofing project, a subcontractor's employee falls through an unguarded skylight. The general contractor (GC) did not create the hazard but had authority over the site and conducted weekly inspections. Under OSHA's multi-employer policy, what role does the GC have?

A. Creating employer
B. Correcting employer
C. Controlling employer
D. Exposing employer

2.9-Q4. The primary purpose of a contractor prequalification process is to:

A. Reduce total project cost
B. Verify contractor insurance coverage only
C. Screen contractors for trade union membership
D. Evaluate contractor safety performance and capability

2.9-Q5. Which of the following should be included in a contractor safety orientation?

A. Company history and financial goals
B. Site hazards, emergency procedures, and reporting expectations
C. Vendor payment process
D. Individual employee HR policies

2.9-Q6. According to OSHA, a competent person is knowledgeable about applicable standards and can identify workplace hazards relating to the specific operation. In addition, OSHA expects which of the following from a competent person?

A. Has the authority to correct hazards immediately
B. Has the budget funds to eliminate the hazard
C. Is an officer of the company
D. Can hire, fire, or effectively recommend a competent person

2.9-Q7. Under OSHA's Recommended Practices for Safety and Health Programs and ISO 45001, employees should be:

A. Told only about major incidents
B. Excluded from EHS planning decisions
C. Consulted and allowed participation in hazard identification and program review
D. Notified of changes after implementation

2.9-Q8. Site-Specific Safety Plan (SSSP) Development A general contractor is preparing a Site-Specific Safety Plan for a new commercial build. The safety professional is asked to define scope, site conditions, and subcontractor safety responsibilities. Which section of the SSSP should this information be included in?

A. Emergency response plan section
B. Project-specific hazard analysis
C. Recordkeeping and documentation
D. Project overview and responsibility assignment

2.9-Q9. While reviewing a subcontractor's safety program, you find that it only partially addresses fall protection requirements. The subcontractor argues it is sufficient based on past performance. What should you do?

A. Approve the program because the subcontractor has a good safety record.
B. Require the program to be revised to fully align with OSHA standards before approving it.
C. Modify the program yourself to meet the requirements and then approve it.
D. Allow the program as is and monitor performance during the project.

2.9-Q10. A chemical processing plant stores large quantities of flammable liquids under pressure. Which of the following best describes the primary objective of a Process Hazard Analysis (PHA)?

A. To identify, evaluate, and control hazards associated with highly hazardous processes
B. To document safety performance indicators for quarterly OSHA reporting
C. To determine the financial impact of loss-of-containment scenarios
D. To verify the competency of emergency response contractors

Answer key and explanations

Q	Ans	Why
1	A	A safety and health policy must include a written commitment from top management; it establishes leadership accountability and supports legal defensibility.
2	B	The HSG65 model (POPMAR): Policy, Organizing, Planning and implementing, Measuring performance, and Auditing and reviewing. Options missing auditing/review are incomplete.
3	C	The GC had general supervisory authority over the site and the ability to correct hazards, which makes it the controlling employer under OSHA's multi-employer citation policy, even though it did not create the hazard.
4	D	Contractor prequalification evaluates safety performance and capability (rates, EMR, programs, training) before award; insurance is only one part.
5	B	Contractor orientation should cover site-specific hazards, emergency procedures and reporting expectations before work begins.
6	A	OSHA defines a competent person as one who can identify existing and predictable hazards and has authorization to take prompt corrective measures to eliminate them.
7	C	Worker participation (OSHA Recommended Practices; ISO 45001 clause 5.4) means employees are consulted and participate in hazard identification and program review, not just informed after the fact.
8	D	Project scope, site conditions and subcontractor responsibilities belong in the project overview and responsibility-assignment section of a site-specific safety plan.
9	B	The host/GC should require the subcontractor's program to be revised to meet OSHA requirements before approval. Past performance does not substitute for compliance, and rewriting it yourself shifts responsibility.
10	A	A PHA under PSM (1910.119(e)) systematically identifies, evaluates and controls hazards of highly hazardous chemical processes.

DOMAIN 2 · COMPETENCY 2.10

Document Retention and Management

CSP11 blueprint: Utilize document retention or management principles (e.g., incident investigation, training records, exposure records, maintenance records, environmental management system, audit results, privacy, trade secrets, personal information)

1. Definition

Document management ensures EHS records are created, controlled, protected, retrievable and kept for the required time, while respecting privacy, confidentiality and trade-secret requirements. Records provide legal evidence, support analysis and demonstrate compliance.

2. Explanation in simple words

If it is not written down, it did not happen. Keep the right records for the right number of years, protect personal and medical information, and make sure you can find the latest version quickly.

3. Key points to remember

- OSHA 300/300A/301 injury and illness records: 5 years after the end of the calendar year; 300A posted Feb 1 - Apr 30.
- Employee exposure and medical records (1910.1020): exposure records 30 years; medical records duration of employment + 30 years; employees and representatives have access rights (usually within 15 working days).
- Other examples: PSM incident investigation reports 5 years; PHA for life of the process; confined-space cancelled permits 1 year; noise exposure measurements 2 years and audiograms duration of employment; respirator fit tests until the next test; bloodborne pathogens training 3 years, medical records employment + 30 years; RCRA hazardous waste manifests 3 years; crane inspection records; forklift training certification.
- Privacy: privacy concern cases on the 300 Log (sexual assault, mental illness, HIV/hepatitis/TB, needlestick from contaminated sharps, others on request) - write 'privacy case' instead of the name. Medical records kept confidential and separate from personnel files; HIPAA (healthcare), GDPR (EU personal data), PII protection (e.g., SSN, financial data).
- Trade secrets: HCS allows withholding a chemical's specific identity on the SDS, but it must be disclosed immediately to a treating health professional in an emergency (confidentiality agreement afterwards); non-emergency requests in writing.
- Document control: version control, approval, distribution, obsolete document removal, backups (cloud, off-site), access control, retention schedule, legal hold during litigation. Never alter or backdate records; if data are missing, redo the measurement and document the failure.
- Records are discoverable in litigation; audit findings that are documented must be acted on - known but uncorrected findings increase liability.

4. Real work scenario

> A noise dosimeter fails during a compliance survey just before a deadline. Instead of estimating readings, the safety professional documents the equipment failure, re-samples the next day and files both records. Months later an attorney requests the records in a hearing-loss claim, and the complete, honest documentation supports the company.

5. CSP exam tips and traps

- 5 years (OSHA logs) vs 30 years (exposure) vs employment + 30 years (medical) is a classic exam trio.
- Ethics plus records: never falsify or 'adjust' data.
- Trade secret + medical emergency = disclose immediately to the treating professional.

6. Every blueprint point (9 blueprint items + 1 more)

Incident records BLUEPRINT

Injury and illness records document what happened and support trend analysis and compliance.

- OSHA 300, 300A, 301: keep 5 years following the end of the calendar year; update the 300 log during the 5 years.
- Employee or representative may request 300 log copy - provide by end of next business day.
- Investigation reports: keep per company policy; PSM incident investigation reports kept 5 years.
- Electronic submission (1904.41): establishments with 20-249 employees in designated industries submit 300A; 100+ employees in high-hazard industries also submit 300 and 301 data.

> **Exam tip:** 300 logs = 5 years; exposure records = 30 years - don't mix them up.

Training records BLUEPRINT

Training records prove competence and compliance; retention periods vary by standard.

- Record: name, date, content/topic, trainer, method of evaluation.
- Bloodborne pathogens training records: 3 years from training date.
- Asbestos training: 1 year beyond last date of employment.
- Forklift: certification of training and evaluation; evaluate every 3 years.
- Many standards require 'certification' records (LOTO periodic inspection, PRCS) - keep at least as long as the standard specifies; keep longer by policy for defense.

Exposure records BLUEPRINT

Employee exposure records (monitoring results, biological monitoring, SDS/chemical identity data) must be kept long-term because occupational diseases have long latency (29 CFR 1910.1020).

- Exposure records: at least 30 years.
- Medical records: duration of employment plus 30 years (except employees employed less than 1 year who are given their records at termination).
- Background data (lab reports, field notes) may be kept 1 year if sampling results, methods and summary are kept 30 years.
- SDSs need not be kept if a record of chemical identity, where and when used is kept for 30 years (the guide states SDSs must be kept 30 years).
- Noise: exposure measurements 2 years; audiograms for duration of employment (1910.95(m)).
- Access: employees and designated representatives within 15 working days.

> **Exam tip:** 1910.1020 = 30 years exposure; employment + 30 years medical.

Maintenance records BLUEPRINT

Maintenance and inspection records prove equipment is safe and support mechanical integrity, warranty and investigation.

- PSM mechanical integrity: document each inspection and test (date, inspector, equipment ID, test performed, results).
- Cranes: monthly inspection records of critical items (hooks, ropes) kept; certification records.
- Fire extinguishers: annual maintenance record kept 1 year after last entry or life of shell, whichever is less (1910.157).
- LOTO periodic inspection certified at least annually.

Environmental records BLUEPRINT

Environmental permits and regulations specify record contents and retention.

- RCRA: hazardous waste manifests, biennial reports and waste analyses - at least 3 years.
- Clean Water Act NPDES monitoring records - at least 3 years; SPCC inspection records - 3 years.
- Clean Air Act Title V monitoring records - 5 years.
- EPCRA Tier II reports filed annually by March 1; TRI (Form R) by July 1.
- Retention clocks are extended during unresolved enforcement actions.

Audit results BLUEPRINT

Audit reports, findings and corrective-action closure records demonstrate continual improvement and due diligence.

- PSM compliance audits at least every 3 years; keep the two most recent audit reports.
- ISO 45001/14001 require retained documented information as evidence of audit programs and results.
- Track each finding to closure with verification evidence.
- Consider attorney-client privilege/self-audit policies (EPA Audit Policy encourages self-disclosure).

> **Exam tip:** Unresolved audit findings repeated in the next audit = failed corrective-action system.

Privacy BLUEPRINT

Injury, medical and exposure records contain sensitive information that must be protected while still meeting access and reporting rules.

- 1904.29 privacy concern cases: write 'privacy case' instead of the name - intimate body part or reproductive system injury, sexual assault, mental illness, HIV/hepatitis/TB, contaminated needlestick or sharps injury, and other illness where the employee requests it.
- Keep medical records separate from personnel files; restrict access to need-to-know (occupational health staff).
- HIPAA generally applies to health-care providers and plans, not to employers acting as employers; ADA requires medical information be kept confidential.
- Electronic data: access controls, encryption, backups.

> **Exam tip:** Needlestick with contaminated sharp is a privacy concern case - no name on the 300 log.

Trade secrets BLUEPRINT

Employers and manufacturers may protect confidential business information, but never at the expense of worker health and safety.

- HazCom 1910.1200(i): specific chemical identity may be withheld from the SDS if the hazards and protective measures are still disclosed.
- In a medical emergency the identity must be disclosed immediately to the treating physician or nurse (confidentiality agreement may follow).
- Non-emergency: written request by health professional with need stated; may require confidentiality agreement; written denial within 30 days.
- PSM: trade secrets cannot be used to withhold PSI needed by PHA teams, procedures, investigations or emergency planning.
- 1910.1020 allows trade-secret protection with similar conditions.

> **Exam tip:** Trade secret protection NEVER overrides emergency medical need.

Personal information BLUEPRINT

Personally identifiable information (PII) - names, SSNs, birth dates, medical details, addresses - must be collected only as needed, protected and disposed of securely.

- Collect minimum needed; limit access; encrypt and password-protect; shred or securely wipe when retention ends.
- Laws: state breach-notification laws, GDPR (EU workers), ADA/GINA confidentiality.
- Do not post incident reports with names and medical detail on bulletin boards.
- Breach of PII requires notification under many state laws.

Document control and retention schedules

A document management system ensures people use current versions and records are retained, retrievable and disposed of on schedule.

- Document (procedure, plan - controlled, revised) vs record (evidence of what happened - not changed).
- Document control: approval, version/revision numbers, distribution, removal of obsolete copies.
- Retention schedule: record type, retention period (longest of legal, contract, company), location, owner, disposal method.
- Litigation hold suspends destruction when litigation is anticipated.

> **Exam tip:** Using an outdated SOP after a revision = document control failure.

Read more: Yates, Safety Professional's Reference and Study Guide (4th ed.): Ch. 3 Recording and Reporting Occupational Injuries and Illnesses, p. 44; Ch. 4 Regulations; Ch. 40 BCSP Code of Ethics

7. Ten CSP-style practice questions

Answer all ten on paper first, then check the key. Score: _____ / 10 Date: _____ Retest: _____ / 10

2.10-Q1. A worker's injury involves a sexual assault at work. How should it be entered on the OSHA 300 Log?

A. Enter the worker's full name and details
B. Enter 'privacy case' instead of the name and keep a separate confidential list
C. Do not record it
D. Record it only on the 301 form

2.10-Q2. How long must an employer retain the OSHA 300 Log, 300A Summary and 301 forms?

A. 1 year
B. 3 years
C. 5 years following the end of the calendar year they cover
D. Duration of employment plus 30 years

2.10-Q3. Under 29 CFR 1910.1020, employee exposure records must be retained for:

A. At least 30 years
B. 50 years
C. 5 years
D. Duration of employment only

2.10-Q4. A supplier withholds a chemical's specific identity on the SDS as a trade secret. A treating physician needs the identity for an exposed worker in a medical emergency. What does the HCS require?

A. The supplier may refuse until a lawsuit is filed
B. Only OSHA can request the identity
C. The employer must guess based on the hazard class
D. The supplier must disclose the identity immediately to the treating health professional, and may later request a confidentiality agreement

2.10-Q5. A document control system should ensure:

A. Unlimited copies of all forms are in circulation
B. Outdated procedures are removed from use
C. All documents remain permanently editable
D. Old records are deleted immediately after revision

2.10-Q6. A safety professional oversees noise monitoring and discovers that the data logger failed to record any readings during the shift. The company is nearing an OSHA deadline for exposure documentation. What is the most appropriate next step?

A. Submit the report anyway and explain the issue if audited
B. Adjust readings using historical data and average estimates
C. Request an exemption from OSHA due to equipment failure
D. Redo the sampling as soon as possible and document the equipment failure

2.10-Q7. Which is an example personally identifiable information (PII)?

A. Credit Card Number
B. Race of the person.
C. City of Residence.
D. Marital status

2.10-Q8. For A Large Quantity Generator Of Hazardous Waste, How Long Must You Retain Copies Of Annual Training Provided To Those Workers Who Are Currently Employed By The Company?

A. Three years from the date of training
B. Duration of employment plus 30 years
C. Until the facility closes
D. Fifteen years from the date of training

2.10-Q9. An internal audit identifies recurring deficiencies in training records. What should management do first?

A. Assign blame to individual trainers
B. Review and correct the system process for training record control
C. Delete all incomplete records
D. Extend the audit schedule

2.10-Q10. A plant's occupational health nurse keeps employee audiograms and respirator medical evaluations. Where should these records be maintained?

A. In a separate, confidential medical file with access limited to authorized persons
B. In each employee's personnel file so supervisors can review fitness for duty
C. In the supervisor's office for quick reference
D. In the training records database with the employee's course history

Answer key and explanations

Q	Ans	Why
1	B	1904.29(b)(7): for privacy concern cases (sexual assault, mental illness, HIV/hepatitis/TB, contaminated needlestick, or on employee request for other illnesses), enter 'privacy case' and keep a separate confidential list of names.
2	C	29 CFR 1904.33: OSHA 300, 300A and 301 forms are retained 5 years following the end of the calendar year they cover. Trap: 30 years applies to exposure/medical records.
3	A	29 CFR 1910.1020: exposure records are kept at least 30 years; medical records for duration of employment plus 30 years.
4	D	1910.1200(i): in a medical emergency the specific chemical identity must be disclosed immediately to the treating physician or nurse; a confidentiality agreement may be requested afterward.
5	B	Document control ensures only current, approved revisions are in use and obsolete versions are withdrawn (and archived/marked if retained).
6	D	Exposure records must be accurate. Redo the sampling promptly and document the equipment failure; never estimate, fabricate or submit incomplete data.

7	**A**	A credit card number uniquely identifies a person's account and is PII. Race, city or marital status alone generally do not identify an individual.
8	**C**	40 CFR 262.17(a)(7)(v): training records for current personnel are kept until facility closure; for former employees, at least 3 years from the last date worked.
9	**B**	Recurring record deficiencies indicate a system problem; correct the record-control process and verify it is sustained rather than blaming individuals or deleting records.
10	**A**	Employee medical records must be kept confidential and separate from general personnel files (ADA; 1910.1020 access rules). Supervisors may be told work restrictions, not diagnoses.

DOMAIN 2 · COMPETENCY 2.11

Budgeting, Finance and Economic Analysis

CSP11 blueprint: Apply budgeting, finance, and economic analysis techniques and principles (e.g., timelines, budget development, resourcing, return on investment, cost/benefit analysis, role in procurement process)

1. Definition

Engineering economics and budgeting allow the safety professional to justify, plan and fund EHS investments in business terms: costs, benefits, time value of money, return on investment and payback, and to influence procurement so that safe equipment and services are purchased.

2. Explanation in simple words

Managers decide with money. Show what a safety project costs, what it saves (including hidden indirect costs of accidents), how fast it pays back and whether the benefit is bigger than the cost.

3. Key points to remember

- Direct (insured) costs: medical, indemnity/compensation. Indirect (uninsured) costs: lost time of injured and coworkers, supervisor time, investigation, retraining, replacement workers, production delays, damaged equipment, reputation. Indirect costs are often several times the direct costs (iceberg).
- Simple payback = initial cost ÷ annual savings.
- ROI (%) = (benefit − cost) ÷ cost × 100 (some questions define ROI as annual savings ÷ investment × 100 - use the formula given).
- Benefit-cost ratio = PV of benefits ÷ PV of costs; > 1 is favorable. Net present value (NPV) = PV of benefits − PV of costs; > 0 accept.
- Time value of money: $F = P(1 + i)^n$; $P = F(1 + i)^{-n}$; uniform series $P = A[(1 + i)^n - 1] \div [i(1 + i)^n]$; $F = A[(1 + i)^n - 1] \div i$.
- Life-cycle cost: purchase + installation + operation + maintenance + disposal − salvage; the cheapest purchase is not always the lowest life-cycle cost.
- Sales needed to cover a loss = loss ÷ profit margin (e.g., $137,000 ÷ 5% = $2.74 million in sales).
- Budget types: capital vs operating; zero-based, incremental, activity-based. Budget development: needs/risk assessment, prioritization, cost estimates, timeline, approval, tracking variance.
- Procurement: specify safety requirements in purchase specs (guarding, noise levels, CE/UL listing), prequalify contractors (EMR, TRIR, programs, training), include safety terms and indemnity in contracts.

4. Real work scenario

> An $80,000 automated palletizer will eliminate manual lifting that causes about four back injuries a year at $44,000 each. Annual savings ≈ $176,000; simple payback = 80,000 ÷ 176,000 ≈ 0.45 years (5.5 months); first-year ROI = (176,000 − 80,000) ÷ 80,000 = 120%. The CFO approves the project.

5. CSP exam tips and traps

- Cost-benefit analysis comparing control cost with injury costs is the MOST persuasive business case on the exam.
- Read whether the question wants payback in months or years.
- NPV and B/C require discounting; simple ROI does not.
- Safety's role in procurement is to build requirements in before purchase (PtD for buying).

6. Every blueprint point (5 blueprint items + 4 more)

Budget development BLUEPRINT

A safety budget allocates money and people to planned activities and is justified with risk and financial analysis aligned to business goals.

- Operating budget (recurring: training, PPE, salaries, consultants) vs capital budget (long-life assets: ventilation, guarding, sprinklers).
- Incremental budgeting adjusts last year's budget; zero-based budgeting justifies every item from zero each cycle.
- Build from the action plan/risk register: prioritize by risk and regulatory requirement.
- Track variance = actual - budgeted; explain significant variances.

> **Exam tip:** Capital projects are justified with ROI, payback or NPV; operating items are justified by risk and compliance need.

Resources BLUEPRINT

Resources include people, equipment, time, money, information and outside expertise; management commitment is shown by allocating them.

- ISO 45001 7.1 and Z10 require management to determine and provide resources.
- Options: in-house staff, consultants, contractors, shared services, insurance carrier loss-control services.
- Allocate resources where risk reduction per dollar is greatest.

Return on investment (ROI) BLUEPRINT

ROI expresses the net benefit of a safety investment as a percentage of its cost.

- ROI (%) = (benefits - cost) / cost x 100.
- Example: $50,000 program avoids $120,000 in losses -> ROI = (120,000 - 50,000)/50,000 = 140%.
- Benefits include direct costs avoided (medical, comp) and indirect costs avoided (lost production, overtime, investigation, hiring, damage).
- Simple ROI ignores time value of money - use NPV for multi-year projects.

> **Exam tip:** Read whether the question wants net ROI (subtract cost) or a simple benefit/cost ratio.

Cost/benefit analysis BLUEPRINT

Cost-benefit analysis compares the (present) value of benefits with costs to decide whether a control is worth funding.

- Benefit-cost ratio = PV benefits / PV costs; >1 = justified.
- Example: benefits $90,000, costs $60,000 -> B/C = 1.5.
- Expected annual loss = probability (frequency) x consequence; benefit = reduction in expected loss.
- Include life-cycle costs: purchase, install, maintenance, training, disposal.
- Cost-effectiveness compares cost per unit of risk reduced when benefits are hard to price.

Payback period

Payback is how long it takes for savings to repay the initial investment.

- Payback (years) = initial cost / annual net savings.
- Example: $40,000 lift assist saves $16,000/yr -> 2.5 years.
- Shorter is better; ignores savings after payback and the time value of money.

> **Exam tip:** Payback is the simplest and most commonly tested measure.

Time value of money and NPV

Money now is worth more than money later; present value converts future savings to today's dollars.

- FV = PV x (1 + i)^n; PV = FV / (1 + i)^n.
- Example: $10,000 in 3 years at 10% -> PV = 10,000 / 1.331 = $7,513.
- NPV = sum of CF_t/(1 + i)^t - initial cost; NPV > 0 = accept.
- Example: cost $20,000; savings $10,000/yr for 3 yrs at 10% -> PV = 9,091 + 8,264 + 7,513 = 24,868; NPV = $4,868.
- Annuity PV = A x [1 - (1 + i)^-n] / i.

Direct and indirect costs of incidents

Direct (insured) costs are medical and compensation; indirect (uninsured) costs are often several times larger.

- Indirect: lost productivity, supervisor time, investigation, training replacements, damaged equipment, legal, morale, reputation.
- Heinrich estimated indirect:direct about 4:1; OSHA Safety Pays uses multipliers that shrink as direct cost rises.
- Sales needed to pay for a loss = loss cost / profit margin. Example: $30,000 loss at 5% margin -> $600,000 in sales.
- Straight-line depreciation = (cost - salvage) / useful life.

> **Exam tip:** The 'sales to cover loss' calculation is a favorite way to show management the value of prevention.

Procurement BLUEPRINT

Procurement controls what hazards enter the workplace; specifying safety requirements at purchase is a prevention-through-design opportunity.

- Write safety, ergonomic and noise specifications into purchase orders; require guarding and lockable isolation devices.
- Review SDSs before approving new chemicals (safer substitutes).
- Contractor prequalification: EMR, TRIR/DART, written programs, training, OSHA citation history.
- Consider total cost of ownership (life-cycle), not lowest bid.
- ISO 45001 8.1.4 requires controls for procurement, contractors and outsourcing.

> **Exam tip:** Cheapest-bid purchasing that ignores safety is a root cause in many case studies.

Insurance and risk financing costs

Insurance costs connect safety performance to finance.

- Workers' comp premium = (payroll/100) x manual rate x EMR (simplified, before other credits).
- Example: $2,000,000 payroll, $5/$100 rate, EMR 0.8 -> 20,000 x 5 x 0.8 = $80,000.
- Deductibles/self-insured retention keep small losses in-house (retention).

Read more: Yates, Safety Professional's Reference and Study Guide (4th ed.): Ch. 20 Engineering Economics, p. 495; Ch. 23 Workers' Compensation, p. 538

7. Ten CSP-style practice questions

Answer all ten on paper first, then check the key. Score: _____ / 10 Date: _____ Retest: _____ / 10

2.11-Q1. An investment of $20,000 earns an annual interest rate of 2.5%, compounded annually. Using F = P(1 + i)^n, what is the approximate future value after 3.5 years?

A. $23,610.37
B. $20,902.69
C. $21,805.37
D. $25,000.69

2.11-Q2. A safety project will save $36,000 per year for 5 years. The company requires a 10% return. What is the maximum investment that can be justified (present value of the annuity)?

A. $108,000
B. $136,468
C. $180,000
D. $215,912

2.11-Q3. A new ventilation system will cost $1,200,000 in five years. How much must the company invest today at 4.5% annual interest to have that amount? Use F = P(1 + i)^n.

A. $120,000
B. $1,052,631
C. $1,114,857
D. $962,964

2.11-Q4. A company invests $10,000 in exoskeletons for workers who install overhead bolts. The exoskeletons save $15,000 in injury costs in the first year. What is the first-year ROI?

A. 50%
B. 150%
C. 15%
D. 5%

2.11-Q5. A company spends $300,000 on a fall protection program that prevents $100,000 in accident-related costs each year. What is the ROI after three years?

A. 0%
B. 66%
C. 100%
D. 150%

2.11-Q6. An amputation cost the company $137,000 in direct and indirect costs. A laser scanner guard that would have prevented it costs $4,000. If the company operates on a 5% profit margin, how much additional sales revenue is needed to pay for the accident?

A. $548,000,000
B. $8,000,000
C. $2,000,000
D. $2,740,000

2.11-Q7. As a safety manager, which approach is MOST effective in demonstrating to senior management the value of investing in additional safety training and staff?

A. Present an ROI figure of 15% without supporting data
B. Compare the cost of safety equipment with the cost of training and staffing
C. Use a cost/benefit analysis that compares these costs with the injury and incident costs they will avoid
D. Forecast where the investment would be most visible to employees

2.11-Q8. During budget planning, a safety director must choose among ergonomics training ($15k), a noise control system ($45k) and a forklift camera system ($30k). Incident data show most losses come from back injuries and musculoskeletal claims. Which project should be prioritized?

A. Noise control system
B. Ergonomics program
C. Forklift camera system
D. Delay all projects until more funds are available

2.11-Q9. A $60,000 automated lift-assist system is expected to save $24,000 per year in injury and productivity costs. What is the simple payback period?

A. 0.4 years
B. 2.5 years
C. 3.0 years
D. 4.0 years

2.11-Q10. Purchasing is about to issue a request for proposals for new powered industrial trucks. What is the safety professional's MOST valuable role in the procurement process?

A. Write safety, ergonomic and training requirements into the specifications and help evaluate bids before selection
B. Inspect the trucks after delivery and reject unsafe units
C. Recommend the lowest-cost bid so savings can fund training
D. Leave procurement to purchasing and train operators after delivery

Answer key and explanations

Q	Ans	Why
1	C	F = 20,000 x 1.025^3.5 = 20,000 x 1.0903 = about $21,805. Trap: using simple interest or rounding n to 4 or more.
2	B	P = A[(1+i)^n - 1]/[i(1+i)^n] = 36,000 x 3.7908 = $136,468. $180,000 ignores the time value of money.
3	D	P = F/(1+i)^n = 1,200,000 / 1.045^5 = 1,200,000 / 1.2462 = about $962,960. Present value is always less than the future amount.
4	A	ROI = (benefit - cost)/cost x 100 = (15,000 - 10,000)/10,000 = 50%. Trap: 150% is benefit/cost, not net return.
5	A	Three years of savings = $300,000; ROI = (300,000 - 300,000)/300,000 = 0%. The program has just broken even (payback = 3 years).
6	D	Sales needed = loss / profit margin = 137,000 / 0.05 = $2,740,000. The $4,000 guard cost is a distractor (though it makes a strong business case).
7	C	A cost/benefit analysis ties the spending to the losses it prevents, which is the language management decides in. Visibility and internal cost comparisons do not show value.
8	B	Budget priority should follow the loss data; MSDs drive the losses, so the ergonomics initiative gives the best risk reduction per dollar.
9	B	Payback = initial cost / annual savings = 60,000 / 24,000 = 2.5 years. Trap: 0.4 is the inverted ratio.
10	A	Safety is cheapest to build in at the specification stage (prevention through design). Post-delivery inspection is reactive and costly.

DOMAIN 2 · COMPETENCY 2.12

Management and Leadership

CSP11 blueprint: Differentiate management leadership techniques (e.g., management theories, leadership theories, motivation, discipline, authority, responsibility, accountability, communication styles)

1. Definition

Management is planning, organizing, directing and controlling resources to achieve objectives; leadership is influencing people to achieve goals willingly. Safety professionals need both to drive change, motivate behavior and hold people accountable.

2. Explanation in simple words

Managers make sure the work is organized; leaders make people want to do it right. Know the main theories, match your style to the people and situation, and remember that authority can be delegated but accountability stays with you.

3. Key points to remember

- Classical/scientific management: Taylor (time studies, one best way), Fayol (14 principles incl. unity of command, span of control), Weber (bureaucracy). Human relations: Mayo (Hawthorne effect - people perform better when observed/valued).
- Motivation: Maslow hierarchy (physiological → safety → belonging → esteem → self-actualization); Herzberg two-factor (motivators: achievement, recognition, responsibility, growth; hygiene: pay, conditions, policy, supervision); McGregor Theory X (people dislike work, need control) vs Theory Y (people seek responsibility); Ouchi Theory Z (Japanese, long-term employment, consensus); McClelland acquired needs (achievement, affiliation, power); Vroom expectancy (effort → performance → reward valued); Adams equity theory; Skinner reinforcement.
- Leadership: trait theories; behavioral (Blake-Mouton managerial grid); situational (Hersey-Blanchard: directing, coaching, supporting, delegating by follower readiness); contingency (Fiedler); path-goal (directive, supportive, participative, achievement-oriented); transformational (inspires, vision) vs transactional (rewards/punishment); servant leadership; Likert's four systems (exploitive-authoritative to participative).
- Quality management: Deming (PDCA, 14 points, TQM), Juran (quality trilogy), Crosby (zero defects).
- Authority (right to decide), responsibility (duty to perform), accountability (answerable for results). Authority should equal responsibility; responsibility and authority can be delegated, accountability cannot.
- Discipline: progressive, consistent, fair, documented, timely (hot-stove rule: immediate, consistent, impersonal, with warning); use as a last resort after removing system barriers.
- Communication: sender → message → channel → receiver → feedback; active listening (paraphrasing, summarizing); assertive vs passive vs aggressive styles; conflict resolution styles (Thomas-Kilmann: competing, collaborating, compromising, avoiding, accommodating); face-to-face is best for important safety messages.

4. Real work scenario

> A new supervisor manages a crew of highly experienced, motivated electricians and tries to direct every step; morale drops. Using situational leadership, the safety manager coaches him to delegate planning to the crew while staying involved in risk reviews. Participation in JHAs and near-miss reporting increases.

5. CSP exam tips and traps

- Recognition is a Herzberg motivator; pay and working conditions are hygiene factors.
- Hawthorne effect = performance changes because people know they are being observed.
- 'Leading by example' is the best answer for how safety professionals should lead.
- Accountability cannot be delegated - a frequent distractor.

6. Every blueprint point (8 blueprint items + 1 more)

Management theories BLUEPRINT

Management theories explain how managers plan, organize, lead and control; each fits different situations.

- Scientific management (Taylor): time study, standardization, task specialization, monetary incentives.
- Bureaucratic (Weber): hierarchy, formal rules, defined roles, impersonality.
- Administrative (Fayol): plan, organize, command, coordinate, control; unity of command, division of work, discipline.
- Human relations (Mayo, Hawthorne studies): social factors, recognition and attention raise productivity.
- Contingency/situational: no one best way; systems theory: organization as open system with feedback; TQM (Deming): continuous improvement, all employees involved.

> **Exam tip:** Hawthorne effect: performance changes because people know they are being observed/attended to.

Leadership theories BLUEPRINT

Leadership theories explain what makes leaders effective - traits, behaviors, situation fit, or the relationship with followers.

- Trait theory: inherent characteristics (integrity, confidence, charisma).
- Behavioral: task-oriented vs people-oriented; Ohio State (initiating structure vs consideration), Michigan (production- vs employee-centered), Blake-Mouton managerial grid.
- Situational (Hersey-Blanchard): directing, coaching, supporting, delegating based on follower readiness; Fiedler contingency: style fit to situational control.
- Transformational: vision, charisma, intellectual stimulation, individualized consideration - drives culture change.
- Transactional: rewards and corrective action for performance - suits routine compliance; servant leadership: leader serves team needs.

> **Exam tip:** Changing safety culture -> transformational; new, inexperienced worker -> directing (Hersey-Blanchard S1).

Motivation BLUEPRINT

Motivation theories explain what drives people to perform and how to design rewards and jobs.

- Maslow hierarchy (bottom to top): physiological, safety, social/belonging, esteem, self-actualization.
- Herzberg two-factor: hygiene factors (pay, conditions, policies) prevent dissatisfaction; motivators (achievement, recognition, responsibility, work itself) create satisfaction.
- McGregor Theory X (dislike work, need control) vs Theory Y (self-motivated, seek responsibility); Ouchi Theory Z (long-term employment, collective decisions).
- McClelland: needs for achievement, affiliation, power; Vroom expectancy: Motivation = Expectancy x Instrumentality x Valence; Adams equity: fairness compared to others.
- Skinner reinforcement: positive reinforcement strengthens behavior; punishment suppresses it only temporarily.

> **Exam tip:** Pay and working conditions are Herzberg HYGIENE factors, not motivators.

Discipline BLUEPRINT

Discipline enforces rules to maintain order and correct conduct; it should be fair, consistent, documented and focused on improvement.

- Progressive discipline: verbal warning -> written warning -> suspension -> termination.
- Positive discipline: collaborative problem solving, clear expectations, coaching, support.
- Preventive discipline: clear policies, consistent enforcement, recognition of good behavior.
- Hot stove rule (McGregor): immediate, with warning, consistent, impersonal.
- Discipline should follow a just-culture review - punish reckless behavior, not system-induced error.

> **Exam tip:** Disciplining a worker when the root cause was a system failure is the wrong answer.

Authority BLUEPRINT

Authority is the formal right to make decisions, give orders and allocate resources.

- Line authority: direct supervisory control over subordinates (production managers).
- Staff authority: advisory - provides expertise (typically the safety professional's role).
- Functional authority: right to direct specific activities in other departments (e.g., safety may stop unsafe work).
- Authority must match responsibility; authority can be delegated.

> **Exam tip:** A safety manager usually has staff (advisory) authority - line managers own safety in their areas.

Responsibility BLUEPRINT

Responsibility is the duty to perform assigned tasks and meet objectives.

- Tasks can be delegated, but ultimate responsibility cannot be fully delegated - the manager remains responsible.
- Must be clearly assigned and communicated (job descriptions, RACI).
- Line management is responsible for safety of their people; the safety professional advises and supports.

Accountability BLUEPRINT

Accountability is the obligation to answer for results and accept consequences; it flows upward through the organization.

- Authority is given, responsibility is accepted, accountability is enforced.
- Authority without responsibility -> misuse; responsibility without authority -> frustration.
- Accountability tools: performance reviews tied to safety activities, metrics, recognition and consequences.
- The single 'A' in RACI is the accountable person.

Communication BLUEPRINT

Effective leaders adapt communication style to the audience, context and goal.

- Autocratic/directive: one-way, fast - best in emergencies or with inexperienced staff.
- Democratic/participative: two-way, builds buy-in - best for problem solving; slower.
- Laissez-faire: minimal direction - only for highly skilled, self-motivated teams.
- Assertive: clear, honest, respectful - generally most effective.
- Avoid passive, aggressive and passive-aggressive styles; use active listening and feedback; nonverbal cues carry much of the message.

> **Exam tip:** Emergency = directive; developing a new procedure with workers = participative.

Organizational structure principles

Classic principles help structure safety roles.

- Unity of command: each person reports to one supervisor.
- Span of control: number of direct reports a supervisor can effectively manage (narrower for complex/hazardous work).
- Delegation: assign authority with responsibility.
- Safety professional position: reporting to senior management gives influence and independence.

Read more: Yates, Safety Professional's Reference and Study Guide (4th ed.): Ch. 21 Management Theories, p. 505

7. Ten CSP-style practice questions

Answer all ten on paper first, then check the key. Score: _____ / 10 Date: _____ Retest: _____ / 10

2.12-Q1. In McGregor's Theory X and Theory Y, a supervisor who believes employees seek responsibility and are capable of self-direction follows:

A. Theory X
B. Theory Y
C. Theory Z
D. Expectancy theory

2.12-Q2. Which statement BEST differentiates transactional from transformational leadership?

A. Transactional focuses on compliance through rewards and penalties; transformational focuses on inspiration and shared vision
B. Transactional is people-oriented; transformational is task-driven
C. Both use identical approaches with different terminology
D. Transactional leadership eliminates formal authority

2.12-Q3. According to Herzberg's two-factor theory, which factor is a true motivator that increases job satisfaction?

A. Company policy
B. Salary and benefits
C. Recognition and achievement
D. Working conditions

2.12-Q4. Which leadership theory states that no single best style exists and that effectiveness depends on leader-member relations, task structure and position power?

A. Trait theory
B. Transactional theory
C. Path-goal theory
D. Fiedler's contingency theory

2.12-Q5. A newly hired, highly enthusiastic technician has little experience with the plant's permit system. Under Hersey-Blanchard situational leadership, which style is MOST appropriate?

A. Delegating
B. Supporting
C. Coaching
D. Directing (telling)

2.12-Q6. A plant manager assigns a supervisor to implement a new lockout/tagout program. Which statement is MOST accurate?

A. The manager can delegate authority and accountability, but not responsibility
B. The manager can delegate authority and responsibility for the task, but remains accountable for the outcome
C. Nothing can be delegated
D. Only accountability can be delegated

2.12-Q7. Which motivational approach aligns with Vroom's expectancy theory?

A. Reward behavior immediately after every safe act
B. Rely mainly on punishment for unsafe acts
C. Ensure employees believe their effort will lead to performance and a reward they value
D. Remove hygiene factors from the workplace

2.12-Q8. Senior management asks the safety manager to sign a confined space entry permit before atmospheric testing is done, calling it "just a formality" so work can start on schedule. What is the MOST ethical and professionally responsible action?

A. Sign the permit and perform testing later
B. Let the supervisor sign on the manager's behalf
C. Document the request, refuse to sign until testing is completed, and escalate if pressured further
D. Give verbal approval and note the delay at the next safety meeting

2.12-Q9. Which of the following BEST describes active listening in a conflict resolution conversation?

A. Restating what was said to confirm understanding
B. Listening for weaknesses in the other person's argument
C. Agreeing with everything to avoid further conflict
D. Taking notes to use against the other person later

2.12-Q10. An experienced operator is seen bypassing a machine guard. The company has a progressive discipline policy that has been applied to others for the same violation. Which response is MOST consistent with effective discipline principles?

A. Wait and raise it at the annual performance review
B. Terminate the operator immediately to set an example
C. Overlook it because the operator is productive and experienced
D. Address it immediately, apply the policy as it was applied to others, focus on the behavior, and document it

Answer key and explanations

Q	Ans	Why
1	B	Theory Y assumes workers are self-motivated, supporting participative management. Theory X assumes workers dislike work and need control.
2	A	Transactional leaders manage by exchange (rewards/discipline); transformational leaders build commitment through vision, trust and individual consideration.
3	C	Motivators are achievement, recognition, responsibility and growth. Pay, policy and conditions are hygiene factors that only prevent dissatisfaction.
4	D	Fiedler's contingency model matches the leader's style to those three situational factors. Path-goal focuses on clearing obstacles to goals.
5	D	Low competence with high commitment calls for directing: clear instructions and close supervision. Move to coaching, supporting and delegating as competence grows.
6	B	Authority and responsibility can be delegated (and should be matched), but accountability stays with the delegator.
7	C	Expectancy theory: motivation = expectancy (effort leads to performance) x instrumentality (performance leads to reward) x valence (reward is valued).
8	C	The BCSP Code of Ethics puts protection of people first; a permit certifies conditions that have not been verified. Delegating or verbally approving is the same violation.
9	A	Active listening means paraphrasing and checking understanding, which builds trust and surfaces the real issue.
10	D	The 'hot stove' rule: discipline should be immediate, consistent, impersonal and with prior warning. Inconsistency destroys credibility and legal defensibility.

DOMAIN 2 · COMPETENCY 2.13

Project Management

CSP11 blueprint: Apply project management principles and techniques (e.g., RACI charts, project timelines)

1. Definition

Project management applies knowledge, tools and techniques to deliver a defined scope on time, within budget and to quality and safety requirements. Key tools include the work breakdown structure, Gantt charts, critical path method (CPM), PERT and RACI responsibility matrices.

2. Explanation in simple words

A project is a one-time job with a start and an end. Break it into tasks, put them on a timeline, find which tasks set the finish date, and make sure everyone knows who does what and who signs off.

3. Key points to remember

- Project life cycle: initiating → planning → executing → monitoring and controlling → closing. Triple constraint: scope, time, cost (with quality and safety).
- Work breakdown structure (WBS): decomposes deliverables into manageable work packages.
- Gantt chart: bar chart of tasks against time; shows sequence and overlaps (useful to spot simultaneous high-risk activities).
- Critical path: the longest path through the network; tasks on it have zero float - any delay delays the project. Float (slack) = latest start − earliest start.
- PERT expected time = (optimistic + 4 × most likely + pessimistic) ÷ 6.
- RACI: Responsible (does the work), Accountable (owns the outcome and signs off - only ONE per task), Consulted (two-way input), Informed (kept up to date, one-way).
- Integrate safety: safety requirements in scope and contracts, risk register, pre-task planning, sequence high-risk tasks to avoid conflicts, stakeholder communication, lessons learned at closeout.

4. Real work scenario

> For a plant shutdown, the safety professional reviews the Gantt chart and sees crane lifts scheduled directly above an area where welders will work on the same day. She asks the project manager to resequence the lifts to the night shift and records it in the RACI as 'Accountable: shutdown manager; Responsible: lift supervisor; Consulted: safety; Informed: operations'.

5. CSP exam tips and traps

- Exactly one 'A' per task in a RACI chart.
- Critical path = longest duration path; tasks off the path have float.
- Gantt charts visualize timelines; they do not calculate injury rates or risk.

6. Every blueprint point (2 blueprint items + 5 more)

RACI BLUEPRINT

A RACI chart (responsibility assignment matrix) lists tasks against people and assigns roles so everyone knows who does what.

- R = Responsible: does the work (can be several).
- A = Accountable: owns the outcome, final decision - only ONE per task.
- C = Consulted: provides input (two-way).
- I = Informed: kept up to date (one-way).
- Prevents overlap and gaps; especially useful for cross-functional teams.

> **Exam tip:** Two 'A's on one task is an error - only one person is accountable.

Timelines BLUEPRINT

Project timelines (Gantt charts) show each task as a bar across a calendar with start/finish dates, dependencies, milestones and percent complete.

- Gantt chart = visual schedule; easy progress tracking and resource adjustment.
- Milestones mark key events with zero duration (e.g., permit approved).
- Dependencies: finish-to-start most common.
- Updated regularly during monitoring and controlling.

> **Exam tip:** Gantt shows schedule over time; PERT/CPM shows network dependencies and the critical path.

Project life cycle and triple constraint

Projects deliver objectives within constraints of scope, time, cost and quality.

- Phases: initiation (objectives, business case, approval) -> planning (tasks, schedule, budget, risks, communication) -> execution -> monitoring and controlling -> closing (acceptance, lessons learned).
- Triple constraint: scope, time, cost (quality in the middle) - changing one affects the others.
- Scope creep = uncontrolled scope growth; manage through formal change control.

Work Breakdown Structure (WBS)

A hierarchical decomposition of the project into deliverables and work packages.

- Foundation for scheduling, budgeting, resource assignment and RACI.
- Lowest level = work package that can be estimated and assigned.
- Example: new ventilation project -> design, procurement, installation, testing, training.

PERT

Program Evaluation and Review Technique models tasks as a network and uses three time estimates to handle uncertainty.

- Expected time te = (O + 4M + P) / 6.
- Example: O = 4, M = 6, P = 14 days -> te = (4 + 24 + 14)/6 = 7 days.
- Standard deviation = (P - O) / 6 -> (14 - 4)/6 = 1.67 days.
- Nodes = activities/events; arrows = dependencies.

> **Exam tip:** Remember the 4 multiplier on the most likely estimate.

Critical Path Method (CPM)

The critical path is the longest sequence of dependent tasks; it sets the minimum project duration and its tasks have zero slack (float).

- Example: path A-B-D = 3 + 5 + 4 = 12 days; path A-C-D = 3 + 2 + 4 = 9 days -> critical path A-B-D, 12 days; task C has 3 days of float.
- Delay of any critical task delays the whole project.
- Crashing = adding resources to shorten critical tasks; fast-tracking = overlapping tasks.

> **Exam tip:** Critical path is the LONGEST path, not the shortest.

Project risk management

Identify, assess and respond to risks that could affect project success.

- Risk register: risk, likelihood, impact, owner, response, status.
- Responses: avoid, mitigate, transfer, accept (and exploit/enhance for opportunities).
- Review risks continuously; communicate with stakeholders; document lessons learned.

Read more: Yates, Safety Professional's Reference and Study Guide (4th ed.): Ch. 21 Management Theories; Ch. 20 Engineering Economics (timelines)

7. Ten CSP-style practice questions

Answer all ten on paper first, then check the key. Score: _____ / 10 Date: _____ Retest: _____ / 10

2.13-Q1. A Gantt chart for a large construction project shows scaffolding erection, crane lifts and electrical installation scheduled in the same area at the same time. What should the safety professional do?

A. Let the schedule proceed because the crews are competent
B. Rely on inspection checklists once work begins
C. Coordinate with the project manager to sequence the high-risk activities so the hazards do not overlap
D. Hold a safety stand-down before the schedule starts

2.13-Q2. Project tasks: A (3 days) then B (5 days) then D (2 days); and A then C (4 days) then D. B and C run in parallel. What is the project duration and the float on task C?

A. 9 days; 0 days float
B. 10 days; 1 day float
C. 10 days; 0 days float
D. 14 days; 1 day float

2.13-Q3. In project management, the Critical Path Method (CPM) is used to determine:

A. The financial cost of controls
B. The number of employees needed per task
C. Only the safety milestones
D. The longest sequence of dependent tasks, which sets the total project duration

2.13-Q4. A safety project manager creates a Work Breakdown Structure (WBS). Its main purpose is to:

A. Define and organize the deliverables into manageable work packages
B. Schedule inspections
C. Document corrective actions
D. Identify safety risks only

2.13-Q5. On a construction project, an EHS deliverable such as a Job Hazard Analysis (JHA) should be tied to which project phase?

A. Initiation
B. Planning
C. Execution
D. Closure

2.13-Q6. During a plant expansion, a RACI chart is developed for a confined space entry program. The site safety professional is marked "A" for approving the entry permit process. What does "A" mean?

A. Accountable - the one person who owns the outcome and signs off
B. Assisting - provides help when requested
C. Acting - performs the hands-on work
D. Advisor - provides general guidance only

2.13-Q7. A safety manager is reviewing a Gantt chart. What is this chart used for?

A. Visualizing project tasks and timelines
B. Tracking hazard severity
C. Displaying noise levels
D. Calculating injury rates

2.13-Q8. In project management, Earned Value (EV) is calculated as:

A. Actual cost x percent complete
B. Planned value / time elapsed
C. Budget at completion x percent complete
D. Actual cost / planned value

2.13-Q9. After completing a major safety improvement project, the closeout report should include:

A. Financial records only
B. Marketing summaries
C. Contractor invoices
D. Lessons learned, residual risks and recommendations

2.13-Q10. While reviewing a RACI chart for a hot-work permit program, you find that three different managers are each marked "A" for permit approval. What is the BEST recommendation?

A. Leave it; more approvers provides more safety oversight
B. Assign a single Accountable person and reclassify the others as Responsible, Consulted or Informed
C. Change all three to Informed
D. Discard the RACI chart and rely on job descriptions

Answer key and explanations

Q	Ans	Why
1	C	Resolving simultaneous operations at the planning stage removes the interaction hazard. Inspections and stand-downs do not fix the schedule conflict.
2	B	Path A-B-D = 10 days (critical path); path A-C-D = 9 days, so C has 1 day of float. 14 days wrongly adds the parallel tasks.
3	D	Critical-path tasks have zero float; any delay to them delays the whole project.
4	A	The WBS decomposes scope into work packages so each can be scheduled, costed and assigned.
5	B	JHAs are developed in planning so controls are built into the work before execution. Doing them during execution is reactive.
6	A	In RACI, A = Accountable: one owner per task who approves the work. The doer is R (Responsible); advisors are C (Consulted).
7	A	A Gantt chart shows tasks as bars against a calendar, including durations, overlaps and dependencies.
8	C	EV = BAC x % complete: the budgeted value of the work actually done. Comparing EV to actual cost and planned value gives cost and schedule variance.
9	D	Project closure captures lessons learned and remaining risks so they feed future planning (continual improvement).
10	B	Each task should have exactly one Accountable person; multiple A's create confusion where everyone assumes someone else approved.

DOMAIN 2 · COMPETENCY 2.14

Data Analysis

CSP11 blueprint: Analyze and/or interpret data (e.g., exposure, release concentrations, sampling data, mean, median, mode, confidence intervals, probabilities, Pareto analysis)

1. Definition

Data analysis uses descriptive and inferential statistics to summarize safety and exposure data, test whether differences are real, estimate uncertainty and prioritize action. Industrial hygiene data are often lognormally distributed.

2. Explanation in simple words

Turn numbers into decisions: what is typical, how spread out is the data, how sure are we, and which few problems cause most of the losses.

3. Key points to remember

- Central tendency: mean (average, sensitive to outliers), median (middle value, best for skewed data), mode (most frequent, best for categories such as injury type).
- Spread: range (max − min), variance, standard deviation ($s = \sqrt{[\Sigma(x - \bar{x})^2 \div (n - 1)]}$), coefficient of variation.
- Normal distribution: ±1σ ≈ 68.3%, ±2σ ≈ 95.4% (±1.96σ = 95%), ±3σ ≈ 99.7%. $z = (x - \mu) \div \sigma$.
- Confidence interval for a mean: $\bar{x} \pm z(s/\sqrt{n})$ (use t for small samples); a larger sample gives a narrower interval.
- Exposure data: geometric mean and geometric standard deviation for lognormal data; 95th percentile compared with the OEL; upper confidence limit for compliance decisions.
- Probability: AND (independent) P(A and B) = P(A) × P(B); OR P(A or B) = P(A) + P(B) − P(A)P(B); complement 1 − P; series reliability = R1 × R2; parallel failure = F1 × F2.
- Pareto analysis (80/20): sort categories by frequency or cost; the 'vital few' causes produce most of the losses.
- Correlation (r from −1 to +1) does not prove causation; control charts show special-cause variation; release concentrations and dispersion depend on wind speed and distance.
- Qualitative data: interviews, focus groups, observations analyzed by thematic coding.

4. Real work scenario

> Ten noise samples average 88 dBA, but two readings at 101 dBA come from a single grinding task. The median is 85 dBA. Instead of averaging the problem away, the hygienist splits the data into task-based groups and finds the grinding task needs engineering controls while the rest of the area is below the action level.

5. CSP exam tips and traps

- When data are skewed or have outliers, the median is the best 'typical' value.
- 'Most common injury type' = mode.
- Pareto questions: add categories in descending order until you reach about 80%.
- Read whether a probability question asks for reliability or failure.

6. Every blueprint point (9 blueprint items + 3 more)

Exposure data BLUEPRINT

Exposure data are concentrations or doses measured on workers, used to compare against OELs and decide on controls.

- 8-h TWA = sum(Ci x Ti) / 8. Example: 100 ppm for 2 h, 50 ppm for 4 h, 0 for 2 h -> (200 + 200)/8 = 50 ppm.
- Mixtures (additive effects): Em = C1/L1 + C2/L2 + ...; > 1 = overexposure. Example: 40/100 + 30/50 = 1.0 -> at limit.
- OSHA action level is usually 1/2 the PEL (e.g., noise 85 dBA vs 90 dBA PEL).
- Exposure data are usually lognormally distributed - summarize with geometric mean (GM) and geometric standard deviation (GSD).
- Compliance: compare result and its confidence limit (UCL/LCL using sampling and analytical error) to the PEL.

> **Exam tip:** TWA for an 8-hour shift always divides by 8 h, even if sampling covered fewer hours (unsampled time assumed zero only if no exposure).

Release concentrations BLUEPRINT

Release calculations estimate airborne concentration from a leak or evaporation, and convert between units.

- ppm to mg/m3 = ppm x MW / 24.45 (25 C, 1 atm). Example: 50 ppm toluene (MW 92) -> 50 x 92/24.45 = 188 mg/m3.
- 1% by volume = 10,000 ppm. Example: LEL 1.2% = 12,000 ppm.
- Steady-state room concentration: C (ppm) = G / Q x 10^6, where G = vapor generation (cfm) and Q = ventilation (cfm). Example: G = 0.05 cfm, Q = 1,000 cfm -> 50 ppm.
- Purge (decay): C = C0 x e^(-Qt/V). Example: V = 10,000 ft3, Q = 1,000 cfm, t = 10 min -> C = C0 x e^-1 = 0.37 C0.
- Use mixing factor K (1-10) to correct for imperfect mixing: Q actual = Q x K.

Sampling data BLUEPRINT

Sampling data turn collected mass and air volume into a concentration; quality depends on calibration, blanks and representative sampling.

- Air volume = flow rate x time. Example: 2.0 L/min x 240 min = 480 L = 0.48 m3.
- Concentration = mass / volume. Example: 0.12 mg / 0.48 m3 = 0.25 mg/m3.
- 1 m3 = 1,000 L; 1 mg/m3 = 1 ug/L.
- Calibrate pumps pre- and post-sampling (use average flow); submit field blanks.
- Sample types: personal (breathing zone) for compliance vs area; full-shift vs short-term (STEL 15 min); grab vs integrated.

> **Exam tip:** Convert liters to cubic meters (divide by 1,000) before computing mg/m3 - a classic unit trap.

Mean BLUEPRINT

The arithmetic mean is the sum of values divided by the number of values - the most common measure of central tendency, but sensitive to outliers.

- Mean = sum(x) / n.
- Example data (ppm): 2, 4, 4, 5, 7, 9, 11 -> sum 42, n = 7 -> mean = 6.
- Geometric mean = nth root of product (or antilog of mean of logs) - preferred for lognormal exposure data. Example: GM of 2 and 8 = sqrt(16) = 4.
- Weighted mean is used for TWA (weighted by time).

> **Exam tip:** One very high outlier pulls the mean up but not the median.

Median BLUEPRINT

The median is the middle value when data are ordered; with an even count, average the two middle values.

- Example: 2, 4, 4, 5, 7, 9, 11 -> median = 5 (4th of 7 values).
- Even count: 3, 5, 8, 10 -> (5 + 8)/2 = 6.5.
- Robust to outliers; good for skewed data such as lost-workday counts or claim costs.

> **Exam tip:** Always sort the data first.

Mode BLUEPRINT

The mode is the most frequently occurring value; a data set may have no mode or more than one (bimodal).

- Example: 2, 4, 4, 5, 7, 9, 11 -> mode = 4.
- Only measure of central tendency usable for categorical data (most common injury type, body part).
- Range = max - min (11 - 2 = 9).

Standard deviation and variance

Dispersion measures show how spread out data are around the mean.

- Sample SD: s = sqrt[sum(x - mean)^2 / (n - 1)]; variance = s^2.
- Example: 4, 6, 8 -> mean 6; deviations^2 = 4 + 0 + 4 = 8; s^2 = 8/2 = 4; s = 2.
- Population SD divides by N instead of n - 1.
- Coefficient of variation = s / mean x 100 (%).
- Normal curve: mean +/- 1 SD = 68%, +/- 2 SD = 95%, +/- 3 SD = 99.7%.

Confidence intervals BLUEPRINT

A confidence interval gives a range likely to contain the true population mean at a chosen confidence level.

- CI = mean +/- z x (s / sqrt(n)); s/sqrt(n) = standard error.
- z: 90% = 1.645, 95% = 1.96, 99% = 2.576.
- Example: mean 40 ppm, s = 8, n = 16 -> SE = 2; 95% CI = 40 +/- 3.92 = 36.08 to 43.92 ppm.
- Small samples (n < 30) use t instead of z (wider interval).
- Larger n or smaller s -> narrower interval; higher confidence -> wider interval.
- If the 95% UCL of the exposure is below the OEL, compliance can be stated with 95% confidence (one-sided UCL uses z = 1.645).

> **Exam tip:** Watch whether the question gives SD or standard error - don't divide by sqrt(n) twice.

Probability BLUEPRINT

Probability (0 to 1) quantifies the chance of events; it underpins FTA, ETA, reliability and risk calculations.

- Independent AND: P(A and B) = P(A) x P(B). Example: 0.1 x 0.2 = 0.02.
- OR: P(A or B) = P(A) + P(B) - P(A and B) = 0.1 + 0.2 - 0.02 = 0.28. Mutually exclusive events: just add.
- At least one in n trials: 1 - (1 - p)^n. Example: p = 0.05 per day, 10 days -> 1 - 0.95^10 = 0.40.
- Reliability in series: R = R1 x R2 (0.9 x 0.9 = 0.81); parallel (redundant): R = 1 - (1 - R1)(1 - R2) = 0.99.
- Complement: P(not A) = 1 - P(A).

> **Exam tip:** Redundant (parallel) components increase reliability; components in series reduce it.

Pareto analysis BLUEPRINT

Pareto analysis ranks categories by frequency or cost to show the 'vital few' that cause most of the problem (about 80/20).

- Steps: categorize, count/cost each, sort descending, compute percent and cumulative percent, plot bars with cumulative line.
- Example: 200 injuries - strains 90 (45%), cuts 50 (70% cumulative), falls 30 (85%), other 30 -> focus on strains and cuts first.
- Rank by cost or severity when frequency alone could mislead.
- Used in gap analysis, incident trending and resource allocation.

> **Exam tip:** Pareto tells you where to focus FIRST - largest bar(s) on the left.

Incidence rates and normalizing data

Rates normalize counts to exposure hours so different-size operations can be compared.

- Rate = N x 200,000 / hours worked (100 workers x 2,000 h).
- Example: 3 recordables in 150,000 h -> 4.0.
- Hours from employees: 250 workers x 2,000 h = 500,000 h.
- Epidemiology: incidence rate = new cases / population at risk over time; prevalence = existing cases / population at a point in time.

Correlation, trend and control charts

Statistical tools show relationships and whether changes are real or random variation.

- Correlation coefficient r ranges -1 to +1; correlation does not prove causation.
- Regression fits a trend line (y = a + bx) for forecasting.
- Control charts: upper/lower control limits at mean +/- 3 SD; points outside = special-cause variation.
- Small data sets (few injuries) fluctuate randomly - avoid overreacting to one month.

> **Exam tip:** A strong correlation between near-miss reporting and fewer injuries does not by itself prove cause and effect.

Read more: Yates, Safety Professional's Reference and Study Guide (4th ed.): Ch. 15 Statistics for the Safety Professional, p. 408; Ch. 5 Math Review, p. 211; Ch. 8 Industrial Hygiene Air Sampling

7. Ten CSP-style practice questions

Answer all ten on paper first, then check the key. Score: _____ / 10 Date: _____ Retest: _____ / 10

2.14-Q1. Twenty-five personal samples have a mean of 2.0 mg/m3 and a standard deviation of 0.5 mg/m3. What is the approximate 95% confidence interval for the true mean (z = 1.96)?

A. 1.02 - 2.98 mg/m3
B. 1.50 - 2.50 mg/m3
C. 1.80 - 2.20 mg/m3
D. 1.90 - 2.10 mg/m3

2.14-Q2. Industrial hygiene data are usually lognormally distributed. Three samples are 1, 4 and 16 ppm. What is the geometric mean?

A. 4 ppm
B. 7 ppm
C. 10 ppm
D. 16 ppm

2.14-Q3. While plotting airborne benzene concentrations over time, a hygienist notices extreme outliers on two days. Which statistic is LEAST affected by these outliers?

A. Mean
B. Maximum
C. Standard deviation
D. Median

2.14-Q4. Reliability data for a chemical transfer system show a 0.5 probability of pump seal failure and a 0.3 probability of pipe gasket leak over 1,000,000 hours. Assuming the events are independent, what is the probability that BOTH failures occur?

A. 0.15
B. 0.60
C. 0.80
D. 1.00

2.14-Q5. A lot of 100 items contains 3% defectives. If a sample of 4 items is drawn without replacement, what is the probability that none is defective?

A. 98.4%
B. 88.4%
C. 96.0%
D. 87.0%

2.14-Q6. A safety manager wants to identify the most common type of injury at a plant last year. Which statistical measure should be used?

A. Mean
B. Mode
C. Median
D. Range

2.14-Q7. In a normal distribution, approximately what percentage of values fall within one standard deviation of the mean?

A. 34%
B. 50%
C. 68%
D. 95%

2.14-Q8. What does a 95% confidence interval for a population mean indicate?

A. The population mean is 95% larger than the sample mean
B. 95% of the sample data fall within the interval
C. Any single interval has exactly 95% of the population values
D. If sampling were repeated many times, about 95% of the intervals built this way would contain the true mean

2.14-Q9. An industrial hygienist measures respirable dust (mg/m3) at five locations: 0.45, 0.50, 0.55, 0.60 and 0.65. What is the arithmetic mean concentration?

A. 0.50 mg/m3
B. 0.55 mg/m3
C. 0.60 mg/m3
D. 0.65 mg/m3

2.14-Q10. Last year's 100 injuries were: lacerations 46, strains 30, contusions 12, burns 7, eye injuries 5. Using Pareto analysis, which categories should be targeted first?

A. Lacerations and strains
B. Lacerations only
C. Contusions, burns and eye injuries, because they are easiest to fix
D. All five categories equally

Answer key and explanations

Q	Ans	Why
1	C	CI = mean +/- z(s/sqrt n) = 2.0 +/- 1.96(0.5/5) = 2.0 +/- 0.196. Trap: using s instead of the standard error gives 1.02-2.98.
2	A	GM = (1 x 4 x 16)^(1/3) = 64^(1/3) = 4 ppm. 7 ppm is the arithmetic mean.
3	D	The median depends only on the middle rank, so extreme values barely move it; mean, maximum and SD are all pulled by outliers.
4	A	AND for independent events = P1 x P2 = 0.5 x 0.3 = 0.15. Trap: adding gives 0.8 (that is not even the correct OR, which is 0.65).
5	B	P = (97/100)(96/99)(95/98)(94/97) = 0.884. Using 0.97^4 gives 88.5%, about the same; 96% wrongly subtracts 4 x 1%.
6	B	The mode is the most frequent value and is the only central-tendency measure for categorical data such as injury type.
7	C	Empirical rule: 68% within 1 SD, 95% within 2 SD, 99.7% within 3 SD. 34% is only one side of the mean.

8	**D**	Confidence describes the method's long-run capture rate of the true mean, not the spread of individual data points.
9	**B**	Sum = 2.75; 2.75 / 5 = 0.55 mg/m3.
10	**A**	Pareto (80/20) focuses on the vital few: lacerations and strains are 76% of injuries. Lacerations alone leave a large share untouched.

DOMAIN 3

Risk Management

15% of the exam · 4 competencies

3.1	Safety Risk Evaluation Process Apply general principles of the safety risk evaluation process (i.e., identifying, analyzing, evaluating, monitoring, and communicating risk affecting an organization)
3.2	Risk Management Strategies and Hierarchy of Controls Apply risk management strategies to identify and mitigate EHS hazards (e.g., risk analysis, job hazard analysis, process hazard analysis, hierarchy of controls)
3.3	Financial Risk Mitigation Strategies Differentiate financial risk mitigation strategies as they relate to risk avoidance, risk retention, risk sharing, risk transfer, loss prevention and reduction
3.4	Risk Analysis: Identify, Rank and Monitor Apply risk analysis process of identifying, ranking, and monitoring (e.g., disasters/emergency preparedness, fire prevention, occupational health, hazardous materials management/environmental compliance)

DOMAIN 3 · COMPETENCY 3.1

Safety Risk Evaluation Process

CSP11 blueprint: Apply general principles of the safety risk evaluation process (i.e., identifying, analyzing, evaluating, monitoring, and communicating risk affecting an organization)

1. Definition

Risk is the combination of the likelihood of an event and the severity of its consequences. The risk evaluation process (ISO 31000, ANSI/ASSP Z690.2, ANSI/ASSP Z590.3) identifies hazards, analyzes and evaluates the resulting risk against criteria, treats it, and continuously monitors and communicates it.

2. Explanation in simple words

Find what can hurt you, judge how likely and how bad it is, decide whether that is acceptable, fix what is not, keep checking, and tell everyone who needs to know.

3. Key points to remember

- ISO 31000 process: establish scope/context/criteria → risk assessment (identification → analysis → evaluation) → risk treatment → monitoring and review; communication and consultation throughout; recording and reporting.
- Identification: inspections, JHAs, incident data, SDSs, process reviews, employee input, regulatory review. Hazard = source of potential harm; risk = likelihood × severity of harm from that hazard.
- Analysis: qualitative (risk matrix: e.g., 5 × 5), semi-quantitative (scored), quantitative (frequency and consequence, QRA).
- Evaluation: compare analyzed risk with risk criteria/tolerability - intolerable, ALARP (reduce unless cost grossly disproportionate), broadly acceptable. Decide priority and need for treatment.
- Risk appetite = amount of risk an organization is willing to pursue; risk tolerance = acceptable variation around objectives; risk register = central list of risks, owners, controls and status.
- Monitoring and review: verify controls are effective, re-assess residual risk after changes, track leading indicators.
- Risk communication: tailored to audience, two-way; in crises be first, be right, be credible, show empathy (CDC CERC). Risk perception differs from technical risk (dread, unfamiliar, involuntary risks feel larger).

4. Real work scenario

> A warehouse identifies forklift-pedestrian interaction as a hazard. Analysis rates it Likely × Major = High. Evaluation against the company criteria says High risks need controls within 30 days. Treatment: separate walkways with barriers and blue spot lights. Monitoring: monthly near-miss tracking. Communication: toolbox talks and signage explain the new traffic rules.

5. CSP exam tips and traps

- Analysis estimates the risk; evaluation compares it with criteria - a classic confusion.
- The FIRST step is hazard identification.
- After controls are installed, the next step is to verify effectiveness and re-rate residual risk.
- Risk communication answers stress clarity, audience understanding and transparency.

6. Every blueprint point (5 blueprint items + 4 more)

Identify (risk identification) BLUEPRINT

Find and describe hazards and risks that could affect people, assets, the environment or objectives. Identification must be proactive and continuous, covering routine and non-routine work and internal and external factors.

- Sources: inspections, audits, incident and near-miss investigations, toolbox talks, surveys, safety observations, SDS and process data.
- Structured tools: JHA (task level), FMEA (component failures), HAZOP (process deviations), What-If, checklists.
- ANSI/ASSP Z690.1 (ISO 31000): identification is proactive and continuous.
- OSHA PSM (1910.119) requires PHAs for covered highly hazardous chemical processes.

> **Exam tip:** You cannot analyze or control a hazard that has not been identified - if a scenario skips identification, that is the gap.

Analyze (risk analysis) BLUEPRINT

Understand the nature, causes and possible consequences of each identified risk and determine its level by estimating likelihood and severity.

- Qualitative: expert judgment, historical comparison, risk matrix (low/medium/high).
- Quantitative: failure-rate data, statistical probabilities, loss data.
- Analytical tools: risk matrices, Fault Tree Analysis (traces top event to causes), bow-tie diagrams (preventive barriers left, mitigative barriers right).
- ISO 45001 requires routine AND non-routine operations to be considered; NFPA 551 covers fire risk analysis (ignition, fuel load, detection/suppression).

> **Exam tip:** Distractors jump straight from identification to controls - analysis must come between them.

Risk estimation (Likelihood x Severity x Exposure)

Risk estimation puts a value on each risk so risks can be compared and prioritized. The guide uses Risk Score = Likelihood x Severity x Exposure, using qualitative, semi-quantitative or quantitative scales.

- Likelihood scale (guide): rare 1%, unlikely 10%, possible 50%, likely 75%, certain 100%.
- Severity scale: negligible (first aid), minor (medical treatment), moderate (lost workdays), severe (permanent disability/hospitalization), critical (fatality/catastrophe).
- Exposure scale: rarely (yearly), occasionally (monthly), frequently (weekly), constantly (daily).
- Worked example: L = 3, S = 4, E = 2 gives risk score 3 x 4 x 2 = 24; after a control drops L to 1, score = 8 (residual).
- Same idea as the Fine-Kinney method (Probability x Exposure x Consequence).

> **Exam tip:** Read which factors the question uses - a 2-factor matrix (L x S) and a 3-factor score (L x S x E) give different numbers.

Detectability and the FMEA Risk Priority Number

FMEA replaces exposure with detectability: a failure that is hard to detect before it causes harm is ranked higher. RPN = Severity x Occurrence x Detection.

- Detection is reverse-scored: easily detectable = low score, undetectable = high score (typically 1-10).
- RPN ranges 1-1,000 on 1-10 scales.
- Example: S = 8, O = 3, D = 7 gives RPN = 168.
- High-severity items (e.g., S = 9-10) usually need action regardless of total RPN.

> **Exam tip:** If a question mentions 'detectability' or 'RPN', it is FMEA, not a standard risk matrix.

Evaluate (risk evaluation) BLUEPRINT

Compare analyzed risk levels against defined risk criteria (tolerance, legal requirements, risk appetite) to decide which risks need treatment and in what priority.

- ALARP (as low as reasonably practicable): reduce risk until further reduction cost is grossly disproportionate to the benefit.
- Rank risks by severity, frequency and detectability.
- Corporate risk appetite is the benchmark for decisions.
- Z690.1 (ISO 31000): decisions must be based on defined evaluation criteria; PSM: prioritize correction of high-severity process hazards.

> **Exam tip:** Evaluation = comparing to criteria. Analysis = determining the level. Know the difference.

Monitor (monitoring and review) BLUEPRINT

Check over time that risks remain controlled, that controls work, and that changes (new processes, people, materials) are captured.

- Methods: trend analysis of incidents and near misses, inspections, compliance audits, real-time monitoring (gas detectors, vibration sensors).
- ISO 45001: ongoing performance evaluation and verification of hazard controls.
- NFPA 551: continuous review of fire safety systems and inspection intervals.
- Reassess after any change - link to Management of Change.

Communicate (risk communication and consultation) BLUEPRINT

Make sure stakeholders understand the risks and the controls. Under ISO 31000 communication is two-way - informing people and gathering feedback - and runs through every step.

- Tools: risk registers (central tracking), dashboards (visual, real-time performance), safety briefings and training tailored to the audience.
- Tailor content: executives (cost, exposure, decisions), supervisors (actions, owners), operators (task controls).
- OSHA PSM requires communication and training on process hazards for all affected employees.
- Convert rankings into a prioritized action plan with owners and due dates.

> **Exam tip:** One-way 'tell them' answers are weaker than two-way consultation answers.

Risk register

A centralized, structured log that documents, tracks and manages identified risks through their life cycle - the core tool of an ISO 31000-aligned system.

- Purpose: single point of truth, prioritization, ownership, monitoring of mitigation, communication.
- Typical fields: risk ID, description, source/cause, consequences, likelihood, severity, risk score, current controls, residual risk, recommended actions, action owner, due date, status (open/in progress/closed), review date.
- Residual risk is rated after current controls are applied.

> **Exam tip:** A risk register with no owner or due date is the classic deficiency in scenario questions.

ISO 31000 / ANSI Z690 risk management process

ANSI/ASSP Z690.1 is the U.S. adoption of ISO 31000 (principles and guidelines); Z690.2 = vocabulary (ISO Guide 73); Z690.3 / IEC 31010 = risk assessment techniques.

- ISO 31000 process: scope, context and criteria -> risk assessment (identification, analysis, evaluation) -> risk treatment, with communication and consultation plus monitoring and review running throughout, and recording and reporting.
- Risk assessment = identification + analysis + evaluation.
- The guide's teaching sequence: Identify -> Analyze -> Evaluate -> Monitor -> Communicate.

> **Exam tip:** The guide states the sequence Identify-Analyze-Evaluate-Monitor-Communicate; in ISO 31000 monitoring and communication are continuous, not only final steps. If options show a misordered sequence, the first three must be identify, analyze, evaluate.

Read more: Yates, Safety Professional's Reference and Study Guide (4th ed.): Ch. 26 Risk Assessment and Management, p. 604

7. Ten CSP-style practice questions

Answer all ten on paper first, then check the key. Score: _____ / 10 Date: _____ Retest: _____ / 10

3.1-Q1. Which sequence correctly represents the safety risk evaluation process described in the CSP blueprint?

A. Evaluate -> Identify -> Monitor -> Communicate
B. Monitor -> Identify -> Evaluate -> Analyze -> Communicate
C. Analyze -> Evaluate -> Identify -> Monitor -> Communicate
D. Identify -> Analyze -> Evaluate -> Monitor -> Communicate

3.1-Q2. In the ISO 31000 process, comparing the results of risk analysis with established risk criteria to decide whether treatment is needed is called:

A. Risk identification
B. Risk analysis
C. Risk evaluation
D. Risk treatment

3.1-Q3. During a risk assessment for a new warehouse conveyor, the team determines that a mechanical failure could cause packages to fall and strike workers. Which step does this represent?

A. Risk evaluation
B. Risk communication
C. Risk control
D. Hazard identification

3.1-Q4. In an industrial hygiene risk assessment, exposure sampling and ventilation performance checks belong to which phase?

A. Risk analysis
B. Risk identification
C. Risk evaluation
D. Risk treatment

3.1-Q5. An organization conducts regular trend analysis of incident and near-miss data. This activity belongs to which risk management step?

A. Risk identification
B. Risk evaluation
C. Risk monitoring
D. Risk treatment

3.1-Q6. According to ISO 31000, effective risk communication and consultation should be:

A. One-way and management-driven
B. Two-way, involving stakeholders, feedback and discussion
C. Only for reporting to regulators
D. Limited to top management

3.1-Q7. A review finds the likelihood of a chemical spill has decreased because of improved storage, but the severity remains high. What should be done NEXT?

A. Stop monitoring because likelihood is low
B. Re-evaluate and update the residual risk rating
C. Remove the hazard from the risk register
D. Classify it as acceptable risk automatically

3.1-Q8. A task with medium likelihood and major severity is rated High Risk on the company's 4x4 risk matrix. What should be the next step?

A. Eliminate the hazard or substitute the task
B. Accept the risk and monitor the activity
C. Proceed with caution and add signage
D. Document the risk and assign it to a supervisor

3.1-Q9. During a PHA, a hazard is rated severity 4 and likelihood 3 on 1-5 scales. The company matrix defines scores 1-4 as Low, 5-9 as Medium, 10-16 as High and 20-25 as Extreme. What is the risk level?

A. Low
B. Medium
C. High
D. Extreme

3.1-Q10. Risk tolerance differs from risk appetite because risk tolerance:

A. Applies to specific objectives, projects or activities
B. Represents the total company-wide risk
C. Is set only by regulators
D. Cannot vary within an organization

Answer key and explanations

Q	Ans	Why
1	D	You must find the hazard before analyzing likelihood/consequence, then evaluate against criteria; monitoring and communication continue throughout.
2	C	Analysis estimates likelihood and consequence; evaluation compares that result to criteria and prioritizes. Trap: confusing analysis with evaluation.
3	D	Recognizing what could go wrong and who could be hurt is identification; no likelihood or ranking has been done yet.
4	A	Sampling quantifies the likelihood and magnitude of exposure, which is analysis. Comparing results to the OEL is the next step (evaluation).
5	C	Monitoring and review confirm that controls remain effective and detect new or changing risks.
6	B	ISO 31000 treats communication and consultation as a continuous two-way process with stakeholders at every step.
7	B	Any change in likelihood or severity requires re-evaluating residual risk against criteria; high-severity hazards stay on the register.
8	A	A High rating means the risk is not tolerable; treatment should start at the top of the hierarchy of controls, not with signage or acceptance.
9	C	Risk = 4 x 3 = 12, which falls in the 10-16 High band. Always use the matrix the question gives you.
10	A	Risk appetite is the overall amount of risk the organization will pursue; tolerance is the acceptable variation for a specific objective or activity.

DOMAIN 3 · COMPETENCY 3.2

Risk Management Strategies and Hierarchy of Controls

CSP11 blueprint: Apply risk management strategies to identify and mitigate EHS hazards (e.g., risk analysis, job hazard analysis, process hazard analysis, hierarchy of controls)

1. Definition

Risk management strategies are the tools used to identify hazards at the task and process level (JHA, PHA, HAZOP, What-If) and to choose controls in order of effectiveness using the hierarchy of controls.

2. Explanation in simple words

Break the job or the process into steps, ask what can go wrong at each step, and then pick the strongest control that is practical - remove the hazard first and rely on PPE last.

3. Key points to remember

- Hierarchy of controls (NIOSH): elimination → substitution → engineering controls → administrative controls → PPE. ANSI Z10 adds 'warnings' between engineering and administrative controls.
- Engineering controls: isolate people from the hazard (guards, interlocks, ventilation, enclosures, guardrails). Administrative: procedures, permits, training, rotation, signage (warnings), scheduling. PPE: last line of defense, does not remove the hazard, depends on fit and behavior.
- PPE first is justified only as an interim measure while higher controls are installed, for emergencies, or when other controls are not feasible.
- Job Hazard Analysis (JHA/JSA): select job (prioritize by injury history, severity, new/changed or non-routine jobs) → break into steps → identify hazards per step → determine controls → implement, review periodically and after changes. Involve workers who do the job.
- Process Hazard Analysis (PHA, PSM): What-If, Checklist, HAZOP, FMEA, FTA; team with process, engineering and operations expertise; recommendations must be resolved and tracked.
- Preliminary Hazard Analysis: early design screening of hazards.
- Hazard mitigation: reduce likelihood (prevention) and/or severity (protection, emergency response); layers of protection (LOPA).

4. Real work scenario

> Silica exposure during abrasive blasting is currently controlled by supplied-air respirators and toolbox talks. Applying the hierarchy, the safety professional substitutes steel grit for silica sand (substitution), adds a blast room with local exhaust (engineering) and keeps respirators as backup. Measured exposures drop below the action level.

5. CSP exam tips and traps

- 'MOST effective' questions: the highest level of the hierarchy that actually addresses the hazard wins.
- Classify correctly: two-person lift = administrative; guardrails = engineering; moving work to a safer place may be called substitution or elimination depending on the source - read the options.
- JHA questions: break the job into steps first; involve the workers who perform it.
- Interlocks and light curtains are engineering controls; signs and labels are warnings/administrative.

6. Every blueprint point (5 blueprint items + 3 more)

Risk analysis BLUEPRINT

The process of understanding the nature of risk and determining its level: identify sources and causes, understand existing controls, estimate likelihood and consequences.

- Qualitative: risk matrix scoring (low/medium/high), 3x3 or 5x5.
- Quantitative: failure probability modeling, statistical loss data, FTA with failure rates.
- OSHA PSM: PHAs updated and revalidated at least every 5 years.
- Output is a risk level used for evaluation and treatment.

Job Hazard Analysis (JHA) BLUEPRINT

A systematic, step-by-step method to identify and control hazards of a specific job or task before they cause injury, illness or damage.

- Steps: (1) break the job into tasks in sequence, (2) identify hazards for each step (physical, chemical, biological, ergonomic, psychosocial), (3) recommend controls using the hierarchy.
- Prioritize jobs with highest injury rates, severe potential, new or changed jobs, and infrequent/non-routine jobs (OSHA 3071).
- Involve the workers who perform the task.
- Example: welding steel frames - UV, fumes, sparks -> screens, LEV, filter-lens helmet, FR clothing.
- ISO 45001 link: JHA supports hazard identification under clause 6.1.2.

> **Exam tip:** Too broad (whole job in one step) or too detailed steps are classic JHA errors; first step is always breaking the job into steps.

Process Hazard Analysis (PHA) BLUEPRINT

A team-based, systematic evaluation of hazards in a process involving highly hazardous chemicals, required by OSHA 1910.119(e) for covered processes.

- Methods: HAZOP (guide words: no, more, less, reverse, other than, part of, as well as), What-If, Checklist, What-If/Checklist, FMEA, FTA.
- Team includes someone with process engineering knowledge, an operator/employee with process experience, and a person knowing the PHA method.
- Revalidated at least every 5 years; recommendations resolved promptly and documented.
- NFPA 551 fire risk evaluation can be integrated for flammable processes.

> **Exam tip:** HAZOP = guide words applied to process deviations at nodes; What-If = brainstorming; FMEA = component level.

Hierarchy of Controls BLUEPRINT

The NIOSH/OSHA order of preference for controls: elimination, substitution, engineering, administrative, PPE. Higher levels are more effective and less dependent on human behavior.

- Elimination: remove the hazard (discontinue the process).
- Substitution: less hazardous material/equipment (non-flammable solvent instead of flammable).
- Engineering: isolate people from the hazard (guards, ventilation, enclosures).
- Administrative: procedures, training, rotation, signage, permits.
- PPE: last line of defense.
- ISO 45001 clause 8.1.2 requires the hierarchy when determining controls; ANSI Z10 also requires it.

> **Exam tip:** When two controls are offered, choose the one higher in the hierarchy - substituting a non-flammable solvent beats issuing FR clothing.

Hazard mitigation BLUEPRINT

Hazard mitigation reduces the likelihood or severity of a hazard by applying preventive and protective measures, combined in layers so no single failure causes harm.

- Preventive barriers reduce likelihood (interlocks, inherently safer design, maintenance).
- Mitigative barriers reduce consequences (sprinklers, containment, emergency response).
- NFPA 551: fire prevention measures (fuel load limits, detection/suppression) belong in the overall hazard mitigation plan.
- PSM: controls for high-hazard processes must be documented, implemented and periodically reviewed.

Integrated (enterprise) risk management approach

Hazard identification, risk assessment and controls are embedded into the organization's overall risk strategy so safety, health, environmental and operational risks are managed together, not in silos.

- Aligns mitigation with organizational objectives, risk appetite and compliance requirements.
- Integrates safety into strategic planning, budgeting and continuous improvement.
- ISO 31000 / Z690.1: risk management integrated into governance, strategy, planning and daily operations.

> **Exam tip:** Combined hazards (chemical + fire) may require citing both PSM and NFPA references.

Bow-tie analysis

A diagram with the hazard/top event in the center, threats and preventive barriers on the left, and consequences and mitigative (recovery) barriers on the right.

- Combines fault tree logic (left) and event tree logic (right).
- Escalation factors show what can defeat a barrier.
- Used to track barrier health (critical controls) on dashboards.
- Good for communicating risk to non-specialists.

> **Exam tip:** Left side = prevention (likelihood); right side = mitigation (consequence).

Layer of Protection Analysis (LOPA)

A semi-quantitative method that estimates scenario frequency by multiplying the initiating event frequency by the probability of failure on demand (PFD) of each independent protection layer (IPL).

- Mitigated frequency = initiating event frequency x PFD1 x PFD2 x ...
- Example: initiating event 0.1/yr, BPCS loop PFD 0.1, relief valve PFD 0.01 -> 0.1 x 0.1 x 0.01 = 1 x 10^-4 per year.
- IPL must be independent, effective (specific to the scenario) and auditable.
- SIL 1: PFD 10^-1 to 10^-2; SIL 2: 10^-2 to 10^-3; SIL 3: 10^-3 to 10^-4 (IEC 61511).
- Procedures and training alone are usually not credited as IPLs.

> **Exam tip:** Weak IPL independence is a reason to escalate to inherently safer or engineering solutions, not PPE.

Read more: Yates, Safety Professional's Reference and Study Guide (4th ed.): Ch. 26 Risk Assessment and Management, p. 604; Ch. 14 Personal Protective Equipment, p. 385; Ch. 35 Prevention through Design

7. Ten CSP-style practice questions

Answer all ten on paper first, then check the key. Score: _____ / 10 Date: _____ Retest: _____ / 10

3.2-Q1. After several near misses on a packaging line, management wants to issue PPE as the immediate solution. According to ANSI/ASSP Z10's hierarchy of controls, what should the safety professional recommend FIRST?

A. Provide upgraded PPE with higher ratings
B. Increase signage and awareness messaging
C. Retrain operators on safe work practices
D. Evaluate and implement higher-level controls, such as elimination or engineering, before relying on PPE

3.2-Q2. A JHA for abrasive blasting finds workers exposed to respirable crystalline silica. Current controls are respirators and daily toolbox talks. What is the BEST action based on the hierarchy of controls?

A. Replace silica sand with a less hazardous blasting medium
B. Conduct annual respirator fit tests
C. Increase the frequency of safety meetings
D. Add a buddy system for air monitoring

3.2-Q3. Before a non-routine lift of an awkward load onto a mezzanine, a supervisor fills out the JHA alone at his truck and says he will "just get the guys to sign it." What is the MOST appropriate immediate action for the safety professional?

A. Let the job proceed and observe the lift for a future JHA update
B. Report the supervisor to the project manager
C. Politely hold the start and gather the crew for a few minutes to review the lift and improve the JHA together
D. Hand the supervisor a generic JHA and ask him to copy it

3.2-Q4. During a walkthrough you observe employees grinding metal parts with sparks and loud noise. What is the FIRST step in conducting a proper PPE hazard assessment?

A. Assess and document the specific hazards of the task
B. Select PPE based on what nearby workers wear
C. Review injury records and order ANSI-rated PPE
D. Distribute earplugs and face shields to all workers

3.2-Q5. Which regulation requires a process hazard analysis (PHA) for processes involving highly hazardous chemicals?

A. ISO 45001
B. OSHA 29 CFR 1910.119
C. ANSI Z690.1
D. NFPA 551

3.2-Q6. Under OSHA's Process Safety Management standard, how often must a PHA be updated and revalidated?

A. Every year
B. Every 3 years
C. At least every 5 years
D. Every 10 years

3.2-Q7. A company identifies a chemical hazard. Until a safer alternative is found, it provides special training, updates work procedures and posts hazard signs. Which control levels are being applied?

A. Elimination and substitution
B. Administrative controls and warnings
C. Engineering controls and PPE
D. Substitution and engineering controls

3.2-Q8. The Job Hazard Analysis (JHA) technique primarily focuses on:

A. Process-level deviations
B. Component-level failure prediction
C. Management decision-making
D. Hazards in each step of a specific task

3.2-Q9. Several loud machines expose nearby workers to noise above the PEL. Which control would be MOST effective in reducing the risk of noise-induced hearing loss?

A. Replace the noisy machines with quieter models
B. Provide earplugs or earmuffs to all workers
C. Start a noise monitoring program
D. Rearrange the workspace to increase distance from the machines

3.2-Q10. The hierarchy of controls, in order of preference, is:

A. PPE -> Administrative -> Engineering -> Substitution -> Elimination
B. Administrative -> PPE -> Engineering -> Substitution -> Elimination
C. Elimination -> Substitution -> Engineering -> Administrative -> PPE
D. Substitution -> Elimination -> PPE -> Administrative -> Engineering

Answer key and explanations

Q	Ans	Why
1	D	Z10 ranks elimination, substitution, engineering, warnings and administrative controls above PPE. PPE can be interim but is not the first recommendation.
2	A	Substitution removes the silica hazard at the source; the other choices are administrative or PPE measures.
3	C	A JHA is only effective when the workers doing the task help identify hazards. Fix it before work starts, constructively; reporting or generic forms do not.
4	A	OSHA 1910.132(d) requires assessing the hazards first and certifying the assessment in writing; selection follows.
5	B	The Process Safety Management standard, 1910.119(e), requires PHAs for covered processes.
6	C	1910.119(e)(6) requires update and revalidation at least every 5 years. Trap: compliance audits are every 3 years.
7	B	Training and procedures are administrative controls; signs are warnings. None change the hazard itself.
8	D	A JHA breaks a job into steps, identifies hazards in each and assigns controls. Process deviations are HAZOP; component failures are FMEA.
9	A	Replacing the source with quieter equipment removes the noise at its origin, the highest control available here. Monitoring is not a control.
10	C	Most effective to least: eliminate, substitute, engineer, administer, then PPE (ANSI Z10 adds warnings before administrative controls).

DOMAIN 3 · COMPETENCY 3.3

Financial Risk Mitigation Strategies

CSP11 blueprint: Differentiate financial risk mitigation strategies as they relate to risk avoidance, risk retention, risk sharing, risk transfer, loss prevention and reduction

1. Definition

Financial risk management (risk financing) decides how an organization pays for losses: avoid the exposure, retain it (self-fund), share it, transfer it (insurance, contracts), or control it through loss prevention (lower frequency) and loss reduction (lower severity).

2. Explanation in simple words

You can stay away from a risk, keep it and pay for losses yourself, split it with a partner, pass it to an insurer or another company by contract, or make losses less likely and less severe. Most organizations use a mix.

3. Key points to remember

- Avoidance: do not undertake the activity (e.g., decline a project, stop using a chemical). Eliminates the exposure entirely.
- Retention: accept and pay for losses - deductibles, self-insurance, self-insured retentions, captive insurance companies (funded retention). Suits frequent, low-severity, predictable losses.
- Sharing: losses split with others - joint ventures, pools, co-insurance. (Some texts treat insurance as sharing among a pool of insureds.)
- Transfer: shift financial consequences - insurance (financial transfer), contractual transfer via hold-harmless/indemnity agreements, additional-insured endorsements, surety bonds, leasing, subcontracting.
- Loss prevention: reduce frequency (inspections, training, guarding, housekeeping, hot-work permits). Loss reduction: reduce severity (sprinklers, first aid, emergency response, backups, business continuity).
- Insurance terms: premium, deductible, EMR (experience modification rate; > 1.0 worse than average → higher premium), subrogation (insurer recovers from a responsible third party), pure risk (loss or no loss - insurable) vs speculative risk (gain or loss - not insurable), moral hazard (dishonesty), morale hazard (carelessness because insured).
- Workers' compensation: no-fault system; benefits include medical, partial wage replacement, rehabilitation; exclusive remedy (employee generally cannot sue the employer); not pain and suffering.
- Expected annual loss = probability × consequence; total cost of risk = retained losses + premiums + risk-control costs + administration.

4. Real work scenario

> A warehouse operator has frequent small forklift-damage claims (about $3,000 each) and a small chance of a major fire. It retains the small losses with a $25,000 deductible and a strong loss-prevention program, buys property insurance for the catastrophic fire exposure, requires tenants to sign hold-harmless agreements, and installs sprinklers (loss reduction).

5. CSP exam tips and traps

- Sprinklers reduce severity (loss reduction), they do not prevent fires.

- Insurance and hold-harmless agreements = transfer; captives and deductibles = retention.
- Video surveillance and guarding are loss control/mitigation, not transfer.
- Know pure vs speculative risk and moral vs morale hazard.

6. Every blueprint point (6 blueprint items + 8 more)

Avoidance (risk avoidance) BLUEPRINT

Removing the hazardous activity, process or substance entirely so no exposure or loss can occur. It is the only strategy that reduces the chance of loss to zero, but it also gives up any benefit of the activity.

- Example (guide): installing a Clean-In-Place system removes the need for manual tank entry, eliminating the confined space hazard.
- Other examples: not entering a high-risk market, discontinuing a hazardous product line.
- Best for catastrophic risks (e.g., handling certain explosives).

> **Exam tip:** Avoidance (financial) parallels elimination (hierarchy of controls).

Retention (risk retention) BLUEPRINT

Accepting a risk and budgeting to pay for losses internally (self-insurance). Retention can be planned (deductibles, self-insured retention, funded reserves) or unplanned (unknown or uninsured risks).

- Appropriate for frequent, low-severity, predictable losses.
- Example (guide): a small firm pays minor property damage claims itself instead of buying expensive insurance.
- Deductibles and self-insured retentions are forms of retention.
- Retaining risk does not remove the duty to control it (General Duty Clause still applies).

> **Exam tip:** Low severity + high frequency = retain (and prevent); high severity + low frequency = transfer (insure).

Sharing (risk sharing) BLUEPRINT

Distributing a risk among several parties so each bears part of the potential loss.

- Example (guide): joint venture partners each assume part of the EHS compliance liability.
- Other examples: consortiums, risk pools, co-insurance.
- ISO 31000 uses 'sharing' to include contracts and insurance.

Transfer (risk transfer) BLUEPRINT

Shifting the financial consequences of a risk to another party through insurance, contracts, warranties, hold-harmless/indemnity clauses or outsourcing.

- Example (guide): buying property and liability insurance for a plant.
- Contractual transfer: hold harmless and indemnity clauses in contractor agreements.
- Transfer moves the financial burden, not the hazard - and not the employer's legal/moral duty of care.
- Best for low-frequency, high-severity losses.

> **Exam tip:** If two answers seem correct, prefer the one that reduces the root cause (prevention) over one that only shifts the cost (transfer).

Loss prevention BLUEPRINT

Measures that reduce the FREQUENCY (likelihood) of losses before they occur.

- Example (guide): interlocks on machinery to prevent unsafe operation.
- Others: inspections, preventive maintenance, training, hot work permits, housekeeping.
- Acts before the event.

> **Exam tip:** Frequency = prevention; severity = reduction.

Loss reduction BLUEPRINT

Measures that reduce the SEVERITY of losses once an incident occurs.

- Example (guide): sprinklers minimize fire damage; compartmentation and fire-rated barriers limit spread.
- Others: emergency response, spill containment, first aid/AEDs, backup data, return-to-work programs.
- Acts during or after the event.

> **Exam tip:** Sprinklers do not prevent fires - they are loss reduction (and fire protection), not loss prevention.

Risk mitigation vs. risk treatment

Risk treatment is the overall selection of options to modify risk (avoid, reduce, share/transfer, retain). Risk mitigation is the subset that actually lowers likelihood, severity or impact through preventive or protective measures.

- ISO 31000 treatment selection depends on: risk appetite and tolerance, cost-benefit analysis, and compliance obligations.
- Strategies are usually combined: avoid catastrophic risks, insure moderate costly risks, prevent and reduce operational hazards.
- Risk control = applying measures to eliminate or reduce risk to acceptable levels.

Risk profiling and sensible risk management

A risk profile describes the types, levels, sources and patterns of risk an organization faces. Risks are then prioritized by probability, severity and cost of control (cost-benefit) before action.

- Sensible risk management: prioritize high probability, high severity, low-cost-to-control risks first.
- Do not ignore low-probability, high-severity risks (e.g., boiler explosion).
- Risk characterization is the final step of risk assessment: describes impact, likelihood and severity (with uncertainties) so decision-makers can act.

Risk appetite vs. risk tolerance

Risk appetite is the overall amount and type of risk the organization is willing to pursue or retain, set by top management. Risk tolerance is the acceptable variation for a specific risk, activity or department within that appetite.

- Appetite: strategic, big picture; can be qualitative ('low tolerance for safety incidents') or quantitative ('no more than 1 lost-time injury per million hours').
- Tolerance: more granular; defines limits before corrective action is triggered.
- Example: low appetite overall, but tolerance of up to two first-aid cases per year in a high-risk maintenance division.

> **Exam tip:** Appetite = big picture; tolerance = limits for specific risks.

Risk terms: acceptable, tolerable, inherent, residual, unacceptable

Exams test precise risk vocabulary using cue words in the scenario.

- Acceptable: reduced to a level satisfactory to the organization, regulators or society (cue: 'meets standard', 'within policy').
- Tolerable: allowed only because further reduction is impracticable or grossly disproportionate - managed under ALARP (e.g., work at height with fall arrest).
- Inherent: raw risk with no controls (solvent room with no safety systems).
- Residual: risk remaining after reasonably practicable controls (small ignition risk in a booth with explosion-proof gear).
- Unacceptable: exceeds legal/organizational thresholds; stop or control immediately (crane with damaged load chain).

> **Exam tip:** Inherent appears before controls in a question; residual appears after controls are implemented.

Pure vs. speculative risk

Pure risk has only the possibility of loss or no loss (no gain); speculative risk can produce gain or loss.

- Workplace hazards and OSHA concerns are pure risks; only pure risks are normally insurable.
- Speculative examples: gambling, stock investment, starting a business, real estate.
- Example: investing in AI safety monitoring is speculative - it may cut losses and raise profit, or lose money.

Legal terms associated with risk

Contract and insurance terms that determine who ultimately pays for a loss.

- Subrogation: insurer steps into the policyholder's shoes to recover what it paid from the at-fault third party.
- Hold harmless: one party agrees not to hold the other responsible for certain risks; indemnity: promise to compensate the other for specified liabilities.
- Limited liability: owners' personal assets protected from business debts.
- Piercing the corporate veil: courts reach owners' personal assets when the corporate form is misused for fraud or to evade obligations.

> **Exam tip:** Insurer recovering from a negligent contractor after paying a claim = subrogation.

Risk homeostasis (risk compensation / Peltzman effect)

People adjust behavior to perceived safety, so added protection may be offset by riskier behavior and the overall risk level stays about the same.

- Example: drivers drive faster after ABS brakes are installed.
- Reason why engineering controls should not depend on behavior and why PPE may increase risk-taking.
- Associated with Wilde (risk homeostasis) and Peltzman (risk compensation).

Cost-benefit, ROI and payback for safety investments

Financial strategies are justified with simple economics: compare the cost of a control with the losses it avoids.

- ROI = (annual savings - annual cost) / investment x 100%.
- Payback period = initial investment / annual savings. Example: $50,000 guard system saving $20,000/yr -> 2.5 years.
- Benefit-cost ratio = benefits / costs (>1 favorable).
- Indirect costs of injuries are often several times the direct (insured) costs - include them in savings.

> **Exam tip:** Expect cost tables; compute payback and ROI carefully and keep annual vs. one-time costs separate.

Read more: Yates, Safety Professional's Reference and Study Guide (4th ed.): Ch. 26 Risk Assessment and Management, p. 604; Ch. 23 Workers' Compensation, p. 538

7. Ten CSP-style practice questions

Answer all ten on paper first, then check the key. Score: _____ / 10 Date: _____ Retest: _____ / 10

3.3-Q1. A large company forms a captive insurance subsidiary to cover its own workers' compensation losses. Which financial risk strategy is this?

A. Avoidance
B. Funded retention (self-insurance)
C. Loss prevention
D. Contractual transfer

3.3-Q2. Management decides to purchase liability insurance for potential chemical spills. Which financial strategy is being applied?

A. Risk retention
B. Risk sharing
C. Risk avoidance
D. Risk transfer

3.3-Q3. A company hires a subcontractor who agrees by contract to carry liability insurance and accept responsibility for injuries caused by its work. What is this an example of?

A. Risk avoidance
B. Risk retention
C. Risk sharing
D. Risk transfer

3.3-Q4. Which of the following is an example of risk sharing?

A. Joint venture partners splitting EHS liabilities and costs
B. Purchasing liability insurance
C. Eliminating flammable materials
D. Installing automatic sprinklers

3.3-Q5. Which statement BEST distinguishes risk mitigation from risk avoidance?

A. Mitigation eliminates the hazard; avoidance reduces exposure
B. Mitigation transfers cost to another party; avoidance shares the risk
C. Mitigation reduces likelihood or impact; avoidance removes the activity or hazard entirely
D. Mitigation retains the hazard; avoidance controls it

3.3-Q6. Which pair correctly matches loss PREVENTION and loss REDUCTION?

A. Prevention: housekeeping and hot-work permits; Reduction: automatic sprinklers
B. Prevention: automatic sprinklers; Reduction: housekeeping
C. Prevention: insurance; Reduction: deductibles
D. Prevention: first aid; Reduction: machine guarding

3.3-Q7. An insurer pays a business for damages and then seeks recovery from the third party who caused the loss. This is called:

A. Hold harmless
B. Limited liability
C. Subrogation
D. Indemnification

3.3-Q8. A company's actual primary losses are $150,000 and expected losses are $120,000. What is the experience modification rate (EMR) and its effect?

A. 0.80; 20% premium credit
B. 1.25; 25% premium surcharge
C. 1.25; 25% premium credit
D. 1.50; 50% premium surcharge

3.3-Q9. A warehouse has a 4% annual probability of a fire causing $250,000 in damage. Insurance costs $14,000 per year. What is the expected annual loss without insurance?

A. $4,000
B. $10,000
C. $14,000
D. $250,000

3.3-Q10. A small contractor chooses not to buy insurance for minor equipment damage and instead budgets for those losses. This demonstrates:

A. Risk retention
B. Risk transfer
C. Risk avoidance
D. Risk sharing

Answer key and explanations

Q	Ans	Why
1	B	The captive is owned by the company, so losses stay in the corporate family: funded retention, not true transfer.
2	D	Insurance transfers the financial consequence of a loss to the insurer in exchange for a premium. It does not reduce the hazard.
3	D	Indemnity and insurance requirements in contracts are contractual (non-insurance) risk transfer.
4	A	Sharing spreads a risk among parties that each keep a portion. Insurance is transfer; eliminating materials is avoidance; sprinklers are loss reduction.
5	C	Avoidance means not doing the activity at all; mitigation (loss prevention/reduction) keeps the activity but lowers likelihood or severity.
6	A	Prevention lowers frequency (stops the fire starting); reduction lowers severity after the event (sprinklers, first aid).
7	C	Subrogation lets the insurer step into the insured's shoes to recover from the responsible party. Hold harmless/indemnity are contract terms.
8	B	EMR = actual / expected = 150,000 / 120,000 = 1.25, so premiums are about 25% above the manual rate. EMR above 1.0 is a debit.
9	B	Expected loss = probability x consequence = 0.04 x 250,000 = $10,000. Premiums exceed expected loss because they include insurer costs and profit; insurance buys certainty.
10	A	Retention means accepting the loss and paying it from internal funds; it suits frequent, low-severity losses.

DOMAIN 3 · COMPETENCY 3.4

Risk Analysis: Identify, Rank and Monitor

CSP11 blueprint: Apply risk analysis process of identifying, ranking, and monitoring (e.g., disasters/emergency preparedness, fire prevention, occupational health, hazardous materials management/environmental compliance)

1. Definition

This competency applies the risk process across specific risk areas: disasters and emergencies (hazard vulnerability analysis), fire, occupational health exposures and hazardous materials/environmental compliance. Risks are identified, ranked so resources go to the highest risks first, and monitored over time.

2. Explanation in simple words

Make a list of everything that could go wrong across the site, score each one, fix the top of the list first, and keep the list current as things change.

3. Key points to remember

- Ranking tools: risk matrix (likelihood × severity), risk priority number (FMEA), risk scores with tie-breaker rules (often higher severity first), expected loss (probability × cost), Pareto of loss data.
- Disaster/emergency: Hazard Vulnerability Analysis (HVA) rates probability, human/property/business impact and preparedness; drives the emergency plan and business continuity priorities.
- Fire risk: fuel load, ignition sources, occupancy, protection systems (sprinklers, detection), housekeeping, hot work; NFPA codes; insurer (FM Global) surveys.
- Occupational health: exposure assessment by similar exposure groups (SEGs); compare exposure (e.g., 95th percentile) with OEL; exposure categories 0-4 (AIHA); prioritize sampling and controls.
- Hazardous materials/environmental compliance: chemical inventory, quantities vs thresholds (EPCRA TPQ, PSM TQ, RMP), compatibility, storage limits, permits, spill potential; compliance calendar and audits.
- Monitoring: risk register reviews, inspections, KPIs, audits, changes in operations (MOC), incident trends; re-rank after controls (residual risk).

4. Real work scenario

> A chemical distributor ranks its risks: ammonia release (likelihood 2, severity 5 = 10), forklift collision (4 × 3 = 12), warehouse fire (2 × 5 = 10). The forklift risk is addressed first by score, and the company's tie-breaker rule ranks the two 10s by severity, so both get specific controls and quarterly monitoring.

5. CSP exam tips and traps

- Calculate every option's score; read any tie-breaker rule in the stem.
- HVA = probability × impact adjusted for preparedness.
- Exposure ranking questions compare the upper percentile with the OEL.

6. Every blueprint point (8 blueprint items + 2 more)

Identification (identifying risks across categories) BLUEPRINT

Turn diverse hazards into one defensible risk picture. ISO 31000 gives the process, ISO 45001 embeds it in the OHSMS, PSM adds process-safety rigor and NFPA 551 guides fire risk methods.

- Tools: Business Impact Analysis, scenario analysis, What-If, bow-tie, ignition source surveys, exposure profiling, compatibility matrices.
- Categories in the blueprint: disasters/emergency preparedness, fire prevention, occupational health, hazardous materials and environmental compliance.
- Document assumptions and data sources so the analysis is defensible.

Ranking (risk ranking toolkit) BLUEPRINT

Rank risks consistently so leadership has a clear decision path.

- Risk matrices (3x3, 5x5) for qualitative ranking; semi-quantitative Severity x Likelihood with calibrated scales.
- Bow-tie with barrier health to weight controls by effectiveness and degradation.
- LOPA for process scenarios with independent protection layers; consequence modeling for fire and release scenarios.
- Define evaluation criteria in writing (ALARP thresholds, legal must-meets) per ISO 31000.
- Mini example (guide): solvent wash line, S = 5, L = 2 -> 10; with weak IPL independence treat as high and escalate to engineering/substitution, not PPE.

> **Exam tip:** Pick the answer that creates the clearest decision path: defined criteria, documented assumptions, verification steps.

Monitoring (leading and lagging indicators for risk) BLUEPRINT

Monitor so you know controls actually work, using leading indicators, lagging indicators, a risk register and management review.

- Leading (PSM): overdue MOCs and PHAs, safety-critical maintenance backlog, bypassed interlocks, IPL proof-test compliance.
- Leading (fire/IH/env): hot work permit quality, impairment counts and duration, dust housekeeping scores, % tasks with current exposure assessment, capture velocity checks, container defects, manifest errors.
- Lagging: TRIR, recordables tied to specific hazards, small fires, near releases, regulatory NOVs, insurance findings.
- Revalidation: PSM PHA at least every 5 years; fire risk after occupancy/process change; IH reassessment when tasks, materials or exposures change.
- ANSI/ASSP Z16.1 covers safety and health metrics.

Disaster/emergency preparedness risk analysis BLUEPRINT

Identify natural, utility, cyber, supply chain and civil unrest threats and rank them for emergency planning.

- Tools: Business Impact Analysis, scenario analysis, What-If, bow-tie for critical scenarios.
- Example (guide): loss of power to critical ventilation - ranked high severity, moderate likelihood without redundancy; monitored by monthly generator tests, transfer switch checks and ERP drill pass rate.
- ISO 45001 clause 8.2 (emergency preparedness and response).

Fire prevention risk analysis BLUEPRINT

Identify ignition sources, fuel loads, ventilation paths, occupancy and impaired protection; rank and monitor fire risk using NFPA 551 methods.

- Tools: ignition source surveys, hot work reviews, combustible dust assessments, bow-tie linking ignition barriers and suppression.
- NFPA 551 = guide for evaluating fire risk assessments; expects transparent assumptions, data sources and sensitivity analysis.
- Example (guide): spray booth with flammable coatings - monitor weekly booth differential pressure, semiannual sprinkler tests, hot work audits.

Occupational health risk analysis BLUEPRINT

Identify chemical, physical, biological and ergonomic exposures, rank them against OELs and task frequency, and monitor trends.

- Tools: exposure profiling, task inventory, SDS review, qualitative and quantitative IH sampling, control banding.
- Example (guide): silica during maintenance chipping - compare samples to OELs, consider frequency and duration; monitor exposure trends, LEV capture velocity, training logs.
- Keep IH data traceable for periodic review.

> **Exam tip:** For IH questions, a sampling strategy plus the hierarchy beats a PPE-only plan.

Hazardous materials risk analysis BLUEPRINT

Identify storage incompatibilities, secondary containment gaps and transport errors, rank by release potential and monitor segregation and containment.

- Tools: compatibility matrices, secondary containment calculations, DOT packaging checks.
- Screen for PSM applicability (highly hazardous chemicals at threshold quantities).
- Example (guide): acids and bases in a shared room with single-wall containment and a drain path = high risk; monitor segregation audits, containment inspections, spill drills.

Environmental compliance risk analysis BLUEPRINT

Identify waste misclassification, spill pathways and permit gaps, and enter compliance exposure into the integrated risk register.

- Tools: cradle-to-grave waste mapping, SPCC alignment, waste stream error rates.
- Leading indicators: manifest errors, SPCC drill performance; lagging: Notices of Violation (NOVs).
- ISO 31000 register entries should include compliance exposure, not only safety consequences.

Communication that leads to action

Risk rankings are only useful when converted into prioritized, owned actions communicated to each audience.

- Two-way communication (ISO 31000); tailor to executives, supervisors and operators.
- Prioritized action plan with owners and due dates.
- Link each high-priority item to the governing standard so the rationale is clear.

> **Exam tip:** Process scenarios cue LOPA, IPL independence, MOC and revalidation; fire scenarios cue NFPA 551 documentation and sensitivity testing.

Risk vocabulary (assessment vs. analysis vs. evaluation)

The guide's memory table distinguishes closely related risk terms that are often used as distractors.

- Risk assessment = identification + analysis + evaluation (overall process).
- Risk analysis = understand nature and determine level of risk; risk estimation = assigning values to likelihood and consequence.
- Risk evaluation = compare results with criteria to decide acceptability.
- Risk characterization = description of nature and magnitude of risk, including uncertainties.
- Risk treatment = selecting and implementing measures to modify risk; risk management = coordinated activities to direct and control an organization with regard to risk.

> **Exam tip:** 'Which step compares risk to criteria?' = evaluation, not analysis.

Read more: Yates, Safety Professional's Reference and Study Guide (4th ed.): Ch. 26 Risk Assessment and Management; Ch. 39 Emergency Management and Preparedness; Ch. 27 Hazardous Materials Management

7. Ten CSP-style practice questions

Answer all ten on paper first, then check the key. Score: _____ / 10 Date: _____ Retest: _____ / 10

3.4-Q1. In a Hazard Vulnerability Analysis (HVA) for emergency planning, hazards are prioritized MAINLY by:

A. Alphabetical order
B. Probability and impact, adjusted for existing preparedness and mitigation
C. Cost of insurance only
D. Number of past OSHA citations

3.4-Q2. In an AIHA-style exposure assessment, a similar exposure group's 95th percentile exposure is estimated to be above the OEL. What category and action are appropriate?

A. Category 1 - no action
B. Category 2 - periodic review
C. Category 4 - unacceptable; implement controls and confirm with monitoring
D. The exposure cannot be categorized without medical data

3.4-Q3. A paint booth has flammable solvent vapors, nearby ignition sources and poor ventilation. Which control should be prioritized?

A. Install properly designed ventilation with explosion-proof (classified) equipment
B. Upgrade worker PPE
C. Train workers on paint handling
D. Add more signage

3.4-Q4. Using Risk = Likelihood x Severity (1-5 scales): Hazard P = L3 x S4, Hazard Q = L4 x S3, Hazard R = L2 x S5. Policy says ties are broken by higher severity. Which hazard is addressed FIRST?

A. P
B. Q
C. R
D. P and Q equally

3.4-Q5. When ranking a facility's environmental compliance risks, an SPCC (Spill Prevention, Control, and Countermeasure) plan primarily addresses:

A. OSHA safety standards
B. Fire suppression codes
C. Industrial hygiene sampling
D. EPA oil spill prevention requirements

3.4-Q6. In emergency preparedness, regular backup generator tests and evaluation of emergency drill performance are examples of:

A. Preventive barriers
B. Risk monitoring activities
C. Loss reduction
D. Administrative risk ranking

3.4-Q7. A company incurred $800,000 in losses over the last 3 years. Assuming the same loss rate and no inflation, what are the expected losses over the next 10 years?

A. $800,000
B. $2,400,000
C. $2,666,667
D. $8,000,000

3.4-Q8. Risk profiling is BEST described as:

A. Listing job steps and hazards for each task
B. Assigning a numerical score to each risk
C. Ranking risks by financial cost only
D. Describing the types, levels and sources of risk an organization faces

3.4-Q9. A fire risk assessment of a distribution warehouse produces four findings. Which should receive the HIGHEST priority?

A. Cardboard stored next to an office copier
B. A closed sprinkler control valve in an area with high-piled plastic commodities and forklift battery charging
C. One burned-out exit sign bulb in a well-lit corridor with a second marked exit
D. A fire extinguisher whose monthly inspection tag is one week overdue

3.4-Q10. An EHS manager must rank hazardous materials risks across a facility's chemical inventory to prioritize controls and compliance work. Which approach is MOST appropriate?

A. Rank each chemical by hazard class, quantity relative to regulatory thresholds and likelihood of release, then monitor changes through inventory updates and MOC
B. Rank chemicals by purchase cost
C. Rank chemicals in the order of the SDS binder
D. Rank only the chemicals that have spilled in the past

Answer key and explanations

Q	Ans	Why
1	B	An HVA scores probability and human/property/business impact, then adjusts for preparedness and response capability to rank relative risk.
2	C	Exposure categories compare the upper percentile with the OEL; above the OEL is Category 4, requiring controls and follow-up sampling.
3	A	An engineering control that keeps vapor below the LEL and removes ignition sources addresses both the fire and the health hazard at the source.
4	A	P = 12, Q = 12, R = 10. P and Q tie, and the tie-breaker (higher severity) selects P. Read the whole stem for decision rules.
5	D	SPCC plans are required by EPA (40 CFR 112) for facilities storing oil above threshold quantities to prevent discharges to navigable waters.
6	B	Testing and drills verify that controls and response capability still work, which is the monitoring part of the risk process.
7	C	Annual rate = 800,000 / 3 = $266,667; x 10 years = $2,666,667. Trap: 2.4 million multiplies by 3.
8	D	A risk profile is the organization-wide picture of its risks, used for strategic prioritization. Listing task steps is a JHA.
9	B	An impaired sprinkler system over a high fuel load with an ignition source combines high likelihood and catastrophic severity; restore protection and use impairment procedures.
10	A	Risk ranking combines hazard severity, quantity (e.g., PSM or EPCRA thresholds) and likelihood, and must be monitored as inventory changes. Spill history alone misses latent risks.

DOMAIN 4

Emergency Management

9% of the exam · 5 competencies

4.1	Emergency Response Plans Create, employ, and maintain an Emergency Response Plan (e.g., fire, severe weather, nuclear incidents, natural disasters, terrorist attacks, chemical spills, utilities systems, cyber security)
4.2	Disaster Response and Recovery Describe the elements in disaster response and recovery (e.g., incident command, business continuity, contingency plans)
4.3	Fire Prevention, Protection and Suppression Identify key components of fire prevention, protection, and suppression systems
4.4	Hazardous Materials Transportation and Security Prepare procedures for the safe transportation and security of hazardous materials
4.5	Workplace Violence Prevention Implement a workplace violence prevention program

DOMAIN 4 · COMPETENCY 4.1

Emergency Response Plans

CSP11 blueprint: Create, employ, and maintain an Emergency Response Plan (e.g., fire, severe weather, nuclear incidents, natural disasters, terrorist attacks, chemical spills, utilities systems, cyber security)

1. Definition

An Emergency Response Plan (ERP) or Emergency Action Plan (EAP) defines how an organization will protect people, property and the environment during emergencies. OSHA 1910.38 sets minimum EAP elements; 1910.120(q) covers emergency response to hazardous substance releases; NFPA 1600 (now NFPA 1660) covers emergency management and business continuity programs.

2. Explanation in simple words

Before the emergency happens, decide who does what, how people are warned, where they go, how you count them, and how you call for help. Then practice it and update it.

3. Key points to remember

- OSHA 1910.38 EAP minimum elements: procedures for reporting fires and emergencies; evacuation procedures and exit route assignments; procedures for employees who remain to operate critical operations; procedures to account for all employees after evacuation; rescue and medical duties; names/titles of people to contact. Written plan required for > 10 employees; alarm system; training and review when the plan changes.
- Development steps: hazard vulnerability analysis → resources and capabilities → plan writing (roles, alarms, evacuation/shelter, accountability, communication, coordination with outside agencies) → training → drills/exercises (tabletop, functional, full-scale) → critique → update.
- Phases: prevention, mitigation, preparedness, response, recovery.
- Priorities: life safety → incident stabilization → property/environmental conservation.
- Severe weather: watch (conditions favorable - prepare) vs warning (occurring/imminent - act); tornado - interior, lowest-level shelter; lightning - 30/30 rule.
- Nuclear/radiological: shelter in place, close HVAC, time-distance-shielding; potassium iodide only when directed (protects thyroid only).
- Terrorism/CBRNE and active shooter: shelter in place for chemical/biological plumes; Run-Hide-Fight; NFPA 3000 (active shooter/hostile event).
- Chemical spills: incidental release (cleaned by trained employees in the area) vs emergency response (HAZWOPER). HAZWOPER training: awareness, operations (8 h), technician (24 h), specialist, incident commander; site workers 40 h.
- Utilities: power loss (generators, emergency lighting), gas leaks (evacuate, no switches), water loss (sprinkler impairment, fire watch). Cyber: incident response plan - contain/isolate, eradicate, recover from clean backups, notify; ICS/OT systems put in a safe state.

4. Real work scenario

> A warehouse fire drill shows employees reached the assembly area quickly, but supervisors could not confirm whether three temporary workers were inside. The plan is updated with a shift roster and a headcount procedure for contractors and temps, and the next drill includes a 'missing person' scenario.

5. CSP exam tips and traps

- Accounting for employees after evacuation is a required EAP element and a common missing-element answer.
- Watch vs warning is a classic vocabulary trap.
- Drills test and improve the plan; the follow-up to a poor drill is training and plan updates.
- Plan may be communicated orally only for employers with 10 or fewer employees.

6. Every blueprint point (8 blueprint items + 6 more)

ERP purpose and legal basis (OSHA 1910.38 EAP, NFPA 1600)

An Emergency Response Plan is a written, site-specific, living document that describes how the organization prepares for, responds to and recovers from emergencies. OSHA 1910.38 sets the minimum Emergency Action Plan; NFPA 1600 is the broader best practice that adds business continuity.

- 1910.38 minimum elements: reporting fires/emergencies; evacuation procedures and exit route assignments; procedures for employees who remain to operate critical operations; accounting for all employees after evacuation; rescue and medical duties; names or job titles to contact for plan information.
- Written plan required if more than 10 employees; 10 or fewer may communicate orally.
- Distinctive employee alarm system required (1910.165).
- Review the plan with each employee when the plan is developed or the employee is assigned, when responsibilities change, and when the plan changes.
- Related: HAZWOPER 1910.120 (site-specific ERP), PSM 1910.119 (emergency planning). NFPA 1600 has been consolidated into NFPA 1660 (2024 edition) - the guide cites NFPA 1600.

> **Exam tip:** Memorize the six 1910.38 minimum elements - a common 'which is NOT required' question.

Identifying and analyzing emergency risks

The ERP is built on a documented hazard identification, risk assessment and impact analysis covering all reasonably expected emergencies - natural, human-caused and biological.

- 1910.38(a): plan must address emergencies reasonably expected in the workplace; NFPA 1600 requires hazard identification, risk assessment and business impact analysis.
- Steps: hazard survey; review incident/loss/insurance records; consult employees, managers and local authorities; risk matrix; prioritize; document and review annually.
- Natural (earthquake, flood, tornado, wildfire) -> evacuation/shelter; human-caused (fire, violence, utility failure, sabotage, cyber) -> lockdown/evacuation/notify; biological (pandemic, bioterrorism, contamination) -> hygiene, isolation, notify.

> **Exam tip:** OSHA expects workplace-specific risk identification, not a generic template.

Elements of an ERP

A complete ERP contains defined purpose and scope, roles, chain of command and communications, risk summary, scenario-specific procedures, resources, contacts, training/drills and a review process.

- Roles: Incident Commander, evacuation/floor wardens, first aiders, communications lead, all employees.
- Chain of command names a successor if the IC is absent; backup contacts for redundancy.
- Procedures: evacuation, lockdown, shelter-in-place, fire/explosion, medical, spills/releases, utility failure.
- Resource inventory (extinguishers, spill kits, first aid, AEDs, radios) with locations and responsible persons; internal and external contact lists (fire, police, EMS, poison control, agencies, insurer, vendors).
- Annual review and after-action reviews after drills or real events (best practice; required by NFPA 1600, not explicitly by 1910.38).

Preparing for emergencies (site readiness)

Preparation covers risk assessment, worksite security, evacuation routes and assembly points, shelter-in-place areas, life safety systems, medical/first aid, drills and training.

- Security: controlled entry, badges, cameras, visitor sign-in/escort, cybersecurity for building controls.
- Evacuation: marked and illuminated exits, posted maps, assembly points at a safe distance, primary and alternate routes (NFPA 101).
- Shelter-in-place rooms: interior, stocked with supplies; used when evacuation is more dangerous (toxic release, tornado, violence).
- Medical: first aid kits and trained responders; 1910.151 requires a trained first aider if no infirmary/clinic is in near proximity (OSHA interprets 3-4 minutes for life-threatening injuries); eyewash where corrosives are used.
- Drills: vary scenarios and shifts; after-action review after each.

> **Exam tip:** OSHA expects you to plan, equip, train and drill for all reasonably anticipated emergencies, not just fire.

Creating, implementing and maintaining the ERP

Development, implementation and maintenance are a continuous cycle: involve stakeholders, train everyone, drill, document and update.

- Develop: involve management, employees, unions, contractors and external responders; base procedures on the risk assessment; include site details (floor plans, chemical inventory, people with disabilities).
- Implement: initial training and refresher after updates; signage; redundant communication channels; document everything.
- Maintain: drills at least annually and after major changes; revise after new hazards, lessons learned, staff/contact changes or regulatory updates; keep current copies accessible.
- Documentation of training and drills is both best practice and proof of compliance.

> **Exam tip:** The ERP is not just a document - it is a living process; 'write it and file it' answers are wrong.

Fire BLUEPRINT

Fire plans focus on detection, alarm, evacuation, extinguishment by trained employees (if permitted) and accounting at assembly points.

- References: OSHA 1910.38 (EAP), 1910.39 (fire prevention plan), 1910.157 (extinguishers), 1910.165 (alarms), NFPA 101, NFPA 72.
- Alarms audible/visible, tested; automatic fire department notification where required.
- Evacuation wardens assist and account for staff and visitors; roll call at assembly point.
- Extinguisher use: PASS (Pull, Aim, Squeeze, Sweep); monthly inspection, annual maintenance.
- Drills at least annually (best practice), more often in high-risk sites.

Severe weather BLUEPRINT

Severe weather plans cover tornadoes, hurricanes, floods, blizzards and extreme heat/cold with monitoring, shelter and shutdown procedures for hazards relevant to the location.

- Tornado: interior rooms away from windows, lowest floor; tornado drills; supplies (water, radios, first aid).
- Hurricane/flood: pre-arranged shutdown and evacuation, secure loose items, protect openings.
- Extreme heat/cold: monitor advisories, hydration/cooling or warming stations.
- Alerts: NOAA weather radios, text alerts; tornado watch = conditions favorable, warning = tornado sighted or on radar - take shelter.

Nuclear incidents BLUEPRINT

Plans for releases of radioactive material from nuclear plants, medical/industrial sources or weapons; required for facilities near nuclear plants and coordinated with NRC, FEMA and local agencies.

- Shelter-in-place: designated rooms, seal openings, shut down HVAC if needed.
- Evacuation routes and transport coordination; follow official advisories.
- Radiation monitoring (dosimeters); potassium iodide (KI) protects only the thyroid from radioactive iodine and is taken only on official instruction.
- Protective principles: time, distance, shielding.
- Planning zones around nuclear plants: about 10-mile plume exposure and 50-mile ingestion pathway zones (FEMA/NRC).

Natural disasters BLUEPRINT

Earthquakes, floods, wildfires and similar events need location-specific procedures, supplies and post-event hazard checks.

- Earthquake: drop, cover and hold on; secure heavy items; afterward check for injuries, gas leaks and fires before re-entry.
- Flood: move critical equipment to higher ground, shut down utilities, use flood maps.
- Wildfire: defensible space, clear combustibles, monitor air quality, respirators for smoke, rapid evacuation.
- Kits: water, food, radios, flashlights, batteries, PPE.

Terrorist attacks and active shooter BLUEPRINT

Deliberate attacks and active shooter events require lockdown, mass notification and Run-Hide-Fight training, coordinated with law enforcement.

- References: OSHA General Duty Clause, DHS guidance, NFPA 3000 (Active Shooter/Hostile Event Response).
- Run, Hide, Fight (DHS) - escape if possible, hide if not, fight only as a last resort.
- Lockdown procedures, silent alarms, mass notification.
- Bomb threats: keep caller talking, use a bomb threat checklist, do not use radios/cell phones near a suspected device.
- Post-incident medical, psychological and legal support; tabletop and live drills with law enforcement.

Chemical spills BLUEPRINT

Chemical spill plans cover immediate evacuation and isolation, containment by trained personnel only, decontamination, medical support and regulatory notification.

- Evacuate, activate alarms, shut down sources only if safe.
- Only trained responders with proper PPE clean up (HAZWOPER 1910.120(q) for emergency response; incidental spills handled by trained area employees).
- Spill kits, neutralizers, booms; decontamination of people and areas; eyewash/showers.
- Notify authorities when a reportable quantity is released (CERCLA RQ -> National Response Center; EPCRA 304 -> state/local).
- Document: incident report, root cause, regulatory notifications.

> **Exam tip:** Untrained employees evacuate and report; they do not attempt cleanup of a non-incidental release.

Utility systems failures BLUEPRINT

Loss of power, water or gas can create safety hazards and stop critical systems, so plans address safe shutdown, backup power and safe restoration.

- Procedures to safely shut down and restart equipment; inspect before re-energizing.
- Emergency lighting and backup power (NFPA 110 generators; NFPA 101 requires 90 min emergency lighting).
- Alternative communications (radios, cell phones with backup power); alternative water for drinking and sanitation.
- Gas leak: evacuate immediately, eliminate ignition sources, call utility provider.

Cybersecurity BLUEPRINT

Cyber incidents can disable safety controls, alarms and communications. There is no OSHA cyber standard, but NFPA 1600 and DHS recommend cyber response planning for safety and operational systems.

- Train employees to recognize phishing, unusual activity and outages; report quickly.
- Activate IT/cyber incident response team and plan.
- Secure offsite backups and restoration protocols.
- Maintain safety monitoring and alarms via manual backups; alternate notification methods if digital systems fail.
- Industrial control systems guidance: ISA/IEC 62443, NIST Cybersecurity Framework.

Pandemic and bioterrorism

Biological emergencies need prevention, detection, communication, medical response, coordination with public health and law enforcement, and business continuity for high absenteeism.

- Pandemic = global outbreak; epidemic = cases above expected in a community or region.
- Pandemic controls: hygiene, ventilation/barriers (engineering), staggered shifts/remote work (administrative), PPE; symptom reporting and return-to-work criteria; cross-training.
- Bioterrorism: recognize suspicious packages/illness clusters, isolate area, notify public health, law enforcement and FBI; decontamination; rumor control point.
- References: OSHA General Duty Clause, 1910.1030 (BBP), 1910.120, CDC/NIOSH guidance, NFPA 1600.

Read more: Yates, Safety Professional's Reference and Study Guide (4th ed.): Ch. 39 Emergency Management and Preparedness, p. 870; Ch. 4 Regulations (HAZWOPER)

7. Ten CSP-style practice questions

Answer all ten on paper first, then check the key. Score: _____ / 10 Date: _____ Retest: _____ / 10

4.1-Q1. A radiological release is expected from a nearby nuclear power plant and officials advise sheltering in place. Which action is MOST appropriate?

A. Open windows to dilute any contamination entering the building
B. Have all employees take potassium iodide to protect against all radiation
C. Shelter indoors, shut down HVAC and close openings, and take potassium iodide only if officials instruct it
D. Drive employees through the plume to reach a shelter farther away

4.1-Q2. During a night shift, a fire breaks out in a section of a warehouse. Several employees evacuate safely, but one forklift operator is unaccounted for. The plan identifies assembly points and alarm systems but has no accountability procedure. What element is missing from the emergency plan?

A. Communication procedures with insurance carriers
B. Coordination procedures with environmental agencies
C. Post-incident counseling for employees
D. A system for accounting for all employees after evacuation

4.1-Q3. The National Weather Service issues a TORNADO WARNING for your county. What should the facility do?

A. Move personnel immediately to designated storm shelters or interior refuge areas
B. Continue normal work but monitor the weather radio
C. Evacuate everyone to their vehicles
D. Wait until the tornado is visible before acting

4.1-Q4. A 1-liter bottle of a known solvent breaks in a lab. Trained lab staff in the immediate area can safely absorb it with a spill kit. Under HAZWOPER 1910.120(q), this is:

A. An emergency response requiring a HAZMAT team
B. An incidental release that may be cleaned up by trained employees in the immediate area
C. A release that automatically requires EPA notification
D. An uncontrolled release requiring evacuation of the building

4.1-Q5. A toxic gas plume is drifting toward a residential area at a steady 10 mph. It takes 15 minutes to mobilize and complete evacuation of the at-risk population. What is the MINIMUM distance the plume must be from the area when the evacuation order is given to ensure safe clearance?

A. 1.5 miles
B. 2.5 miles
C. 3 miles
D. 4 miles

4.1-Q6. What is the MOST important step after conducting an emergency drill?

A. Repeat the same drill immediately until all employees pass
B. Send the drill results to OSHA for recordkeeping
C. Conduct an after-action review and document improvement areas
D. Dismiss employees early as a reward for participation

4.1-Q7. During a drill evaluation, observers note confusion about who should contact external emergency responders. Which plan element MOST likely needs improvement?

A. Command hierarchy and communication chain
B. Evacuation route signage
C. Number and placement of fire extinguishers
D. Sprinkler system activation pressure

4.1-Q8. A ransomware attack encrypts the building automation and process control servers at a chemical plant during operations. According to the facility's emergency response plan, what should be done FIRST?

A. Pay the ransom quickly to restore operations
B. Restore the servers from backup while they remain connected to the network
C. Wait for the IT vendor to diagnose the problem before taking any action
D. Isolate affected systems from the network and activate the cyber incident response plan, placing processes in a safe state

4.1-Q9. Employees report a strong odor of natural gas in an office building after nearby utility work. What is the BEST immediate action?

A. Turn on exhaust fans to ventilate the building
B. Locate the leak by checking appliances and connections
C. Evacuate the building without operating electrical switches, then call the gas utility and 911 from a safe location
D. Have employees stay at their desks while maintenance investigates

4.1-Q10. A receptionist receives a telephone bomb threat. According to a typical emergency response plan, what should the receptionist do?

A. Hang up immediately and continue working
B. Keep the caller talking, record details using a bomb threat checklist, and notify security and police per the plan
C. Personally search the building for suspicious packages
D. Announce over the public address system that there is a bomb in the building

Answer key and explanations

Q	Ans	Why
1	C	Buildings provide shielding and closing HVAC limits infiltration. KI only blocks thyroid uptake of radioiodine and is taken only on official instruction.
2	D	OSHA 1910.38(c) requires procedures to account for all employees after evacuation; without it, missing workers may not be identified for rescue.
3	A	A warning means a tornado is occurring or imminent: shelter now in interior, lowest-level, windowless areas. Trap: treating a warning like a watch (prepare only).
4	B	Releases that employees in the immediate area can absorb or control without significant safety or health risk are incidental, not emergency responses under 1910.120(q).
5	B	Distance = speed x time = 10 mi/h x (15/60) h = 2.5 miles. Trap: forgetting to convert minutes to hours.
6	C	Drills exist to test and improve the plan; the after-action review captures weaknesses and assigns corrective actions. OSHA does not collect drill results.
7	A	The plan must assign clear roles, including who notifies outside responders; confusion here is a chain-of-command and communication gap, not a hardware issue.
8	D	Containment comes first: isolate infected systems to stop spread and move the process to a safe state. Restoring while still connected risks reinfection.
9	C	Utility gas leaks present an explosion hazard: evacuate, avoid ignition sources such as switches and phones, and report from a safe distance.
10	B	Bomb threat procedures emphasize capturing caller details (checklist) and notifying authorities, who decide on search or evacuation; panic announcements and self-searches are inappropriate.

DOMAIN 4 · COMPETENCY 4.2

Disaster Response and Recovery

CSP11 blueprint: Describe the elements in disaster response and recovery (e.g., incident command, business continuity, contingency plans)

1. Definition

Disaster response and recovery covers how an organization organizes the emergency response (Incident Command System), keeps critical functions running or restores them (business continuity), and prepares alternatives for foreseeable disruptions (contingency plans).

2. Explanation in simple words

During a disaster one person is in charge and everyone knows their role (ICS). Afterwards, the business needs a plan to get the most important work running again fast (business continuity).

3. Key points to remember

- Incident Command System (ICS/NIMS): Incident Commander; Command Staff - Safety Officer, Public Information Officer, Liaison Officer; General Staff sections - Operations, Planning, Logistics, Finance/Administration.
- ICS principles: common terminology, modular organization, management by objectives, Incident Action Plan (prepared by Planning), manageable span of control (3-7, ideally 5), unity of command, unified command for multi-agency incidents, formal transfer of command with briefing, accountability, integrated communications.
- Business continuity: business impact analysis (BIA) identifies critical functions and impacts; Recovery Time Objective (RTO) = maximum acceptable downtime; Recovery Point Objective (RPO) = maximum acceptable data loss; strategies include alternate sites, backup suppliers, cloud backups, mutual aid.
- Standards: NFPA 1600/1660, ISO 22301 (business continuity management systems), FEMA guidance.
- Contingency planning: 'what-if' alternatives for supply disruption, utility loss, staffing shortages, pandemic; pre-arranged contracts; succession plans.
- Recovery: damage assessment, restoration, employee assistance (EAP/counseling), insurance claims, lessons learned and plan updates. Disaster recovery (IT) focuses on systems and data.

4. Real work scenario

> A regional power outage hits during peak production. The facility's generators run life-safety systems, but IT and supply links are down. The business continuity plan activates: critical orders move to a sister plant, data are restored from cloud backups within the 4-hour RTO, and the logistics section of the ICS arranges fuel deliveries for the generators.

5. CSP exam tips and traps

- Planning Section prepares the IAP; Logistics provides resources; Operations does the tactical work; Finance tracks costs and claims.
- Safety Officer, PIO and Liaison are Command Staff, not sections.
- RTO = time; RPO = data loss.
- Unified Command keeps each agency's authority while sharing objectives.

6. Every blueprint point (3 blueprint items + 5 more)

Incident Command System (ICS) BLUEPRINT

ICS is the standardized, scalable on-scene management system within the National Incident Management System (NIMS). OSHA HAZWOPER 1910.120(q) requires an ICS for emergency response to hazardous substance releases, led by the senior emergency response official.

- Five major functions: Command, Operations, Planning, Logistics, Finance/Administration (Intelligence/Investigations may be added).
- Command staff: Public Information Officer, Safety Officer (authority to stop unsafe acts), Liaison Officer.
- Span of control: 3 to 7 subordinates, optimum 5.
- Principles: common terminology, modular organization, unity of command, unified command for multi-agency incidents, Incident Action Plan, integrated communications.
- The Incident Commander is responsible for everything until delegated; the IC position always exists.

> **Exam tip:** Safety Officer is command staff (reports to IC), not general staff; unified command = several agencies sharing command, not several ICs giving separate orders.

Business continuity BLUEPRINT

Business continuity keeps critical functions running or restores them quickly after a disruption. It is driven by a Business Impact Analysis (BIA) and covered by NFPA 1600 (now NFPA 1660) and ISO 22301.

- BIA identifies critical processes, dependencies and the impact of disruption over time.
- RTO (recovery time objective) = target time to restore a function; RPO (recovery point objective) = maximum tolerable data loss (how far back backups must go); MTPD = maximum tolerable period of disruption (RTO must be shorter).
- Strategies: alternate sites (hot, warm, cold), backup suppliers, data backups, cross-training, remote work.
- Benefits: less downtime and financial loss, preserved reputation, insurance and regulatory compliance.
- ISO 22301 = business continuity management systems (requirements).

> **Exam tip:** A BIA comes before choosing continuity strategies.

Contingency planning BLUEPRINT

Contingency plans are pre-arranged 'what if' alternatives for specific disruptions - backup resources, alternate procedures and triggers for using them.

- Examples: mutual aid agreements with neighboring plants or fire departments, pre-qualified restoration contractors, backup generators, alternate communications.
- Define activation triggers, decision authority and resources in advance.
- Test through tabletop exercises and update after use.

Phases of emergency management

Emergency management is a cycle of mitigation (and prevention), preparedness, response and recovery.

- Mitigation: reduce the impact before an event (elevating equipment above flood level, seismic bracing).
- Preparedness: plans, training, drills, equipment, agreements.
- Response: protect life, stabilize the incident, protect property and environment (life safety -> incident stabilization -> property conservation).
- Recovery: restore operations, repair, return to normal, lessons learned.

> **Exam tip:** 'Before the event, reducing impact' = mitigation; 'plans and drills' = preparedness.

HAZWOPER emergency response (1910.120(q))

Employers whose employees respond to hazardous substance releases need a written emergency response plan, ICS, training by role and medical surveillance for HAZMAT team members.

- Training levels: First Responder Awareness (recognize, notify), Operations (8 hr, defensive containment), Hazmat Technician (24 hr, offensive stopping of release), Specialist (24 hr), On-Scene Incident Commander (24 hr plus competencies).
- Annual refresher training or demonstrated competency.
- Employers who evacuate all employees and do not respond may use a 1910.38 EAP instead of a (q) ERP.
- Buddy system and backup personnel required in danger areas; positive-pressure SCBA in unknown atmospheres.

> **Exam tip:** Defensive (contain from a distance) = Operations level; plugging/patching the leak = Technician.

Exercises and drills

Exercises test plans, train people and reveal gaps; results feed corrective actions.

- Types (FEMA HSEEP): discussion-based - seminar, workshop, tabletop, game; operations-based - drill, functional exercise, full-scale exercise.
- Tabletop: discussion of a scenario in a conference room, low cost.
- Functional: tests coordination and command/communications without moving people and equipment to the field.
- Full-scale: real-time deployment with multiple agencies; most realistic and costly.
- Follow every exercise or real event with an after-action review and improvement plan.

Recovery and after-action review

Recovery restores safe operations and captures lessons learned so the plan improves.

- Re-entry only after hazards are assessed (structural, electrical, gas, atmosphere) and authorities clear the site.
- Restoration contractors, damage assessment, insurance claims and documentation.
- Psychological support (EAP, critical incident stress management) for affected workers.
- After-action review: what was planned, what happened, why, what to change; update ERP, training and resources.

Crisis and emergency communications

Clear, redundant communication protects people during response and protects the organization's reputation afterward.

- Mass notification and redundant channels (PA, text, email, radios).
- Single designated spokesperson / Public Information Officer; consistent, accurate, timely messages.
- Pre-scripted messages and contact lists; rumor control.
- Notify regulators as required (e.g., OSHA fatality within 8 hours, inpatient hospitalization, amputation or eye loss within 24 hours).

Read more: Yates, Safety Professional's Reference and Study Guide (4th ed.): Ch. 39 Emergency Management and Preparedness, p. 870

7. Ten CSP-style practice questions

Answer all ten on paper first, then check the key. Score: _____ / 10 Date: _____ Retest: _____ / 10

4.2-Q1. Which positions make up the command staff under the Incident Command System (ICS)?

A. Public Information Officer, Chief Executive Officer, and Safety Officer
B. Logistics Section Chief, Safety Officer, and Contracting Officer
C. Liaison Officer, Operations Section Chief, and Finance Section Chief
D. Safety Officer, Public Information Officer, and Liaison Officer

4.2-Q2. Which ICS component is responsible for developing the Incident Action Plan (IAP) and tracking resources?

A. Planning Section
B. Operations Section
C. Logistics Section
D. Finance/Administration Section

4.2-Q3. In business continuity planning, the Recovery POINT Objective (RPO) is:

A. The maximum acceptable time to restore a process
B. The location of the alternate recovery site
C. The maximum acceptable amount of data loss, measured back in time from the disruption
D. The point at which the incident commander declares recovery complete

4.2-Q4. What is the purpose of a Business Impact Analysis (BIA) in business continuity planning?

A. To evaluate the effectiveness of marketing strategies
B. To identify the most critical processes and the potential impact of a disruption on them
C. To determine the organization's key performance indicators
D. To analyze the overall financial health of the organization

4.2-Q5. A plant's emergency coordinator is acting as Incident Commander when the fire department battalion chief arrives. What is the correct ICS practice?

A. Transfer command through a formal face-to-face briefing; the plant coordinator may remain in a supporting or unified command role
B. Both act as independent incident commanders
C. The plant coordinator leaves the scene immediately
D. Command stays with the plant regardless of the incident

4.2-Q6. A post-incident review finds that slow decision-making resulted from responders receiving orders from several supervisors. Which ICS principle was violated?

A. Common terminology
B. Modular organization
C. Span of control
D. Unity of command

4.2-Q7. During an emergency, which ICS function is responsible for obtaining and managing supplies, facilities, and equipment?

A. Operations Section
B. Planning Section
C. Logistics Section
D. Finance/Administration Section

4.2-Q8. A hurricane destroys a regional data center that stores critical safety inspection records and training files. As part of the disaster recovery team, what is the BEST immediate course of action?

A. Rebuild the data center at the same location as quickly as possible
B. Implement the disaster recovery plan by restoring records from off-site or cloud backups to an alternate site
C. Manually recreate records from employee memory and paper notes
D. Wait until insurance claims are processed before attempting recovery

4.2-Q9. Following a severe storm, a company activates its Business Continuity Plan (BCP). Which action should occur FIRST?

A. Conduct a financial impact review
B. File insurance claims
C. Schedule post-incident interviews
D. Restore the critical operations identified as essential in the Business Impact Analysis

4.2-Q10. Which activity represents a continuity (recovery) objective rather than an immediate response activity?

A. Restoring IT systems and production operations
B. Evacuating employees from a fire
C. Coordinating with arriving fire services
D. Activating the alarm system

Answer key and explanations

Q	Ans	Why
1	D	Command staff report directly to the IC: Safety, Public Information and Liaison Officers. Section chiefs (Operations, Planning, Logistics, Finance/Admin) form the general staff.
2	A	Planning collects and evaluates information, tracks resource status, and prepares the IAP. Operations executes it; Logistics obtains resources.
3	C	RPO is how much data you can afford to lose (drives backup frequency); RTO is how long a process can be down. Swapping the two is the classic trap.
4	B	The BIA identifies time-critical functions, their downtime tolerance and dependencies, which sets recovery priorities, RTOs and RPOs.
5	A	ICS requires a formal transfer of command with a briefing on situation, resources and objectives; the facility representative often joins unified command or advises on plant issues.
6	D	Unity of command means each responder reports to only one supervisor. Span of control (3-7, ideally 5) limits how many people one supervisor manages.
7	C	Logistics provides facilities, services and materials (food, fuel, communications, equipment). Finance/Admin tracks costs, Planning tracks resource status.
8	B	Contingency planning pre-arranges backups and alternate sites so critical records can be restored quickly; rebuilding or waiting delays recovery.
9	D	BCP activation prioritizes restoring the time-critical functions ranked in the BIA; financial, insurance and review activities follow.
10	A	Response protects life and stabilizes the incident; recovery and continuity restore critical systems and business functions after stabilization.

DOMAIN 4 · COMPETENCY 4.3

Fire Prevention, Protection and Suppression

CSP11 blueprint: Identify key components of fire prevention, protection, and suppression systems

1. Definition

Fire prevention removes or controls fuel, oxygen and ignition sources so fires do not start; fire protection detects fires and protects occupants and property (alarms, compartmentation, egress); suppression systems control or extinguish fires (sprinklers, standpipes, special agents, extinguishers).

2. Explanation in simple words

Stop fires starting (housekeeping, hot-work permits, safe storage), find them fast (detectors and alarms), and put them out or keep them small (sprinklers and extinguishers) while people get out.

3. Key points to remember

- Fire triangle/tetrahedron: fuel, heat, oxygen + chemical chain reaction. Dust explosion pentagon adds dispersion and confinement.
- Fire classes: A (ordinary combustibles), B (flammable liquids/gases), C (energized electrical), D (combustible metals), K (cooking oils/fats). Lithium-ion battery fires: large amounts of water for cooling; thermal runaway.
- Flammable liquids (OSHA/GHS): Category 1 FP < 73.4 °F and BP ≤ 95 °F; Category 2 FP < 73.4 °F and BP > 95 °F; Category 3 FP 73.4-140 °F; Category 4 FP 140-199.4 °F. Cabinet limit 60 gal Cat 1-3 (120 gal Cat 4), max 3 cabinets per fire area; storage rooms 6 air changes/hour; bonding and grounding during transfer.
- LFL/UFL: flammable range; work limit typically < 10% LFL. Flash point, autoignition temperature, vapor density (> 1 heavier than air).
- Hot work (NFPA 51B, OSHA 1910.252): permit, PAI (permit authorizing individual), fire watch during and at least 30 min after, combustibles 35 ft away or covered.
- Detection: heat (fixed temperature, rate-of-rise), smoke (ionization - fast flaming; photoelectric - smoldering), flame (UV/IR - fast, open areas).
- Sprinklers (NFPA 13; ITM per NFPA 25): wet pipe (most common), dry pipe (freezing areas), pre-action (sensitive areas), deluge (high-hazard, all heads open). $Q = K\sqrt{P}$. Keep 18 in. clearance below deflectors.
- Portable extinguishers (1910.157): monthly visual inspection, annual maintenance, hydrostatic testing per type; training annually if employees are expected to use them; PASS (pull, aim, squeeze, sweep).
- Special agents: CO_2 (no residue, asphyxiation hazard), clean agents (FM-200, Novec 1230), dry chemical, foam (Class B), wet chemical (Class K).
- Combustible dust: NFPA 652/660 dust hazard analysis (DHA), housekeeping (avoid compressed air), explosion venting/isolation/suppression, Class II electrical classification; secondary explosions cause most damage.

4. Real work scenario

> A food plant's corn-starch dust collector is indoors with no explosion protection, and dust layers have built up on rafters. The safety professional orders a dust hazard analysis, moves the collector outdoors with explosion venting and isolation valves, starts a housekeeping schedule using vacuum systems approved for combustible dust, and bans compressed-air cleaning.

5. CSP exam tips and traps

- Know which extinguisher class matches the fire: kitchen grease = K; energized panel = C (no water).
- Fire watch: 30 minutes after hot work (NFPA 51B).
- Prevention vs protection: housekeeping prevents; explosion venting and sprinklers protect.
- Sprinkler K-factor and flammable storage limits are favorite numbers.

6. Every blueprint point (3 blueprint items + 13 more)

Fire prevention BLUEPRINT

Fire prevention keeps fires from starting by controlling ignition sources and fuel loads (NFPA 1 Fire Code). OSHA 1910.39 requires a fire prevention plan when an OSHA standard calls for one.

- 1910.39 plan contents: list of major fire hazards with proper handling/storage, potential ignition sources and their control, type of fire protection equipment for each hazard; names/job titles responsible for maintaining equipment and for controlling fuel source hazards; housekeeping procedures.
- Written if more than 10 employees; oral if 10 or fewer; review with employees upon assignment.
- Controls: hot work permits, housekeeping, electrical maintenance, static control, segregation of incompatibles, smoking rules.
- Employers must provide a workplace free of recognized fire hazards (General Duty Clause).

> **Exam tip:** Prevention = stop ignition (frequency). Sprinklers and alarms are protection, not prevention.

Fire protection BLUEPRINT

Fire protection is the systems and methods that detect, control or extinguish a fire and protect occupants (NFPA 72, NFPA 13, NFPA 101). It includes active systems (alarms, sprinklers, extinguishers) and passive features (fire-rated construction).

- Active: needs detection/activation - sprinklers, extinguishers, alarm and detection systems, clean agent, foam, standpipes; needs ongoing ITM (NFPA 25, 72; OSHA 1910.157).
- Passive: built-in, no action needed - fire-rated walls/doors/floors (NFPA 80, 101), fire dampers, smoke barriers, fireproofing on steel, firestopping, compartmentation, intumescent paint.
- Layered approach: engineering (sprinklers, alarms, ventilation), administrative (prevention plans, permits, drills), protective equipment.
- Annual program review and adjustment after audits, new hazards or code changes.

> **Exam tip:** Fire doors propped open defeat passive protection - a frequent inspection finding.

Fire suppression BLUEPRINT

Suppression systems control or extinguish fire, from portable extinguishers to automatic sprinklers and special agent systems.

- Wet pipe: water in pipes, most common, heated areas.
- Dry pipe: pressurized air, for freezing areas (unheated warehouses, garages).
- Deluge: all heads open, high-hazard areas (aircraft hangars, chemical processing).
- Preaction: requires detection signal before water enters pipes (double interlock = two signals); data centers, museums, libraries.
- Occupancy hazard classes for sprinkler design: light (offices), ordinary (warehouses, manufacturing), extra (flammable liquids); hydraulically calculated (NFPA 13).

> **Exam tip:** Water-damage-sensitive area = preaction; freezing = dry pipe; rapid total flooding of water = deluge.

Fire science: triangle, tetrahedron, pentagon and phases

Fire is rapid exothermic oxidation (combustion). Removing any element of the fire model stops it.

- Triangle: fuel, oxygen (~21% in air), heat. Tetrahedron adds the chemical chain reaction (interrupted by dry chemical and halogenated agents).
- Dust explosion pentagon: fuel, oxygen, ignition, dispersion, confinement.
- Deflagration = subsonic flame front (flash fire); detonation = supersonic.
- Fire phases: incipient, growth, fully developed, decay.
- Incomplete combustion produces more CO and soot.

Flash point, fire point, autoignition and flammable limits

Key properties that classify liquids and define when vapors can ignite.

- Flash point: lowest temperature at which a liquid gives off enough vapor to form an ignitable mixture with an ignition source (closed-cup method for OSHA).
- Fire point: vapors sustain burning for at least 5 s; typically about 10-30 deg C above flash point.
- Autoignition temperature (AIT): ignites with no external ignition source (NFPA 497, SDS).
- LEL/UEL: below LEL too lean, above UEL too rich; flammable range lies between. Gasoline about 1.4%-7.6%.
- Hot work not allowed where atmosphere is within flammable range; confined space entry limit is 10% LEL.

> **Exam tip:** Flash point = vapors CAN ignite; fire point = fire KEEPS burning; AIT = no ignition source needed.

Heat transfer in fires

Fire spreads by conduction, convection and radiation; understanding them predicts fire spread.

- Conduction: through solids by direct contact (steel beam through a fire wall ignites the other side).
- Convection: movement of hot gases/liquids (smoke rising through stairwells, shafts, ducts to upper floors).
- Radiation: electromagnetic (infrared) waves, no medium needed (burning building ignites a neighbor across an alley).

> **Exam tip:** Spread to upper floors via shafts = convection; across a gap to another building = radiation.

Flammable and combustible liquid classification and storage (NFPA 30 / OSHA 1910.106)

Liquids are classed by flash point and boiling point; the lower the class number, the stricter the storage and handling.

- Class IA: FP < 73 deg F, BP < 100 deg F (ethyl ether). IB: FP < 73, BP >= 100 (acetone, gasoline, toluene). IC: FP 73-100 (xylene, turpentine).
- Class II: 100-140 deg F (diesel, mineral spirits); IIIA: 140-200 (the guide lists kerosene here, but kerosene is usually Class II, FP about 100-150 deg F); IIIB: >= 200 (vegetable oil, glycerol).
- OSHA 1910.106 now uses GHS Categories 1-4 (Cat 4 = FP > 140 to 199.4 deg F); NFPA 30 still uses classes.
- Cabinets: no more than 60 gal of Category 1-3 (Class I/II) or 120 gal of Category 4/Class III per cabinet; max 3 cabinets in one fire area.
- Storage rooms: 1-hour (or 2-hour) fire rating, self-closing doors, curbs/sumps; ventilation at least 1 cfm/ft2 of floor area (NFPA 30); keep vapors below 10% LEL.
- Bond and ground containers when transferring Class I liquids (NFPA 77).

> **Exam tip:** Know how to classify a liquid from flash point and boiling point and then pick the control.

Electrical and static ignition sources

Faulty wiring, overloaded circuits, arcing and static discharge are leading ignition sources.

- Grounding connects to earth; bonding equalizes potential between conductive objects - both used for flammable liquid transfer (NFPA 77).
- Static controls also: humidity control, anti-static flooring, non-sparking tools.
- Hazards: overloaded circuits, damaged insulation, loose connections (arcing), unlisted equipment, misused extension cords.
- Controls: AFCIs, explosion-proof equipment in classified locations, infrared thermography, NFPA 70E program.
- Electrical fires: Class C agents (CO2, dry chemical); de-energize before using water. Heat detectors are better than smoke detectors in dusty electrical rooms.

Hot work safety (NFPA 51B / OSHA 1910.252)

Welding, cutting, brazing, grinding and soldering are major ignition sources; outside designated hot work areas they require a permit.

- Permit: date, location, nature of job, authorization signature, fire watch assigned, precautions checklist.
- Move combustibles at least 35 ft away or protect with fire-resistant covers/shields.
- Fire watch during work and at least 30 min after (OSHA 1910.252); NFPA 51B (2019 and later editions) requires at least 1 hour after, followed by further fire monitoring - the guide states 30 minutes.
- Test atmosphere with LEL detection in enclosed or poorly ventilated areas.
- Fire watch must know how to use extinguishers and sound the alarm; designated permanent hot work areas preferred.

> **Exam tip:** Combustibles that cannot be moved 35 ft must be covered - and a fire watch is required.

Combustible dust

Finely divided solids that present a fire or explosion hazard when dispersed in air - even materials not normally flammable (sugar, flour, aluminum, wood).

- Stages: accumulation, dispersion, confinement, ignition, primary explosion, secondary explosion (secondary usually causes most damage and deaths).
- NFPA 652 requires a Dust Hazard Analysis (DHA); NFPA 654 prevention; OSHA uses General Duty Clause and the Combustible Dust NEP (CPL 03-00-008).
- Housekeeping trigger: dust layer over 1/32 in. (paper clip thickness) over 5% of floor area; vacuum with approved equipment, avoid dry sweeping and compressed air.
- Electrical: Class II locations; Group E metal, F carbonaceous, G grain/organic; Div 1 normal operations, Div 2 abnormal.
- Tests: Kst (deflagration index; St1 0-200, St2 200-300, St3 > 300 bar-m/s), MIE, MEC, go/no-go screening.
- Engineering: explosion venting (NFPA 68), isolation and inerting (NFPA 69), spark detection; cases: Imperial Sugar 2008 (14 deaths).

> **Exam tip:** Secondary explosion prevention = housekeeping of settled dust.

Fire detection and alarm systems (NFPA 72)

Detection provides early warning, starts evacuation and can trigger suppression and shutdowns.

- Ionization smoke detectors: fast-flaming fires; photoelectric: smoldering fires.
- Heat detectors: fixed temperature (e.g., 135 or 190 deg F) or rate-of-rise; used where smoke detectors false-alarm (kitchens, dusty areas).
- Flame detectors: UV fastest (also triggered by welding arcs), IR for hydrocarbon fires, UV/IR dual for accuracy.
- Gas detectors alarm before LEL is reached.
- FACP supervises power and circuits; notification appliances meet ADA/NFPA 72; mass notification per NFPA 72 Ch. 24; records available to the AHJ.
- Interlocks: sprinklers, gas shutoff, HVAC smoke dampers, emergency power (NFPA 110).

> **Exam tip:** Smoldering fire = photoelectric; flaming fire = ionization; dusty room = heat detector.

Portable fire extinguishers (NFPA 10 / OSHA 1910.157)

Extinguishers are classed by fire type and must be selected, located, inspected and maintained for the expected hazard.

- Classes: A ordinary combustibles; B flammable liquids/gases; C energized electrical; D combustible metals (dry powder); K cooking oils (wet chemical, potassium acetate).
- Max travel distance: A 75 ft; B 30 or 50 ft depending on rating; D 75 ft; K 30 ft; C based on the A or B hazard.
- Monthly visual inspection; annual maintenance (records kept 1 year or until next maintenance).
- Hydrostatic test every 5 years (water, CO2, wet chemical) or 12 years (dry chemical) - the guide lists '6-year and 12-year'; 6 years is the internal maintenance interval for stored-pressure dry chemical.
- Training on initial hire and annually if employees are expected to use them.

Special suppression systems

Gaseous and foam systems protect areas where water is unsuitable.

- Clean agents (NFPA 2001): FM-200 (HFC-227ea), Novec 1230, inert gases (Inergen, Argonite) - no residue, non-conductive; for data centers, electrical rooms, museums.
- CO2 (NFPA 12): smothers by displacing oxygen; lethal at design concentrations - pre-discharge alarm, delay, lockout for entry; use in normally unoccupied areas (transformers, generators).
- Foam (NFPA 11, 16): AFFF and AR-AFFF for Class B fires (hangars, fuel racks); alcohol-resistant foam for polar solvents. PFAS-containing AFFF is being phased out.
- Clean agent inspection: semiannual cylinder weight/pressure checks and functional tests without discharge.

Sprinkler inspection, testing and maintenance (NFPA 25)

Water-based systems must be inspected and tested on fixed schedules to remain reliable.

- Monthly: control valves (sealed/locked) and gauges (dry, preaction, deluge).
- Quarterly: alarm devices (waterflow, supervisory).
- Annually: visual sprinkler inspection, main drain/flow tests.
- Every 5 years: internal pipe inspection.
- Sprinkler head testing: dry sprinklers every 10 years; standard-response heads at 50 years then every 10; fast-response at 20 years then every 10.
- Keep at least 18 in. clearance below sprinkler deflectors; do not hang items from piping.

> **Exam tip:** Closed control valve = most common reason sprinklers fail - valves must be supervised or locked open.

Segregation and separation

Separate incompatible or flammable materials and control building design and ventilation to prevent fires and limit their spread.

- Oxygen cylinders separated from fuel-gas cylinders or combustibles by 20 ft or a 5-ft-high noncombustible barrier with at least 1/2-hour fire rating (1910.253).
- NFPA 400 segregation by hazard class (oxidizers away from flammables); NFPA 30 Ch. 9 container storage; 1910.1450 lab segregation procedures.
- Separate full and empty containers; store chemicals below eye level; secondary containment; keep fire department access clear.
- Outdoor storage: minimum distances to buildings, property lines and public ways per NFPA 30 tables.

Fire safety housekeeping

Good housekeeping reduces fuel load, prevents ignition and keeps exits and fire equipment accessible (1910.22, NFPA 1, 101, 13).

- Remove trash, scrap and oily rags daily; oily rags in covered metal containers.
- Keep aisles, exits, extinguishers, alarms and electrical panels clear (36 in. working clearance at panels per NEC).
- No storage in stairwells, exits or mechanical rooms; maintain 18 in. below sprinklers.
- Written checklists, assigned duties, documentation and surprise audits.

Read more: Yates, Safety Professional's Reference and Study Guide (4th ed.): Ch. 12 Fire Prevention and Protection, p. 350

7. Ten CSP-style practice questions

Answer all ten on paper first, then check the key. Score: _____ / 10 Date: _____ Retest: _____ / 10

4.3-Q1. A safety professional notes that a Class BC extinguisher has been placed near the range hood of a deep fat fryer in the cafeteria. The manager says a Class BC extinguisher is fine for a kitchen. Who is correct?

A. The manager, because a BC agent both cools grease and displaces oxygen
B. The safety professional, because one extinguisher never holds enough agent for a fryer fire
C. Neither, because a Class A water extinguisher should be used
D. The safety professional, because a Class K extinguisher is required for cooking oil and fat fires

4.3-Q2. A combustible dust explosion requires which two elements in addition to the fire tetrahedron (fuel, heat, oxygen, chain reaction)?

A. Fuel and confinement
B. Dispersion and confinement
C. Dispersion and ignition
D. Oxygen and pressure

4.3-Q3. Which detector is MOST appropriate for fast-developing flammable-liquid fires in an open process area?

A. Optical flame detector (UV/IR)
B. Rate-of-rise heat detector
C. Photoelectric smoke detector
D. Fixed-temperature heat detector

4.3-Q4. A contractor will weld near a combustible wood-framed wall. According to NFPA 51B, what is the BEST course of action before starting hot work?

A. Cover the wall with a plastic tarp to prevent damage
B. Keep a fire extinguisher nearby and start welding immediately
C. Move combustibles at least 35 feet away or protect them with approved fire-resistant shields or blankets
D. Post a 'Caution: Hot Work in Progress' sign and proceed

4.3-Q5. When nearby combustible materials cannot be removed or guarded during welding, a fire watch must be maintained for at least how long after the work is completed?

A. 15 minutes
B. 30 minutes
C. 45 minutes
D. 90 minutes

4.3-Q6. Which NFPA standard specifies the inspection, testing, and maintenance requirements for water-based fire protection systems such as sprinklers?

A. NFPA 25
B. NFPA 13
C. NFPA 72
D. NFPA 70E

4.3-Q7. An employee uses a portable fire extinguisher on an office fire, but the flames continue to grow. What is the MOST appropriate next action?

A. Move flammable materials away from the fire
B. Open windows to allow smoke to escape
C. Retrieve a larger extinguisher from another floor
D. Sound the fire alarm and evacuate immediately

4.3-Q8. A renovated attic is being fitted with fire detection. The space is dusty, has wide temperature swings, and no HVAC. What should the safety engineer recommend?

A. Install combination smoke/CO detectors
B. Defer installation until the dust settles
C. Install heat detectors instead of smoke detectors
D. Calibrate the smoke detectors with open-flame testing

4.3-Q9. Which element was added to the traditional fire triangle to form the fire tetrahedron?

A. Dispersion
B. Chemical chain reaction
C. Confinement
D. Vaporization

4.3-Q10. What is the required inspection and maintenance frequency for portable fire extinguishers under OSHA 1910.157 and NFPA 10?

A. Weekly visual check only
B. Semi-annual inspection only
C. Inspection every 3 years
D. Monthly visual inspection and annual maintenance

Answer key and explanations

Q	Ans	Why
1	D	NFPA 10 requires Class K (wet chemical) extinguishers for cooking media; the agent saponifies and cools hot oil to prevent reignition. Water on grease causes violent spattering.
2	B	The dust explosion pentagon adds dispersion (a dust cloud) and confinement to fuel, ignition and oxygen. Fuel, ignition and oxygen are already part of the base elements.
3	A	UV/IR flame detectors respond in seconds to radiant energy and work outdoors, where smoke and heat dissipate before reaching smoke or heat detectors.
4	C	NFPA 51B and OSHA 1910.252 require removal of combustibles within 35 ft or protection with fire-resistant guards, plus a permit and fire watch. Plastic tarps are combustible.
5	B	OSHA 1910.252(a)(2)(iii) requires the fire watch for at least 30 minutes after hot work ends to detect smoldering fires. NFPA 51B (2019+) extends this to 60 minutes, so read what the question cites.
6	A	NFPA 25 covers ITM of water-based systems; NFPA 13 covers sprinkler installation and design, NFPA 72 fire alarms, NFPA 70E electrical safety.
7	D	Portable extinguishers are for incipient fires only. If the fire grows, the employee should leave, alert others and let trained responders take over.
8	C	Dust and environmental extremes cause nuisance alarms and failures in smoke detectors; heat detectors are suited to dirty or unconditioned spaces.
9	B	The tetrahedron adds the uninhibited chemical chain reaction to fuel, heat and oxygen; agents such as dry chemical and halon work by interrupting it. Dispersion and confinement belong to the dust pentagon.
10	D	OSHA 1910.157(e) requires monthly visual inspection and annual maintenance checks; hydrostatic testing occurs at longer intervals (5 or 12 years).

DOMAIN 4 · COMPETENCY 4.4

Hazardous Materials Transportation and Security

CSP11 blueprint: Prepare procedures for the safe transportation and security of hazardous materials

1. Definition

Hazardous materials transportation is regulated by the U.S. DOT Hazardous Materials Regulations (49 CFR 171-180) and internationally by the UN Model Regulations, IMDG Code (sea), IATA DGR (air) and ADR (road, Europe). Security requirements address theft, sabotage and terrorism risks (49 CFR 172 Subpart I security plans; DHS CFATS for facilities).

2. Explanation in simple words

Before hazmat moves, you must classify it, package it correctly, label and placard it, prepare shipping papers, train the people involved and keep it secure so it cannot be stolen or misused.

3. Key points to remember

- Nine hazard classes: 1 explosives, 2 gases, 3 flammable liquids, 4 flammable solids, 5 oxidizers/organic peroxides, 6 toxic/infectious, 7 radioactive, 8 corrosives, 9 miscellaneous.
- Shipper responsibilities: classify, select UN-specification packaging, mark (proper shipping name, UN number), label, prepare shipping papers with emergency response information and 24-h telephone number, placard information, certify.
- Placards (172.504): Table 1 materials (e.g., Division 1.1-1.3, 2.3, 4.3 dangerous when wet, 6.1 PIH, 7 radioactive yellow III) - any quantity; Table 2 materials - when the aggregate gross weight is 1,001 lb or more.
- Shipping papers must be within the driver's immediate reach (or in the door pocket when out of the vehicle).
- Hazmat employee training (172.704): general awareness, function-specific, safety, security awareness, in-depth security (if a security plan is required) - within 90 days of hire and at least every 3 years.
- Emergency Response Guidebook (ERG): first responders' guide - initial isolation and protective action distances; SDS gives detailed chemical information.
- Security plans: personnel security (background checks), unauthorized access controls, en-route security (routes, stops, GPS tracking).
- Other modes: IMDG Code (sea), IATA (air), pipelines (PHMSA).

4. Real work scenario

> A plant ships drums of flammable solvent weighing 1,200 lb in total by truck. The shipping coordinator prepares shipping papers with the UN number and emergency phone, applies Class 3 labels, provides Class 3 placards because the aggregate weight exceeds 1,000 lb, and confirms that the driver has hazmat training and keeps the papers within reach.

5. CSP exam tips and traps

- Table 1 = placard any quantity; Table 2 = 1,001 lb aggregate.
- Hazmat training every 3 years; new employees within 90 days.
- ERG is for first responders; SDS for detailed chemical data.
- Class 8 = corrosive; Class 2 = gases; Class 3 = flammable liquids.

6. Every blueprint point (2 blueprint items + 4 more)

Hazmat transportation (DOT 49 CFR 171-180) BLUEPRINT

DOT's Hazardous Materials Regulations govern classification, packaging, marking, labeling, placarding, shipping papers, training and incident reporting for hazmat in commerce - from loading through transit to unloading.

- Nine hazard classes: 1 explosives, 2 gases (2.1 flammable, 2.2 non-flammable, 2.3 toxic), 3 flammable liquids (FP <= 140 deg F / 60 deg C), 4 flammable solids, 5 oxidizers (5.1) and organic peroxides (5.2), 6 toxic (6.1) and infectious (6.2), 7 radioactive, 8 corrosives, 9 miscellaneous.
- Packing groups: I great danger, II medium, III minor.
- Shipping paper basic description order: UN ID number, proper shipping name, hazard class, packing group; plus 24-hour emergency response phone number.
- Placards: any quantity of Table 1 materials (e.g., 1.1-1.3, 2.3, 4.3, 7 Yellow-III); Table 2 materials when aggregate gross weight is 1,001 lb or more.
- PHMSA (DOT) enforces; FMCSA covers motor carrier rules; incidents reported by phone to the National Response Center immediately (within 12 hours) and in writing on DOT Form 5800.1 within 30 days.

> **Exam tip:** DOT applies during transport; EPA RCRA applies to hazardous waste management and disposal; OSHA HazCom applies in the workplace.

Hazmat security BLUEPRINT

Hazmat security prevents theft, sabotage and terrorist misuse of hazardous materials in transport and at facilities.

- DOT security plans (49 CFR 172.800, Subpart I) required for higher-risk shipments (e.g., placarded quantities, Division 1.1-1.3, select agents, large bulk toxic inhalation hazards).
- Plan must address personnel security, unauthorized access and en-route security, based on a security risk assessment.
- Facility security: access control, inventory reconciliation, cameras, lighting, background checks, cybersecurity of process controls.
- DHS CFATS (6 CFR 27) covered high-risk chemical facilities; its statutory authority lapsed in July 2023. EPA RMP (40 CFR 68) and PSM still apply.
- Integrate security planning with emergency response planning.

DOT hazmat employee training (49 CFR 172.704)

Every hazmat employee (anyone who loads, unloads, packages, marks, prepares shipping papers or transports hazmat) must be trained, tested and certified.

- Content: general awareness/familiarization, function-specific, safety, security awareness, and in-depth security training if a security plan is required.
- Initial training within 90 days of employment or job change (may work under direct supervision meanwhile).
- Recurrent training at least every 3 years; records kept for the preceding 3 years and 90 days after employment ends.
- Driver training also required for those operating motor vehicles carrying hazmat.

> **Exam tip:** DOT = every 3 years; contrast with HAZWOPER annual refresher.

Marking, labeling, placarding and the Emergency Response Guidebook (ERG)

Communication of hazard during transport lets responders identify materials quickly.

- Labels go on packages (4-in. diamonds); placards go on vehicles, bulk containers and rail cars (10.8-in. diamonds) on all four sides.
- UN/NA number displayed on placards or orange panels for bulk shipments.
- ERG: yellow pages (ID number index), blue pages (material name index), orange pages (response guides), green pages (initial isolation and protective action distances for toxic inhalation hazards).
- Shipper certifies shipment is properly classified, described, packaged, marked and labeled.
- CHEMTREC and other services can provide the required 24-hour emergency response number.

> **Exam tip:** Green ERG pages = isolation and protective action distances.

Packaging and loading/segregation in transport

Packaging must be authorized for the material and packing group, and incompatible classes must be segregated in vehicles.

- UN specification packaging is performance-tested (drop, stack, leakproofness) and marked with the UN packaging code.
- Segregation table (49 CFR 177.848) prohibits loading certain classes together (e.g., oxidizers with flammables, cyanides with acids).
- Secure packages against movement; bulk containers need closures and venting as specified.
- Hazardous waste shipments also need an EPA uniform hazardous waste manifest (cradle-to-grave tracking).

Hazmat transportation incident response

Transport incidents require isolation, identification and notification before action.

- Approach from upwind, uphill and upstream; isolate the area; identify the material from placards, shipping papers (in the cab within reach of the driver) and the ERG.
- Use ERG initial isolation distance first, then protective action (evacuate or shelter in place).
- Only trained HAZWOPER responders take offensive action.
- Report to NRC and DOT as required; CERCLA reportable quantities trigger federal notification.

> **Exam tip:** Shipping papers are kept within the driver's reach (or in the door pocket) so responders can find them.

Read more: Yates, Safety Professional's Reference and Study Guide (4th ed.): Ch. 27 Hazardous Materials Management, p. 616; Ch. 32 Site Security, p. 731

7. Ten CSP-style practice questions

Answer all ten on paper first, then check the key. Score: _____ / 10 Date: _____ Retest: _____ / 10

4.4-Q1. A company driver is transporting several 55-gallon drums of flammable liquid. At a roadside inspection, the officer notes that a placard is missing and the shipping papers are in the glove compartment under personal items. What is the MOST appropriate corrective action?

A. Place the shipping papers within the driver's immediate reach and replace the missing placard before proceeding
B. Continue the trip because the drums themselves are sealed and labeled
C. Have the shipper send new documents while the trip continues
D. Replace the placard with a handwritten sign

4.4-Q2. A truck transporting compressed gas would display which DOT hazard class placard?

A. Class 1
B. Class 3
C. Class 2
D. Class 8

4.4-Q3. A new employee will serve as a hazardous materials technician, responding to uncontrolled releases to stop them but not doing post-emergency cleanup. Under 29 CFR 1910.120(q), what is the MINIMUM initial training?

A. 8 hours
B. 24 hours
C. 40 hours
D. 48 hours

4.4-Q4. A HAZMAT team member enters an area to retrieve soil samples wearing safety glasses, steel-toe boots, coveralls, gloves, and a hard hat, with no respiratory protection. What EPA protection level is this?

A. Level D
B. Level A
C. Level B
D. Level C

4.4-Q5. A tanker truck overturns near a facility gate, spilling an unknown liquid. Employees smell a strong chemical odor, and one feels dizzy. The facility has a trained HAZWOPER team on-site. What is the MOST appropriate action?

A. Call the fire department while employees wait near the spill
B. Have all employees approach the spill wearing available PPE
C. Attempt to contain the spill with sandbags right away
D. Evacuate the area upwind and allow only trained HAZWOPER responders to assess the spill

4.4-Q6. A truck carries Table 2 hazardous materials (for example Class 3 flammable liquids) in non-bulk packages. When are placards required?

A. For any quantity
B. Only above 10,000 lb
C. When the aggregate gross weight of Table 2 materials is 1,001 lb or more
D. Never for non-bulk packages

4.4-Q7. Where can first responders find generic hazard and initial response information for materials regulated in transportation?

A. Incident Command Guide
B. DOT Emergency Response Guidebook (ERG)
C. The shipper's corporate website
D. OSHA Hazard Communication standard

4.4-Q8. What is the primary purpose of the International Maritime Dangerous Goods (IMDG) Code?

A. To establish rules for preventing collisions at sea
B. To regulate the discharge of pollutants from ships
C. To set standards for ship construction and design
D. To provide requirements for safely transporting dangerous goods by sea

4.4-Q9. A company ships placarded quantities of hazardous materials by highway. Under 49 CFR 172 Subpart I, its written security plan must address which elements?

A. Personnel security, unauthorized access, and en route security
B. Fire extinguisher placement and sprinkler coverage
C. Driver hours-of-service logs and vehicle fuel economy
D. Insurance coverage and customer billing

4.4-Q10. Under DOT hazmat training rules (49 CFR 172.704), which training must ALL hazmat employees receive in addition to general awareness and function-specific training?

A. 40-hour HAZWOPER training
B. Commercial driver's license training
C. Safety and security awareness training
D. Incident commander training

Answer key and explanations

Q	Ans	Why

1	A	49 CFR 177.817 requires shipping papers within the driver's immediate reach (or in the door pouch), and required placards must be displayed before transport continues.
2	C	DOT Class 2 = gases (2.1 flammable, 2.2 non-flammable, 2.3 toxic). Class 1 = explosives, Class 3 = flammable liquids, Class 8 = corrosives.
3	B	1910.120(q)(6)(iii): HAZMAT technicians need at least 24 hours of training equal to the operations level plus specific competencies. 40 hours applies to general site workers at cleanup sites.
4	A	Level D is a work uniform with no respiratory protection and minimal skin protection. Level C adds an air-purifying respirator; B adds SCBA; A is fully encapsulating.
5	D	With an unknown material and symptoms already present, untrained workers must evacuate; only trained responders with proper PPE should identify and control the release.
6	C	49 CFR 172.504: Table 1 materials (e.g., Division 1.1-1.3 explosives, poison inhalation hazards) need placards in any quantity; Table 2 materials at 1,001 lb or more aggregate gross weight.
7	B	The ERG gives generic guidance by UN/NA number and guide page (isolation distances, fire and spill response). The SDS gives product-specific detail.
8	D	The IMDG Code (IMO) governs sea transport of dangerous goods; IATA DGR covers air, and 49 CFR covers US ground transport.
9	A	DOT security plans must include an assessment of transportation security risks and measures for personnel security, unauthorized access and en route security, plus security training.
10	C	All hazmat employees need general awareness, function-specific, safety and security awareness training, with recurrent training at least every 3 years. In-depth security training applies when a security plan is required.

DOMAIN 4 · COMPETENCY 4.5

Workplace Violence Prevention

CSP11 blueprint: Implement a workplace violence prevention program

1. Definition

Workplace violence is any act or threat of physical violence, harassment, intimidation or other threatening disruptive behavior at the work site. A prevention program combines management commitment, worksite analysis, hazard prevention and control, training and recordkeeping/evaluation (OSHA 3148 guidelines; ANSI/ASIS/SHRM WVPI standard).

2. Explanation in simple words

Violence at work usually has warning signs. Assess where and how it could happen, design the workplace to deter it, train people to recognize and report warning signs, respond quickly to threats and support those affected.

3. Key points to remember

- Types: Type I criminal intent (robbery, no relationship - leading cause of workplace homicide), Type II customer/client/patient (healthcare, social services), Type III worker-on-worker, Type IV personal relationship (domestic violence spilling into work).
- Risk factors: handling cash, working alone or at night, contact with the public, delivering services in community settings, high-crime areas, guarding valuables.
- Program elements: written policy (zero tolerance, no retaliation for reporting), threat assessment team (HR, security, legal, EHS, management), worksite analysis and security survey, controls, training, reporting, post-incident response (medical and psychological support, EAP, investigation), evaluation.
- Controls: engineering (access control, lighting, cameras, panic alarms, drop safes, barrier windows, metal detectors), administrative (two-person staffing, cash-handling limits, visitor policies, de-escalation procedures), CPTED (crime prevention through environmental design).
- Warning signs: threats, talk of weapons or violence, intimidation, escalating anger, sudden behavior change, isolation, substance abuse. Aggression continuum: anxiety → intimidation → crisis → violence (stages vary by model).
- Response: report concerns immediately to supervisor/HR/security; for an imminent threat, call security/911 and follow emergency procedures (Run-Hide-Fight for active shooter). Do not confront or physically intervene.
- Security concepts: deter, detect, delay, respond.

4. Real work scenario

> A former employee sends emails hinting that people 'will understand' his anger. The threat assessment team reviews the messages the same day, notifies security and local police, updates the visitor and badge system, briefs reception staff with a photo, and offers EAP support to the targeted supervisors.

5. CSP exam tips and traps

- Immediate response to threats: report to HR/security; do not ignore, confront or forward to coworkers.
- Know the four violence types and the main risk factors.
- Engineering vs administrative: metal detectors and drop safes = engineering; weapon bans and visitor rules = administrative.

- Post-incident: medical and psychological support first, then investigation and review of controls.

6. Every blueprint point (1 blueprint items + 5 more)

Workplace violence prevention program BLUEPRINT

A written program that commits management, involves employees, analyzes the worksite, controls hazards, trains staff and records and evaluates incidents. OSHA has no specific standard but enforces under the General Duty Clause (directive CPL 02-01-064).

- OSHA program elements (OSHA 3148): management commitment and employee participation; worksite analysis; hazard prevention and control; safety and health training; recordkeeping and program evaluation.
- Written policy: zero tolerance, clear reporting procedures, no reprisal for reporting.
- Threat assessment team (HR, security, safety, legal, management, EAP) to assess and manage threats.
- California requires most employers to have a written Workplace Violence Prevention Plan (Labor Code 6401.9, from July 2024).
- Consensus standard: ANSI/ASIS/SHRM WVPI.1 (workplace violence and active assailant prevention, intervention and response).

> **Exam tip:** Management commitment and employee involvement come first; 'install cameras' alone is not a program.

Types of workplace violence

Workplace violence is classified by the relationship between the perpetrator and the workplace (NIOSH / UIIPRC typology).

- Type I criminal intent: no legitimate relationship (robbery) - most workplace homicides.
- Type II customer/client: patients, customers, students, inmates - most common in healthcare and social services.
- Type III worker-on-worker: current or former employee.
- Type IV personal relationship: domestic violence that spills into the workplace.

> **Exam tip:** Patient assaulting a nurse = Type II; ex-spouse attacking at work = Type IV.

Risk factors and high-risk occupations

Certain job conditions increase the likelihood of violence and should drive the worksite analysis.

- Risk factors: exchanging money with the public, working alone or in small numbers, late night/early morning hours, high-crime areas, guarding valuables, contact with unstable or volatile persons, alcohol served, delivering passengers/goods/services.
- High-risk sectors: healthcare and social assistance, retail/late-night retail, taxi/rideshare, law enforcement and corrections.
- Worksite analysis: records review (OSHA 300, incident and security reports), employee surveys, security audits.

Engineering and administrative controls for workplace violence

Apply the hierarchy: design out opportunity, then procedures, training and staffing.

- Engineering: controlled access, badge systems, bullet-resistant barriers, good lighting, cameras, panic/duress alarms, drop safes and limited cash, curved mirrors, furniture arrangement preventing entrapment, escape routes.
- Administrative: adequate staffing, no working alone at night, visitor management, cash-handling limits, check-in procedures for field staff, flagging of violent patients/clients.
- Training: recognizing warning signs, de-escalation, reporting, emergency procedures, Run-Hide-Fight.

Warning signs and threat assessment

Most perpetrators show observable concerning behavior beforehand; a trained team assesses and manages threats.

- Warning signs: verbal threats, intimidation, fascination with weapons or prior violent incidents, sudden behavior changes, substance abuse, severe stress, paranoia, violation of protective orders.
- Threat assessment: gather facts, assess credibility and capability, decide interventions (EAP referral, discipline, security, law enforcement, protective orders).
- Treat all threats seriously and document them.

> **Exam tip:** Report and assess threats early - waiting for an overt act is the wrong answer.

Active shooter response and post-incident response

Employees must know what to do during an attack and the organization must support victims afterward.

- DHS Run, Hide, Fight; when police arrive, keep hands visible and follow commands.
- NFPA 3000 (ASHER) - integrated active shooter/hostile event preparedness, response and recovery with unified command.
- Post-incident: medical care, report and investigate, critical incident stress debriefing, EAP/counseling, legal support.
- Record injuries from workplace violence on the OSHA 300 log if recordable criteria are met; evaluate the program after each incident.

Read more: Yates, Safety Professional's Reference and Study Guide (4th ed.): Ch. 32 Site Security, p. 731

7. Ten CSP-style practice questions

Answer all ten on paper first, then check the key. Score: _____ / 10 Date: _____ Retest: _____ / 10

4.5-Q1. A hospital emergency department nurse is assaulted by a patient's relative. Which type of workplace violence is this?

A. Type I - criminal intent (no relationship)
B. Type II - customer/client/patient
C. Type III - worker-on-worker
D. Type IV - personal relationship

4.5-Q2. An employee reports that a co-worker has been making aggressive comments about supervisors and slammed equipment after a performance review. The company has a written workplace violence prevention program. What should the supervisor do FIRST?

A. Report the concern to HR or the safety/threat assessment team for immediate assessment under the program
B. Ignore the behavior until it escalates into a physical incident
C. Confront the worker publicly to discourage further aggression
D. Tell employees to avoid the worker but take no formal action

4.5-Q3. A recently terminated employee is sending hostile, vaguely threatening emails to your team, saying he might do something to 'make everyone understand.' How should the team respond?

A. Reply to explain the reasons for his termination
B. Ignore the emails, assuming he will cool down
C. Forward the emails to coworkers to gather opinions
D. Report the emails to security and human resources as soon as possible

4.5-Q4. Before construction begins, management convenes architects, safety experts, police, and fire officials to design security into a new facility (lighting, sight lines, controlled entrances). Which approach is this?

A. Safe construction through environmental design
B. Planned protection through environmental engineering
C. Crime Prevention Through Environmental Design (CPTED)
D. Planned construction through environmental engineering

4.5-Q5. In a workplace violence prevention program, which element BEST demonstrates management commitment?

A. Installing CCTV cameras without any policy updates
B. A written zero-tolerance policy signed by top leadership and communicated to all employees
C. Conducting unannounced drills only
D. Providing counseling only after incidents

4.5-Q6. What is the MOST appropriate post-incident action following a workplace violence event?

A. Provide medical and psychological support, then investigate and review controls
B. Return operations to normal immediately
C. Focus only on legal reporting requirements
D. Suspend all security staff involved

4.5-Q7. Which four basic security concepts should be included in a facility security program?

A. Deter, detect, train, respond
B. Read, react, identify, respond
C. Deter, detect, delay, dial
D. Deter, detect, delay, respond

4.5-Q8. According to NSC data, what is the greatest risk factor for homicide at work?

A. A disgruntled coworker
B. A random shooter
C. Robbery
D. A recently fired worker

4.5-Q9. What can a company do to help employees avoid the bystander effect if an active shooter enters the workplace?

A. Permit employees to carry concealed firearms
B. Train employees in advance on what to do (e.g., run, hide, fight) and how to report
C. Establish a clear hierarchy of communication
D. Develop stress management programs

4.5-Q10. A worksite analysis of a 24-hour convenience store shows that most incidents occur late at night during robberies. Which control is MOST effective at reducing this risk?

A. Require employees to work faster during night shifts
B. Post a zero-tolerance policy in the break room
C. Limit cash on hand with a drop safe, keep windows clear for visibility, and provide bright lighting with posted signs
D. Provide annual training only for day-shift workers

Answer key and explanations

Q	Ans	Why

1	B	Type II involves customers, clients, patients, students or their families and is most common in healthcare. Type I is criminal intent, such as robbery by a stranger.
2	A	Warning signs should be reported early so a trained team can assess the threat; public confrontation can escalate and ignoring it misses the chance for prevention.
3	D	Veiled threats from a former employee are a recognized risk factor; preserve the emails and report them to security and HR for threat assessment and protective measures.
4	C	CPTED uses natural surveillance, access control, territorial reinforcement and maintenance to reduce crime opportunities; it is the security analog of prevention through design.
5	B	OSHA's program elements begin with management commitment and employee participation, shown by a visible, leadership-endorsed written policy backed by resources.
6	A	OSHA guidance calls for immediate care, trauma counseling and debriefing, followed by investigation and corrective action to prevent recurrence.
7	D	Layered security aims to deter attempts, detect intrusions, delay the intruder long enough for a response, and respond.
8	C	Most workplace homicides are Type I, committed during robberies, which is why late-night retail and taxi work are high-risk. Coworker violence is far less frequent.
9	B	Prior training and drills reduce freezing and diffusion of responsibility, so people act quickly instead of waiting for others.
10	C	Engineering and administrative controls that reduce the reward and raise visibility (drop safes, low cash, clear sight lines, lighting) are proven robbery deterrents. A policy alone does not deter outsiders.

DOMAIN 5

Environmental Management

6% of the exam · 5 competencies

5.1	Environmental Protection and Pollution Prevention Describe environmental protection and pollution prevention programs (e.g., spill containment, abatement, best practices)
5.2	Hazardous Materials Management Identify procedures used to manage hazardous materials (e.g., GHS classification system, storage and handling, policy, security, hazardous waste storage and disposal)
5.3	Waste Management Identify procedures used to manage waste (e.g., universal, recycling, spill clean-up, labeling, remediation)
5.4	Sustainability Determine sustainability principles and practices (e.g., supply chain; reduce, reuse, recycle)
5.5	Environmental Issues and ESG Describe the impact of environmental issues (e.g., aging infrastructure, asbestos, air pollution, climate change, environmental, social, and governance)

DOMAIN 5 · COMPETENCY 5.1

Environmental Protection and Pollution Prevention

CSP11 blueprint: Describe environmental protection and pollution prevention programs (e.g., spill containment, abatement, best practices)

1. Definition

Environmental protection programs prevent or control releases to air, water and land. Pollution prevention (P2) reduces pollution at its source rather than treating it after it is created (Pollution Prevention Act of 1990). Key U.S. laws include the Clean Air Act, Clean Water Act (NPDES, SPCC), RCRA, CERCLA, EPCRA and TSCA.

2. Explanation in simple words

It is cheaper and safer not to make pollution in the first place. When you must store oils or chemicals, put a second container around them so a leak cannot reach drains, soil or rivers.

3. Key points to remember

- P2 hierarchy: source reduction → recycling/reuse → treatment → disposal (last resort).
- SPCC (40 CFR 112): facilities with > 1,320 gal aggregate aboveground oil storage capacity (containers ≥ 55 gal) or > 42,000 gal underground that could discharge to navigable waters; secondary containment for the largest container plus freeboard for precipitation; plan certified by a PE (or self-certified if qualified); annual training for oil-handling personnel.
- Secondary containment design: 110% of the largest tank is a common rule; RCRA container storage: 10% of total volume or 100% of the largest container, whichever is greater.
- Clean Water Act: NPDES permits for point-source discharges; stormwater (SWPPP), construction site erosion and sediment controls (silt fence, inlet protection).
- Clean Air Act: NAAQS for six criteria pollutants (CO, lead, NO_2, ozone, PM, SO_2); NESHAP for hazardous air pollutants; Title V operating permits; abatement equipment - baghouses (particulates), cyclones, electrostatic precipitators, wet scrubbers (acid gases), thermal oxidizers (VOCs), carbon adsorption.
- EPCRA (SARA Title III): emergency planning (Section 302 threshold planning quantities of extremely hazardous substances), release notification (304), Tier II inventory reporting (311/312), Toxics Release Inventory (313). CERCLA (Superfund): cleanup of contaminated sites and release reporting of reportable quantities (RQ).
- Best practices: inventory minimization, closed-loop systems, leak detection and repair, housekeeping, spill kits and drills, environmental aspects/impacts review (ISO 14001).

4. Real work scenario

> A trucking terminal stores three 1,000-gallon diesel tanks. Its aggregate capacity (3,000 gal) exceeds 1,320 gal, so it needs an SPCC plan regardless of how much fuel is actually in the tanks. Double-walled tanks, a curbed fueling pad draining to an oil-water separator, and annual spill-response training are added.

5. CSP exam tips and traps

- SPCC applicability is based on storage CAPACITY, not the amount on hand.
- Source reduction beats recycling, treatment and disposal.

- Match law to topic: EPCRA = community right-to-know; CERCLA = Superfund; RCRA = hazardous waste cradle to grave; TSCA = chemical inventory (PCBs, asbestos, lead).

6. Every blueprint point (3 blueprint items + 9 more)

Spill containment BLUEPRINT

Spill containment prevents the uncontrolled release of oil, fuel, chemicals or waste into soil, water or air. It combines primary containment (tanks, drums, pipes), secondary containment (berms, double walls, spill pallets), floor and drain protection, and procedures and training.

- Spills come from overfilling, equipment failure, corrosion, improper handling and transfer accidents.
- Primary containment = the container holding the material under normal conditions; must be sound and compatible with contents.
- Secondary containment = catches a primary failure: concrete/earthen berms, double-walled tanks, spill pallets, portable trays.
- Floors in spill-prone areas: chemical-resistant, non-slip coating, sloped to sumps; storm drains covered or plugged during transfers.
- Overfill prevention: high-level alarms, automatic shutoff valves, backup float systems (tank filling is a common spill source).

> **Exam tip:** Containment is a system (design + technology + procedures + training), not just a berm.

SPCC rule (40 CFR 112)

EPA's Spill Prevention, Control and Countermeasure rule requires a written plan at non-transportation facilities that store oil above threshold amounts and could reasonably discharge to navigable waters.

- Thresholds: more than 1,320 gal aggregate aboveground oil storage (counting containers 55 gal or larger) or more than 42,000 gal completely buried.
- Plan includes site diagrams, drain maps, containment, inspection checklists, emergency contacts, training records.
- Certified by a Professional Engineer and reviewed at least every 5 years (qualified facilities with 10,000 gal or less may self-certify).
- Oil-water separators and secondary containment are typical SPCC controls.

> **Exam tip:** SPCC is about OIL (EPA, water protection); HAZWOPER is about worker protection during response.

Secondary containment sizing (calculation)

Containment volume is sized from the largest container and the total stored volume. The guide states the rule as 110% of the largest container or 10% of total volume, whichever is greater.

- RCRA container storage (40 CFR 264.175): 10% of total volume or 100% of the largest container, whichever is greater - the guide states 110%.
- SPCC: hold the largest container plus freeboard for precipitation; 110% of the largest tank is the common design practice for outdoor dikes.
- Example: four 55-gal drums = 220 gal total; 10% = 22 gal; largest = 55 gal (110% = 60.5 gal) -> size for the larger value (55-60.5 gal).
- Example: 500-gal diesel tank -> 110% = 550 gal minimum; a 1,100-gal berm complies.
- Dike volume = L x W x H (ft3) x 7.48 gal/ft3; subtract volume displaced by other tanks inside the dike.

> **Exam tip:** Always compute BOTH values (percent of largest and percent of total) and choose the greater.

Spill response equipment and readiness

Spill kits must be located near potential spill points, matched to the hazard, and fully stocked.

- Oil-only kits: hydrophobic pads/socks (repel water). Universal kits: water-based and non-aggressive chemicals. Hazmat kits: acids, bases, unknowns.
- Contents: pads, socks, pillows, neutralizers, gloves/goggles/aprons, disposal bags and labels, emergency contacts.
- Absorbent must be compatible - oil absorbents may be unsuitable for strong acids.
- SDS Section 6 (accidental release measures) gives spill guidance.

> **Exam tip:** Wrong absorbent for the chemical (e.g., combustible absorbent on an oxidizer) is a classic distractor.

Spill-related regulatory frameworks

Several overlapping rules govern spill prevention and response.

- RCRA tank rules (40 CFR 264/265 Subpart J): secondary containment, leak detection, daily inspections with records.
- SWPPP (40 CFR 122.26): stormwater controls such as berms, drain protection and covers.
- OSHA HAZWOPER 1910.120: tiered responder training (awareness, operations - defensive, technician - offensive/hands-on, specialist, incident commander).
- OSHA HazCom 1910.1200: SDS Section 6 and label/SDS training.
- NFPA 30: storage spacing, containment and drainage for flammable/combustible liquids; ISO 14001 clause 8.2: emergency preparedness including environmental releases.

> **Exam tip:** Operations level = defensive actions from a safe distance; technician level = approach the release to stop it.

Abatement (pollution abatement overview) BLUEPRINT

Pollution abatement captures, treats or reduces pollutants before they reach air, water or soil, ideally at the point of generation. It uses engineering controls plus administrative procedures to meet CAA, CWA, NESHAP and SWPPP requirements.

- Proactive - controls pollution at the source rather than reacting to effects.
- Media: air emissions, water discharges, stormwater runoff, soil.
- Benefits: legal compliance, lower operational risk, worker safety, public image, sustainability metrics.
- Multistage control is common (e.g., cyclone pre-cleaner then baghouse).

> **Exam tip:** Memory set: Baghouse = dust; Scrubber = acid gas; VRU = VOC; Separator = oil; pH system = acid/base; SWPPP = sediment.

Air regulations: CAA, NAAQS, Title V, NESHAP

The Clean Air Act authorizes EPA to regulate air pollutants from stationary and mobile sources.

- NAAQS criteria pollutants: PM (PM10, PM2.5), SO2, NOx (NO2), CO, ozone, lead.
- Title V operating permits (40 CFR 70) required for major sources.
- New Source Review: BACT for new or modified major sources (PSD areas); LAER in nonattainment areas.
- NESHAP (40 CFR 63) controls hazardous air pollutants (benzene, formaldehyde, methylene chloride) by industry sector using MACT.
- Nonattainment area = region not meeting a NAAQS; stricter controls for ozone precursors (VOC, NOx).

> **Exam tip:** BACT = new/modified sources under NSR; MACT = HAPs under NESHAP.

Air pollution control devices

Device selection depends on pollutant type (particulate vs gas), particle size, industry and permit limits.

- Baghouse (fabric filter): fine PM10/PM2.5, over 99% efficiency; watch bag rupture/bypass.
- Wet scrubber: acid gases (SO2, HCl, HF) with alkaline liquid; needs mist eliminator; makes wastewater (CWA).
- Dry scrubber/dry sorbent injection: acid gases with lime or sodium bicarbonate; no wastewater, lower cost, good for arid areas and retrofits; solids caught by baghouse/ESP.
- Cyclone: coarse particles by centrifugal force, 70-90% for particles over 10 microns; pre-cleaner only.
- Electrostatic precipitator (ESP): fine/submicron particles, up to 99.9%; affected by moisture and sticky particles.
- Vapor recovery unit (VRU): VOCs at tank vents and loading; condenser, adsorber or compressor; required under NESHAP for bulk terminals.

> **Exam tip:** Submicron particles -> ESP or baghouse; coarse only -> cyclone; acid gas -> scrubber; VOC -> VRU/carbon/thermal oxidizer.

Water pollution abatement (CWA, NPDES, pretreatment)

The Clean Water Act regulates discharges to waters of the U.S. Direct dischargers need NPDES permits; indirect dischargers to a POTW must meet pretreatment standards.

- NPDES (40 CFR 122): effluent limits, monitoring, stormwater controls.
- Pretreatment (40 CFR 403) for industrial users of municipal sewers; effluent guidelines 40 CFR 400-471.
- Oil-water separator: oil rises and is skimmed, solids settle; sized for flow and oil type; required under SPCC/many NPDES permits.
- pH neutralization: automated dosing (lime, caustic, acid) to typical discharge range 6.0-9.0.
- Clarifiers/settling tanks: gravity removal of suspended solids, often with coagulants/flocculants.
- Activated carbon: adsorbs trace organics, VOCs, chlorine, color; polishing step; monitor breakthrough; spent carbon may be hazardous waste.

> **Exam tip:** Electroplating acidic rinsewater to a city sewer -> pH neutralization + industrial pretreatment permit.

Stormwater controls (SWPPP BMPs)

Stormwater Pollution Prevention Plans under 40 CFR 122.26 use best management practices to keep sediment and pollutants out of runoff.

- BMPs: silt fences, retention/detention ponds, inlet protection (filter socks, drain baskets), vegetative swales and buffers.
- Controls must be maintained; clogged filters and damaged fences are ineffective.
- Visual monitoring and sampling during rain events are often required.
- Prevent illicit discharges of chemicals or wash water to storm drains.

Soil pollution abatement

Soil abatement mitigates or removes hazardous substances (metals, hydrocarbons, pesticides, solvents, lead paint dust) under CERCLA, RCRA corrective action, TSCA and brownfield programs.

- Excavation and off-site disposal: fast and definitive but costly; RCRA waste classification and DOT 49 CFR transport.
- Stabilization/solidification: cement, fly ash or lime binders immobilize metals; does not remove contamination; verify with TCLP (Method 1311).
- Bioremediation: microbes degrade hydrocarbons; needs oxygen, nutrients, proper pH; months to years (land farming, biopiles).
- Soil vapor extraction: vacuum removes VOCs from dry permeable soil; off-gas treatment required (CAA); often paired with air sparging.
- Phytoremediation: plants (sunflowers, willows, mustard) for low-level metals; cheap but slow.

> **Exam tip:** Heavy metals -> stabilization; VOCs in dry soil -> SVE; fuel spills -> bioremediation; urgent/near receptors -> excavation.

Best practices in pollution prevention BLUEPRINT

Best practices combine the pollution prevention hierarchy with administrative controls: plans, training, inspections, compatibility checks and drills.

- Pollution Prevention Act 1990 hierarchy: source reduction > recycling > treatment > disposal/release.
- Train spill responders per HAZWOPER with hands-on drills; awareness training for all employees.
- Weekly inspection of tanks, hoses, valves and containment (best practice); log findings and track corrective actions.
- Keep drain valves on containment closed and locked; inspect rainwater before release.
- Integrate into an ISO 14001 EMS for continual improvement.

> **Exam tip:** Source reduction (prevent at origin) beats any end-of-pipe abatement on 'BEST' questions.

Read more: Yates, Safety Professional's Reference and Study Guide (4th ed.): Ch. 34 Environmental Management, p. 754; Ch. 33 Sustainability

7. Ten CSP-style practice questions

Answer all ten on paper first, then check the key. Score: _____ / 10 Date: _____ Retest: _____ / 10

5.1-Q1. Which regulation requires facilities with more than 1,320 gallons of aboveground oil storage capacity to prepare a Spill Prevention, Control, and Countermeasure (SPCC) Plan?

A. Clean Air Act
B. Resource Conservation and Recovery Act
C. EPA Oil Pollution Prevention rule (40 CFR 112)
D. Toxic Substances Control Act

5.1-Q2. Under EPA's SPCC rule, secondary containment for bulk oil storage containers must hold:

A. The capacity of the largest single container plus sufficient freeboard for precipitation
B. 50% of the total stored volume
C. Exactly 100% of all containers combined
D. 10% of the largest container

5.1-Q3. A hazardous waste container storage area holds twenty 55-gallon drums of liquid waste. Under RCRA (40 CFR 264.175), what is the MINIMUM secondary containment capacity?

A. 55 gallons
B. 60.5 gallons
C. 1,100 gallons
D. 110 gallons

5.1-Q4. What is the main goal of a pollution abatement program?

A. To record environmental data for audits
B. To capture or reduce pollutants before they are released to the environment
C. To clean up chemical spills after release
D. To eliminate the need for any waste treatment

5.1-Q5. Which control is a Best Management Practice (BMP) for stormwater pollution prevention at a construction site?

A. Silt fences and sediment traps
B. HEPA filtration
C. Baghouse filters
D. Cyclone separators

5.1-Q6. Which action ranks HIGHEST in the EPA pollution prevention hierarchy?

A. Treating waste before discharge
B. Recycling spent solvent off-site
C. Disposing of waste in a permitted landfill
D. Changing the process to use a water-based coating that produces no hazardous waste

5.1-Q7. Which air pollution control device is MOST suitable for removing coarse particles (greater than 10 microns), often as a pre-cleaner?

A. Baghouse filter
B. Electrostatic precipitator
C. Cyclone separator
D. Wet scrubber for acid gases

5.1-Q8. Under the Clean Air Act, what type of permit is required for major stationary emission sources?

A. RCRA Part B permit
B. Title V operating permit
C. CERCLA clearance certificate
D. NPDES permit

5.1-Q9. A facility neutralizes acidic process wastewater to meet pH limits before discharging it to the municipal sewer (POTW). Which pollution-control approach is this?

A. Pretreatment
B. Solidification
C. Recycling
D. Filtration

5.1-Q10. Under EPCRA Section 302, what is a 'threshold planning quantity' (TPQ)?

A. The minimum quantity of any chemical that can be stored without reporting
B. The quantity of an extremely hazardous substance (EHS) on site that triggers emergency planning notification
C. The maximum amount of any chemical that may be released without consequence
D. The quantity of waste a company must dispose of yearly to avoid EPA oversight

Answer key and explanations

Q	Ans	Why
1	C	The SPCC rule (40 CFR 112, issued under the Clean Water Act) applies to non-transportation facilities with over 1,320 gal aboveground (or 42,000 gal buried) oil capacity that could reach navigable waters.
2	A	40 CFR 112.8(c)(2): containment for the entire capacity of the largest single container plus freeboard for precipitation (commonly designed as 110%).
3	D	Containment must hold 10% of total volume or 100% of the largest container, whichever is greater: 10% x (20 x 55 = 1,100) = 110 gal, which exceeds 55 gal. Trap: using only the largest container.
4	B	Abatement means controlling or reducing emissions and discharges (e.g., scrubbers, filters, treatment) before release; cleanup after release is remediation.
5	A	Stormwater BMPs such as silt fences, sediment basins and inlet protection keep sediment out of runoff. The other choices are air pollution controls.
6	D	The Pollution Prevention Act hierarchy is source reduction > recycling > treatment > disposal. Eliminating the waste at the source is best.
7	C	Cyclones use centrifugal force and are efficient and cheap for large particles; baghouses and ESPs are used for fine particulates.

8	**B**	Title V operating permits consolidate all CAA requirements for major sources. NPDES permits cover point-source water discharges under the CWA.
9	**A**	Industrial users must pretreat wastewater to meet POTW and CWA pretreatment standards; pH adjustment is a common pretreatment step.
10	**B**	When an EHS is present at or above its TPQ, the facility must notify the state and local emergency planning bodies and take part in local emergency planning.

DOMAIN 5 · COMPETENCY 5.2

Hazardous Materials Management

CSP11 blueprint: Identify procedures used to manage hazardous materials (e.g., GHS classification system, storage and handling, policy, security, hazardous waste storage and disposal)

1. Definition

Hazardous materials management covers the whole life of hazardous chemicals at a site: purchasing approval, hazard classification and communication (GHS/HCS), safe storage and handling, security, and the storage and disposal of hazardous waste under RCRA.

2. Explanation in simple words

Know what chemicals you have, label them clearly, store incompatible ones apart, keep them secure, and when they become waste, follow the RCRA rules from the moment the waste is generated until final disposal.

3. Key points to remember

- GHS/HCS 2012 (1910.1200, updated 2024): classification into physical, health (and environmental) hazard classes and categories; labels with product identifier, signal word (Danger or Warning), hazard statements, precautionary statements, pictograms, supplier information; 16-section SDS (Sec. 2 hazard ID, Sec. 7 handling and storage, Sec. 8 exposure controls/PPE, Sec. 10 stability and reactivity).
- Nine HCS pictograms (eight OSHA-required): flame, flame over circle (oxidizer), exploding bomb, gas cylinder, corrosion, skull and crossbones (acute toxicity severe), exclamation mark (irritant, sensitizer, less severe), health hazard (carcinogen, mutagen, reproductive, respiratory sensitizer), environment (not required by OSHA).
- Category 1 is the most severe in GHS (opposite of NFPA 704, where 4 is most severe).
- Written HazCom program: chemical inventory list, labeling, SDS access, training (initially and when new hazards are introduced), non-routine tasks, contractor communication.
- Storage and handling: segregation by compatibility (oxidizers away from flammables, acids away from bases, water-reactives dry), secondary containment, ventilation, flammable cabinets, quantity limits, inspections.
- Security: chemical security (CFATS), inventory control, locked storage, access lists.
- RCRA hazardous waste: listed (F - non-specific sources such as spent solvents; K - specific industries; P - acutely hazardous discarded commercial chemicals; U - toxic discarded commercial chemicals) or characteristic (ignitable, corrosive, reactive, toxic - TCLP).
- Generator categories: VSQG ≤ 100 kg/month; SQG > 100 to < 1,000 kg/month (180 days storage, 270 if > 200 miles); LQG ≥ 1,000 kg/month or > 1 kg acute (90 days). Satellite accumulation up to 55 gal (1 qt acute). Containers closed, labeled 'Hazardous Waste' with hazard indication and accumulation start date; weekly inspections; uniform hazardous waste manifest; LQG annual training.

4. Real work scenario

> An inspector finds unlabeled spray bottles of a solvent cleaner and containers of waste solvent with no dates. The safety professional labels the bottles with GHS workplace labels, adds the solvent to the chemical inventory, dates the waste containers with the accumulation start date, moves them to the central accumulation area, and schedules weekly inspections.

5. CSP exam tips and traps

- SDS Section 8 = PPE and exposure limits; Section 2 = hazard identification.
- GHS Category 1 = most hazardous.
- Generator category thresholds and accumulation times are high-yield numbers.
- Flame over circle = oxidizer.

6. Every blueprint point (5 blueprint items + 8 more)

GHS classification system BLUEPRINT

The UN Globally Harmonized System standardizes chemical classification, labels and SDSs. OSHA adopted it in the Hazard Communication Standard, 29 CFR 1910.1200 (2012, updated 2024).

- Three hazard groups: physical, health, environmental (OSHA does not enforce environmental classes).
- Each hazard class is split into categories; Category 1 is the most severe (Flammable Liquid Cat 1 is worse than Cat 3).
- Physical: flammables, gases under pressure, explosives, oxidizers, pyrophorics, self-heating.
- Health: acute toxicity, skin corrosion, respiratory sensitization, carcinogenicity, reproductive toxicity, STOT.
- EPA uses hazard categories for EPCRA Tier II; DOT uses its own 9 classes; NFPA 704 is a separate 0-4 system (4 = most severe).

> **Exam tip:** GHS: Category 1 = MOST hazardous. NFPA 704: 4 = MOST hazardous. The numbers run opposite.

GHS labels: signal words, pictograms and statements

A shipped-container label has six elements: product identifier, signal word, pictogram(s), hazard statements, precautionary statements and supplier information.

- Only two signal words: Danger (severe) and Warning (less severe); only one per label - Danger overrides.
- Nine pictograms (red diamond border); OSHA requires eight - the environment pictogram is not mandatory in U.S. workplaces.
- Flame = flammables; flame over circle = oxidizers; exploding bomb = explosives/self-reactives; gas cylinder = gases under pressure; corrosion = skin burns/metal corrosion; skull and crossbones = acute toxicity (fatal/toxic); health hazard = carcinogen, mutagen, respiratory sensitizer; exclamation mark = irritant, skin sensitizer, narcotic effects.
- Hazard statements are standardized and cannot be altered (e.g., 'May cause cancer').
- Precautionary statements give handling, storage, PPE and emergency instructions.

> **Exam tip:** Respiratory sensitizer = health hazard pictogram; skin sensitizer = exclamation mark.

Safety Data Sheets (16 sections)

The SDS replaced the MSDS and must follow a fixed 16-section format. Employers must keep SDSs readily accessible to workers on every shift and update them when new hazard information arrives.

- 1 Identification, 2 Hazard identification, 3 Composition, 4 First aid, 5 Fire-fighting, 6 Accidental release, 7 Handling and storage, 8 Exposure controls/PPE.
- 9 Physical/chemical properties, 10 Stability and reactivity, 11 Toxicological information.
- 12 Ecological, 13 Disposal, 14 Transport, 15 Regulatory - included for GHS consistency but not enforced by OSHA.
- 16 Other information, including date of last revision.
- Electronic access is acceptable if there are no barriers to immediate access.

> **Exam tip:** Spill guidance = Section 6; PPE and exposure limits = Section 8; incompatibles = Section 10.

Storage of hazardous materials BLUEPRINT

Storage must prevent fires, toxic releases and reactions by using proper cabinets/rooms, segregation, ventilation and containment, under OSHA 1910.106, NFPA 30/400/45 and RCRA.

- Flammable cabinets (NFPA 30/OSHA): max 60 gal Class I/II (120 gal Class III) per cabinet; labeled 'FLAMMABLE - KEEP FIRE AWAY'; doors closed; away from ignition sources.
- Corrosives: chemically resistant (polyethylene or epoxy-coated) cabinets with trays; acids and bases separate.
- Toxics/poisons: locked, access-restricted, inventoried.
- Compressed gas cylinders: upright, capped, secured; full and empty separated; oxygen and fuel gas separated by 20 ft or a 5-ft-high, 1/2-hour fire-rated wall (1910.253).
- Inside storage rooms for flammables: ventilation of at least six air changes per hour; explosion-proof electrical where required.

> **Exam tip:** Oxygen cylinders stored next to acetylene with no 20 ft / 5 ft wall separation is a classic violation.

Segregation of incompatible materials

Incompatible chemicals must be physically separated so a leak cannot cause fire, explosion or toxic gas.

- Acids away from bases; oxidizers away from flammables/combustibles.
- Cyanides never near acids (hydrogen cyanide gas); hypochlorite away from acids (chlorine gas).
- Water-reactives isolated from sinks, sprinklers and humid areas.
- Segregate with separate cabinets, distance or fire-rated barriers - not just different shelves of one cabinet.
- Check SDS Section 10 and compatibility charts.

Handling and transfer of hazardous materials BLUEPRINT

Transfers are high-risk; controls focus on static, splashes and container breakage.

- Bond and ground flammable liquid transfers to prevent static ignition (NFPA 77).
- Use closed dispensing systems (pumps, nozzles) instead of open pouring.
- Never carry breakable containers by hand - use carriers, spill trays and carts.
- Select PPE from the SDS; eyewash/shower within 10 seconds travel (ANSI Z358.1) where corrosives are handled.
- Containers kept closed, compatible and in good condition; secondary containers labeled.

> **Exam tip:** Bonding equalizes potential between containers; grounding drains charge to earth - both are needed for Class I transfers.

Security of hazardous materials BLUEPRINT

Hazmat security protects against intentional misuse - theft, sabotage or terrorism - through access control, physical barriers, information protection and emergency integration.

- Access control: authorized trained staff only, key cards/PINs/biometrics, sign-in logs, restricted-area signage.
- Physical: locked cabinets/cages, steel doors, fencing, CCTV, motion detectors, lighting, intrusion alarms.
- Bulk tanks: locked hatches and remote fill points, screened vents, fenced dikes, tamper-evident seals.
- Protect inventory data: limit database access, secure Tier II reports and chemical maps, no public posting of tank contents.
- Plan for theft, sabotage and suspicious items; drill with police, fire and LEPC.

> **Exam tip:** Insider threat and missing inventory scenarios: first secure/control access and investigate, then report.

CFATS (DHS Chemical Facility Anti-Terrorism Standards)

CFATS covered facilities holding screening threshold quantities of Chemicals of Interest (e.g., ammonium nitrate, chlorine, hydrogen peroxide).

- Top-Screen survey submitted to DHS; high-risk facilities assigned Tiers 1-4 (Tier 1 = highest risk).
- Tiered facilities implement a Site Security Plan: perimeter security, personnel vetting, cybersecurity, incident response coordination.
- Note: CFATS statutory authority lapsed in July 2023; verify current status. EPA RMP and DOT security plans still apply.
- DOT security plans (49 CFR 172 Subpart I) apply to shippers/carriers of certain hazmat quantities.

Hazardous materials policy and program

A written, management-approved hazmat program defines how chemicals are acquired, stored, used and disposed of, integrating HazCom, EPA emergency planning and NFPA codes.

- Chemical inventory: names, quantities, container sizes, locations, linked SDSs; supports EPCRA Tier II (40 CFR 370) and emergency planning.
- Policy elements: procurement controls, labeling, handling/PPE, spill and first aid, storage assignments, inspections.
- SOPs for receiving, transfers, disposal/decontamination, equipment cleanup.
- HazCom training at initial assignment and whenever a new hazard is introduced; document it.
- Inspections (weekly/monthly) for container condition, labels, containment, expired/unknown materials. Note: the guide cites 3-year retention under 1910.1020, but exposure records under 1910.1020 must be kept 30 years.

> **Exam tip:** 'When is retraining required?' - when a NEW hazard is introduced, not when a new product with the same hazards arrives.

DOT hazard classes (49 CFR 171-180)

DOT's Hazardous Materials Regulations govern hazmat in transport (loading, transit, unloading) using nine hazard classes that drive packaging, labels and placards.

- 1 Explosives; 2 Gases (2.1 flammable, 2.2 non-flammable non-toxic, 2.3 toxic).
- 3 Flammable liquids (flash point not over 60 C / 140 F); 4 Flammable solids (4.1, 4.2 spontaneously combustible, 4.3 dangerous when wet).
- 5 Oxidizers (5.1) and organic peroxides (5.2); 6 Toxic (6.1) and infectious (6.2).
- 7 Radioactive; 8 Corrosives; 9 Miscellaneous (lithium batteries, dry ice, magnetized materials).
- Hazmat employees need DOT training (49 CFR 172.704) at least every 3 years.

> **Exam tip:** GHS governs the workplace label; DOT governs the transport label/placard - they can legitimately differ on the same drum.

Hazardous waste BLUEPRINT

Once a hazardous material is no longer usable and is discarded, it becomes waste governed cradle-to-grave by RCRA Subtitle C (40 CFR 260-273), with DOT for transport and HAZWOPER for workers.

- Hazardous materials are regulated in use; hazardous waste is regulated after use or disposal.
- Store on impervious surfaces with secondary containment, in central or satellite accumulation areas.
- Containers: compatible, closed, undamaged, labeled 'Hazardous Waste', hazards indicated, accumulation start date (central areas).
- Inspect container storage areas weekly; keep spill kits, PPE and emergency info nearby.
- States (e.g., California DTSC) may be more stringent than federal RCRA.

Hazardous waste identification

Generators must determine whether a waste is hazardous by listing or by characteristic, using testing or process knowledge.

- Ignitability D001: liquid flash point below 140 F (60 C).
- Corrosivity D002: aqueous pH 2 or less, or 12.5 or more (or corrodes steel).
- Reactivity D003: unstable, water-reactive, explosive, generates toxic gas.
- Toxicity D004-D043: TCLP extract exceeds limits (e.g., lead, benzene, mercury).
- Listed wastes: F (non-specific sources, e.g., F003 spent solvents), K (specific industries), P (acutely hazardous discarded chemicals), U (toxic discarded chemicals).

> **Exam tip:** pH 2 or less OR 12.5 or more = corrosive (D002). pH 12 is NOT a characteristic hazardous waste.

Generator categories, manifests and TSDFs

Generator status depends on monthly quantity and sets accumulation time and paperwork.

- VSQG: 100 kg/month or less (and 1 kg or less acute); no federal time limit, but 1,000 kg on-site cap.
- SQG: more than 100 and less than 1,000 kg/month; 180 days (270 if TSDF is over 200 miles); 6,000 kg cap.
- LQG: 1,000 kg/month or more (or over 1 kg acute); 90 days; contingency plan and annual training.
- Ship to a permitted TSDF via registered transporter (EPA ID) with the Uniform Hazardous Waste Manifest (EPA Form 8700-22) / e-Manifest; keep copies 3 years.
- Example: 1,200 kg/month spent solvent -> LQG -> 90-day limit.

> **Exam tip:** The guide's table lists SQG as 100-1,000 kg; federal rule is more than 100 and less than 1,000 kg; 1,000 kg exactly is LQG.

Read more: Yates, Safety Professional's Reference and Study Guide (4th ed.): Ch. 4 Regulations (HazCom/GHS comparison), p. 55; Ch. 27 Hazardous Materials Management, p. 616

7. Ten CSP-style practice questions

Answer all ten on paper first, then check the key. Score: _____ / 10 Date: _____ Retest: _____ / 10

5.2-Q1. A facility generates 600 kg of hazardous waste per month (no acute hazardous waste). What is its RCRA generator category?

A. Very Small Quantity Generator
B. Small Quantity Generator
C. Large Quantity Generator
D. Treatment, Storage and Disposal Facility

5.2-Q2. A supplier's SDS for a new solvent shows a flame pictogram and the hazard statement 'Flammable liquid, Category 2.' What is your FIRST action before allowing the material on site?

A. Verify that proper flammable storage and ventilation controls are available
B. Require the supplier to train all employees in SDS reading
C. Approve the purchase since the supplier provided an SDS
D. Have employees store it in the nearest open area

5.2-Q3. An international project site receives a drum of toluene marked only with US DOT hazard labels. A European subcontractor refuses to use it, citing labeling noncompliance. What is the MOST appropriate corrective action?

A. Repackage the drum into smaller unlabeled containers
B. Explain that DOT labels are sufficient for workplace use
C. Provide an SDS only if a worker asks for it
D. Apply a GHS label with the product identifier, pictograms, signal word, and hazard and precautionary statements

5.2-Q4. Under GHS, which elements are required on a shipped container label?

A. Chemical formula, warnings, and pictograms only
B. pH, boiling point, flash point, and pictograms
C. Product identifier, pictogram, signal word, hazard statement, precautionary statement, and supplier information
D. SDS number, pictograms, and training date

5.2-Q5. During an inspection, you find a drum of acid cleaner (corrosive pictogram) stored next to a pallet of bleach. What is your MOST urgent concern?

A. The chemicals could react to release toxic chlorine gas
B. The drum label is missing the manufacturer's phone number
C. The pictogram is somewhat faded
D. The bleach lacks a transport placard

5.2-Q6. An SDS for a parts-cleaning solvent lists: 'Keep away from heat/sparks/open flame. Use explosion-proof equipment.' In which SDS section are these precautionary statements found?

A. Section 2 - Hazard Identification
B. Section 4 - First-Aid Measures
C. Section 7 - Handling and Storage
D. Section 10 - Stability and Reactivity

5.2-Q7. A safety professional is reviewing an SDS before training employees. Which section contains personal protective equipment recommendations?

A. Section 2 - Hazard Identification
B. Section 4 - First-Aid Measures
C. Section 8 - Exposure Controls/Personal Protection
D. Section 10 - Stability and Reactivity

5.2-Q8. Which area may accumulate up to 55 gallons of non-acute hazardous waste at or near the point of generation without the full central storage requirements?

A. Central Accumulation Area (CAA)
B. Universal Waste Area
C. Hazardous Storage Room
D. Satellite Accumulation Area (SAA)

5.2-Q9. During a spill response, a new worker reaches for a container labeled with a flame-over-circle pictogram. What should the supervisor remind the worker?

A. This symbol means the chemical is flammable
B. This symbol identifies an oxidizer, which can cause or intensify a fire
C. This symbol identifies explosive chemicals
D. This symbol warns of corrosion of skin and metals

5.2-Q10. A facility stores ammonium nitrate and other theft-attractive hazardous materials. Which measure BEST addresses hazardous materials security?

A. Store the materials near the loading dock for faster shipping
B. Restrict access to authorized personnel, conduct regular inventory reconciliation, and report unexplained losses
C. Rely on GHS labels to deter theft
D. Keep the storage area unlocked so emergency responders can enter

Answer key and explanations

Q	Ans	Why
1	B	VSQG: 100 kg/month or less; SQG: more than 100 and less than 1,000 kg/month; LQG: 1,000 kg/month or more (or more than 1 kg acute). 600 kg falls in the SQG range.
2	A	Pre-purchase review should confirm that storage (cabinets, quantity limits), ventilation and ignition controls exist before the material arrives.
3	D	DOT labels communicate transport hazards; workplace use requires GHS/HazCom labeling and an accessible SDS.

4	**C**	HazCom 1910.1200(f) requires product identifier, signal word, hazard statements, pictograms, precautionary statements, and the supplier's name, address and phone.
5	**A**	Acids mixed with sodium hypochlorite liberate chlorine gas; incompatible materials must be segregated in storage (SDS Sections 7 and 10).
6	**A**	Precautionary statements are label elements and are listed in Section 2 along with signal word, hazard statements and pictograms. Section 7 gives detailed handling practices.
7	**C**	Section 8 lists exposure limits (PEL, TLV), engineering controls and PPE.
8	**D**	SAAs at or near the point of generation, under operator control, may hold up to 55 gal (or 1 qt acute); containers must be closed and marked 'Hazardous Waste' with the hazards indicated.
9	**B**	Flame over circle = oxidizer; plain flame = flammable; exploding bomb = explosive; corrosion = skin and metal damage. Keep oxidizers away from combustibles.
10	**B**	Hazmat security combines access control, inventory accountability and reporting of losses so diversion is prevented or detected quickly. Labels communicate hazards but do not secure them.

DOMAIN 5 · COMPETENCY 5.3

Waste Management

CSP11 blueprint: Identify procedures used to manage waste (e.g., universal, recycling, spill clean-up, labeling, remediation)

1. Definition

Waste management covers how all wastes are characterized, segregated, labeled, stored, recycled, transported and disposed of, how spills are cleaned up, and how contaminated sites are remediated.

2. Explanation in simple words

Sort waste correctly, label it, recycle what you can, clean up spills safely, and fix contaminated soil and water using the right technology.

3. Key points to remember

- Universal waste (40 CFR 273): batteries, pesticides (recalled), mercury-containing equipment, lamps (and aerosol cans federally since 2019). Simplified rules: label 'Universal Waste - [type]', accumulate up to 1 year, train handlers, send to a destination facility. Used oil is managed under 40 CFR 279, not as universal waste.
- Other categories: non-hazardous solid waste (municipal landfill), medical/regulated waste (state rules, sharps containers), radioactive waste (NRC; decay-in-storage for short half-lives), e-waste.
- Recycling and reuse: solvent reclamation, scrap metal, pallets; legitimate recycling vs 'sham recycling'.
- Spill cleanup: assess (SDS, size, hazards) → protect people (evacuate if needed, PPE) → stop the source if safe → contain (dikes, absorbents, drain covers) → clean up and decontaminate → dispose of cleanup materials as waste → report if above a reportable quantity → review. Incidental vs emergency spills (HAZWOPER).
- Labeling: hazardous waste labels ('Hazardous Waste', hazard indication, start date), DOT labels for transport, universal waste labels.
- Remediation technologies: pump-and-treat (groundwater), soil vapor extraction (VOCs in unsaturated soil), air sparging, bioremediation, phytoremediation, excavation and disposal, capping/containment (does not remove contamination), in-situ chemical oxidation, monitored natural attenuation.
- CERCLA: potentially responsible parties (strict, joint and several liability), National Priorities List; brownfield redevelopment; Phase I/II environmental site assessments.

4. Real work scenario

> A facility generates spent fluorescent lamps, used batteries, used oil and waste paint. The safety professional sets up universal-waste containers for the lamps and batteries (labeled, dated, kept less than one year), sends used oil to a used-oil recycler under Part 279, and characterizes the paint waste as ignitable hazardous waste.

5. CSP exam tips and traps

- Used oil is NOT universal waste - a common 'which is NOT' trap.
- Capping limits migration but does not remove contaminants.
- Universal waste accumulation limit: one year.

6. Every blueprint point (5 blueprint items + 3 more)

Waste management regulatory overview

Waste management covers identifying, segregating, labeling, storing and disposing of all waste streams (hazardous and non-hazardous) from operations, spills, expired chemicals and maintenance.

- RCRA 40 CFR 260-273: definitions, generator rules, labeling, storage, manifests.
- DOT 49 CFR 171-180: packaging, marking and transport of hazardous waste.
- OSHA HAZWOPER 1910.120: workers handling hazardous waste or cleaning up releases.
- State programs may be stricter than federal rules.
- Non-hazardous solid waste falls under RCRA Subtitle D.

Universal waste BLUEPRINT

Universal waste (40 CFR 273) is a streamlined category for common, lower-risk hazardous wastes so they are easier to collect and recycle.

- Federal types: batteries, recalled/unused pesticides, mercury-containing equipment (thermostats, thermometers), lamps (fluorescent, HID), aerosol cans (added 2019); some states add electronics.
- Label 'Universal Waste - [type]' (or 'Waste/Used [type]') and show accumulation start date.
- Accumulation limit: 1 year.
- Containers closed, intact and compatible; employees trained on handling and emergency procedures.
- Universal waste does not count toward generator status.

> **Exam tip:** Fluorescent lamps in a fiber drum labeled 'Universal Waste - Lamps' with a start date = compliant. 'Hazardous Waste' label is not required.

Recycling of industrial waste BLUEPRINT

Recycling reduces disposal cost and liability while supporting sustainability, provided materials are kept clean and recyclers are legitimate.

- Common streams: used oil (40 CFR 279), scrap metal, distillable solvents, cardboard and plastic drums.
- Label containers (e.g., 'Used Oil'), avoid contamination, verify recycler licensing and keep documentation proving recycling.
- Closed-loop recycling: material reclaimed and returned to the same process may be excluded from RCRA.
- Used oil mixed with hazardous waste can become hazardous waste - keep streams separate.

> **Exam tip:** Documentation proving material was actually recycled (not sham recycling) is the CSP's protection.

Spill cleanup BLUEPRINT

Spills must be handled quickly and safely using a formal response plan. HAZWOPER applies when cleanup involves an uncontrolled release of a hazardous substance.

- Sequence: 1 recognize the material (SDS/label), 2 alert and evacuate (EAP), 3 select PPE, 4 contain (absorbents, booms, dikes), 5 clean and decontaminate/neutralize, 6 dispose of debris (often as hazardous waste), 7 document and report.
- Incidental spills handled by trained area employees are not HAZWOPER emergency response.
- Oil spill planning: SPCC (40 CFR 112).
- Cleanup debris from a hazardous chemical spill is generally hazardous waste.

> **Exam tip:** FIRST step questions: identify the material and protect people before containment.

Release reporting (CERCLA and EPCRA)

Releases of hazardous substances at or above a Reportable Quantity must be reported to the National Response Center.

- CERCLA RQ release: notify the NRC immediately upon knowledge (the guide says within 24 hours).
- The RQ is measured as the amount released within any 24-hour period.
- EPCRA Section 304 also requires notice to state and local emergency planning bodies (SERC/LEPC).
- Oil discharges producing a sheen on water are reportable under the CWA.
- Document the release and follow-up written reports as required.

Labeling BLUEPRINT

Labeling is one of the most-cited EPA/OSHA areas; containers must be labeled, compatible and closed at all times.

- Hazardous waste: 'Hazardous Waste', indication of hazards, EPA waste codes (e.g., D001) before shipment, accumulation start date, clear contents.
- Universal waste: type and accumulation start date.
- Used oil: 'Used Oil'.
- Shipping: DOT marking, labels and placards plus the manifest.

Satellite accumulation areas

An SAA lets generators accumulate waste at or near the point of generation under reduced requirements.

- Up to 55 gal of hazardous waste (or 1 qt liquid / 1 kg solid acute) per SAA.
- Must be under control of the operator; containers closed, labeled 'Hazardous Waste' with hazard indication.
- Once the limit is exceeded, excess moves to the central area within 3 consecutive days and is dated.
- Central accumulation area clock (90/180 days) starts when waste enters it.

Remediation BLUEPRINT

Remediation cleans up contaminated soil, water or structures under CERCLA (Superfund) or RCRA corrective action.

- Techniques: excavation, solidification/stabilization, bioremediation, soil vapor extraction, pump and treat (groundwater).
- Phase I ESA: records review, site visit, interviews (no sampling; ASTM E1527).
- Phase II ESA: sampling and analysis to confirm contamination.
- Brownfield programs support redevelopment of contaminated land.
- CERCLA liability is strict, joint and several, and retroactive for potentially responsible parties.

> **Exam tip:** Phase I = no sampling; Phase II = sampling.

Read more: Yates, Safety Professional's Reference and Study Guide (4th ed.): Ch. 34 Environmental Management, p. 754; Ch. 27 Hazardous Materials Management

7. Ten CSP-style practice questions

Answer all ten on paper first, then check the key. Score: _____ / 10 Date: _____ Retest: _____ / 10

5.3-Q1. A consultant proposes installing a cap over contaminated soil on company property. What is a key LIMITATION of capping as a remediation method?

A. A cap increases water infiltration and speeds plume movement
B. A cap makes the property unattractive
C. A cap does not remove the contamination; it only minimizes exposure and migration
D. A cap prevents any reuse of the property

5.3-Q2. Under 40 CFR 279, containers and tanks of used oil stored for recycling must be labeled:

A. 'Used Oil'
B. 'Hazardous Waste - Oil'
C. 'Universal Waste - Oil'
D. 'Recyclable Waste Only'

5.3-Q3. What marking is required on a container of universal waste lamps?

A. 'Hazardous Waste - Lamps' with an EPA waste code
B. 'Used Oil for Recycling'
C. 'Flammable - Keep Fire Away'
D. 'Universal Waste - Lamp(s)' (or 'Waste Lamps'/'Used Lamps'), with the accumulation start date demonstrated

5.3-Q4. A worker must segregate four waste streams into 55-gallon drums: wastewater, lithium metal, toluene, and halogenated hydrocarbons. What is the GREATEST concern?

A. Mixing lithium metal and toluene
B. Mixing lithium metal and water
C. Mixing halogenated hydrocarbons and toluene
D. Mixing wastewater and toluene

5.3-Q5. Volatile organic compounds are found in unsaturated (vadose zone) soil above the water table. Which remediation technology is MOST suitable?

A. Pump and treat
B. Phytoremediation of groundwater
C. Capping only
D. Soil vapor extraction

5.3-Q6. While inspecting a small facility, you find several containers labeled 'corrosive waste' in the central accumulation area with no dates. What is your BEST course of action?

A. Notify the facility manager that the containers are noncompliant and must be marked with the accumulation start date
B. Remove the containers immediately without telling anyone
C. Have workers put on PPE and move the waste elsewhere
D. Call the fire department to dispose of the waste

5.3-Q7. How long may a universal waste handler accumulate universal waste (e.g., spent lamps, batteries) on site?

A. 90 days
B. 180 days
C. 1 year from the date it became a waste
D. Indefinitely

5.3-Q8. Which action represents the LOWEST (least preferred) level of the waste management hierarchy?

A. Redesigning a process to eliminate hazardous chemical use
B. Disposing of hazardous waste in a permitted landfill
C. Reusing scrap material in the same process
D. Sending waste solvent off-site for recycling

5.3-Q9. Which characteristic identifies a waste as EPA hazardous waste code D001?

A. Ignitability (e.g., liquid with flash point below 140 F / 60 C)
B. Corrosivity (pH 2 or less, or 12.5 or more)
C. Reactivity (unstable or explosive)
D. Toxicity (TCLP metals such as lead or mercury)

5.3-Q10. Employees clean up a spill of a listed hazardous solvent using absorbent pads. How should the used absorbent be managed?

A. In the regular trash once it has dried
B. By rinsing it in a floor drain and reusing the pads
C. As universal waste, since it came from a spill
D. As hazardous waste, properly containerized, labeled, and disposed of through a permitted facility

Answer key and explanations

Q	Ans	Why

1	C	Capping is containment: it reduces infiltration and direct contact but leaves contaminants in place, requiring long-term monitoring and maintenance.
2	A	Used oil managed for recycling falls under the used oil standards, not hazardous waste rules; containers must be labeled with the words 'Used Oil'.
3	D	40 CFR 273 requires the universal waste marking and a way to show how long it has been accumulated (one-year limit).
4	B	Lithium is a water-reactive metal that releases flammable hydrogen and heat on contact with water; waste segregation must keep reactives apart.
5	D	SVE applies a vacuum to the unsaturated zone to remove volatile contaminants. Pump and treat addresses groundwater, and capping does not remove contaminants.
6	A	Generators must mark containers 'Hazardous Waste', indicate the hazards, and record the accumulation start date to track time limits (40 CFR 262.16/262.17).
7	C	Universal waste may be accumulated up to one year (longer only if needed to collect quantities for proper recovery). 90 and 180 days are LQG and SQG hazardous waste limits.
8	B	Hierarchy: source reduction > reuse/recycling > treatment > disposal. Landfill disposal is the last resort.
9	A	D001 = ignitable, D002 = corrosive, D003 = reactive, D004-D043 = toxicity characteristic.
10	D	Spill cleanup materials contaminated with a listed hazardous waste carry that listing (contained-in principle) and must be managed as hazardous waste. Absorbents are not a universal waste category.

DOMAIN 5 · COMPETENCY 5.4

Sustainability

CSP11 blueprint: Determine sustainability principles and practices (e.g., supply chain; reduce, reuse, recycle)

1. Definition

Sustainability means meeting the needs of the present without compromising the ability of future generations to meet their own needs (Brundtland, 1987). In business it is often framed as the triple bottom line - people, planet, profit - and includes supply-chain responsibility, resource efficiency and life-cycle thinking.

2. Explanation in simple words

Use less, reuse what you can, recycle the rest, and choose suppliers who also work safely and responsibly. Sustainability and safety support each other because waste, inefficiency and unsafe work all cost money and harm people.

3. Key points to remember

- Waste hierarchy: refuse/reduce → reuse → recycle → recover (energy) → dispose.
- Life-cycle assessment (ISO 14040/44): impacts from raw material extraction through manufacturing, use and end of life ('cradle to grave' or 'cradle to cradle').
- Supply chain: supplier codes of conduct, audits, conflict minerals, responsible sourcing, diversified suppliers to reduce disruption risk, green procurement (recycled content, low-VOC materials, energy-efficient equipment).
- Green building: LEED (Leadership in Energy and Environmental Design) - energy, water, materials, indoor environmental quality.
- Sustainability reporting: GRI, SASB, integrated reporting; safety metrics (fatalities, TRIR) appear in sustainability reports.
- Links to safety: substitution of less hazardous chemicals reduces both exposure and hazardous waste; energy efficiency reduces emissions; worker well-being is the 'social' pillar.

4. Real work scenario

> A construction company changes its purchasing policy to require recycled concrete aggregate, low-VOC paints and LED lighting. The safety professional notes that low-VOC paint also reduces worker solvent exposure, and the switch is reported in both the safety and the sustainability report.

5. CSP exam tips and traps

- Source reduction is the best sustainability answer.
- LEED = Leadership in Energy and Environmental Design.
- Diversifying suppliers = supply-chain risk reduction.

6. Every blueprint point (4 blueprint items + 8 more)

Sustainability frameworks

Sustainability means operations that are environmentally sound, socially responsible and economically viable across the product life cycle and supply chain.

- ISO 14001: EMS with legal compliance, continual improvement and a life-cycle perspective.
- EPA: CAA, CWA, RCRA and the Pollution Prevention Act (source reduction first), green chemistry, LCA support.
- OSHA: green jobs and renewable-technology hazards (solar, battery storage), IAQ, ergonomics.
- Triple bottom line: people, planet, profit.

UN Sustainable Development Goals (SDGs)

The 17 UN SDGs are non-binding global goals; several align directly with EHS work.

- Goal 3 Good health and well-being (injury/illness prevention).
- Goal 6 Clean water and sanitation; Goal 7 Affordable clean energy.
- Goal 9 Resilient infrastructure; Goal 12 Responsible consumption and production; Goal 13 Climate action.
- Not legally binding - a voluntary vision.

Supply chain sustainability BLUEPRINT

Sustainable supply chain management extends EHS expectations upstream and downstream to suppliers, contractors and logistics partners.

- Supplier audits cover environmental compliance (ISO 14001), labor practices (ILO, SA8000), resource efficiency and hazmat handling.
- Conflict minerals (3TG: tin, tantalum, tungsten, gold) - Dodd-Frank Section 1502.
- SA8000: social accountability standard - health and safety, working hours, compensation, freedom of association.
- Integrate procurement into ISO 14001 clause 8.1 (operational planning and control).
- Supply chain emissions = Scope 3.

> **Exam tip:** Auditing a supplier's labor and safety conditions in a developing region -> SA8000.

Green procurement and responsible sourcing

Green procurement adds environmental and social criteria to purchasing, not just cost and availability.

- Specify recycled content, low-VOC/non-toxic materials, minimal/reusable packaging.
- Prefer vendors with ISO 14001 or CDP ratings and low-carbon logistics.
- EPA Environmentally Preferable Purchasing (EPP) is a model program.
- Example: solvents in reusable stainless drums cut hazardous waste about 40%.

Reduce BLUEPRINT

Reduce (source reduction) eliminates waste and pollution before it is created and is the most effective of the 3Rs.

- Strategies: lean process design, energy audits, material substitution (water-based for solvent), Design for Environment.
- Matches the top of the hierarchy of controls (eliminate/substitute).
- Pollution Prevention Act: source reduction is the preferred strategy.

> **Exam tip:** Upstream redesign that removes a hazardous input = 'Reduce'.

Reuse BLUEPRINT

Reuse keeps products and materials in service without significant reprocessing, cutting raw material demand and disposal volume.

- Examples: reusable totes, refillable chemical drums, launderable rags/absorbents, upcycled scrap, internal asset recovery.
- Safety check: reused containers must be clean, compatible and not degraded (especially after corrosives/flammables).
- Example: switching to reusable PPE totes cut plastic waste 60% and procurement cost 30%.

Recycle BLUEPRINT

Recycling turns waste into new material; it ranks below reduce and reuse but is vital when those are not practical.

- Examples: cardboard baling, e-waste management, metal recovery, closed-loop recycling.
- Can lower hazardous waste quantities and generator status (e.g., solvent recycling).
- Hierarchy: Reduce > Reuse > Recycle > Treat/energy recovery > Dispose.

> **Exam tip:** If a question asks which 3R is BEST, choose Reduce.

Life cycle assessment (LCA)

LCA quantifies environmental impacts across every stage of a product's life, cradle-to-grave or cradle-to-cradle, per ISO 14040/14044.

- Stages: raw material extraction, manufacturing, distribution/transport, use, end of life.
- ISO 14040/14044 phases: goal and scope, inventory analysis (LCI), impact assessment (LCIA), interpretation.
- Tools: SimaPro, GaBi, OpenLCA, EPA Sustainable Materials Management tools.
- Differs from a waste audit or carbon footprint - it covers all impacts across all stages.

Life cycle costing and total cost of ownership

Life cycle costing (LCC) weighs purchase price against long-term environmental, social and operational costs.

- Include disposal fees, emissions penalties, water charges, reputational and supply-continuity risk.
- TCO can justify paying more for a durable, certified product to avoid fines, failures and injuries.
- LCA measures environmental impact; LCC measures money.

Sustainable design and green engineering

Design for Environment (DfE) and green engineering build environmental and safety considerations into the earliest design stage.

- Aligns with the EPA P2 hierarchy, ISO 14001 clause 6.1.2 (environmental aspects) and Prevention through Design (ANSI/ASSP Z590.3; the guide cites Z10).
- DfE: fewer raw materials, easy disassembly and recycling, low energy/water use, less waste.
- Energy-efficient equipment: VFDs, compressed-air leak repair, LED lighting with motion sensors (also improves visibility).
- Modular equipment and non-toxic paints/adhesives reduce IAQ problems and hazardous waste.

Safer chemical substitution

Substitution improves worker safety and environmental performance at the same time when evaluated holistically.

- Use low-VOC solvents/coatings; lower GHS hazard categories (remove sensitizers and carcinogens).
- Avoid substances restricted under TSCA or EU REACH.
- Tools: EPA Safer Chemical Ingredients List (SCIL), GreenScreen.
- Check new waste profile, process compatibility and PPE needs before switching.

> **Exam tip:** A substitute that introduces a new, worse hazard is not a valid improvement.

Sustainability metrics and KPIs

Sustainability progress is tracked with environmental, economic and social indicators and benchmarked against recognized programs.

- Environmental: water per unit, GHG (CO2e), hazardous waste per product.
- Economic: life-cycle cost, return on sustainability investment, resource productivity.
- Social: injury rates, engagement in green initiatives, fair labor certifications.
- Benchmarking tools: ENERGY STAR, EPA WasteWise, GRI.

Read more: Yates, Safety Professional's Reference and Study Guide (4th ed.): Ch. 33 Sustainability and Sustainability Principles, p. 743

7. Ten CSP-style practice questions

Answer all ten on paper first, then check the key. Score: _____ / 10 Date: _____ Retest: _____ / 10

5.4-Q1. Which is the MOST effective step in the 'Reduce-Reuse-Recycle' hierarchy?

A. Reuse
B. Recycle
C. Reduce
D. Dispose

5.4-Q2. Which sustainability practice manages suppliers and vendors based on their ethical, safety, and environmental performance?

A. Sustainable supply chain management
B. Energy audits
C. Waste minimization
D. Pollution abatement

5.4-Q3. A construction company changes its purchasing policy to require recycled concrete aggregate, energy-efficient lighting, and low-VOC materials on all projects. Which sustainability principle does this BEST represent?

A. Waste isolation
B. Pollution control after production
C. Compliance with local code minimums only
D. Source reduction and sustainable material selection

5.4-Q4. A company replaces solvent-based paints with water-based coatings to lower VOC emissions. Which sustainability principle does this demonstrate?

A. Reuse
B. Source reduction (Reduce)
C. Recycling
D. Waste diversion

5.4-Q5. What is the primary focus of LEED certification?

A. Sustainable building performance, such as energy efficiency and water conservation
B. Building aesthetics and architectural style
C. Minimizing short-term construction costs
D. Urban zoning and development approvals

5.4-Q6. A Life-Cycle Assessment (LCA) evaluates:

A. Worker performance over a career
B. Product pricing over its sales life
C. Regulatory compliance at a single facility
D. Environmental impacts of a product from raw material extraction through disposal

5.4-Q7. A manufacturer replaces single-use cardboard shipping boxes with durable plastic totes that suppliers return for refilling. Which principle does this represent?

A. Recycle
B. Reuse
C. Source reduction through product redesign
D. Energy recovery

5.4-Q8. A company finds that most of its environmental footprint comes from raw materials made by overseas suppliers. What is the BEST first step to improve supply chain sustainability?

A. Stop tracking supplier impacts because they are outside the company's control
B. Switch all suppliers immediately regardless of cost or quality
C. Establish a supplier code of conduct with sustainability criteria and assess suppliers against it
D. Publish a sustainability statement without supplier engagement

5.4-Q9. The 'triple bottom line' approach to sustainability evaluates an organization's performance in which three areas?

A. People, planet, and profit
B. Quality, cost, and schedule
C. Plan, do, and check
D. Reduce, reuse, and recycle

5.4-Q10. A plant collects scrap aluminum trimmings, sends them to be remelted, and buys back the resulting ingots for production. This is BEST described as:

A. Reuse
B. Source reduction
C. Landfill diversion by incineration
D. Closed-loop recycling

Answer key and explanations

Q	Ans	Why
1	C	Reducing eliminates waste and resource use at the source, so no handling, reprocessing or disposal is needed; recycling still consumes energy.
2	A	Sustainable supply chain management sets supplier codes of conduct, screens and audits suppliers, and extends EHS expectations beyond the company's own sites.
3	D	Specifying recycled, efficient and low-toxicity inputs at procurement reduces impacts before they occur (green purchasing).
4	B	Substituting a less hazardous input prevents the pollution from being generated at all, which is source reduction.
5	A	LEED (Leadership in Energy and Environmental Design) rates buildings on energy, water, materials, indoor environmental quality and site sustainability.
6	D	LCA is a cradle-to-grave (or cradle-to-cradle) analysis of materials, energy, emissions and waste across all life-cycle stages (ISO 14040/14044).
7	B	Using the same item again for its original purpose without reprocessing is reuse. Recycling would break the material down to make new products.
8	C	Setting clear expectations, then assessing and engaging suppliers, is the foundation of supply chain sustainability; ignoring or making unfounded claims (greenwashing) is weak.
9	A	The triple bottom line measures social (people), environmental (planet) and economic (profit) performance, a precursor to ESG reporting.
10	D	The material is reprocessed (remelted) and returned to the same product system, which is closed-loop recycling, a circular economy practice. Reuse would involve no reprocessing.

DOMAIN 5 · COMPETENCY 5.5

Environmental Issues and ESG

CSP11 blueprint: Describe the impact of environmental issues (e.g., aging infrastructure, asbestos, air pollution, climate change, environmental, social, and governance)

1. Definition

This competency covers major environmental issues that affect organizations and communities - deteriorating infrastructure, legacy hazards such as asbestos and lead, air pollution, climate change - and ESG (Environmental, Social and Governance) frameworks used by investors and stakeholders to judge corporate performance.

2. Explanation in simple words

Old pipes, tanks and buildings can leak or contain hazardous materials; dirty air and a changing climate affect health and operations; and investors now look at how companies manage environment, people and governance, not just profit.

3. Key points to remember

- Aging infrastructure: corroded pipelines, tanks, sewers and bridges increase release and failure risk; manage with risk-based inspection, integrity management programs and planned replacement.
- Asbestos: friable vs non-friable; asbestosis, mesothelioma, lung cancer; OSHA PEL 0.1 f/cc (8-h TWA), excursion limit 1.0 f/cc (30 min); building survey before renovation/demolition; NESHAP notification 10 working days before work when RACM thresholds are exceeded (260 linear ft, 160 ft^2, 35 ft^3); removal with containment, negative pressure, wet methods and HEPA; never dry-sweep suspect material.
- Lead: renovation (EPA RRP rule for pre-1978 housing), OSHA lead standards (PEL 50 $\mu g/m^3$, action level 30 $\mu g/m^3$).
- Air pollution: criteria pollutants, ozone, PM2.5, VOCs; health effects (respiratory, cardiovascular); indoor air quality links.
- Climate change: greenhouse gases (CO_2, CH_4, N_2O, fluorinated gases); scope 1 (direct), scope 2 (purchased energy), scope 3 (value chain) emissions; physical risks to operations (heat stress, flooding, severe weather) and transition risks (regulation, markets).
- ESG: Environmental (emissions, waste, water), Social (worker safety and health, human rights, community), Governance (board oversight, ethics, transparency). Frameworks: GRI, SASB/ISSB, TCFD. Safety performance is a core 'S' metric.

4. Real work scenario

> Before renovating a 1960s office, the safety professional orders an asbestos survey. It finds 400 linear feet of friable pipe insulation, so the company files a NESHAP notification 10 working days ahead, hires a licensed abatement contractor using negative-pressure containment, and performs clearance air sampling before re-occupancy.

5. CSP exam tips and traps

- Any suspect insulation in an old building: stop, do not sweep, assume asbestos until tested.
- Scope 2 = purchased electricity.
- ESG 'S' includes occupational safety and health.

6. Every blueprint point (5 blueprint items + 8 more)

Aging infrastructure BLUEPRINT

Many facilities run past their design life; pipelines, tanks, structures and electrical systems 40+ years old face corrosion, fatigue and overload, raising failure risk under modern demands and extreme weather.

- Common failures: corroded USTs/transfer lines (40 CFR 280 requires leak detection, corrosion protection, secondary containment for new USTs).
- Fire sprinklers: scale-clogged galvanized pipe; inspect/test/maintain per NFPA 25; design to NFPA 13.
- Structures: rusted fasteners, rotted supports (1910 Subpart D).
- Electrical: degraded insulation, overloaded panels, no arc-flash labels, not lockable (1910.147, NFPA 70E).
- HVAC: decayed ducts, mold - review ASHRAE 62.1.

> **Exam tip:** Aging infrastructure is multi-disciplinary: compliance, life safety and business continuity.

Inspecting and managing aging assets

Risk-based inspection and asset management prioritize work by consequence and likelihood of failure.

- Risk-based inspection: API 580/581.
- NDT: ultrasonic thickness, radiography, magnetic particle, dye penetrant.
- Infrared thermography (electrical hot spots, bearings), vibration and acoustic monitoring.
- ISO 55000 asset management; CMMS, SCADA, structural health monitoring sensors.
- Administrative: aging infrastructure register, retrofit timelines in capital plans, pre-task checks. Example: ultrasonic testing of tanks over 20 years old after a corrosion leak.

> **Exam tip:** Wall thinning from corrosion -> ultrasonic thickness testing.

Asbestos BLUEPRINT

Asbestos is a group of fibrous silicate minerals once used in insulation, pipe wrap, flooring, roofing, gaskets and fireproofing. Disturbed fibers cause mesothelioma, asbestosis and lung cancer decades later; there is no known safe level.

- OSHA PEL 0.1 f/cc 8-hr TWA; excursion limit 1.0 f/cc over 30 minutes (1910.1001; construction 1926.1101).
- Above the PEL: regulated areas, respirators, training, signage, medical surveillance.
- Buildings built before 1981 - presume thermal insulation and surfacing materials are ACM unless tested.
- Latency of 20-40+ years; smoking multiplies lung cancer risk.

> **Exam tip:** Exceeding the excursion limit (1.0 f/cc for 30 min) is a violation even if the 8-hr TWA is under 0.1.

Asbestos regulations, abatement and disposal

Several agencies regulate asbestos during work, demolition, schools and transport.

- EPA NESHAP (40 CFR 61 Subpart M): survey before renovation/demolition; notify 10 working days before work involving at least 260 linear ft, 160 sq ft or 35 cu ft of regulated ACM (demolition always needs notice).
- AHERA (schools): inspections, management plans, re-inspection every 3 years (surveillance every 6 months).
- Controls: licensed contractors, negative-pressure enclosures, wet methods, glove bags, HEPA vacuums, decon units.
- Clearance: PCM or TEM air sampling; often under 0.01 f/cc.
- Waste: wetted, double-bagged in 6-mil leak-tight bags, labeled, transported to approved landfill with records.

> **Exam tip:** Disturbed suspect ceiling insulation: FIRST stop work, evacuate and isolate the area, then sample.

Other legacy contaminants (lead, PCBs, USTs)

Legacy contaminants in pre-1980s buildings become hazardous when disturbed during maintenance, renovation or demolition.

- Lead-based paint: OSHA 1926.62 (PEL 50 ug/m3, action level 30); EPA RRP rule for target housing.
- PCBs (old transformers, ballasts, caulk): TSCA regulated.
- Leaking underground storage tanks: 40 CFR 280.
- Failure to manage can bring civil penalties, criminal prosecution or fatal exposures.

Air pollution BLUEPRINT

Industrial air pollution affects worker health, the community, regulatory standing and ESG scores.

- Combustion: NOx and SO2 (acid rain and smog precursors).
- Particulate matter: PM2.5 penetrates deep into lungs (diesel exhaust, grinding).
- VOCs from paints, solvents and fuels form ground-level ozone.
- Fugitive dust may carry silica or heavy metals.
- Health effects: short-term irritation, headaches; long-term asthma, bronchitis, cardiovascular disease, lung cancer.

Air quality regulation, reporting and monitoring

EPA, OSHA and states regulate air emissions and exposures.

- CAA: NAAQS, NSR/PSD permits, Title V permits when potential to emit exceeds major-source thresholds; state SIPs may be stricter.
- Title V reports semiannually with CEMS data for key sources.
- TRI (EPCRA 313): annual Form R when a listed chemical exceeds 25,000 lb manufactured/processed or 10,000 lb otherwise used (the guide cites only 10,000 lb).
- Penalties: the guide states up to $50,000/day; inflation-adjusted CAA civil penalties now exceed $100,000 per day per violation.
- Monitoring: fence-line stations, DustTrak (PM), PID (VOCs), MIRAN/infrared analyzers.

> **Exam tip:** PID measures VOCs; DustTrak measures particulate - know which instrument fits the pollutant.

Climate change BLUEPRINT

Climate change is now a safety, health, operational and regulatory issue: hotter summers, extreme weather, shifting disease vectors and utility disruptions.

- Heat stress for outdoor and hot indoor work; OSHA heat National Emphasis Program (launched 2022; the guide says 2023).
- Extreme weather: floods, hurricanes, wildfires disrupt facilities and supply chains; EAPs (1910.38) must reflect new risks.
- Vectors: ticks and mosquitoes spreading (Lyme, West Nile, dengue).
- Grid strain and blackouts affect ventilation, cooling and communications; floods can release chemicals.
- ISO 14001 (2024 amendment) requires considering climate change in context of the organization; TCFD for financial disclosure.

Heat illness prevention

A heat illness prevention program is expected under the General Duty Clause and the NEP.

- Water: about 1 quart per hour (1 cup every 15-20 minutes).
- Rest and shade on work-rest cycles based on heat index or WBGT (NIOSH RELs, ACGIH TLVs).
- Acclimatization: new and returning workers ramp up over 7-14 days (20% first day rule).
- Train workers and supervisors on symptoms; emergency plan for heat stroke (medical emergency - cool immediately, call 911).
- Humidity raises heat index even at moderate air temperature.

> **Exam tip:** Most heat deaths occur in the first days on the job - acclimatization is the key control.

Climate resilience and emergency planning

Facilities adapt to climate risk through resilience audits, updated ERPs and forward-looking climate risk assessment.

- ERP updates: long blackouts, wildfire smoke (N95s, HVAC sealing), shelter-in-place for hurricanes/floods, resilient communications, drills with local agencies.
- Resilience: backup power (generators, UPS), drainage capacity, anchored rooftop equipment, fire-resistant materials, supplier redundancy.
- Climate risk assessment: identify regional trends (NOAA, IPCC), overlay on critical assets, score and control, add to aspects register or BCP.
- Use ISO 31010 risk techniques; results inform capital projects and insurance.

ESG (Environmental, Social and Governance) BLUEPRINT

ESG evaluates how responsibly and sustainably an organization operates; it affects investment, insurance and public trust. Safety performance feeds every pillar.

- Environmental: GHG emissions, resource efficiency, compliance record (fines, spills).
- Social: worker health and safety (TRIR, DART, near misses), DEI, community engagement and environmental justice.
- Governance: transparency and reporting, regulatory compliance, whistleblower protection, ethics.
- A strong safety culture underpins the Social score.

GHG scopes and ESG reporting frameworks

ESG data is reported through recognized frameworks, and emissions are split into three scopes.

- Scope 1: direct emissions from owned sources (boilers, fleet, forklifts, refrigerants).
- Scope 2: indirect emissions from purchased electricity, steam, heat.
- Scope 3: other value chain emissions (suppliers, transport, product use).
- Frameworks: GRI, SASB (now under IFRS/ISSB), CDP, TCFD.
- Management systems supporting ESG: ISO 14001, ISO 45001.

> **Exam tip:** Purchased electricity = Scope 2; supplier emissions = Scope 3.

ESG KPIs

CSPs prove value by supplying measurable indicators for each pillar.

- Environmental: % GHG reduction per revenue unit, tons diverted from landfill, water intensity.
- Social: TRIR/DART trends, % workforce trained, safety perception survey results.
- Governance: near misses reported, % audit findings closed on time, inspections without citations.
- KPIs may link to bonuses, public reports and stock value.

Read more: Yates, Safety Professional's Reference and Study Guide (4th ed.): Ch. 34 Environmental Management, p. 754; Ch. 33 Sustainability; Ch. 4 Regulations (asbestos)

7. Ten CSP-style practice questions

Answer all ten on paper first, then check the key. Score: _____ / 10 Date: _____ Retest: _____ / 10

5.5-Q1. A renovation will remove 300 linear feet of friable asbestos pipe insulation from a commercial building. Under the asbestos NESHAP, the owner/operator must notify the regulatory agency:

A. At least 10 working days before the work begins
B. After the work is complete
C. Only if workers complain
D. 24 hours before the work begins

5.5-Q2. A crew will remove damaged pipe insulation confirmed to contain asbestos. What is the MOST effective method to prevent workers from inhaling asbestos fibers?

A. Provide N95 respirators and ask workers to minimize disturbance
B. Cover the insulation with duct tape before removal
C. Use full containment with negative air pressure, wet methods, and appropriate respirators such as P100
D. Limit exposure time and ventilate the room with fans

5.5-Q3. According to OSHA's asbestos standard (1910.1001(k)(7)), which statement about sanding asbestos-containing flooring is correct?

A. Sanding is allowed if local exhaust ventilation captures fibers above the PEL
B. Sanding is allowed if the surface is first wetted or enclosed
C. Sanding is allowed if it is done in a wet state
D. Sanding of asbestos-containing flooring material is prohibited

5.5-Q4. A CSP leads ISO 14001 and ISO 45001 programs, reports sustainability metrics to the board, and improves safety culture. Which ESG pillar does this work strengthen?

A. Environmental only
B. All three: Environmental, Social, and Governance
C. Social only
D. Governance only

5.5-Q5. In greenhouse gas reporting, emissions from electricity a company PURCHASES from the utility are classified as:

A. Scope 2
B. Scope 1
C. Scope 3
D. Not reportable

5.5-Q6. What is the MOST common cause of chemical releases from aging infrastructure?

A. Software failure
B. Overtraining of personnel
C. Corrosion and metal fatigue of tanks, piping, and valves
D. Excessive electrical grounding

5.5-Q7. What is the OSHA Permissible Exposure Limit (PEL) for airborne asbestos?

A. 0.5 fiber/cc as an 8-hour TWA
B. 1.0 fiber/cc as an 8-hour TWA
C. 0.05 fiber/cc as an 8-hour TWA
D. 0.1 fiber/cc as an 8-hour TWA

5.5-Q8. In ESG frameworks, the 'S' (Social) pillar MOST directly includes which CSP-related metric?

A. GHG intensity (tons CO2e per unit)
B. TRIR and DART incident rates
C. Energy efficiency ratio
D. Hazardous waste recycling rate

5.5-Q9. A plant has 40-year-old underground process piping with no inspection history. What is the BEST approach to manage environmental release risk?

A. Wait for a leak to identify weak sections
B. Apply risk-based inspection: prioritize segments by likelihood and consequence of failure and inspect or replace accordingly
C. Replace all piping immediately regardless of cost
D. Increase insurance coverage

5.5-Q10. Electrostatic precipitators (ESPs) are primarily used to control air pollution by:

A. Separating oil from water
B. Neutralizing acid gases
C. Removing fine particulates from exhaust gas using an electric field
D. Adsorbing VOCs on activated carbon

Answer key and explanations

Q	Ans	Why
1	A	NESHAP (40 CFR 61 Subpart M) requires written notice 10 working days in advance when regulated ACM meets or exceeds 260 linear ft, 160 sq ft, or 35 cu ft.

2	**C**	Regulated asbestos removal relies on engineering controls (negative-pressure enclosure, HEPA, wet methods) supplemented by proper respirators; fans spread fibers.
3	**D**	OSHA prohibits sanding asbestos-containing flooring outright; floor care must use wet methods and low-abrasion pads at low speeds.
4	**B**	ISO 14001 supports E, worker safety and culture support S, and board-level reporting and management systems support G.
5	**A**	Scope 1 = direct emissions from owned sources (boilers, fleet); Scope 2 = indirect emissions from purchased energy; Scope 3 = other value-chain emissions.
6	**C**	Over time corrosion, fatigue and wear thin walls and weaken joints, leading to leaks; mechanical integrity programs address this.
7	**D**	OSHA's asbestos PEL is 0.1 f/cc (8-hr TWA) with an excursion limit of 1.0 f/cc over 30 minutes. Trap: confusing the excursion limit with the PEL.
8	**B**	Worker health and safety performance is a core Social metric; GHG, energy and waste metrics are Environmental.
9	**B**	Risk-based inspection and integrity management focus resources where failure is most likely and most harmful. Waiting is reactive; insurance transfers cost but not the hazard.
10	**C**	ESPs charge particles and collect them on plates, achieving high efficiency for fine particulates at power plants and smelters.

DOMAIN 6

Occupational Health and Applied Science

10% of the exam · 6 competencies

6.1	Occupational Exposure: Anticipate, Recognize, Evaluate, Control Anticipate, recognize, evaluate, and control occupational exposures by implementing techniques for measurement, sampling, and analysis (e.g., hazardous chemicals, SDS, radiation, noise, biological hazards, heat/cold, indoor air quality, ventilation, nanoparticles, combustible dust, heat systems, high pressure, silica, powder and spray applications, blasting, molten metals, hot work, cold and heat stress, laser)
6.2	Public Health and Epidemiology Understand principles of public health as applicable (i.e., fundamentals of epidemiology, infectious disease, risk factors, statistics to interpret data)
6.3	Toxicology Apply toxicology principles to create exposure control plans and develop risk mitigation plans (e.g., using sampling equipment, symptoms of an exposure, LD50, LC50, mutagens, carcinogens, teratogens, ototoxins)
6.4	Ergonomics and Human Factors Evaluate principles related to ergonomics and human factors (e.g., visual acuity, body mechanics, lifting, vibration, anthropometrics, fatigue management)
6.5	Chemistry: Containment Volume and Storage Requirements Apply chemistry principles to calculate required containment volumes and hazardous materials storage requirements
6.6	Physics: Energy, Weight, Force and Stress Apply core concepts in physics (e.g., forms of energy, weights, forces, stresses)

DOMAIN 6 · COMPETENCY 6.1

Occupational Exposure: Anticipate, Recognize, Evaluate, Control

CSP11 blueprint: Anticipate, recognize, evaluate, and control occupational exposures by implementing techniques for measurement, sampling, and analysis (e.g., hazardous chemicals, SDS, radiation, noise, biological hazards, heat/cold, indoor air quality, ventilation, nanoparticles, combustible dust, heat systems, high pressure, silica, powder and spray applications, blasting, molten metals, hot work, cold and heat stress, laser)

1. Definition

Industrial hygiene is the science of anticipating, recognizing, evaluating and controlling workplace health hazards (chemical, physical, biological and ergonomic) that can cause illness or impaired well-being. Evaluation compares measured exposures with occupational exposure limits (OSHA PELs, ACGIH TLVs, NIOSH RELs).

2. Explanation in simple words

Expect where exposures will be, spot them, measure how much people breathe, hear or absorb, compare that with the safe limit, and then reduce it - using engineering controls first.

3. Key points to remember

- Exposure limits: PEL (OSHA, legal), TLV (ACGIH, guideline, updated annually), REL (NIOSH); TWA (8-h), STEL (15-min), Ceiling (never exceed), IDLH (immediately dangerous to life or health). Action level usually ½ PEL.
- Sampling: personal samples in the breathing zone; active (pump, calibrated before/after) vs passive (diffusion badges); media - filters (particulates), sorbent tubes (vapors), impingers; direct-reading instruments (4-gas meters, PIDs). Sample volume = flow rate × time. Concentration = mass ÷ volume.
- Chemicals: ppm = mg/m³ × 24.45 ÷ MW (25 °C, 1 atm). TWA = Σ(C × T) ÷ 8. Mixtures with additive effects: Σ(C/T) must be ≤ 1.
- Noise (1910.95): PEL 90 dBA (5-dB exchange rate); action level 85 dBA (50% dose) triggers hearing conservation program - monitoring, annual audiometric testing, hearing protectors, training, records. Standard threshold shift = average 10 dB shift at 2,000, 3,000 and 4,000 Hz. T = 8 ÷ 2^((L − 90)/5). Adding sources: L = 10 log Σ10^(L/10); doubling distance −6 dB (point source).
- Radiation: ionizing (alpha - stopped by paper; beta - plastic/aluminum; gamma/X-ray - lead/concrete; neutron - water/hydrogen-rich materials). ALARA with time, distance (inverse square law $I_2 = I_1(d_1/d_2)^2$), shielding. Dosimeters measure cumulative dose. OSHA/NRC occupational limit 5 rem (50 mSv)/year whole body. Half-life: $A = A_0(½)^{(t/T½)}$. Non-ionizing: UV, IR, RF/microwave, lasers.
- Lasers (ANSI Z136.1): Class 1, 2, 3R, 3B, 4; Class 3B and 4 require a Laser Safety Officer, nominal hazard zone, interlocks, eyewear with correct optical density.
- Heat stress: WBGT outdoors = 0.7 NWB + 0.2 GT + 0.1 DB; indoors = 0.7 NWB + 0.3 GT. Heat stroke (core > 104 °F, confusion, may stop sweating) is a medical emergency; heat exhaustion (heavy sweating, weakness). Acclimatization: new workers 20% day 1, +20%/day. Cold stress: hypothermia, frostbite, trench foot; wind chill.
- Ventilation: Q = V × A; V = 4005√VP; TP = SP + VP; local exhaust (capture at source - preferred for toxics) vs general/dilution ventilation (low-toxicity, low-volume, dispersed sources). Dilution Q = 403 × 10^6 × SG × ER × K ÷ (MW × C). Air changes per hour = Q × 60 ÷ volume.
- Respirators (1910.134): written program, medical evaluation, fit test (annually for tight-fitting), training; APFs - half-mask 10, full-face APR 50, PAPR loose hood 25, SAR/SCBA pressure-demand full face

1,000/10,000; MUC = PEL × APF; atmosphere-supplying respirators for IDLH or oxygen deficient (< 19.5%).

- Biological hazards: bloodborne pathogens (1910.1030 - exposure control plan, universal precautions, hepatitis B vaccine within 10 working days), Legionella (cooling towers), molds, TB (BSL-3 labs).
- Indoor air quality: CO_2 as a ventilation indicator (> ~700 ppm above outdoor), ASHRAE 62.1, sources (VOCs, mold, outdoor air intake location).
- Specific hazards: respirable crystalline silica (PEL 50 µg/m³, AL 25; wet methods, LEV; Table 1 for construction), combustible dust (DHA, housekeeping), nanoparticles (high surface area, HEPA effective), abrasive blasting (substitute non-silica media, blast enclosures, Type CE respirators), molten metal (moisture → steam explosions, aluminized PPE), welding fumes (manganese, hexavalent chromium from stainless steel, zinc → metal fume fever), high-pressure systems (stored energy, injection injuries).

4. Real work scenario

> Workers polishing stainless steel complain of sinus irritation. The hygienist first reviews SDSs, then takes task-based personal breathing-zone samples for hexavalent chromium and nickel. Results exceed the action level, so a downdraft table with local exhaust is installed, respirators are used while it is built, and follow-up sampling confirms exposures below the action level.

5. CSP exam tips and traps

- Always start with SDS review and hazard recognition, then personal breathing-zone sampling.
- Noise and radiation calculations (dose, adding dB, inverse square, half-life) are high-yield.
- Local exhaust is preferred over dilution ventilation for toxic contaminants.
- Mixture exposure can exceed the limit even if each chemical is individually below its own limit.

6. Every blueprint point (22 blueprint items + 7 more)

OSHA hazard categories

Workplace health hazards are grouped into four categories: physical, chemical, biological and ergonomic. Recognizing the category tells you which standards, sampling methods and controls apply.

- Physical: noise, vibration, heat/cold, ionizing and non-ionizing radiation, electricity, pressure.
- Chemical: solvents, metals, dusts (silica, asbestos), gases, corrosives - governed by HazCom 1910.1200 and substance-specific standards.
- Biological: bacteria, viruses, fungi, parasites, prions/toxins - e.g., Bloodborne Pathogens 1910.1030.
- Ergonomic: force, repetition, awkward posture, contact stress, vibration - no OSHA ergonomics standard; enforced via the General Duty Clause 5(a)(1).

> **Exam tip:** IH sequence: anticipate, recognize, evaluate, control (plus confirm). Recognition comes before measurement.

Occupational exposure limits (PEL, TLV, REL, STEL, Ceiling, IDLH)

Exposure limits set the airborne concentration most workers can tolerate without adverse effect. Know who publishes each and whether it is enforceable.

- PEL = OSHA, legally enforceable (1910.1000 Z-tables and substance standards); many date from 1971 and are outdated.
- TLV = ACGIH, advisory, updated yearly; REL = NIOSH, recommended (10-h TWA basis); WEL = UK; IOELV = EU.
- TWA = 8-h average; STEL = 15-min average, max 4 times/day, at least 60 min apart; Ceiling (C) = never exceed.
- Action level (AL) is usually 1/2 the PEL and triggers monitoring and medical surveillance.
- IDLH (NIOSH) = concentration from which escape within 30 min is possible without irreversible effects; used for respirator selection (e.g., CO IDLH 1,200 ppm).

> **Exam tip:** 'Legally enforceable limit' = PEL; 'best practice' or 'most protective' usually = TLV or REL.

Measurement BLUEPRINT

Measurement quantifies exposure with direct-reading instruments or integrated sampling so it can be compared with an exposure limit. Instruments must be appropriate for the agent, calibrated and used by a competent person.

- Direct-reading: multigas meters (O2, LEL, CO, H2S), PID for VOCs, FID, electrochemical sensors, colorimetric detector tubes, real-time aerosol monitors.
- Combustible gas indicators are calibrated to a specific gas (e.g., methane or pentane); readings for other gases need a response/correction factor.
- Bump test before each day's use; full calibration per manufacturer; pre- and post-calibrate sampling pumps (accept if within about 5%).
- Noise: sound level meter (Type 1 precision, Type 2 general survey); dosimeter for personal dose; octave band analyzer for frequency content.
- Heat: WBGT monitor; radiation: GM counter (survey), TLD/film badge (personal dose).

> **Exam tip:** Direct-reading = immediate decision (entry, alarm); integrated personal sampling = compliance with a TWA.

Sampling BLUEPRINT

Air sampling collects contaminant from the worker's breathing zone over a known air volume. The strategy - who, when, how long, how many - matters as much as the method.

- Personal breathing-zone samples (within about a 1-ft radius of nose/mouth) are required for compliance; area samples characterize sources.
- Media: charcoal/sorbent tubes for vapors (about 0.01-0.2 L/min), filters/cassettes for particulates and metals (about 1-4 L/min), cyclone for respirable dust (10-mm nylon about 1.7 L/min), PCM cassette for asbestos fibers, bags/canisters for grab samples.
- Concentration = mass collected / air volume. Example: 0.5 mg at 2.0 L/min for 240 min = 480 L = 0.48 m3, so C = 0.5/0.48 = 1.04 mg/m3.
- Full-shift single sample or consecutive samples are preferred over grab samples for TWA comparisons; STEL samples are 15 min.
- Include field blanks (commonly 10% of samples, at least 1-2 per set); sample similar exposure groups (SEGs) and the maximum-risk employee first.
- Other types: bulk samples (identify material, e.g., asbestos in insulation), wipe/surface samples (lead, beryllium, hygiene), biological monitoring (BEIs).

> **Exam tip:** Unit trap: 1 m3 = 1,000 L. Forgetting this gives answers off by 1,000.

Analysis BLUEPRINT

Analysis turns sample results into decisions: the lab quantifies the agent, and the safety professional compares the result with the limit using sound statistics.

- Lab methods: gas chromatography (organics), AAS/ICP (metals), X-ray diffraction (crystalline silica), PCM (asbestos fibers, TEM to confirm type), gravimetric (total/respirable dust). Use an AIHA-accredited lab and NIOSH/OSHA methods.
- OSHA compliance test: Y = result/PEL; UCL = Y + SAE, LCL = Y - SAE. UCL <= 1 compliance; LCL > 1 violation; otherwise 'possible overexposure'.
- Example: result 105 ppm, PEL 100 ppm, SAE 0.10 -> Y = 1.05, LCL = 0.95 -> possible overexposure, not a citable violation.
- Exposure data are usually lognormal: describe with geometric mean (GM) and geometric standard deviation (GSD) and judge the 95th percentile against the OEL.
- Extended shifts (Brief and Scala): RF = (8/h) x ((24 - h)/16). 12-h shift: RF = (8/12)(12/16) = 0.5, so a 100 ppm limit becomes 50 ppm.
- AIHA exposure categories compare the 95th percentile with the OEL (e.g., <10% highly controlled ... >100% poorly controlled).

> **Exam tip:** One result just under the PEL does not prove compliance - variability means more samples or a lower AL are needed.

TWA and mixture calculations

The 8-hour time-weighted average combines different concentrations over a shift; mixtures of chemicals with similar effects are assessed with the additive formula.

- TWA = (C1T1 + C2T2 + ... + CnTn) / 8 h. Unsampled time is assumed zero only if no exposure is known to occur.
- Example: 3 h at 50 ppm, 2 h at 100 ppm, 3 h at 20 ppm -> (150 + 200 + 60)/8 = 51.25 ppm.
- Additive mixture: Em = C1/L1 + C2/L2 + ... + Cn/Ln; Em > 1 means the mixture limit is exceeded.
- Example: 50 ppm (L 100) + 40 ppm (L 100) + 15 ppm (L 50) -> 0.5 + 0.4 + 0.3 = 1.2 -> exceeds.
- Use the additive formula only for agents with the same target organ/effect (e.g., CNS-depressant solvents); independent effects are judged separately.

> **Exam tip:** Always divide by 8 h for an 8-hr TWA, even if the worker was exposed for only 6 h.

Chemicals BLUEPRINT

Chemical hazards include toxic, flammable, corrosive and reactive substances in gas, vapor, mist, fume, dust or fiber form. OSHA's Hazard Communication standard (1910.1200, aligned with GHS) is the core program.

- HazCom elements: written program, chemical inventory, labels, SDSs, employee training (before initial assignment and when a new hazard is introduced).
- Shipped-container label elements: product identifier, signal word, hazard statements, pictograms, precautionary statements, supplier name/address/phone.
- Signal words: 'Danger' (more severe) or 'Warning'; only one is used on a label.
- Physical forms: gases/vapors (molecular), aerosols - dusts (mechanical), fumes (condensed metal vapor, < 1 um), mists (liquid droplets), fibers.
- Substance-specific standards (1910.1001-1053) add AL, exposure monitoring, regulated areas, medical surveillance and hygiene rules (e.g., lead, benzene, Cr(VI), silica).

> **Exam tip:** If a question gives a new chemical, the first step is to obtain and review the SDS.

SDS (Safety Data Sheet) BLUEPRINT

The SDS is the 16-section GHS document from the manufacturer that describes a chemical's hazards and safe handling. It must be readily accessible to workers on every shift.

- Sections: 1 Identification, 2 Hazard(s), 3 Composition, 4 First aid, 5 Fire-fighting, 6 Accidental release, 7 Handling and storage, 8 Exposure controls/PPE, 9 Physical/chemical properties, 10 Stability and reactivity, 11 Toxicological, 12-15 Ecological, Disposal, Transport, Regulatory (not enforced by OSHA), 16 Other (revision date).
- Section 8 = PEL/TLV and PPE; Section 10 = incompatibilities; Section 11 = LD50/LC50 and carcinogenicity; Section 9 = flash point, vapor pressure, vapor density.
- Employers can rely on the supplier's SDS; electronic access is fine if no barriers exist (power outage backup).
- 1910.1020 treats SDSs (or a record of chemical identity, where used and when) as exposure records kept 30 years.

> **Exam tip:** 'Which SDS section lists incompatible materials?' = Section 10 (handling/storage segregation guidance is in 7).

Radiation BLUEPRINT

Radiation is energy emitted as particles or waves. Ionizing radiation (alpha, beta, gamma/X-ray, neutrons) can remove electrons and damage DNA; non-ionizing radiation (UV, visible, IR, microwave, RF) mainly causes heating, burns and eye damage.

- Shielding: alpha - paper/skin (internal hazard if inhaled/ingested); beta - plastic, glass, aluminum (use low-Z first to limit bremsstrahlung); gamma/X-ray - lead, concrete, water; neutrons - water, polyethylene, concrete.
- Protection = time, distance, shielding (ALARA). Dose = dose rate x time: 50 mrem/h for 30 min = 25 mrem.
- Inverse square law: I1 x d1^2 = I2 x d2^2. Example: 100 mR/h at 2 ft -> at 10 ft: 100 x (2/10)^2 = 4 mR/h. Doubling distance cuts intensity to 1/4.
- Half-life: A = A0 x (1/2)^(t/T). 100 mCi of I-131 (T = 8 d) after 24 d -> 12.5 mCi.
- Non-ionizing: UV - photokeratitis (welder's flash), skin cancer, cataracts; IR - cataracts, burns (glassblowers, furnace work); RF/microwave - tissue heating.
- Regulators: NRC (licensed nuclear material, 10 CFR 20), states (Agreement States), OSHA 1910.1096 (other sources, e.g., X-ray machines), EPA/DOE environmental and defense.

> **Exam tip:** Inverse square law applies to point sources; for 'move from 3 ft to 6 ft' the answer is always 1/4 the original intensity.

Radiation units, dose limits and area classifications

Radiation quantities have traditional and SI units, and dose limits apply to workers, lens of eye, skin/extremities and the public.

- Activity: curie (Ci) / becquerel (Bq; 1 Ci = 3.7 x 10^10 Bq). Exposure: roentgen (R). Absorbed dose: rad / gray (1 Gy = 100 rad). Dose equivalent: rem / sievert (1 Sv = 100 rem).
- Dose equivalent = absorbed dose x quality (weighting) factor: alpha 20, neutrons 5-20, beta/gamma/X-ray 1.
- Annual occupational limits (NRC 10 CFR 20): 5 rem (50 mSv) whole body TEDE, 15 rem lens of eye, 50 rem skin/extremities; public 0.1 rem (1 mSv); declared pregnant worker 0.5 rem over the pregnancy. The guide labels these OSHA limits; OSHA 1910.1096 itself uses 1.25 rem per calendar quarter whole body.
- Areas (NRC): Radiation Area > 5 mrem/h at 30 cm; High Radiation Area > 100 mrem/h at 30 cm; Very High Radiation Area > 500 rad/h at 1 m.
- Monitoring: TLD/OSL badges and film badges measure accumulated dose; GM counters survey contamination.
- Overexposure reporting under 1910.1096: immediate/24-h notification depending on dose, written report within 30 days.

> **Exam tip:** Gray/rad = energy absorbed; sievert/rem = biological effect. Pick rem/Sv when the question mentions 'biological damage'.

Radiation emergency response

Radiation incidents follow life-safety-first priorities, isolation, notification and ALARA-based protection of responders.

- Rescue the injured first if safe - medical care outranks decontamination for serious injuries.
- Isolate and post the area; notify NRC/state, DOE or EPA as applicable, and OSHA if limits are exceeded.
- Responders use time, distance, shielding and worker rotation; wear dosimeters and respirators for airborne alpha/beta contamination.
- Personal decon: remove clothing (removes about 90% of contamination), wash with mild soap and water, do not abrade skin.
- Potassium iodide (KI) blocks thyroid uptake of radioactive iodine only; shelter-in-place if evacuation is unsafe.
- Contaminated PPE is radioactive waste; store until decayed per half-life or ship to licensed disposal.

> **Exam tip:** 'First action' at a radiation spill with an injured worker = life-safety rescue/medical care, then control contamination.

Noise BLUEPRINT

Noise is unwanted sound; prolonged exposure above 85 dBA destroys cochlear hair cells and causes permanent noise-induced hearing loss (NIHL). OSHA 1910.95 sets the PEL and Hearing Conservation Program triggers.

- OSHA PEL 90 dBA 8-h TWA (5-dB exchange rate); Action Level 85 dBA (dose 50%) triggers the HCP; impulse noise should not exceed 140 dB peak.
- NIOSH REL and ACGIH TLV: 85 dBA 8-h with 3-dB exchange rate (more protective).
- Human hearing 20-20,000 Hz; most sensitive 1,000-4,000 Hz; NIHL first appears as a notch at 3,000-6,000 Hz (classic 4 kHz notch).
- dBA weighting mimics the ear and is used for compliance; dBC/peak (flat) for impulse and low-frequency noise.
- Instruments: SLM for area spot checks, integrating SLM for variable noise, dosimeter for personal dose (compliance).

Exam tip: PEL 90 vs AL 85: at 87 dBA TWA the employer is under the PEL but must run a hearing conservation program.

Noise calculations (dB addition, dose, distance)

Decibels are logarithmic, so sound levels cannot be added arithmetically. Dose and TWA calculations follow OSHA 1910.95 Appendix A.

- Adding levels: L = 10 log10(sum of 10^(Li/10)). Two equal sources = +3 dB (90 + 90 = 93 dB); 90 + 95 = 96.2 dB.
- Shortcut: difference 0-1 dB add 3; 2-3 add 2; 4-9 add 1; 10+ add 0 to the higher level.
- OSHA allowed time T = 8 / 2^((L - 90)/5): 90 dBA 8 h, 95 4 h, 100 2 h, 105 1 h, 110 0.5 h, 115 0.25 h max (85 dBA = 16 h). NIOSH: T = 8 / 2^((L - 85)/3).
- Dose D = 100 x (C1/T1 + C2/T2 + ...). Example: 2 h at 95 (T 4), 4 h at 90 (T 8), 2 h at 85 (T 16) -> D = 100 x (0.5 + 0.5 + 0.125) = 112.5%.
- TWA = 16.61 x log10(D/100) + 90. D = 112.5% -> 90.85 dBA; D = 200% -> 95 dBA; D = 50% -> 85 dBA.
- Distance (point source, free field): L2 = L1 - 20 log10(d2/d1); drops 6 dB per doubling. 100 dB at 3 ft -> 88 dB at 12 ft.

Exam tip: Halving exposure time under OSHA allows +5 dB; under NIOSH only +3 dB. Do not mix exchange rates.

Hearing conservation program and hearing protectors

1910.95(c)-(o) requires a hearing conservation program when the 8-h TWA reaches 85 dBA. Engineering and administrative controls are required above 90 dBA; HPDs are interim/last line of defense.

- Elements: monitoring, audiometric testing, HPDs at no cost (mandatory over 90 dBA or with an STS), training annually, recordkeeping.
- Baseline audiogram within 6 months of first exposure at/above AL (1 year with mobile van, HPDs required meanwhile); 14 h without workplace noise before baseline; annual audiograms thereafter.
- Standard threshold shift (STS) = average change of 10 dB or more at 2,000, 3,000 and 4,000 Hz in either ear; OSHA-recordable if also 25 dB or more above audiometric zero. Notify employee in writing within 21 days.
- NRR derating (OSHA App. B, dBA): effective level = TWA - (NRR - 7). OSHA FOM also applies a 50% safety factor: TWA - (NRR - 7)/2. Example: 100 dBA, NRR 29 -> 100 - 11 = 89 dBA.
- Records: noise exposure measurements 2 years; audiograms for duration of employment.
- Hearing loss types: conductive (outer/middle ear, often treatable), sensorineural (cochlea/nerve - NIHL, permanent), mixed, central (brain). TTS is temporary; PTS and presbycusis (age) are permanent.

Exam tip: Calculation trap: subtract 7 from NRR before derating when the exposure is in dBA.

Biological hazards BLUEPRINT

Biohazards are microorganisms, their toxins and biological materials that cause infection, allergy or poisoning. Exposure is significant in healthcare, labs, agriculture, animal handling, sewage and waste work.

- Categories: bacteria (TB, anthrax, Legionella), viruses (HBV, HCV, HIV, rabies, influenza), fungi (Aspergillus, Histoplasma, Candida), parasites (malaria, Giardia), prions and toxins (CJD, botulinum, aflatoxin).
- Bloodborne Pathogens 1910.1030: exposure control plan reviewed annually, universal/standard precautions, engineered sharps, HBV vaccine offered within 10 working days at no cost (signed declination if refused), annual training, sharps injury log, post-exposure evaluation.
- BBP records: medical records duration of employment + 30 years; training records 3 years.
- Biosafety levels: BSL-1 teaching labs (E. coli K-12); BSL-2 clinical labs, Class II BSC (Salmonella, HBV, HIV samples); BSL-3 negative pressure, HEPA exhaust (TB, SARS-CoV-2, Coxiella); BSL-4 positive-pressure suits, isolated building (Ebola, Marburg).
- Controls: elimination/substitution, engineering (BSCs, negative-pressure rooms, ventilation), vaccination, administrative (screening), PPE (N95 or higher for airborne agents).

> **Exam tip:** Match the containment clue: 'negative pressure' = BSL-3; 'full-body positive-pressure suit' = BSL-4.

Heat/cold BLUEPRINT

The thermal environment (heat or cold) is assessed by measuring air temperature, humidity, air movement and radiant heat, together with work rate and clothing. The WBGT index is the standard heat-stress metric.

- Outdoor with sun: WBGT = 0.7 NWB + 0.2 GT + 0.1 DB. Indoor/no solar load: WBGT = 0.7 NWB + 0.3 GT (NWB = natural wet bulb, GT = globe, DB = dry bulb).
- Example: NWB 25 C, GT 40 C, DB 30 C -> outdoor WBGT = 17.5 + 8 + 3 = 28.5 C; indoor = 17.5 + 12 = 29.5 C.
- Compare WBGT (adjusted for clothing) with ACGIH TLV/Action Limit tables by metabolic rate and work-rest regimen (e.g., continuous moderate work, acclimatized TLV about 28 C).
- Heat index (temperature + humidity) is used for outdoor work planning; OSHA relies on the General Duty Clause and a heat NEP (a federal heat rule was proposed in 2024).
- Cold: wind chill (air temperature + wind speed) drives frostbite risk; ACGIH cold TLV aims to keep core temperature at or above 36 C (96.8 F).

> **Exam tip:** NWB carries 70% of the WBGT weight - humidity and evaporative cooling dominate heat stress.

Heat/cold stress BLUEPRINT

Heat and cold illnesses range from mild skin conditions to life-threatening emergencies. Recognize the signature symptom and the correct first response.

- Heat rash - blocked sweat ducts; heat cramps - sodium/potassium loss (electrolyte drinks); heat syncope - blood pooling in legs (lie down, elevate legs).
- Heat exhaustion - heavy sweating, nausea, rapid pulse, core temperature below 104 F (40 C); reversible with rest, cooling and fluids.
- Heat stroke - core above 104 F (40 C), confusion/seizures, often hot dry skin; MEDICAL EMERGENCY: call EMS and cool rapidly (cold-water immersion).
- Prevention: acclimatization (NIOSH: no more than 20% of usual exposure on day 1 for new workers, increasing about 20%/day), water about 1 cup every 15-20 min, shade, work-rest cycles, buddy system.
- Cold: hypothermia (core below 95 F / 35 C, shivering then confusion), frostbite (frozen tissue, pale waxy skin), chilblains (non-freezing, red itchy patches), trench/immersion foot (cold and wet, 32-60 F), reduced dexterity from vasoconstriction.

> **Exam tip:** Confusion/altered mental state + heat = heat stroke, regardless of whether the worker is still sweating.

Indoor air quality BLUEPRINT

IAQ problems arise from inadequate ventilation, indoor sources (VOCs, mold, combustion products) and outdoor air intake contamination. Investigations combine occupant surveys, walkthroughs and targeted measurements.

- Tools: CO2 monitoring (surrogate for ventilation adequacy), direct-reading CO/VOC/particle meters, smoke tubes and anemometers for airflow, culture plates and spore traps for mold.
- CO2 above about 1,000 ppm (or more than about 700 ppm above outdoor, ~400 ppm) suggests inadequate outdoor air.
- Sick building syndrome (SBS): non-specific symptoms (headache, irritation, fatigue) that improve after leaving the building; no single cause.
- Building-related illness (BRI): a diagnosable illness from an identified contaminant (Legionnaires' disease, hypersensitivity pneumonitis, CO poisoning); OSHA-recordable if work-related.
- ASHRAE 55 = thermal comfort (about 68-76 F, RH about 30-60%); ASHRAE 62.1 = ventilation for acceptable IAQ.
- Moisture control is key to mold; remediate water damage within 24-48 h.

> **Exam tip:** SBS vs BRI: 'symptoms disappear when they leave' = SBS; 'confirmed diagnosis linked to a contaminant' = BRI.

Ventilation BLUEPRINT

Ventilation controls airborne contaminants either by diluting them with clean air (general/dilution ventilation) or by capturing them at the source (local exhaust ventilation, LEV). LEV is preferred for toxic contaminants and point sources.

- Q = V x A (cfm = fpm x ft2). Example: 12-in duct (A = 0.785 ft2) at 3,000 fpm -> Q = 2,356 cfm.
- Air changes per hour: ACH = Q (cfm) x 60 / room volume (ft3). Example: 2,000 cfm in 50 x 40 x 10 ft room (20,000 ft3) -> 6 ACH.
- Plain-opening hood capture (DallaValle): Q = V(10X^2 + A); flanged hood Q = 0.75V(10X^2 + A). Example: 100 fpm at X = 1 ft, A = 0.5 ft2 -> Q = 1,050 cfm.
- Velocity from velocity pressure (standard air): V = 4,005 x sqrt(VP in. w.g.); VP = 1 in. -> 4,005 fpm.
- Dilution: Q = (403 x 10^6 x SG x ER x K)/(MW x C), ER in pints/min evaporated, K = mixing safety factor 1-10; only for low-toxicity contaminants with dispersed sources.
- LEV components: hood, ductwork, air cleaner, fan, exhaust stack; provide make-up air; typical dust transport velocity about 3,500-4,500 fpm. ASHRAE 62.1 and ACGIH Industrial Ventilation manual are references.

> **Exam tip:** Capture velocity falls off roughly with the square of distance - moving the hood closer is the cheapest improvement.

Nanoparticles BLUEPRINT

Engineered nanoparticles are smaller than 100 nm. Their very high surface-area-to-mass ratio makes them more reactive and lets them reach the alveoli and possibly the bloodstream.

- Sources: nanotech labs, coatings/paints, electronics, drug delivery, carbon nanotubes/nanofibers, 3D-printing powders.
- Hazards: lung inflammation and fibrosis, possible cancer, cardiovascular effects (translocation), skin/eye irritation; long-term toxicity data are limited - apply precautionary controls.
- NIOSH RELs: carbon nanotubes/nanofibers 1 ug/m3 (respirable elemental carbon); ultrafine TiO2 0.3 mg/m3.
- Controls: enclosure and LEV with HEPA filtration (HEPA captures nanoparticles efficiently by diffusion), wet methods, HEPA vacuums, no dry sweeping or compressed air, respirators with P100 filters.
- References: NIOSH Approaches to Safe Nanotechnology; ISO/TR 12885.

> **Exam tip:** Mass-based limits under-represent nano hazards - surface area and particle count matter more.

Combustible dust BLUEPRINT

Finely divided combustible solids can deflagrate when dispersed in air in a confined space. Secondary explosions from accumulated dust disturbed by a primary event cause most deaths.

- Dust explosion pentagon: fuel, oxygen, ignition, dispersion, confinement (fire triangle + 2).
- Materials: wood, grain/sugar/flour, plastics, coal, metals (aluminum, magnesium), pharmaceuticals.
- Standards: NFPA 652 (fundamentals, Dust Hazard Analysis required), 654 (general), 61 (agricultural/food), 484 (metals), 664 (wood); OSHA Combustible Dust NEP; 1910.272 grain handling.
- Kst (deflagration index) classes: St 1 up to 200, St 2 201-300, St 3 above 300 bar-m/s; higher = more violent.
- Housekeeping: remove accumulations (historic NFPA 654 benchmark 1/32 in. over 5% of floor area); use approved vacuums; avoid compressed-air blowdown; inspect hidden surfaces.
- Engineering: dust collection, explosion venting (NFPA 68), suppression/isolation (NFPA 69), Class II hazardous location equipment, bonding/grounding, spark detection.

> **Exam tip:** The most effective prevention is controlling dust accumulation (housekeeping/containment), not just ignition control.

Heat systems (boilers, steam, thermal fluids, furnaces) BLUEPRINT

Process heating systems store large amounts of thermal and pressure energy. Failures cause burns, scalds, fires and boiler or furnace explosions.

- Boilers and pressure vessels: ASME Boiler and Pressure Vessel Code, National Board inspection, state jurisdictional inspections.
- Safeguards: safety relief valves, low-water fuel cutoffs, pressure/temperature limits, burner management systems and flame safeguards (NFPA 85/86), purge cycles before ignition.
- Steam: scalding, water hammer, flashing condensate; lock out and depressurize/cool before opening; use proper steam-rated hoses.
- Thermal fluid (hot oil) systems: leaks into insulation can autoignite below the fluid's normal autoignition temperature.
- Controls also include insulation/guarding of hot surfaces (burn threshold about 140 F/60 C contact), heat-resistant PPE and trained operators.

> **Exam tip:** Low-water condition in a boiler: never add cold water to a hot dry boiler - shut down the burner and let it cool.

High pressure (compressed air, hydraulics, gas cylinders, pressure testing) BLUEPRINT

High-pressure systems release stored energy violently through hose whip, projectiles, injection injuries and vessel rupture.

- Compressed air for cleaning must be reduced below 30 psi with chip guarding and PPE (1910.242(b)); never use on skin or clothing.
- Hydraulic/high-pressure fluid injection injuries look minor but are surgical emergencies; never locate leaks with hands - use cardboard.
- Cylinders (1910.101, CGA P-1): store upright, secured, valve caps on; oxygen 20 ft from fuel gases or separated by a 5-ft, 1/2-hour fire-rated barrier; move with carts, not by rolling or lifting by caps.
- Pressure testing: hydrostatic is preferred over pneumatic because gas stores far more energy; establish exclusion zones.
- Whip checks on air hoses; relief valves on receivers; LOTO includes bleeding stored pressure.
- Water jetting (very high pressure) requires trained operators, deadman controls and specific PPE.

> **Exam tip:** Pneumatic test vs hydrostatic test: hydrostatic is safer because liquids are nearly incompressible.

Silica BLUEPRINT

Respirable crystalline silica (quartz, cristobalite) from cutting, grinding, drilling, blasting and foundry work causes silicosis, lung cancer, COPD, kidney and autoimmune disease.

- OSHA 1910.1053 / 1926.1153: PEL 50 ug/m3 8-h TWA; action level 25 ug/m3. NIOSH REL 50 ug/m3; ACGIH TLV 25 ug/m3 (stricter, best practice).
- Construction: follow Table 1 control methods (water, LEV, respirators) or do exposure assessment; competent person and written exposure control plan required.
- Medical surveillance: general industry - exposure at/above AL for 30+ days/yr; construction - required respirator use 30+ days/yr; exams every 3 years.
- Prohibited: dry sweeping and compressed-air cleaning where alternatives exist.
- IARC Group 1 carcinogen; 'pneumoconiosis + autoimmune disease' points to silica.
- Controls: substitution (non-silica abrasives), wet methods, LEV/vacuum shrouds, enclosed cabs, respirators.

> **Exam tip:** Best practice limit = ACGIH TLV 25 ug/m3; enforceable limit = OSHA PEL 50 ug/m3.

Powder/spray applications (spray finishing, powder coating) BLUEPRINT

Spray painting and powder coating create flammable vapors, combustible dusts and toxic or sensitizing aerosols (isocyanates, Cr(VI) primers, solvents).

- Standards: OSHA 1910.107 (spray finishing) and 1910.94(c) (spray booth ventilation); NFPA 33.
- Booth interior and exhaust plenum are hazardous (classified) locations; no open flames or spark sources within 20 ft of spraying unless separated.
- Keep vapor below 25% of LFL; exhaust fans interlocked with spray equipment; clean filters and residue regularly.
- Electrostatic equipment and powder coating require bonding/grounding of parts and workers; powder coatings are combustible dusts.
- Health: isocyanate (2-part urethane) asthma - supplied-air respirators often required; hexavalent chromium mist in aircraft painting.
- Controls: high-transfer-efficiency (HVLP) guns, water-based coatings, enclosure, PPE (respirators, coveralls, gloves).

> **Exam tip:** 'Aircraft painting + chromate primer' = hexavalent chromium; 'auto-body clearcoat + asthma' = isocyanates.

Blasting (abrasive blasting) BLUEPRINT

Abrasive blasting cleans and prepares surfaces by propelling media at high pressure. It exposes workers to silica and toxic metals, noise, high-pressure injury and fire/explosion hazards.

- Respirable silica (silicosis, lung cancer) - substitute non-silica media (steel grit, garnet, crushed glass, slag - check metals content).
- Toxic metals from coatings: lead, cadmium, chromium, beryllium.
- Noise often exceeds 100 dBA; high-pressure stream can inject or lacerate (deadman controls on nozzles).
- OSHA 1910.94(a): blast cleaning enclosures with exhaust; abrasive-blasting respirators (Type CE supplied-air) for operators inside enclosures or where silica/toxic dust exists.
- Dust-laden confined spaces raise fire/explosion risk; ground nozzles and hoses to prevent static.
- Explosive blasting (demolition, quarrying) is separate: 1926 Subpart U and 1910.109 - blaster in charge, misfire procedures, radio-frequency precautions.

> **Exam tip:** First control for silica in blasting is substitution of the abrasive, not a better respirator.

Molten metals BLUEPRINT

Foundry, smelting, casting and welding operations combine thermal, chemical and explosion hazards.

- Burns from splashes and spills; aluminized/FR clothing, face shields, spats and leggings.
- Steam explosions: molten metal contacting water or moisture (wet scrap, tools, molds, floors) expands violently - preheat and dry everything that contacts the melt.
- Metal fumes: zinc/copper - metal fume fever; manganese - neurotoxic (manganism); Cr(VI) and others - respiratory/systemic toxicity.
- Infrared radiation damages eyes (cataracts) and skin; use IR-rated eyewear and shielding.
- Radiant heat causes heat stress; apply WBGT monitoring, work-rest cycles and cooling.
- Additional: silica in foundry sand, CO from cupolas, noise, crane and ladle handling.

> **Exam tip:** Most catastrophic molten-metal incidents involve water/moisture - the key preventive control is keeping everything dry.

Hot work BLUEPRINT

Hot work is any welding, cutting, brazing, grinding or other spark- or flame-producing work. It requires a permit system, fire prevention measures and control of welding fumes and radiation.

- Standards: OSHA 1910.252 (and 1926 Subpart J), NFPA 51B. Prefer moving the work to a designated safe area.
- Remove combustibles within 35 ft or protect them with covers/shields; cover floor and wall openings.
- Fire watch during work and at least 30 min after (OSHA); NFPA 51B (2019 and later) requires 60 min followed by additional monitoring.
- Permit authorizing individual inspects the area and issues a written permit; test atmosphere for flammables in confined spaces and around tanks (do not weld on containers that held flammables until cleaned/purged/inerted).
- Health: welding fumes (Mn, Cr(VI) from stainless, Zn from galvanized), ozone, NOx, UV (welder's flash) - use LEV and correct filter shade numbers.
- Cylinders, hoses and flashback arrestors inspected; separate oxygen and fuel gas in storage.

> **Exam tip:** Where the work can be relocated to a shop, that beats any permit or fire-watch answer.

Lasers BLUEPRINT

Lasers emit coherent, highly collimated optical radiation that can damage the eye (retina for visible/near-IR, cornea for UV/far-IR) and skin. ANSI Z136.1 classifies lasers by hazard.

- Classes: 1 and 1M safe (1M unless viewed with optics); 2 visible, protected by blink reflex (0.25 s); 2M hazardous with optics; 3R low risk; 3B direct and specular beam eye hazard; 4 eye, skin, fire and diffuse-reflection hazard.
- A Laser Safety Officer (LSO) is required for Class 3B and 4.
- Engineering controls first: beam enclosures, interlocks, key switches, beam stops, shutters; administrative: SOPs, training, 'Laser On' illuminated signs, restricted access, nominal hazard zone (NHZ).
- Eyewear must match the laser's wavelength and have adequate optical density: OD = log10(H0/MPE). Example: exposure 10 W/cm2, MPE 0.001 W/cm2 -> OD = log10(10,000) = 4.
- Class 4 lasers also create fire and laser-generated air contaminants (plume) - need FR materials and smoke evacuation.

> **Exam tip:** The blink reflex protects only against Class 2 visible beams - never assume it protects against 3B or 4.

Read more: Yates, Safety Professional's Reference and Study Guide (4th ed.): Ch. 6 Particulates and Gases, p. 228; Ch. 8 Industrial Hygiene Air Sampling, p. 273; Ch. 9 Ventilation, p. 294; Ch. 10 Noise, p. 314; Ch. 11 Biological Hazards, p. 332; Ch. 13 Thermal Stressors, p. 367; Ch. 14 PPE, p. 385; Ch. 28 Radiation Safety, p. 636

7. Ten CSP-style practice questions

Answer all ten on paper first, then check the key. Score: _____ / 10 Date: _____ Retest: _____ / 10

6.1-Q1. A worker's acetone exposure over an 8-hour shift was: 2 h at 300 ppm, 3 h at 200 ppm, and 3 h at 100 ppm. What is the 8-hour TWA?

A. 187.5 ppm
B. 202 ppm
C. 163 ppm
D. 221 ppm

6.1-Q2. A worker is exposed to 95 dBA for 2 hours, 90 dBA for 4 hours, and 85 dBA for 2 hours. What is the worker's OSHA noise dose?

A. 81.25%
B. 100%
C. 92%
D. 112.5%

6.1-Q3. A noise dosimeter reports a daily dose of 200% (OSHA criteria). What is the equivalent 8-hour TWA? (TWA = 16.61 log10(D/100) + 90)

A. 90 dBA
B. 93 dBA
C. 95 dBA
D. 100 dBA

6.1-Q4. A jackhammer produces 98 dB at 3 ft. Using the inverse square law (L2 = L1 - 20 log(d2/d1)), what is the approximate sound level at 12 ft?

A. 86 dB
B. 88 dB
C. 92 dB
D. 94 dB

6.1-Q5. A radiography crew performs weld inspections with a gamma source. A helper normally works 4 ft from the source but steps back to 12 ft during exposures. How does the radiation intensity at 12 ft compare with that at 4 ft?

A. 1/2 as much
B. 1/9 as much
C. 1/3 as much
D. The same amount

6.1-Q6. Outdoors in direct sun, a heat stress survey records: natural wet bulb 25 C, globe 40 C, dry bulb 30 C. What is the WBGT? (Outdoor: WBGT = 0.7 NWB + 0.2 GT + 0.1 DB)

A. 29.5 C
B. 28.5 C
C. 31.7 C
D. 25.0 C

6.1-Q7. Under OSHA's respirable crystalline silica standard, what are the PEL and action level (8-hour TWA)?

A. PEL 100 ug/m3; AL 50 ug/m3
B. PEL 50 ug/m3; AL 25 ug/m3
C. PEL 25 ug/m3; AL 10 ug/m3
D. PEL 10 mg/m3; AL 5 mg/m3

6.1-Q8. A metal finishing shop uses a nickel-containing polishing compound. Workers complain of sinus irritation during buffing. No prior exposure data exists and ventilation effectiveness is unknown. What is the MOST appropriate first step in the exposure assessment?

A. Conduct full-shift dosimetry on all workers in the department
B. Rely on SDS information to estimate exposures without sampling
C. Assume exposures are below limits because symptoms are mild
D. Perform task-based personal air sampling on workers performing the buffing activity

6.1-Q9. Ralph is assigned to sweep a dusty floor and asks for a filtering facepiece respirator (dust mask). Dust levels are well below regulatory limits. The supervisor refuses because Ralph has not been fit tested. Was the supervisor correct?

A. Yes, anyone wearing a tight-fitting respirator must be medically evaluated and fit tested prior to wearing one.
B. Yes, Ralph should not have asked for the respirator due to budget constraints.
C. No, fit testing is not required in this specific case.
D. No, the safety supervisor should have performed a fit test and issued the respirator.

6.1-Q10. An employee with facial hair that prevents a half-mask seal is assigned a powered air-purifying respirator (PAPR) with a loose-fitting hood. Which type of fit testing is required?

A. Fit testing is not required
B. Medical fit testing
C. Qualitative fit testing
D. Quantitative fit testing

Answer key and explanations

Q	Ans	Why
1	**A**	TWA = sum(C x T)/8 = (600 + 600 + 300)/8 = 187.5 ppm. Trap: averaging the three concentrations without time weighting gives 200 ppm.
2	**D**	Allowed times (OSHA 5-dB rule): 95 dBA = 4 h, 90 dBA = 8 h, 85 dBA = 16 h. D = 2/4 + 4/8 + 2/16 = 0.5 + 0.5 + 0.125 = 112.5%, which exceeds the PEL.
3	**C**	TWA = 16.61 x log10(200/100) + 90 = 16.61 x 0.301 + 90 = 95 dBA. With a 5-dB exchange rate, doubling the dose adds 5 dB; 93 dBA is the 3-dB (NIOSH) trap.
4	**A**	Inverse square law for a point source: L2 = L1 - 20 log(d2/d1) = 98 - 20 log(12/3) = 98 - 12 = 86 dB. Each doubling of distance drops about 6 dB.
5	**B**	Radiation intensity follows the inverse square law: I2/I1 = (d1/d2)^2 = (4/12)^2 = 1/9. Trap: 1/3 assumes intensity drops linearly with distance.
6	**B**	WBGT = 0.7(25) + 0.2(40) + 0.1(30) = 17.5 + 8 + 3 = 28.5 C. Trap: 29.5 C uses the indoor/no-solar-load formula (0.7 NWB + 0.3 GT).
7	**B**	OSHA 1910.1053/1926.1153 set the respirable crystalline silica PEL at 50 ug/m3 and the action level at 25 ug/m3 (8-h TWA). The old 100 ug/m3 value is the trap.
8	**D**	With symptoms, no data, and unknown ventilation, the first step is task-based personal breathing-zone sampling of the buffing workers. Department-wide dosimetry is unfocused; SDS data alone cannot quantify exposure.
9	**C**	Voluntary use of a filtering facepiece where exposures are below limits falls under 1910.134 Appendix D: give the user Appendix D information; fit testing and medical evaluation are not required.
10	**A**	Fit testing under 1910.134 applies to tight-fitting facepieces. A loose-fitting PAPR hood does not need a facial seal, so no fit test is required (medical evaluation is still required).

DOMAIN 6 · COMPETENCY 6.2

Public Health and Epidemiology

CSP11 blueprint: Understand principles of public health as applicable (i.e., fundamentals of epidemiology, infectious disease, risk factors, statistics to interpret data)

1. Definition

Epidemiology is the study of the distribution and determinants of health-related events in populations and the application of that study to control health problems. Public health principles help the safety professional recognize disease patterns, infectious disease risks and risk factors in the workforce.

2. Explanation in simple words

Epidemiology asks who gets sick, where, when and why. Comparing exposed and unexposed groups shows whether a workplace exposure raises the risk of disease.

3. Key points to remember

- Measures: incidence (new cases ÷ population at risk over time), prevalence (all existing cases ÷ population at a point/period), attack rate, mortality rate.
- Study designs: cohort (follows exposed vs unexposed forward; relative risk), case-control (starts with cases vs controls, looks back at exposure; odds ratio), cross-sectional (snapshot; prevalence), ecological, randomized controlled trial.
- Relative risk RR = incidence exposed ÷ incidence unexposed (RR > 1 = increased risk). Odds ratio OR = (a × d) ÷ (b × c). Attributable risk = Ie − Iu.
- Causation criteria (Bradford Hill): strength, consistency, specificity, temporality (exposure before disease), dose-response, plausibility, coherence, experiment, analogy. Correlation is not causation; watch for confounding and bias (e.g., healthy worker effect).
- Infectious disease chain of infection: agent → reservoir → portal of exit → mode of transmission (contact, droplet, airborne, vector, common vehicle) → portal of entry → susceptible host. Break the chain with vaccination, hygiene, ventilation, isolation, PPE.
- Occupational infectious risks: bloodborne pathogens (HBV, HCV, HIV), TB, Legionella, zoonoses (brucellosis from animals/milk, anthrax, histoplasmosis), pandemic influenza/COVID-19 (pandemic plans).
- Risk factors: modifiable (smoking, obesity, inactivity) vs non-modifiable (age, genetics); occupational vs non-occupational; Total Worker Health (NIOSH) integrates protection and health promotion.
- Screening tests: sensitivity (true positive rate), specificity (true negative rate).

4. Real work scenario

> A plant notices several cases of a rare respiratory disease. A cohort analysis shows 30 cases per 1,000 among workers exposed to a metalworking fluid versus 10 per 1,000 among unexposed workers - a relative risk of 3.0. The fluid system is found contaminated with bacteria; the system is cleaned, a biocide program and mist collectors are installed, and new cases stop.

5. CSP exam tips and traps

- Cohort → relative risk; case-control → odds ratio.
- Incidence = new cases; prevalence = existing cases.
- Temporality (exposure before disease) is the one causation criterion that is required.

6. Every blueprint point (4 blueprint items + 3 more)

Epidemiology BLUEPRINT

Epidemiology studies the distribution (who, where, when) and determinants (why) of disease in populations and applies that knowledge to control health problems. In occupational health it links exposures to outcomes such as cancer or hearing loss.

- Functions: prevention of disease, measuring frequency (incidence and prevalence) and describing distribution by person, place and time.
- Descriptive studies generate hypotheses; analytic studies (cohort, case-control) test them.
- Cohort study: start with exposed and unexposed groups and follow for disease (prospective or retrospective/historical); gives incidence and relative risk.
- Case-control study: start with cases and controls and look back at exposures; good for rare diseases; gives odds ratio.
- Cross-sectional: snapshot of prevalence; ecological: group-level data (ecological fallacy risk).
- Bradford Hill causation criteria: strength, consistency, specificity, temporality (must have), dose-response, plausibility, coherence, experiment, analogy.

> **Exam tip:** Rare disease, limited budget, look back at exposure = case-control. Following workers forward = cohort.

Infectious disease BLUEPRINT

Infectious disease spreads through the chain of infection. Breaking any link - at the agent, reservoir, transmission route or host - prevents spread.

- Chain of infection: infectious agent, reservoir, portal of exit, mode of transmission, portal of entry, susceptible host.
- Modes: contact (direct/indirect), droplet (large particles, about 3-6 ft), airborne (droplet nuclei - TB, measles), vehicle (food/water - Hep A), vector (ticks, fleas, mosquitoes).
- Standard precautions apply to all patients; transmission-based precautions add contact, droplet or airborne measures (airborne infection isolation rooms: negative pressure, 12+ ACH, N95 or higher).
- Host defenses: vaccination (HBV, Hep A, rabies pre-exposure, Tdap), post-exposure prophylaxis (HBIG, rabies PEP, TIG).
- Outbreak response: case definition, confirm diagnosis, count cases, describe by person/place/time, hypothesize, test, control, communicate.
- Pandemic planning ties to business continuity: absenteeism planning, ventilation, PPE stockpiles, remote work.

> **Exam tip:** TB transmission is airborne (droplet nuclei), not surface contact - controls are ventilation, isolation and respirators.

Risk factors BLUEPRINT

A risk factor is any attribute or exposure that increases the probability of disease or injury. Public-health thinking considers agent, host and environment together (the epidemiologic triangle).

- Host factors: age, sex, genetics, immune status, pregnancy, pre-existing disease (diabetes, asthma), lifestyle (smoking, alcohol, obesity).
- Agent factors: toxicity, infectivity, dose, duration; environmental factors: temperature, ventilation, crowding, sanitation, work organization.
- Modifiable (smoking, PPE use, exposure level) vs non-modifiable (age, genetics).
- Confounder: a factor linked to both exposure and outcome (e.g., smoking in a lung-cancer study of miners) - control by matching, stratification or adjustment.
- Effect modification: smoking multiplies asbestos lung-cancer risk (synergy); healthy worker effect makes worker mortality look lower than the general population.

> **Exam tip:** When a study ignores smoking in a lung disease outcome, the weakness is confounding.

Statistics (epidemiologic measures) BLUEPRINT

Public-health statistics express how often disease occurs and how strongly it is associated with an exposure.

- Incidence rate = new cases / population at risk over a time period; prevalence = existing cases / total population at a point in time. Prevalence roughly = incidence x duration.
- Relative risk RR = incidence exposed / incidence unexposed. Example: 30/1,000 vs 10/1,000 -> RR = 3. RR = 1 no association; > 1 increased risk.
- Attributable risk = Ie - Iu = 0.03 - 0.01 = 20 per 1,000 exposed.
- Odds ratio (case-control) = ad/bc. Example: exposed cases 40, unexposed cases 60, exposed controls 20, unexposed controls 80 -> (40 x 80)/(20 x 60) = 2.67.
- SMR = observed deaths / expected deaths x 100. 15 observed vs 10 expected = 150 (50% excess).
- Screening: sensitivity = TP/(TP + FN); specificity = TN/(TN + FP). 95% CI = mean +/- 1.96 x SD/sqrt(n); if a CI for RR/OR includes 1 the result is not statistically significant (p >= 0.05).

> **Exam tip:** Odds ratio comes from case-control studies; relative risk comes from cohort studies.

Zoonotic and occupational bacterial diseases

Several bacterial diseases are acquired from animals, animal products, soil or water in specific occupations. Exam questions pair an occupation and a signature symptom.

- Anthrax (Bacillus anthracis): wool, hides, livestock; cutaneous = black eschar (95% of cases); inhalation = 'woolsorter's disease' (most lethal); not person-to-person; CDC Category A agent.
- Leptospirosis (Leptospira): animal urine in water/soil - rice-field, sewage, flood workers; severe form (Weil's disease) = jaundice + kidney failure.
- Brucellosis (Brucella): meat packing, dairy, unpasteurized milk; 'undulant fever'; no human vaccine - pasteurization is key.
- Q fever (Coxiella burnetii): inhalation of aerosols from animal birth products; pneumonia + hepatitis; chronic = endocarditis; BSL-3.
- Psittacosis (Chlamydia psittaci): bird droppings/feather dust - pet shops, poultry; atypical pneumonia.
- Tularemia (rabbit/deer-fly fever), plague (Yersinia pestis; rodent-flea-human; pneumonic form spreads person-to-person), tetanus (Clostridium tetani; lockjaw; Td booster every 10 years), cat scratch fever (Bartonella), RMSF (Rickettsia; tick-borne; rash starts at wrists/ankles).

> **Exam tip:** Category A bioterrorism agents in the guide: anthrax, plague, tularemia.

Viral and fungal occupational diseases

Viral and fungal infections in the workplace follow distinct transmission routes that determine the control.

- Hepatitis A: fecal-oral (food handlers, daycare, sewage); self-limiting, no chronic infection; vaccine available.
- Hepatitis B: blood/body fluids (needlesticks, splashes); can become chronic (cirrhosis, liver cancer); 3-dose vaccine; HBIG + vaccine post-exposure. HBV survives on surfaces up to 7 days.
- Hepatitis C and HIV: bloodborne; no vaccine; needlestick transmission risk roughly HBV 6-30% (unvaccinated) > HCV about 1.8% > HIV about 0.3%.
- Rabies: bite/saliva of dogs, bats, raccoons; hydrophobia; nearly always fatal once symptoms appear - immediate wound washing and PEP.
- Orf (sheep/goats - painful hand nodules); Histoplasmosis (bird/bat droppings - demolition, bridge work); Aspergillosis (construction dust, compost - HEPA in hospitals during renovation); Candidiasis (opportunistic, moist PPE, ICU).
- Legionella: aerosols from cooling towers, hot water systems; prevented by water management programs (ASHRAE 188).

> **Exam tip:** Hep A = fecal-oral and acute only; Hep B = bloodborne and can be chronic.

Total Worker Health

Total Worker Health (NIOSH) integrates protection from work-related hazards with promotion of injury and illness prevention and worker well-being.

- Hazard prevention comes first; health promotion supplements, never replaces, controls.
- Addresses work organization: scheduling, stress, fatigue, workload, supervision.
- Integrates programs such as fitness, nutrition, smoking cessation, mental health and financial well-being.
- Participatory approach - workers help design programs; evidence-based evaluation.
- Reference: NIOSH Fundamentals of Total Worker Health Approaches.

> **Exam tip:** Question stresses 'integration of OSH with health promotion' = Total Worker Health.

Read more: Yates, Safety Professional's Reference and Study Guide (4th ed.): Ch. 11 Biological Hazards, p. 332; Ch. 15 Statistics for the Safety Professional, p. 408

7. Ten CSP-style practice questions

Answer all ten on paper first, then check the key. Score: _____ / 10 Date: _____ Retest: _____ / 10

6.2-Q1. Which statement correctly distinguishes incidence from prevalence?

A. Incidence counts all existing cases; prevalence counts new cases
B. Incidence counts NEW cases over a period; prevalence counts ALL existing cases at a point or period in time
C. They are the same
D. Prevalence applies only to infectious disease

6.2-Q2. In a cohort study, 30 of 1,000 exposed workers and 10 of 1,000 unexposed workers develop a disease. What is the relative risk?

A. 0.33
B. 2.0
C. 3.0
D. 20

6.2-Q3. In a case-control study, 40 cases were exposed and 60 were not; 20 controls were exposed and 80 were not. What is the odds ratio?

A. 0.38
B. 1.5
C. 2.0
D. 2.67

6.2-Q4. A cohort study reports a relative risk of 1.4 for solvent exposure and liver disease, with a 95% confidence interval of 0.8 to 2.4. What is the BEST interpretation?

A. The association is not statistically significant at the 0.05 level because the interval includes 1.0
B. Solvent exposure is proven to cause a 40% increase in liver disease
C. Solvent exposure is protective against liver disease
D. Solvent exposure reduces liver disease risk by 40%

6.2-Q5. Foundry workers show a higher lung cancer rate than office workers, but the foundry workers also smoke at a much higher rate. In this study, smoking is BEST described as a:

A. Outcome variable
B. Primary exposure
C. Random sampling error
D. Confounding variable

6.2-Q6. At the start of a year, 25 of a plant's 400 workers already have occupational dermatitis. During the year, 15 new cases occur. What is the annual incidence rate per 100 workers at risk?

A. 3.75
B. 4.0
C. 10.0
D. 6.25

6.2-Q7. What is the primary route of transmission for Tuberculosis (TB)?

A. Airborne droplets
B. Skin contact
C. Blood transfusion
D. Contaminated water

6.2-Q8. Which indoor contaminant is primarily responsible for Legionnaires' disease outbreaks?

A. Asbestos
B. Molds
C. Legionella bacteria in cooling towers and water systems
D. Ozone

6.2-Q9. Brucellosis is commonly known as:

A. Undulant fever
B. Blackwater fever
C. Weil's disease
D. Rabbit fever

6.2-Q10. Which occupation is at highest risk of Leptospirosis infection?

A. Electricians
B. Rice-field workers
C. Office staff
D. Painters

Answer key and explanations

Q	Ans	Why
1	B	Incidence = new cases over a period among those at risk (measures risk); prevalence = all existing cases at a point or period (measures burden).
2	C	RR = incidence in exposed / incidence in unexposed = (30/1,000)/(10/1,000) = 3.0. Exposed workers are three times as likely to develop the disease.
3	D	OR = (a x d)/(b x c) = (40 x 80)/(60 x 20) = 3,200/1,200 = 2.67. Case-control studies yield odds ratios; cohort studies yield relative risks.
4	A	A 95% CI for a ratio that includes 1.0 (no effect) means the result is not statistically significant. Trap: reading the point estimate of 1.4 as proof of causation.
5	D	A confounder is linked to both the exposure group and the outcome and can distort the apparent association. It must be controlled by stratification, matching or adjustment.
6	B	Incidence = new cases / population at risk = 15/(400 - 25) = 15/375 = 4.0 per 100. Trap: 3.75 wrongly includes existing cases in the denominator; 10.0 counts all 40 cases (prevalence).
7	A	Tuberculosis (Mycobacterium tuberculosis) spreads by inhaling airborne droplet nuclei expelled when an infected person coughs or sneezes, so controls focus on airborne isolation and respirators.
8	C	Legionella pneumophila grows in poorly maintained warm-water systems such as cooling towers and is spread by inhaled aerosols. Water management programs are the main control.

9	**A**	Brucellosis, spread through contact with infected animal tissues and fluids (meat packers, dairy and abattoir workers), is called undulant fever because the fever comes and goes. Weil's disease is leptospirosis and rabbit fever is tularemia.
10	**B**	Leptospira spreads through water and soil contaminated with animal urine, so rice-field, sewer and farm workers are at highest risk.

DOMAIN 6 · COMPETENCY 6.3

Toxicology

CSP11 blueprint: Apply toxicology principles to create exposure control plans and develop risk mitigation plans (e.g., using sampling equipment, symptoms of an exposure, LD50, LC50, mutagens, carcinogens, teratogens, ototoxins)

1. Definition

Toxicology is the study of the adverse effects of chemical, physical and biological agents on living organisms. The dose makes the poison (Paracelsus): the effect depends on dose, route of entry, duration and frequency of exposure and individual susceptibility.

2. Explanation in simple words

Every chemical can be harmful at a high enough dose. Know how it gets into the body, what it attacks, how much it takes, and the signs that someone has been exposed - then plan controls to keep the dose low.

3. Key points to remember

- Routes of entry: inhalation (most important occupationally), skin absorption, ingestion, injection.
- Dose-response: NOAEL/LOAEL, threshold; LD50 (lethal dose for 50% of test animals, mg/kg, oral/dermal) and LC50 (lethal concentration in air, ppm or mg/m^3, over a stated time); lower LD50/LC50 = more toxic. Most OELs are derived from animal toxicology studies.
- Acute (short-term, high exposure, immediate effects) vs chronic (long-term, low exposure, delayed effects); local vs systemic effects.
- Interactions: additive (1 + 1 = 2), synergistic (1 + 1 = 5, e.g., asbestos + smoking), potentiation (0 + 1 = 3, non-toxic agent increases another's effect), antagonism (1 + 1 < 2).
- Target effects: carcinogens (IARC Group 1 = carcinogenic to humans, 2A probably, 2B possibly, 3 not classifiable), mutagens (change DNA), teratogens (birth defects), reproductive toxicants, sensitizers (allergic reaction after repeated exposure - isocyanates, epoxies), neurotoxins, hepatotoxins (liver), nephrotoxins (kidney), ototoxins (hearing - solvents like styrene, toluene; lead, CO), asphyxiants (simple - displace O_2, e.g., N_2, CO_2; chemical - interfere with O_2 use, e.g., CO, H_2S, cyanide), irritants.
- Particulate lung diseases: silicosis (silica), asbestosis, siderosis (iron oxide), byssinosis (cotton dust), black lung/anthracosis (coal), berylliosis; fibrosis = scarring of lung tissue.
- Symptoms of exposure: solvents - dizziness, headache, narcosis, defatting dermatitis; CO - headache, cherry-red skin; H_2S - rotten egg odor then olfactory fatigue; metal fume fever (zinc) - flu-like.
- Exposure control plan: identify agents and SEGs, sampling strategy and equipment, compare with OELs, controls by hierarchy, medical surveillance, training, reassessment.

4. Real work scenario

> A furniture plant introduces a new isocyanate-based coating. The safety professional reviews the SDS (respiratory sensitizer, carcinogen category 2), sets up personal sampling with the correct coated filters, installs a ventilated spray booth, adds supplied-air respirators for spraying, starts medical surveillance for sensitization, and trains workers to report wheezing early.

5. CSP exam tips and traps

- Lower LD50/LC50 = MORE toxic.
- Synergism vs potentiation is a common definition trap.
- Teratogen = birth defects; mutagen = DNA changes; carcinogen = cancer; ototoxin = hearing.

- IARC Group 1 = known human carcinogen.

6. Every blueprint point (9 blueprint items + 6 more)

Routes of entry and dose

Toxic substances enter by inhalation, ingestion, skin/eye absorption or injection. The effect depends on the dose that reaches the target organ.

- Inhalation is the most common and significant occupational route (gases, vapors, dusts, fumes, mists).
- Ingestion from poor hygiene (lead dust on hands/food); absorption through intact skin for lipid-soluble chemicals (solvents, pesticides, phenol - 'Skin' notation); injection by needlesticks or high-pressure fluid.
- Dose = concentration x time (Haber's rule); Paracelsus: 'the dose makes the poison'.
- Inhaled dose = C x breathing volume x time (x retention). Light work about 10 m3 per 8-h shift: 2 mg/m3 x 10 m3 = 20 mg inhaled.
- Resting minute ventilation about 5 L/min (guide: 4.3-5.7 L/min) and tidal volume about 500 mL; work raises both.
- Acute vs chronic exposure and effects; local (at contact site) vs systemic (distributed to target organs).

> **Exam tip:** 'Most significant occupational route of entry' = inhalation.

Dose-response (NOAEL, LOAEL, threshold)

Dose-response describes how the magnitude of effect changes with dose. Thresholds and no-effect levels are used to set exposure limits.

- Threshold: dose below which no effect occurs (assumed for most non-carcinogens; not for genotoxic carcinogens).
- NOAEL = highest dose with no observed adverse effect; LOAEL = lowest dose with an observed adverse effect (always above NOAEL).
- Limits are derived from NOAEL/LOAEL divided by uncertainty (safety) factors, e.g., 10 for animal-to-human, 10 for human variability.
- Example (guide): benzene animal NOAEL 10 ppm and LOAEL 25 ppm, but OSHA PEL 1 ppm and NIOSH REL 0.1 ppm after safety factors and cancer data.
- Potency: a steeper curve or lower ED50 means more potent; TD50/ED50 relate toxic to effective dose (therapeutic index).

> **Exam tip:** NOAEL is the regulatory benchmark; LOAEL is used when no NOAEL could be determined.

LD50 BLUEPRINT

LD50 is the dose that kills 50% of a test-animal population by a non-inhalation route (oral, dermal, injection), expressed in mg/kg of body weight. Lower LD50 means higher acute toxicity.

- Units mg/kg body weight; always state species and route (e.g., oral rat).
- GHS acute oral categories: Cat 1 <= 5, Cat 2 > 5-50, Cat 3 > 50-300, Cat 4 > 300-2,000, Cat 5 > 2,000-5,000 mg/kg. Dermal: 50 / 200 / 1,000 / 2,000 / 5,000.
- Examples (guide): sodium cyanide oral LD50 about 5.7 mg/kg (Cat 1/2 border); caffeine about 190 mg/kg (Cat 3); table salt about 3,000 mg/kg (Cat 5 range).
- Signal words: Cat 1-2 'Danger', Cat 3-4 'Warning'; skull-and-crossbones for Cat 1-3, exclamation mark for Cat 4. OSHA HazCom does not adopt Cat 5.
- Scale-up: lethal dose for a 70-kg person at LD50 50 mg/kg = 3,500 mg (3.5 g) - animal data extrapolate only approximately.

> **Exam tip:** Keywords oral, dermal, injection, mg/kg = LD50. A chemical with LD50 4 mg/kg is MORE toxic than one with 400 mg/kg.

LC50 BLUEPRINT

LC50 is the airborne concentration that kills 50% of test animals over a stated exposure time (usually 4 h), used for gases, vapors, dusts, mists and fumes.

- Units: ppm (gases/vapors) or mg/L (vapors, dusts/mists); exposure duration must be stated.
- GHS inhalation (gases, ppmV, 4 h): Cat 1 <= 100, Cat 2 <= 500, Cat 3 <= 2,500, Cat 4 <= 20,000.
- Lower LC50 = more toxic. Example (guide): solvent with LC50 500 ppm (4-h rat) and LD50 200 mg/kg oral.
- LC50 is an acute measure; it says nothing about chronic, carcinogenic or reproductive effects.
- Related: LCLo / LDLo = lowest reported lethal concentration/dose; TCLo = lowest toxic concentration.

> **Exam tip:** Keyword 'inhalation' or 'airborne' = LC50, not LD50.

IDLH and acute exposure guidance

IDLH values (NIOSH) mark atmospheres that pose an immediate threat to life, irreversible health effects or impaired escape. They drive respirator selection and emergency planning.

- IDLH basis: escape within 30 minutes without escape-impairing or irreversible effects (e.g., CO 1,200 ppm, benzene 500 ppm, H2S 50 ppm).
- Oxygen-deficient atmospheres (< 19.5%) are treated as IDLH under 1910.134 unless the employer shows otherwise.
- IDLH entry requires pressure-demand full-facepiece SCBA (APF 10,000) or supplied-air with auxiliary escape SCBA, plus standby persons outside.
- Emergency planning also uses ERPGs (AIHA) and AEGLs (EPA) for community exposure tiers.
- H2S causes olfactory fatigue at about 100 ppm - smell is not a warning.

> **Exam tip:** Any unknown concentration or IDLH atmosphere = SCBA (pressure-demand), never an air-purifying respirator.

Exposure control BLUEPRINT

Exposure control applies the hierarchy of controls to chemical, physical and biological agents; respirators are the last line of defense and require a full program.

- Order: elimination, substitution, engineering (enclosure, isolation, LEV, wet methods), administrative (rotation, scheduling, hygiene, training), PPE.
- Respiratory protection program (1910.134): written program, program administrator, medical evaluation before fit testing, annual fit test, training, cartridge change schedule, voluntary-use provisions.
- APFs: half-mask APR 10; full-facepiece APR 50; loose-fitting PAPR 25; full-face PAPR 1,000; supplied-air pressure-demand full face 1,000; SCBA pressure-demand 10,000.
- Maximum use concentration MUC = APF x PEL. Example: half-mask APF 10, PEL 50 ppm -> MUC 500 ppm (but never above IDLH).
- Required protection factor = measured concentration / PEL: 300 ppm with PEL 25 ppm needs APF >= 12 -> full-face APR (50).
- Personal hygiene: no eating/drinking in work areas, change rooms, showers (lead, arsenic, cadmium).

> **Exam tip:** Hazard factor = C/PEL; select the respirator whose APF is at least that number.

Sampling (biological monitoring and toxicological exposure assessment) BLUEPRINT

Toxicological sampling measures how much of an agent the body has absorbed from all routes. Biological monitoring complements air sampling, especially when skin absorption or ingestion matters.

- Biological Exposure Indices (BEIs, ACGIH) = reference values in blood, urine or exhaled air corresponding to exposure at the TLV.
- Examples: blood lead (OSHA medical removal at 60 ug/dL or average of last 3 >= 50 ug/dL in general industry; 50 ug/dL in construction), urinary cadmium and beta-2 microglobulin, urinary metabolites (hippuric acid for toluene, phenol/S-PMA for benzene), carboxyhemoglobin for CO, cholinesterase for organophosphates.
- Sample timing matters: end of shift, end of work week, or prior to shift depending on the agent's half-life.
- Medical surveillance (pre-placement, periodic, exit) is required by substance standards (lead, cadmium, benzene, silica, asbestos, beryllium).
- Exposure and medical records retained duration of employment + 30 years (1910.1020); employees have access within 15 working days.

> **Exam tip:** Air sampling misses skin absorption and ingestion - biological monitoring captures total dose.

Exposure symptoms (signature effects of common agents) BLUEPRINT

Many exam questions describe a worker's symptoms and occupation and ask for the likely agent. Learn the signature effects.

- Benzene - bone marrow suppression, aplastic anemia, leukemia (IARC Group 1). Lead - anemia, wrist/foot drop, slowed reflexes, kidney damage, hypertension, reproductive harm.
- Arsenic - skin lesions (hyperpigmentation, keratosis) + peripheral neuropathy; skin/lung/bladder cancer. Mercury - tremor, erethism, Minamata disease (methylmercury).
- Cadmium - acute pulmonary edema; chronic kidney damage (proteinuria) and osteomalacia (Itai-itai). Cr(VI) - nasal septum perforation ('chrome holes'), skin ulcers, lung cancer.
- Manganese - Parkinson-like manganism in welders (mask face, micrographia). Zinc/copper - metal fume fever (flu-like, resolves 24-48 h). Cobalt (carbide tool grinding) - occupational asthma.
- Formaldehyde - eye/airway irritation, nasopharyngeal cancer (mortuaries, labs). Antimony - GI effects (diarrhea, vomiting). Aluminum - pulmonary fibrosis.
- CO - headache, cherry-red skin, hypoxia; H2S - rotten eggs then olfactory paralysis; cyanide - cellular asphyxiant; n-hexane - peripheral neuropathy; methanol - blindness.

> **Exam tip:** Read the occupation clue: 'welding stainless' = Cr(VI) or Mn; 'battery plant' = lead or cadmium; 'galvanized' = zinc.

Pneumoconioses and occupational lung disease

Pneumoconioses are lung diseases caused by inhaled mineral or organic dust. Some are fibrotic and disabling; others are benign.

- Silicosis - crystalline silica (nodular fibrosis, autoimmune link, lung cancer). Asbestosis - asbestos (also mesothelioma and lung cancer; synergistic with smoking).
- Berylliosis (chronic beryllium disease) - granulomas that mimic sarcoidosis; sensitization test (BeLPT); P100 filtration; aerospace, electronics, ceramics, nuclear.
- Coal workers' pneumoconiosis ('black lung'); siderosis - iron oxide (welders), benign/non-fibrotic.
- Byssinosis ('brown lung') - cotton, flax, hemp dust; chest tightness on the first day back after a break ('Monday fever').
- Hypersensitivity pneumonitis (farmer's lung - moldy hay); occupational asthma from isocyanates, cobalt, flour, latex.

> **Exam tip:** 'Cotton dust' = byssinosis; 'iron oxide, benign' = siderosis; 'granulomas, aerospace' = berylliosis.

Target organ toxicity (STOT)

Toxicants often act on a specific organ. GHS calls this specific target organ toxicity (single or repeated exposure).

- Nephrotoxins (kidney): lead, cadmium (proximal tubule, proteinuria), mercury, halogens, ethylene glycol (oxalate crystals), high-dose NSAIDs.
- Hepatotoxins (liver): carbon tetrachloride (fatty liver, centrilobular necrosis), chloroform, vinyl chloride (angiosarcoma), acetaminophen overdose (NAPQI), ethanol, aflatoxin (Group 1 liver carcinogen).
- Neurotoxins: lead (cognition, neuropathy), methylmercury (Minamata disease), arsenic (nerve conduction), n-hexane, organophosphates (cholinesterase inhibition), manganese.
- Hematotoxins: benzene (marrow), CO (carboxyhemoglobin), nitrites/aniline (methemoglobinemia); pulmonary toxins: silica, asbestos, NO2, phosgene (delayed edema).
- Sensitizers (skin/respiratory): isocyanates, nickel, chromium, epoxy, latex - once sensitized, very low exposure triggers reaction.

> **Exam tip:** Ethylene glycol = oxalate crystals; CCl4 = liver; cadmium = kidney - classic exam pairs.

Chemical interactions (additive, synergistic, potentiation, antagonism)

When two or more chemicals are present, their combined effect may differ from each acting alone.

- Additive: 2 + 2 = 4 - e.g., toluene + xylene CNS depression (basis of the mixture formula).
- Synergistic: 2 + 2 > 4 - smoking + asbestos multiplies lung-cancer risk (roughly 50-90x combined vs about 5x asbestos alone and 10x smoking alone).
- Potentiation: 0 + 2 > 2 - isopropanol (non-hepatotoxic) increases CCl4 liver toxicity.
- Antagonism: 4 + 4 < 8 or 3 + (-2) = 1 - chelating agents binding heavy metals, antidotes.
- Ototoxic interaction: solvents (toluene, styrene) + noise cause more hearing loss than either alone.

> **Exam tip:** 'One chemical has no effect alone but increases the other's toxicity' = potentiation, not synergism.

Mutagens BLUEPRINT

Mutagens damage DNA and cause heritable genetic changes. Germ-cell mutations can pass to offspring; somatic mutations can lead to cancer.

- GHS germ cell mutagenicity: Category 1A (known human evidence), 1B (presumed), 2 (suspected).
- Examples: benzene (bone marrow, leukemia), ethylene oxide (sterilizers), vinyl chloride (liver angiosarcoma), ionizing radiation.
- Tests: Ames test (bacterial reverse mutation) screens for mutagenicity; many mutagens are also carcinogens.
- Control as non-threshold agents: minimize exposure (ALARA), closed systems, substitution.

> **Exam tip:** Keywords 'DNA damage' and 'heritable' = mutagen; 'fetus during pregnancy only' = teratogen.

Carcinogens BLUEPRINT

Carcinogens cause or promote cancer. Several bodies classify them by strength of evidence; only OSHA's list is legally enforceable.

- IARC: Group 1 carcinogenic (asbestos, benzene, silica, arsenic, formaldehyde, diesel exhaust since 2012), 2A probably (glyphosate, night-shift work), 2B possibly (RF-EMF, gasoline engine exhaust, chloroform), 3 not classifiable (caffeine). The guide lists Group 4 'probably not' (caprolactam) and diesel as 2A; IARC removed Group 4 in 2019 and diesel exhaust has been Group 1 since 2012.
- ACGIH: A1 confirmed human, A2 suspected human, A3 confirmed animal (unknown human relevance), A4 not classifiable, A5 not suspected.
- NTP Report on Carcinogens: 'known to be' vs 'reasonably anticipated to be' human carcinogens.
- OSHA regulates specific carcinogens with substance standards (asbestos, benzene, formaldehyde, Cr(VI), vinyl chloride, coke oven emissions, 13 carcinogens standard 1910.1003).
- GHS carcinogenicity Category 1A/1B (Danger, health hazard pictogram) and 2 (Warning).
- Latency is long (10-40 years for asbestos-related cancers); most are treated as non-threshold.

> **Exam tip:** Group 1 / A1 / 'known' = confirmed in humans; 2A = probable; 2B = possible. Enforceable = OSHA.

Teratogens BLUEPRINT

Teratogens harm the developing embryo or fetus during pregnancy, causing birth defects; unlike mutagens, their effects are not passed to later generations.

- Examples: thalidomide (limb malformations), ethanol (fetal alcohol syndrome), methylmercury (fetal brain), lead, ionizing radiation, some anticancer drugs and anesthetic gases.
- Most sensitive period: organogenesis, roughly weeks 3-8 of gestation - often before pregnancy is known.
- Reproductive toxicants also include agents affecting fertility (lead, DBCP, ethylene glycol ethers).
- Program approach: hazard communication, exposure minimization for all workers (not exclusion of women), NRC declared-pregnant-worker limit 0.5 rem over the pregnancy.

> **Exam tip:** Fetal protection policies that bar women from jobs are illegal (UAW v. Johnson Controls, 1991) - control the exposure instead.

Ototoxins BLUEPRINT

Ototoxic chemicals damage the cochlea or auditory nerve and can cause hearing loss on their own or worsen noise-induced loss.

- Solvents: toluene, styrene, xylene, ethylbenzene, n-hexane, trichloroethylene.
- Asphyxiants: carbon monoxide, hydrogen cyanide; metals: lead, mercury, organotins.
- Combined solvent + noise exposure produces greater-than-additive hearing loss.
- OSHA/NIOSH (2018 SHIB) recommend audiometric testing for workers exposed to ototoxicants even below 85 dBA; ACGIH notes ototoxic agents in the noise TLV.
- Controls: reduce both noise and chemical exposure; hearing protectors do not prevent ototoxic damage from inhaled/absorbed chemicals.

> **Exam tip:** Painter exposed to toluene + noise with unexpected hearing loss = ototoxic synergy.

Read more: Yates, Safety Professional's Reference and Study Guide (4th ed.): Ch. 7 Toxicology, p. 251; Ch. 6 Particulates and Gases, p. 228

7. Ten CSP-style practice questions

Answer all ten on paper first, then check the key. Score: _____ / 10 Date: _____ Retest: _____ / 10

6.3-Q1. Chemical A has a 4-hour rat inhalation LC50 of 50 ppm; Chemical B has an LC50 of 5,000 ppm. Which is correct?

A. B is more acutely toxic because its number is larger
B. A is more acutely toxic by inhalation
C. LC50 measures carcinogenic potency
D. Neither can be compared without an LD50

6.3-Q2. A chemical has an oral LD50 of 4 mg/kg. What is its GHS acute toxicity category?

A. Category 1
B. Category 2
C. Category 3
D. Category 4

6.3-Q3. Isopropyl alcohol, which is not itself hepatotoxic, greatly increases the liver toxicity of carbon tetrachloride. What is this interaction called?

A. Additive
B. Synergistic
C. Antagonistic
D. Potentiation

6.3-Q4. Smoking combined with asbestos exposure increases lung cancer risk far beyond the sum of the individual risks. What interaction is this?

A. Additive
B. Antagonistic
C. Synergistic
D. Potentiation

6.3-Q5. A worker is exposed to acetone at 300 ppm (TLV 500 ppm) and MEK at 100 ppm (TLV 200 ppm). Both affect the same target organs (additive). Is the mixture limit exceeded?

A. No - each is below its own TLV
B. Yes - the sum of ratios is 1.1
C. No - the sum of ratios is 0.8
D. Yes - the sum of ratios is 2.0

6.3-Q6. What does a Group 1 classification by the International Agency for Research on Cancer (IARC) indicate about a substance?

A. The substance is not classifiable as to its carcinogenicity to humans.
B. The substance is probably not carcinogenic to humans.
C. The substance is possibly carcinogenic to humans.
D. The substance is carcinogenic to humans.

6.3-Q7. A welder cutting galvanized steel develops fever, chills, a metallic taste and muscle aches several hours after the shift. Symptoms resolve within two days. Which fume is the MOST likely cause?

A. Iron oxide
B. Manganese
C. Tin oxide
D. Zinc oxide

6.3-Q8. Area noise levels average 82 dBA (below the 85 dBA action level), yet several workers in a fiberglass boat lay-up area that uses styrene resin show standard threshold shifts. What is the MOST likely contributing factor?

A. Normal age-related hearing loss only
B. An uncalibrated audiometer
C. Ototoxic styrene exposure acting together with noise
D. Overuse of hearing protection

6.3-Q9. Section 11 of an SDS states that a solvent caused malformations in the offspring of pregnant test animals but showed no germ-cell mutations. How is this solvent BEST classified?

A. Teratogen
B. Mutagen
C. Sensitizer
D. Carcinogen

6.3-Q10. An industrial hygienist must measure a painter's 8-hour exposure to toluene vapor. Which sampling method is MOST appropriate?

A. Activated charcoal sorbent tube on a low-flow personal pump in the breathing zone
B. Mixed cellulose ester (MCE) filter cassette
C. PVC filter with a cyclone pre-selector
D. Area grab sample taken at the room exhaust fan

Answer key and explanations

Q	Ans	Why
1	B	LC50 is the air concentration lethal to 50% of test animals over a stated time. A LOWER LC50 means GREATER acute inhalation toxicity, so A (50 ppm) is more toxic.
2	A	Under GHS, acute oral toxicity Category 1 is LD50 <= 5 mg/kg, so 4 mg/kg is Category 1 (the most severe). Lower category numbers mean higher toxicity.
3	D	Potentiation: a substance with no toxicity of its own (for example isopropanol) increases another's toxicity (carbon tetrachloride liver damage): 0 + 2 > 2. Synergism requires both agents to be toxic.
4	C	Synergism: two toxic agents together produce an effect much greater than the sum of their individual effects. Smoking plus asbestos multiplies lung cancer risk.
5	B	For additive mixtures, Em = C1/T1 + C2/T2 = 300/500 + 100/200 = 0.6 + 0.5 = 1.1. Because this exceeds 1.0, the mixture limit is exceeded even though each chemical alone is below its TLV.
6	D	IARC Group 1 = carcinogenic to humans (sufficient human evidence). 2A = probably, 2B = possibly, 3 = not classifiable.
7	D	Inhaling freshly formed zinc oxide fume (for example when welding galvanized steel) causes metal fume fever: flu-like fever and chills that resolve within 24-48 h.
8	C	Ototoxins such as styrene, toluene, lead and carbon monoxide can damage hearing on their own and add to the effect of noise, causing hearing loss below noise limits.
9	A	A teratogen causes birth defects in a developing embryo or fetus. A mutagen alters genetic material (DNA), and a carcinogen causes cancer. Exposure control plans should address pregnant workers.
10	A	Organic vapors such as toluene are collected on sorbent media (charcoal tube or passive badge) in the breathing zone. Filters collect particulates, and a cyclone is for respirable dust.

DOMAIN 6 · COMPETENCY 6.4

Ergonomics and Human Factors

CSP11 blueprint: Evaluate principles related to ergonomics and human factors (e.g., visual acuity, body mechanics, lifting, vibration, anthropometrics, fatigue management)

1. Definition

Ergonomics (human factors engineering) is the science of fitting the job, equipment and environment to the worker's capabilities and limitations to prevent musculoskeletal disorders (MSDs), reduce error and improve performance.

2. Explanation in simple words

Design the work to fit the person, not the person to fit the work. Reduce force, awkward postures, repetition, vibration and fatigue, and design controls and displays that people can see and use without mistakes.

3. Key points to remember

- MSD risk factors: force, repetition, awkward/static postures, contact stress, vibration, cold. Common MSDs: carpal tunnel syndrome (median nerve at wrist), lateral epicondylitis ('tennis elbow'), De Quervain's tenosynovitis (thumb-side wrist), rotator cuff injuries, low back pain, trigger finger.
- Ergonomics program elements: management commitment, employee involvement, job analysis (RULA, REBA, NIOSH lifting equation, Strain Index, Snook tables), controls (engineering first), training, medical management, evaluation.
- Lifting: Revised NIOSH equation RWL = 51 × HM × VM × DM × AM × FM × CM (HM = 10/H in inches); LI = load ÷ RWL. Best lifting zone between knuckle and shoulder height, close to the body. Back belts are not proven effective (NIOSH).
- Neutral posture: wrists straight, elbows close to body at about 90°, shoulders relaxed, knees about hip height, feet flat, lumbar support; monitor top at or slightly below eye level about an arm's length away.
- Vibration: hand-arm (HAVS, Raynaud's phenomenon - white finger) and whole-body (low back, vehicles); A(8) = a × √(T ÷ 8); controls - low-vibration tools, reduced exposure time, warm hands.
- Anthropometrics: body dimensions by percentile; design reach for the 5th percentile (small) user, clearance for the 95th percentile (large) user, adjustability for the 5th-95th range.
- Visual acuity and displays: Snellen 20/40 = sees at 20 ft what normal sees at 40 ft; lighting levels, glare control, contrast; display/control compatibility (population stereotypes - clockwise to increase).
- Human factors: human error types (slips, lapses, mistakes, violations), error-proofing (poka-yoke), workload, situational awareness.
- Fatigue management: shift length and rotation (forward rotation), circadian rhythm, sleep debt, fatigue risk management systems, rest breaks, limits on consecutive night shifts.

4. Real work scenario

> Warehouse workers lift 40-lb cases from floor-level pallets to a 36-in. conveyor 300 times a shift and report back pain. A NIOSH analysis gives LI ≈ 2.0. The company installs pallet lift tables that keep the top layer at waist height and a vacuum lift assist for heavy items; the LI falls below 1.0 and back complaints drop.

5. CSP exam tips and traps

- Engineering redesign beats training, stretching and back belts in ergonomics questions.
- LI = load ÷ RWL; LI > 1 means increased risk.
- Reach → 5th percentile; clearance → 95th percentile.
- Know where MSDs occur: epicondylitis = elbow; De Quervain's = thumb side of wrist; Raynaud's = fingers (vibration/cold).

6. Every blueprint point (6 blueprint items + 3 more)

Visual acuity BLUEPRINT

Visual acuity is the sharpness of vision needed for accurate, safe work. Lighting, glare, contrast, display design and the aging workforce all affect it.

- Snellen 20/20 = normal acuity; aging (presbyopia) reduces near focus and increases light needs - a 60-year-old needs about 2-3 times the light of a 20-year-old.
- IES recommended illuminance (guide): offices/VDT 300-500 lux (30-50 fc); drafting 750-1,000 lux; labs 500-1,000; precision assembly/inspection 1,000-2,000 lux (100-200 fc); active warehouses 200-300 lux.
- 1 footcandle = 10.76 lux. Illuminance from a point source follows the inverse square law: E = I/d^2.
- Glare: direct (light source in field of view) and reflected (screen, glossy surfaces); control with diffusers, indirect lighting, screen positioning perpendicular to windows.
- Controls: task lighting, adjustable contrast, anti-glare screens, regular vision testing, prescription safety eyewear.
- OSHA 1926.56 sets minimum illumination for construction (e.g., 5 fc general construction areas).

> **Exam tip:** Increasing overall room light can worsen screen glare - task lighting and positioning are often the better answer.

Body mechanics BLUEPRINT

Good body mechanics keep the body in a neutral posture - straight spine, elbows close to the body, wrists straight - and minimize twisting, bending, overhead work and long reaches.

- Neutral posture minimizes muscle effort and joint stress; awkward postures multiply the force on the spine and tendons.
- Spinal disc compression rises sharply with load distance from the body (moment arm); L5/S1 is the most stressed disc. NIOSH action limit for disc compression is 3,400 N (770 lb); 6,400 N is the maximum permissible limit.
- Use power zone: between mid-thigh and shoulder height, close to the body.
- Pushing is preferred over pulling; avoid twisting - turn with the feet.
- Controls: adjustable work heights, tool redesign, training on posture, job design to reduce static and repetitive positions.

> **Exam tip:** A load held 20 in. from the spine stresses the back far more than the same load held at 10 in. - reduce horizontal distance.

Lifting (NIOSH lifting equation) BLUEPRINT

The revised NIOSH Lifting Equation computes a Recommended Weight Limit (RWL) for two-handed manual lifts; the Lifting Index compares the actual load to the RWL.

- RWL = LC x HM x VM x DM x AM x FM x CM, with load constant LC = 51 lb (23 kg).
- HM = 10/H (in.) [25/H cm], H 10-25 in.; VM = 1 - 0.0075|V - 30| (in.); DM = 0.82 + 1.8/D (in.); AM = 1 - 0.0032A (degrees); FM from frequency/duration table; CM from coupling (good/fair/poor) table.
- Lifting Index LI = load / RWL. LI <= 1 nominal risk; LI > 1 increased risk; LI > 3 high risk for most workers.
- Example: H = 20 in. (HM 0.5), V = 30 in. (VM 1.0), D = 10 in. (DM 1.0), A = 0, FM = CM = 1 -> RWL = 25.5 lb. A 30-lb box -> LI = 1.18 (increased risk).
- Biggest lever: horizontal distance - halving H from 20 to 10 in. doubles the RWL.
- Equation does not apply to one-handed, seated, kneeling, team lifts, high-speed lifting or unstable loads; consider Snook/Liberty Mutual tables or ACGIH lifting TLV.

Exam tip: If V is above 30 in. or below, VM drops; any multiplier is <= 1, so the RWL can never exceed 51 lb.

Vibration BLUEPRINT

Vibration transmitted to the hands (hand-arm) or whole body damages vessels, nerves and the spine. OSHA has no vibration PEL; ACGIH TLVs and ISO standards apply.

- Hand-arm vibration (HAV) about 8-1,000 Hz (guide: 50-1,000 Hz) from grinders, chainsaws, jackhammers -> HAVS / vibration white finger (Raynaud's phenomenon): blanching in cold, numbness, loss of dexterity.
- Whole-body vibration (WBV) about 0.5-80 Hz from trucks, forklifts, mining and construction vehicles -> low back pain, spinal degeneration, fatigue.
- ACGIH (guide values): HAV TLV 5 m/s2 A(8), action level 2.5 m/s2; WBV about 0.5-1.0 m/s2 A(8). ISO 5349 = HAV; ISO 2631 = WBV.
- Measured with accelerometers (frequency-weighted acceleration, m/s2).
- Controls: low-vibration tools, tool maintenance/balancing, isolation mounts and suspension seats, limiting exposure time, keeping hands warm and dry, lighter grip; anti-vibration gloves have limited value.

Exam tip: 'White fingers in winter + power tools' = HAVS; 'truck driver back pain' = WBV.

Anthropometrics BLUEPRINT

Anthropometrics is the measurement of human body dimensions used to design workplaces, tools and equipment that fit the user population.

- Common design range: 5th percentile female to 95th percentile male (about 90-95% of users).
- Reach distances and control forces: design for the smallest/weakest (5th percentile female).
- Clearances (doorways, aisles, headroom) and tool handles/openings: design for the largest (95th percentile male).
- Work surfaces and seats: make adjustable across the 5th-95th range; standing work height about elbow height (lower for heavy work, higher for precision).
- Designing for the 'average' person fits almost nobody; use adjustability and universal design.

Exam tip: Reach = smallest user; clearance = largest user; height = adjustable.

Fatigue BLUEPRINT

Fatigue is reduced physical and mental capability caused by long hours, shift work, inadequate sleep, heavy or monotonous work. It increases errors, slower reactions and incidents.

- Hazards: night and rotating shifts, extended overtime, long commutes, sleep disorders, circadian low point roughly 2-6 a.m.
- About 17-19 hours awake impairs performance comparably to a BAC of about 0.05%; 24 hours about 0.10%.
- IARC classifies night shift work as Group 2A (probably carcinogenic).
- Controls: fatigue risk management system, forward-rotating shifts (day -> evening -> night), limit consecutive nights and 12-h shifts, guaranteed rest days, micro-breaks, task rotation, fatigue monitoring.
- Transport: DOT/FMCSA hours-of-service and NRC 10 CFR 26 fatigue rules for regulated industries.

Exam tip: 'Night shift' in a question should make you think of both performance decline and IARC 2A.

Musculoskeletal disorders and risk factors

Work-related MSDs (cumulative trauma disorders) are injuries to muscles, tendons, nerves and joints that develop over time from repeated stress.

- Primary risk factors: force, repetition, awkward/static posture; secondary: contact stress, vibration, cold, inadequate recovery time; personal factors: diabetes, obesity, pregnancy, arthritis.
- Carpal tunnel syndrome = median nerve compression at the wrist: numbness/tingling in thumb, index, middle and half of ring finger, worse at night, thenar atrophy when chronic.
- Others: tendinitis, tenosynovitis, De Quervain's (thumb), trigger finger, epicondylitis (tennis elbow), rotator cuff injury, low back pain.
- Assessment tools: RULA, REBA, Strain Index, ACGIH HAL TLV, Snook tables, checklists, symptom surveys and OSHA 300 log trends.
- Controls: engineering (tool redesign, lift assists, workstation adjustment, automation), administrative (job rotation, rest breaks, pacing), training on neutral posture, early medical management.

> **Exam tip:** Early reporting and medical management is a core ergonomics-program element - it catches MSDs before they become lost-time cases.

Ergonomics program and workstation design

An effective ergonomics program follows a systematic process and is enforced in the US through the General Duty Clause and industry guidelines.

- Elements: management commitment, employee involvement, job hazard analysis, hazard prevention/control, training, medical management, program evaluation.
- Computer workstation: monitor about an arm's length away with the top at or slightly below eye level; elbows about 90 degrees; wrists neutral; feet flat or on a footrest; ANSI/HFES 100.
- Human factors design principles: follow population stereotypes (clockwise/up = increase, red = stop), make controls distinguishable, design out error (poka-yoke), keep displays simple.
- Human error types: slips and lapses (skill-based), mistakes (rule- or knowledge-based), violations (intentional) - each needs a different control.
- OSHA has industry guidelines (poultry, nursing homes, retail grocery, shipyards) but no general ergonomics standard (the 2000 rule was repealed in 2001).

> **Exam tip:** Retraining fixes knowledge-based mistakes, not slips - slips need design changes.

Fitness for duty and return to work

Fitness for duty confirms a worker can safely perform the job; return-to-work programs bring injured or ill workers back gradually and safely.

- Fitness for duty covers medical fitness (vision, hearing, strength), substance use testing (DOT, NRC 10 CFR 26) and psychological fitness (fatigue, stress).
- Evaluations must be job-related and consistent with business necessity (ADA); use essential functions from job descriptions.
- Return to work: medical clearance, transitional/light duty, written work restrictions, coordination among employer, provider, insurer and employee, follow-up monitoring.
- Early RTW reduces lost workdays, claim costs and disability duration; modified duty converts lost-time cases to restricted cases on the OSHA 300 log.

> **Exam tip:** 'Worker capable of performing safely' = fitness for duty; 'gradual, monitored reintegration' = return to work.

Read more: Yates, Safety Professional's Reference and Study Guide (4th ed.): Ch. 24 Ergonomics, p. 546; Ch. 13 Thermal Stressors (fatigue/heat)

7. Ten CSP-style practice questions

Answer all ten on paper first, then check the key. Score: _____ / 10 Date: _____ Retest: _____ / 10

6.4-Q1. A forklift driver's vision test shows 20/40 in each eye. What does this mean?

A. The driver sees at 40 ft what normal vision sees at 20 ft
B. The driver sees at 20 ft what a person with normal vision sees at 40 ft
C. The driver has double normal acuity
D. The driver is legally blind

6.4-Q2. When designing a control panel, which anthropometric rule is MOST appropriate for REACH distances?

A. Design for the 5th percentile (small) user so nearly everyone can reach
B. Design for the 95th percentile male
C. Design for the 50th percentile (average) user
D. Anthropometrics do not apply to reach

6.4-Q3. In anthropometric design, clearances such as door height are based on which percentile?

A. 5th percentile female
B. 50th percentile male
C. 95th percentile male
D. 100th percentile male

6.4-Q4. For a lifting task the load weight is 70 lb and the NIOSH Recommended Weight Limit (RWL) is 20 lb. What is the Lifting Index (LI = LW/RWL)?

A. 1.00
B. 2.50
C. 4.00
D. 3.50

6.4-Q5. A NIOSH lifting analysis gives these multipliers: HM 0.625, VM 0.93, DM 0.91, AM 1.0, FM 0.88, CM 0.95. The load is 30 lb. What are the RWL and Lifting Index? (RWL = 51 lb x HM x VM x DM x AM x FM x CM)

A. RWL 22.6 lb; LI 0.75
B. RWL 22.6 lb; LI 1.33
C. RWL 51 lb; LI 0.59
D. RWL 30 lb; LI 1.00

6.4-Q6. At a fitness equipment plant, four back injuries this year involved lifting a 110-lb box from a 36-in-high conveyor to a 4-in-high pallet 36 in away. What is the BEST solution?

A. Rotate the task so each worker lifts the product only two hours a day
B. Lower the conveyor to 28 inches and ensure two workers lift the boxes
C. Provide the worker with safe lifting training and implement a stretching program
D. Provide a vacuum lift-assist device to lift the box to the pallet

6.4-Q7. On an assembly line, workers repeatedly reach more than 20 in in front of their bodies to grab parts from a supply bin, and several report shoulder fatigue late in the shift. Which improvement would MOST effectively reduce the shoulder strain?

A. Increase job rotation so workers switch positions every two hours
B. Provide anti-fatigue mats to reduce discomfort while standing
C. Move the parts bin closer to the worker to reduce extended forward reaches
D. Require workers to stretch before starting their shift

6.4-Q8. Hand-Arm Vibration Syndrome (HAVS) is also known as:

A. Raynaud's Phenomenon / Vibration White Finger
B. Siderosis
C. Carpal Tunnel Syndrome
D. Berylliosis

6.4-Q9. Which fatigue-management measure is MOST effective in reducing the risk of human error in shift work?

A. Long shifts with fewer breaks
B. Rotating shifts forward with regular rest days
C. Fixed night shifts every week
D. Increasing overtime hours

6.4-Q10. Workers in an assembly line are reporting back pain and discomfort. What initial step should you take to address potential ergonomic issues?

A. Conduct an ergonomic assessment to identify risk factors contributing to discomfort.
B. Provide training on proper lifting techniques and posture.
C. Redesign the workstation layout to minimize awkward postures and repetitive motions.
D. Implement job rotation to reduce the physical strain on workers.

Answer key and explanations

Q	Ans	Why
1	B	Snellen 20/40 means the person must be at 20 ft to see what a person with normal vision sees at 40 ft, which is worse than normal. 20/200 is the usual legal-blindness threshold.
2	A	Reach is designed for small users (5th percentile female) so about 95% of people can reach. Clearances are designed for large users (95th percentile male).
3	C	Clearances such as door height and legroom use the largest expected user (95th percentile male) so nearly everyone fits. A 100th percentile does not exist in design practice.
4	D	LI = LW/RWL = 70/20 = 3.5. An LI above 1.0 means increased risk, and above 3.0 most workers are at high risk, so redesign is needed.
5	B	RWL = 51 x 0.625 x 0.93 x 0.91 x 1.0 x 0.88 x 0.95 = 22.6 lb; LI = 30/22.6 = 1.33 (above 1.0, so the task needs redesign). Trap: dividing RWL by load gives 0.75.
6	D	Four injuries and a 110-lb box need an engineering control. A vacuum lift assist removes the manual lift. Rotation and training are administrative controls that leave the hazard in place.
7	C	Reducing the reach distance (moving the bin into the primary reach zone) removes the risk factor at its source. Rotation, mats and stretching do not change the reach.
8	A	Hand-arm vibration syndrome (vibration white finger, a secondary Raynaud's phenomenon) comes from prolonged use of vibrating hand tools. It causes finger blanching, numbness and loss of dexterity.
9	B	Forward-rotating shifts (day to evening to night) with regular rest days fit the circadian rhythm and limit cumulative fatigue. Long shifts, fixed nights and overtime increase fatigue and error.
10	A	The FIRST step is an ergonomic assessment (job hazard / risk-factor analysis) to identify causes. Controls such as redesign or rotation should follow from what it finds.

DOMAIN 6 · COMPETENCY 6.5

Chemistry: Containment Volume and Storage Requirements

CSP11 blueprint: Apply chemistry principles to calculate required containment volumes and hazardous materials storage requirements

1. Definition

Applied chemistry for the CSP covers the properties of chemicals (states of matter, molecular weight, vapor pressure, flash point, pH, reactivity), gas laws, concentration conversions and the calculations needed to size secondary containment and comply with storage limits.

2. Explanation in simple words

Know how chemicals behave - how fast they evaporate, whether vapors sink, whether they burn or react - and be able to calculate how big a dike must be and how much you are allowed to store.

3. Key points to remember

- Containment volume: required volume = % × largest container (e.g., 110%); convert gallons to ft^3 (÷ 7.48) or liters to m^3 (÷ 1,000); wall height = volume ÷ floor area; subtract volume displaced by other tanks inside the dike.
- Geometry: cylinder $V = \pi r^2 h$; rectangle $V = L \times W \times H$; 1 ft^3 = 7.48 gal; 1 gal = 3.785 L; 1 m^3 = 264.2 gal.
- Gas laws (use absolute temperature and pressure): Boyle $P_1V_1 = P_2V_2$; Charles $V_1/T_1 = V_2/T_2$; Gay-Lussac $P_1/T_1 = P_2/T_2$; combined $P_1V_1/T_1 = P_2V_2/T_2$; ideal gas PV = nRT (R = 0.0821 L·atm/mol·K or 8.314 J/mol·K). K = °C + 273; °R = °F + 460; psia = psig + 14.7. 1 mole of gas = 24.45 L at 25 °C, 22.4 L at 0 °C.
- Concentration: ppm = $mg/m^3 \times 24.45 \div MW$; % to ppm: 1% = 10,000 ppm; % LEL readings convert using the LEL (e.g., 10% of methane LEL 5% = 0.5% = 5,000 ppm).
- pH = $-\log[H^+]$; 7 neutral; < 7 acid; > 7 base/caustic; each unit is a 10× change. Neutralization of strong acids and bases is highly exothermic.
- Properties: vapor density (> 1 heavier than air, collects low), specific gravity (< 1 floats on water), vapor pressure (higher = evaporates faster), flash point, boiling point, LEL/UEL, autoignition temperature.
- Storage requirements: flammable cabinets (60 gal Cat 1-3, 120 gal Cat 4, max 3 per fire area), container size limits by category, storage rooms (ventilation 6 ACH), separation of compressed gas oxygen and fuel gas cylinders by 20 ft or a 5-ft-high ½-hour fire-rated barrier, maximum allowable quantities per control area (IFC/NFPA 400), incompatibility segregation.
- Balanced equations and chemical change (e.g., $Fe_2O_3 + 3CO \rightarrow 2Fe + 3CO_2$; rusting is a chemical change).

4. Real work scenario

> A dike must hold 110% of the largest of three tanks (10,000, 8,000 and 5,000 gal). Required volume = 11,000 gal ÷ 7.48 = 1,470.6 ft^3. The dike floor is 40 × 30 ft (1,200 ft^2), so the minimum wall height is 1.23 ft before allowing for the other tanks' displacement; the engineer specifies 1.6 ft to include displacement and freeboard.

5. CSP exam tips and traps

- Containment questions use the LARGEST tank, not the total, unless the rule says otherwise.
- Gas-law problems require absolute temperature and pressure - a frequent trap.

- 1% = 10,000 ppm.
- Higher vapor density means vapors flow to low points (pits, trenches).

6. Every blueprint point (2 blueprint items + 4 more)

Containment-volume calculations BLUEPRINT

Secondary containment must hold the release from the largest container plus allowances. Calculate the required volume, then size the dike or berm.

- Rule set: SPCC (40 CFR 112) - capacity of the largest container plus freeboard for precipitation (110% is the common design value); RCRA container areas (40 CFR 264.175) - 10% of total volume or 100% of the largest container, whichever is greater (the guide states 110% of the largest). NFPA 30 dikes - at least the capacity of the largest tank.
- Conversions: 1 ft3 = 7.48 gal; 1 gal water = 8.34 lb; 1 m3 = 1,000 L = 264.2 gal.
- Rectangular dike volume = L x W x H. Cylinder volume = pi x r^2 x h.
- Example: largest tank 10,000 gal -> 110% = 11,000 gal / 7.48 = 1,470.6 ft3. Dike 30 ft x 25 ft (750 ft2) -> wall height = 1,470.6/750 = 1.96 ft (about 2 ft, plus freeboard).
- Example: ten 55-gal drums (550 gal total) -> 10% = 55 gal vs largest 55 gal -> 55 gal required under RCRA (110% rule gives 60.5 gal).
- Subtract volume displaced by other tanks/foundations inside the dike below the wall height.

> **Exam tip:** Always use the LARGEST container (not the sum) for the 100/110% rule, then compare with 10% of the total.

Hazardous-material storage requirements BLUEPRINT

Storage requirements limit quantities, separate incompatibles, control ignition sources and provide containment, ventilation and fire protection.

- OSHA 1910.106 flammable liquid categories: Cat 1 FP < 73.4 F and BP <= 95 F; Cat 2 FP < 73.4 F and BP > 95 F; Cat 3 FP 73.4-140 F; Cat 4 FP 140-199.4 F. NFPA classes: IA, IB, IC (FP < 100 F), II (100-140 F), IIIA (140-200 F), IIIB (>= 200 F).
- Flammable storage cabinets: max 60 gal Cat 1-3 or 120 gal Cat 4 per cabinet; no more than 3 cabinets in one fire area (more allowed in sprinklered areas per NFPA 30). Safety cans <= 5 gal with flame arrester and spring-closing lid.
- Outside a room or cabinet in one area: max 25 gal Cat 1 in containers, 120 gal Cat 2-4 in containers, 660 gal Cat 2-4 in one portable tank (1910.106(e)).
- Segregate incompatibles: oxidizers from flammables, acids from bases and cyanides/sulfides, water-reactives away from water/sprinklers; NFPA 400 hazardous materials code; SDS Sections 7 and 10.
- Compressed gases: upright, secured, caps on; oxygen 20 ft from fuel gases or 5-ft half-hour barrier. Peroxide formers (ethers): date on receipt/opening, test or dispose per schedule.
- Bond and ground when transferring Class I liquids; ventilation and Class I Div 2 electrical in storage rooms; NFPA 704 diamond (0-4, 4 = most severe; opposite of GHS where Cat 1 is most severe).

> **Exam tip:** NFPA 704: 4 = worst. GHS category: 1 = worst. Know which scale the question uses.

Concentration units and ppm to mg/m3 conversion

Airborne concentrations of gases and vapors are expressed as ppm (by volume) or mg/m3 (by mass). Conversion uses the molar volume of an ideal gas.

- mg/m3 = ppm x MW / 24.45; ppm = mg/m3 x 24.45 / MW (25 C, 1 atm; use 22.4 at 0 C).
- Example: toluene (MW 92.14) at 100 ppm -> 100 x 92.14 / 24.45 = 376.9 mg/m3.
- Example: 50 mg/m3 of acetone (MW 58.08) -> 50 x 24.45 / 58.08 = 21.0 ppm.
- 1% by volume = 10,000 ppm; 1 ppm = 0.0001%. Example: LEL 1.2% = 12,000 ppm.
- Particulates and fibers cannot be expressed in ppm - use mg/m3 or fibers/cc.
- Molar volume adjusts with temperature/pressure: V = 24.45 x (T/298) x (760/P).

> **Exam tip:** 24.45 is the magic number - if a question gives MW and asks to convert, this is the formula.

Flammability properties (flash point, LEL/UEL, vapor pressure, vapor density)

Fire and explosion hazard of a chemical is predicted from its physical properties on SDS Section 9.

- Flash point: lowest temperature at which vapors ignite with an ignition source; fire point slightly higher; autoignition temperature ignites without a spark.
- Flammable range LEL-UEL: gasoline about 1.4-7.6%, methane 5-15%, hydrogen 4-75%, acetylene 2.5-100% (wider range = more hazardous). Confined-space alarm at 10% LEL.
- Mixture LEL (Le Chatelier): LEL_mix = 100 / (P1/LEL1 + P2/LEL2 + ...). Example: 50% methane (5%) + 50% propane (2.1%) -> 100/(10 + 23.8) = 2.96%.
- Vapor pressure: higher = evaporates faster (more airborne exposure). Vapor density (air = 1): > 1 sinks to low areas (propane 1.55, H2S 1.19, gasoline vapors 3-4); < 1 rises (methane 0.55, hydrogen 0.07).
- Specific gravity (water = 1): < 1 floats (most hydrocarbons - water spreads the fire); > 1 sinks. Weight = gallons x 8.34 x SG.
- Water solubility determines whether water dilutes (alcohols) or spreads (hydrocarbons) a spill fire - use alcohol-resistant foam for polar solvents.

> **Exam tip:** Vapor density > 1 means look for vapors at floor level, pits and trenches.

Gas laws and pressure

Gas laws predict how pressure, volume and temperature of a confined gas change. Always use absolute temperature and absolute pressure.

- Combined gas law: P1V1/T1 = P2V2/T2. Boyle (T constant): P1V1 = P2V2. Charles (P constant): V1/T1 = V2/T2. Ideal gas: PV = nRT.
- Absolute temperature: K = C + 273; Rankine R = F + 460. Absolute pressure: psia = psig + 14.7.
- Example: cylinder at 2,000 psig and 70 F heated to 150 F -> P2 = 2,014.7 x (610/530) = 2,318.8 psia = 2,304 psig.
- Liquid-to-gas expansion ratios are large (e.g., liquid nitrogen about 1:694, LPG about 1:270) - oxygen-deficiency and BLEVE hazards.
- Water pressure head: 0.433 psi per ft of water; 1 atm = 14.7 psi = 33.9 ft of water = 760 mm Hg.

> **Exam tip:** Classic error: using F or psig directly in gas law ratios - convert to absolute units first.

Corrosives, reactivity and pH

Corrosives destroy tissue and materials; reactive chemicals can release energy or toxic products when mixed, heated, shocked or exposed to air or water.

- pH scale 0-14: < 7 acidic, > 7 basic; each unit is a 10-fold change. GHS skin corrosion Cat 1 typically pH <= 2 or >= 11.5. RCRA corrosivity characteristic: pH <= 2 or >= 12.5.
- Bases (caustics) penetrate deeper than acids (liquefaction necrosis); hydrofluoric acid needs calcium gluconate first aid.
- Eyewash/shower (1910.151(c); ANSI Z358.1): reachable within 10 s (about 55 ft), flush 15 min, tepid water 60-100 F.
- Dangerous combinations: bleach + acid -> chlorine; bleach + ammonia -> chloramines; acid + cyanide -> HCN; acid + sulfide -> H2S; oxidizer + organic -> fire.
- Reactive classes: water-reactive (sodium, calcium carbide), pyrophoric (spontaneously ignite in air), peroxide formers, polymerizers, oxidizers (NFPA 400 classes 1-4).
- Exothermic runaway reactions are controlled by cooling, inhibitors, relief systems and MOC (link to process safety).

> **Exam tip:** Always add acid to water, never water to acid ('Do what you oughta, add acid to water').

Read more: Yates, Safety Professional's Reference and Study Guide (4th ed.): Ch. 5 Math Review, p. 211; Ch. 6 Particulates and Gases, p. 228; Ch. 12 Fire Prevention (storage); Ch. 27 Hazardous Materials Management

7. Ten CSP-style practice questions

Answer all ten on paper first, then check the key. Score: _____ / 10 Date: _____ Retest: _____ / 10

6.5-Q1. A dike must contain 110% of the largest of three tanks (10,000, 8,000 and 5,000 gal). The dike floor is 40 ft x 30 ft. Ignoring the volume displaced by the other tanks, what minimum wall height is needed? (7.48 gal = 1 ft3)

A. 0.92 ft
B. 1.23 ft
C. 1.91 ft
D. 2.56 ft

6.5-Q2. A RCRA hazardous waste container storage area holds twenty 55-gallon drums. Under 40 CFR 264.175, secondary containment must hold 10% of the total volume or 100% of the largest container, whichever is greater. What minimum containment capacity is required?

A. 55 gal
B. 110 gal
C. 1,100 gal
D. 1,210 gal

6.5-Q3. A cylindrical tank storing a hazardous liquid is 10 ft in diameter and 20 ft high. What is its full capacity? (Use pi = 3.14)

A. 1,570 cubic feet
B. 3,140 cubic feet
C. 6,280 cubic feet
D. 7,850 cubic feet

6.5-Q4. A compressed gas cylinder reads 2,000 psig at 70 F. It is left in the sun and reaches 140 F. Assuming constant volume, what is the new gauge pressure? (Use 14.7 psi for atmospheric and R = F + 460)

A. 2,000 psig
B. 2,400 psig
C. 4,000 psig
D. 2,266 psig

6.5-Q5. An SDS lists the density of petroleum diesel as 6.943 lb/gal. What is the weight of diesel in a full 100,000-gallon tank?

A. 694,000 lb
B. 700,943 lb
C. 700,694 lb
D. 694,300 lb

6.5-Q6. A container holds 0.55 mol of gas at 105.7 kPa and 300 K. What volume does the gas occupy? (PV = nRT; R = 8.31 L kPa/mol K)

A. 42 L
B. 24 L
C. 13 L
D. 4 L

6.5-Q7. At constant pressure, a gas occupies 8.57 L at 273 K. What will its volume be at 355 K?

A. 11.1 L
B. 10.4 L
C. 6.4 L
D. 23.6 L

6.5-Q8. Under OSHA 1910.106, an approved flammable storage cabinet may hold no more than 60 gallons of Category 1, 2 or 3 flammable liquids. How many full 5-gallon safety cans of a Category 2 liquid may be stored in one cabinet?

A. 6
B. 10
C. 12
D. 24

6.5-Q9. Several flammable liquids in a paint shop have a flash point of 58 F (14 C). What does this mean for their handling and storage?

A. It can produce flammable vapors at normal room temperature and should be stored in a flammable cabinet
B. It is not hazardous below 100 F and can be stored with non-flammable liquids
C. Flash point only matters during combustion, not storage
D. Only liquids with a flash point below 32 F require special handling

6.5-Q10. During an inspection, you find containers of nitric acid stored on the same shelf as containers of acetone. What should you do?

A. Approve the setup since both are common industrial chemicals
B. Suggest labeling the containers more clearly
C. Suggest diluting the nitric acid to make it less reactive
D. Move the acetone to a flammable cabinet and segregate it from the nitric acid

Answer key and explanations

Q	Ans	Why
1	B	Required volume = 1.1 x 10,000 = 11,000 gal / 7.48 = 1,470.6 ft3. Height = 1,470.6 / (40 x 30) = 1.23 ft. Trap: 2.56 ft wrongly sums all three tanks.
2	B	Total = 20 x 55 = 1,100 gal; 10% = 110 gal; largest container = 55 gal. The greater value is 110 gal. Trap: 55 gal applies only the largest-container rule.
3	A	V = pi r^2 h = 3.14 x 5^2 x 20 = 1,570 ft3 (about 11,750 gal at 7.48 gal/ft3). Trap: using the diameter (10) as the radius gives 6,280 ft3.
4	D	Gas law calculations need absolute units: P2 = 2,014.7 psia x (600 R / 530 R) = 2,280.8 psia = about 2,266 psig. Trap: using F directly (140/70) doubles the pressure to 4,000 psig.
5	D	Mass = volume x density = 100,000 gal x 6.943 lb/gal = 694,300 lb. This matters for foundation, floor loading and containment design.
6	C	V = nRT/P = (0.55 x 8.31 x 300)/105.7 = 1,371/105.7 = 13 L.
7	A	Charles's law at constant pressure: V2 = V1 x T2/T1 = 8.57 x 355/273 = 11.1 L. Temperatures must be absolute (K).
8	C	60 gal / 5 gal per can = 12 cans. The 120-gal cabinet limit applies only to Category 4 (combustible) liquids, and no more than three cabinets are allowed per fire area.
9	A	A flash point of 58 F is below normal room temperature, so ignitable vapors form at ambient conditions (OSHA Category 2). Store the liquid in approved flammable cabinets or rooms away from ignition sources.
10	D	Nitric acid is a strong oxidizer and acetone is a flammable organic, so they are incompatible and can react violently. Segregate them: put acetone in a flammable cabinet and acids in a separate corrosive cabinet.

DOMAIN 6 · COMPETENCY 6.6

Physics: Energy, Weight, Force and Stress

CSP11 blueprint: Apply core concepts in physics (e.g., forms of energy, weights, forces, stresses)

1. Definition

Applied physics (mechanics, hydrostatics and electricity) explains how energy causes injury and how structures, rigging and equipment resist loads. The CSP exam tests calculations of force, energy, velocity, pressure, stress and electrical quantities.

2. Explanation in simple words

Injuries happen when energy reaches the body faster than it can absorb it. Physics lets you calculate how fast something falls, how hard it hits, how much force a sling or pressure vessel carries, and whether a part is strong enough.

3. Key points to remember

- Forms of energy: kinetic (KE = ½mv² - doubling speed quadruples energy), potential (PE = mgh), elastic (springs), thermal, chemical, electrical, radiant; energy is conserved (PE converts to KE in a fall).
- Newton's laws: 1st - inertia (unsecured loads keep moving when a vehicle stops); 2nd - F = ma; 3rd - action-reaction. Weight W = mg (g = 9.81 m/s² or 32.2 ft/s²). 1 N = 1 kg·m/s².
- Kinematics: $v = \sqrt{(2gh)}$; $d = \frac{1}{2}gt^2$; $v^2 = v_0^2 + 2ad$; stopping distance = reaction distance (v × t) + braking distance (v² ÷ 2a). Momentum p = mv; impact force ≈ change in momentum ÷ time.
- Friction: F = μN; static friction > kinetic friction (harder to start moving); slip resistance measured by coefficient of friction with a tribometer.
- Stress and strength: stress σ = F ÷ A (psi or Pa); factor of safety = ultimate strength ÷ working stress (or breaking strength ÷ working load); tension, compression, shear, bending, torsion; fatigue failure under cyclic loading.
- Pressure: P = F ÷ A, so F = P × A (force on a vessel door = pressure × area); hydrostatic P = ρgh (water: 0.433 psi per ft; 62.4 lb/ft³).
- Mechanical advantage: levers, pulleys (effort = load ÷ number of supporting rope parts), inclined planes.
- Electricity: Ohm's law V = IR; power P = VI = I²R; series and parallel resistance; current (not voltage) causes injury.
- Work = force × distance; power = work ÷ time (1 hp = 550 ft·lb/s = 746 W).

4. Real work scenario

> A 10-lb wrench falls 64 ft from a tower. Using $v = \sqrt{(2gh)} = \sqrt{(2 \times 32.2 \times 64)} \approx 64$ ft/s, it hits the ground at about 44 mph with KE = ½ × (10/32.2) × 64² ≈ 636 ft·lb - far beyond what a hard hat is rated to absorb. The safety professional requires tool tethers and an exclusion zone below.

5. CSP exam tips and traps

- Keep units consistent (pounds-mass vs slugs; convert lb to slugs by ÷ 32.2 when using F = ma in English units).
- Doubling speed quadruples kinetic energy - why speed control matters.
- Force on a door or plate = pressure × area (use radius, not diameter, in πr²).
- Static friction > kinetic friction explains 'hard to start, easy to keep moving'.

6. Every blueprint point (4 blueprint items + 2 more)

Energy (kinetic and potential) BLUEPRINT

Energy is the capacity to do work; injuries occur when energy is released uncontrolled and transferred to the body. Kinetic energy depends on mass and the square of velocity; potential energy on weight and height.

- KE = 1/2 m v^2 (joules with kg and m/s; ft-lb with slugs and ft/s, where mass in slugs = weight lb / 32.2).
- PE = m g h = W x h. Example: 10-lb tool dropped 50 ft -> PE = 500 ft-lb; 1 J = 0.738 ft-lb.
- Velocity after a free fall: v = sqrt(2gh). 16 ft -> sqrt(2 x 32.2 x 16) = 32.1 ft/s (time t = sqrt(2h/g) = 1.0 s).
- Doubling speed quadruples KE: 3,000-lb car (93.2 slugs) at 44 ft/s (30 mph) -> KE = 0.5 x 93.2 x 44^2 = about 90,200 ft-lb; at 60 mph about 361,000 ft-lb.
- Other forms: electrical, chemical, thermal, pressure, radiant - all must be controlled in LOTO (energy isolation and stored energy release).
- Haddon's energy-release theory: prevent the build-up, reduce the amount, prevent release, separate in time/space, put barriers, modify surfaces, strengthen the target.

> **Exam tip:** Speed matters more than mass for vehicle crash energy because velocity is squared.

Weight BLUEPRINT

Weight is the force of gravity on a mass: W = m x g. In US units the pound is a force, so weight in lb divided by g gives mass in slugs.

- W = m g; g = 9.81 m/s2 = 32.2 ft/s2. Example: 100 kg -> 981 N; 1 kg weighs about 2.2 lb; 1 lb = 4.448 N.
- Mass is constant; weight changes with gravity (and with acceleration - an accelerating hoist feels heavier).
- Liquid weight: water 8.34 lb/gal, 62.4 lb/ft3; other liquids = 8.34 x SG. Example: 55 gal of liquid with SG 1.2 = 55 x 8.34 x 1.2 = 550 lb (plus drum).
- Steel about 490 lb/ft3; concrete about 150 lb/ft3 - used to estimate rigging loads (weight = volume x density).
- Center of gravity: the point where the weight acts; for lifting, the hook must be directly above the CG to avoid swing/tilt.

> **Exam tip:** When rigging, estimate weight from dimensions and density if it is not marked - never guess.

Force BLUEPRINT

Force changes an object's motion: F = m x a. Force questions on the CSP exam include accelerations, impacts, fall arrest, friction and rigging tension.

- Newton's 2nd law F = m a; units N (kg m/s2) or lb (slug ft/s2). Example: 2,000-kg forklift decelerating at 3 m/s2 -> 6,000 N.
- Impulse-momentum: F x t = m x delta-v; increasing stopping time or distance reduces force (energy absorbers, airbags).
- Fall arrest average force: F = W(h + d)/d. Example: 220-lb worker, 6-ft free fall, 3.5-ft deceleration -> 220 x 9.5 / 3.5 = 597 lb average (OSHA max arresting force 1,800 lb with harness).
- Friction: F = mu x N. Walking surfaces: static COF about 0.5 commonly cited as slip-resistant (ANSI A1264.2).
- Pressure-force: F = P x A. Example: 1,500 psi on a 4-in2 hydraulic piston -> 6,000 lb.
- Stopping distance: d = v^2 / (2a). 60 mph (88 ft/s) braking at 0.7 g (22.5 ft/s2) -> 172 ft, plus reaction distance (1.5 s x 88 = 132 ft).

> **Exam tip:** Shorter stopping distance = higher impact force: that is why a 2-ft free fall on a positioning device is limited.

Stress BLUEPRINT

Stress is internal force per unit area in a material: stress = F/A. Comparing working stress with material strength gives the safety factor used in rigging, scaffolds and structures.

- Stress (psi or Pa) = force / cross-sectional area. Example: 2,000 lb on a 0.5-in2 rod -> 4,000 psi.
- Types: tensile (pulling), compressive (pushing), shear (sliding), bending and torsion.
- Strain = change in length / original length; Young's modulus E = stress/strain (elastic region). Beyond the yield point deformation is permanent; ultimate strength = failure.
- Safety factor = ultimate (breaking) strength / working load. Wire rope 5:1 -> 10,000-lb breaking strength gives WLL 2,000 lb. Scaffolds: 4 times intended load; guardrail top rail 200 lb.
- Fatigue failure: repeated cyclic stress below yield can crack metal over time - inspect cranes, hooks, pressure vessels.

> **Exam tip:** Doubling the rod diameter quadruples the area and cuts the stress to 1/4.

Rigging and sling angles

Sling leg tension increases as the angle from horizontal decreases. Rigging calculations combine weight, number of legs and sling angle.

- Leg tension T = (W / n) / sin(angle from horizontal). Load angle factor = 1/sin: 90 deg 1.000, 60 deg 1.155, 45 deg 1.414, 30 deg 2.000.
- Example: 2,000-lb load, 2-leg bridle at 60 deg -> each leg 1,000 x 1.155 = 1,155 lb; at 30 deg each leg 2,000 lb.
- Sling angles below 30 deg from horizontal are not recommended (ASME B30.9).
- Choker hitch reduces rated capacity (about 75-80% of vertical); basket hitch with vertical legs = 2 x vertical capacity.
- Assume only 2 legs carry the load on a 3- or 4-leg bridle unless an equalizer is used.

> **Exam tip:** Flatter sling = more tension. At 30 degrees, each leg carries the full load weight.

Work, power, torque and stability

Work and power quantify energy transfer and its rate; torque (moment) explains levers, wrenches, crane and forklift stability.

- Work = force x distance (ft-lb or J). Power = work / time; 1 hp = 550 ft-lb/s = 746 W. Example: lifting 500 lb 10 ft in 5 s = 1,000 ft-lb/s = 1.82 hp.
- Torque/moment = force x perpendicular distance. Example: 50 lb on a 2-ft wrench = 100 ft-lb.
- Equilibrium: sum of moments about the fulcrum = 0. Forklift: load moment (load x distance from front axle) must not exceed counterweight moment - capacity drops as the load center moves out (e.g., rated at 24-in. load center).
- Crane stability: tipping load and rated capacity charts (often 75% of tipping for crawlers, 85% for outriggers); capacity falls with longer radius.
- Electrical: Ohm's law V = I x R; power P = V x I = I^2 x R. Example: 120 V across 1,000 ohm body resistance -> 0.12 A (120 mA) - potentially fatal.

> **Exam tip:** Moving a forklift load forward beyond its rated load center reduces safe capacity even if the weight is unchanged.

Read more: Yates, Safety Professional's Reference and Study Guide (4th ed.): Ch. 17 Mechanics, p. 454; Ch. 18 Hydrostatics and Hydraulics, p. 469; Ch. 16 Electrical Safety (Ohm's law); Ch. 5 Math Review

7. Ten CSP-style practice questions

Answer all ten on paper first, then check the key. Score: _____ / 10 Date: _____ Retest: _____ / 10

6.6-Q1. A steel rod with a cross-sectional area of 0.5 in2 carries a tensile load of 2,000 lb. The material's ultimate strength is 60,000 psi. What is the working stress and factor of safety?

A. 1,000 psi; 60
B. 4,000 psi; 15
C. 4,000 psi; 5
D. 8,000 psi; 7.5

6.6-Q2. A 10-kg toolbox falls from a scaffold platform 2 m high. Ignoring air resistance, what is its approximate velocity just before it hits the ground? (v = sqrt(2gh), g = 9.8 m/s2)

A. 6.3 m/s
B. 4.5 m/s
C. 2.2 m/s
D. 9.8 m/s

6.6-Q3. A 1,500-kg vehicle travels at 10 m/s. What is its kinetic energy? (KE = 1/2 m v^2)

A. 7,500 J
B. 15,000 J
C. 75,000 J
D. 150,000 J

6.6-Q4. A 24-inch-diameter access door on a vessel is under 50 psig. What total force acts on the door?

A. 1,200 lb
B. 7,540 lb
C. 22,620 lb
D. 90,480 lb

6.6-Q5. A steel crane cable has a cross-sectional area of 150 mm2 and an allowable tensile stress of 400 MPa (1 MPa = 1 N/mm2). What is the maximum tensile force it can safely support? (F = stress x A)

A. 4,000 N
B. 60,000 N
C. 600,000 N
D. 6,000 N

6.6-Q6. A delivery truck is traveling through a jobsite at 15 mph when it strikes a pedestrian. If the same truck had been traveling at 30 mph, how would the impact energy compare?

A. It would double
B. It would increase by 1.5 times
C. It would stay the same
D. It would increase by four times

6.6-Q7. A 10-kg rubber block slides on a concrete floor (mu = 0.65). What is the friction force? (F = mu N; g = 9.8 m/s2)

A. 33.7 N
B. 43.7 N
C. 53.7 N
D. 63.7 N

6.6-Q8. A car accelerates uniformly from 10 m/s to 30 m/s over 10 seconds. How far does it travel? (a = (v2 - v1)/t; v2^2 - v1^2 = 2a d)

A. 200 m
B. 300 m
C. 400 m
D. 500 m

6.6-Q9. A worker says a heavy cart is hard to start moving but easier to keep moving. Which physics concept BEST explains this?

A. Kinetic energy
B. Static friction
C. Gravitational potential
D. Thermal expansion

6.6-Q10. A 10-lb wrench slips from a platform 20 ft above a worker who is wearing a hard hat. Which physical principle BEST explains why the wrench still poses a serious injury risk?

A. Gravitational potential energy converting to kinetic energy
B. Frictional resistance
C. Conservation of momentum
D. Electromagnetic interference

Answer key and explanations

Q	Ans	Why
1	B	Stress = F/A = 2,000/0.5 = 4,000 psi. Factor of safety = ultimate strength / working stress = 60,000/4,000 = 15.
2	A	v = sqrt(2gh) = sqrt(2 x 9.8 x 2) = sqrt(39.2) = 6.3 m/s. Mass does not affect free-fall velocity (it does affect impact energy).
3	C	KE = 1/2 mv^2 = 0.5 x 1,500 x 10^2 = 75,000 J. Because KE depends on v^2, speed control is critical in fleet and forklift safety.
4	C	Area = pi r^2 = pi x 12^2 = 452.4 in2; F = P x A = 50 psi x 452.4 = 22,620 lb. Trap: using the diameter as the radius gives 90,480 lb. Always depressurize before opening.
5	B	F = stress x area = 400 N/mm2 x 150 mm2 = 60,000 N (60 kN).
6	D	KE = 1/2 mv^2, so doubling speed from 15 to 30 mph raises impact energy by 2^2 = 4 times. Trap: assuming energy scales linearly (doubles).
7	D	Friction F = mu x N = 0.65 x (10 kg x 9.8 m/s2) = 0.65 x 98 N = 63.7 N.
8	A	a = (30 - 10)/10 = 2 m/s2; distance = (v2^2 - v1^2)/(2a) = (900 - 100)/4 = 200 m. Check: average velocity 20 m/s x 10 s = 200 m.
9	B	Static friction (resistance to starting motion) is greater than kinetic friction (resistance to continued sliding), so a cart takes more force to start than to keep moving.
10	A	The 10-lb wrench's potential energy (mgh) at 20 ft becomes kinetic energy as it falls and can exceed what a hard hat is designed to absorb. Prevent drops with tool tethers, toe boards and barricades.

DOMAIN 7

Training

10% of the exam · 6 competencies

7.1	Training Needs Assessment Describe the needs assessment process to determine worker training, competencies, and qualifications
7.2	Training Program Development Develop training programs with training materials to address various learning styles (e.g., presentation methods and tools)
7.3	Continuous Improvement of Training Describe how to implement training programs utilizing the Continuous Improvement model
7.4	Training Effectiveness Evaluation Determine the effectiveness of training programs (e.g., surveys, on-the-job compliance, feedback, assessments, demonstrations, quizzes)
7.5	Education and Training Methods Demonstrate working knowledge of education and training methods and techniques (e.g., classroom, online, simulation, computer-based, Artificial Intelligence, coaching, on-the-job training)
7.6	Adult Learning Principles Understand adult learning principles (e.g., visual, auditory, reading and writing, kinesthetic)

DOMAIN 7 · COMPETENCY 7.1

Training Needs Assessment

CSP11 blueprint: Describe the needs assessment process to determine worker training, competencies, and qualifications

1. Definition

A training needs assessment (TNA) is a systematic process to identify gaps between current and required knowledge, skills and abilities (KSAs) and to decide whether training is the right solution. It defines the competencies and qualifications required for each job.

2. Explanation in simple words

Before you train, find out what people need to know and do, what they already know, and whether the problem is really a lack of skill. If they can already do the job but don't, more training will not fix it.

3. Key points to remember

- Levels: organizational analysis (goals, regulations, incident trends), task/job analysis (what the job requires - task inventory, JHA), person analysis (who needs training - assessments, observation, performance data).
- Data sources: injury/incident data, audits, regulatory requirements, JHAs, observations, interviews, surveys, focus groups, performance appraisals, new equipment/processes (MOC), task matrix.
- Mager and Pipe performance analysis: 'Could they do it if their life depended on it?' If yes → not a skill deficiency (fix consequences, obstacles, feedback, equipment, supervision). If no → training, practice or job aids.
- Competency = demonstrated ability to perform to a standard (knowledge + skills + behavior). Qualification = formal credential, certificate or license. Competent person (OSHA) = can identify hazards and has authority to correct them; qualified person = has a degree/certificate or extensive knowledge to solve problems.
- Regulatory training requirements: e.g., forklift (evaluation every 3 years), HAZWOPER (40 h/24 h/8 h refresher annually), confined space, LOTO, HazCom (initially and when new hazards), respirators (annually), bloodborne pathogens (annually), hearing conservation (annually), crane operator certification.
- Prioritize training needs by risk (severity and consequence of failure), legal requirements and number of workers affected.
- Output: training plan with objectives, audience, methods, resources, schedule, evaluation.

4. Real work scenario

> Forklift near-misses rise at a warehouse. Operators score 95% on written tests and perform inspections correctly when observed, but three trucks have broken backup alarms and the night-shift supervisor rarely checks pre-use inspections. The needs assessment concludes training is not the main fix: alarms are repaired and supervisors are made accountable for inspection checks.

5. CSP exam tips and traps

- The FIRST step in designing training is the needs assessment.
- If workers know how but don't do it, the answer is not more training.
- Prioritize training by safety impact and consequences of failure.

6. Every blueprint point (3 blueprint items + 5 more)

Worker training BLUEPRINT

Worker training gives employees the knowledge and skills to recognize hazards and perform tasks safely. It is driven by regulatory requirements, task hazards, incidents and changes in the workplace, and it is only the right solution when the gap is a knowledge or skill gap.

- OSHA 1926.21(b)(2): employer must instruct each employee in recognizing and avoiding unsafe conditions.
- Training triggers: new hire, new task or equipment, process change (MOC), incident/near miss, audit finding, regulatory refresher.
- Common refresher frequencies: HAZWOPER 8 h annually; forklift evaluation at least every 3 years; PSM refresher at least every 3 years; bloodborne pathogens annually; respirator annually.
- ANSI/ASSP Z490.1 = accepted practices for SH&E training (needs assessment, design, delivery, evaluation, records).
- Training is NOT the fix when the problem is equipment design, lack of resources or motivation - use engineering, PtD or management actions instead.

> **Exam tip:** If a performance problem exists even though workers can do the task correctly, more training is the wrong answer.

Competencies BLUEPRINT

A competency is a demonstrated combination of knowledge, skills and abilities (KSAs) needed to perform a task safely. Training needs are defined by comparing the competency a job requires with what the worker currently has.

- Knowledge: regulations, SOPs, hazard recognition.
- Skills: equipment operation, emergency response, PPE use.
- Abilities: physical/mental readiness, decision-making under stress.
- Example: a scaffold inspector must know 1926.451, demonstrate inspection skills and be able to stop unsafe work.
- ISO 45001 clause 7.2 requires the organization to determine and ensure worker competence and retain documented evidence.

> **Exam tip:** Competence must be demonstrated, not just attended - a sign-in sheet proves attendance, not competency.

Qualifications (competent vs. qualified person) BLUEPRINT

OSHA uses two distinct roles. A competent person can identify existing and predictable hazards and has authority to take prompt corrective action; a qualified person has a recognized degree, certificate or professional standing, or extensive knowledge, training and experience, and can solve problems in the subject area.

- Competent person: 29 CFR 1926.32(f) - hazard recognition + authority to correct/stop work; no degree required.
- Qualified person: 29 CFR 1926.32(m) - degree/certificate/professional standing or extensive knowledge, training and experience.
- Competent examples: scaffold inspector, trench supervisor (soil classification, daily inspection), fall protection supervisor.
- Qualified examples: PE designing an anchorage or trench protective system, electrician on electrical systems, industrial hygienist.
- Rule of thumb: qualified = designs; competent = inspects and corrects on site.

> **Exam tip:** Key discriminator: AUTHORITY to take corrective action belongs to the competent person definition.

Training needs assessment (ADDIE Analyze phase)

A needs assessment identifies who must be trained, on what, and to what level before any course is designed. It is the Analyze step of ADDIE and answers who the learners are, their current skill level and what gaps exist.

- Sources: JHA/JSA, incident and near-miss data, audit findings, regulatory requirements, new equipment/processes, worker and supervisor input.
- Output: target audience, performance gaps, learning objectives, priorities.
- Prioritize high-risk tasks first (confined spaces, electrical, work at heights).
- Keywords 'needs analysis' or 'gap identification' = Analyze phase.

> **Exam tip:** FIRST step in developing training = needs assessment (not choosing a method or writing slides).

Defining baseline competency

Baseline competency is the minimum level of KSAs a worker must demonstrate to do a task safely. Comparing baseline (current) competency against required competency reveals the training gap.

- Steps: identify job tasks and hazards -> determine required KSAs -> assess current workforce competency -> define competency standards -> use the gap to build the needs assessment.
- Assessing current competency: pre-training tests, interviews, on-the-job observation, performance data (KPIs, incident reports).
- Standards used as benchmarks: OSHA/ANSI requirements, ISO 45001, company SOPs.
- Example: forklift operators - certification plus skill evaluation at least every 3 years (1910.178(l)).

> **Exam tip**: 'How do you decide whether workers need additional training?' = compare baseline (current) competency with required competency.

Tools for needs and competency assessment

Several practical tools link tasks to the skills needed and show where workers fall short.

- Job Hazard Analysis: breaks a job into steps and links each to hazards and required skills.
- Competency matrix: visual grid of required skills vs. employees, showing who is trained/qualified and when refreshers are due.
- Pre-assessment tests: measure baseline knowledge.
- Observation checklists: track safe behaviors vs. shortcuts.
- Task analysis + interviews and surveys of workers and supervisors.

Human risk factors affecting performance

Training fails if human factors are ignored. Behavior, decision making, situational awareness, workload, risk perception and stress all determine whether workers apply what they learned.

- Behavior: observable actions shaped by habits and culture - control with BBS and positive reinforcement.
- Decision making: choices under pressure - scenario training, clear procedures, stop-work authority.
- Situational awareness: perceive, understand, anticipate - hazard recognition training, checklists, pre-task briefings, fatigue management.
- Workload: too much = fatigue/errors, too little = complacency - job rotation, breaks, staffing.
- Risk perception: underestimating hazards leads to shortcuts - case studies, risk matrices.
- Stress: reduces focus and memory - EAP, reasonable scheduling, stress management training.

> **Exam tip:** Match the scenario: overwhelmed by demands = workload; anticipating hazards in dynamic settings = situational awareness; 'this low scaffold is not dangerous' = risk perception.

Who needs training: audience analysis

Different audiences need different content and depth: new hires, experienced workers, supervisors, managers, contractors and temporary workers.

- Supervisors and managers need training on their responsibilities, leading by example and enforcing rules, not just hazard content.
- Host and staffing agency share responsibility for temporary worker training; host provides site-specific training.
- Consider language, literacy and culture - OSHA requires training in a manner and language workers understand.
- Experienced workers still need training when tasks, equipment or rules change.

Read more: Yates, Safety Professional's Reference and Study Guide (4th ed.): Ch. 19 Training, p. 483

7. Ten CSP-style practice questions

Answer all ten on paper first, then check the key. Score: _____ / 10 Date: _____ Retest: _____ / 10

7.1-Q1. What is the primary purpose of conducting a Training Needs Assessment (TNA)?

A. To evaluate the effectiveness of past training programs
B. To identify specific training needs based on current competencies and desired performance levels
C. To assess the financial resources available for training
D. To finalize the training schedule for the year

7.1-Q2. What is the primary purpose of a gap analysis in a training needs assessment?

A. To count the number of employees requiring training
B. To determine the profitability of training programs
C. To identify discrepancies between current and desired performance levels
D. To evaluate the effectiveness of past training programs

7.1-Q3. A needs assessment reveals more training needs than can be addressed in the available time. What is the BEST approach?

A. Focus on the tasks that are performed most frequently
B. Train employees on tasks they already excel at to gain quick wins
C. Delay training until all needs can be addressed at once
D. Prioritize tasks with the highest risk or consequence of failure

7.1-Q4. Baseline competency is BEST defined as:

A. The minimum knowledge, skills, and abilities required to perform a job safely
B. The maximum level of performance expected from a worker
C. A supervisor's subjective rating of worker effort
D. The average score on a post-training quiz

7.1-Q5. Which tool maps each employee's current skills against the competencies required for their role?

A. Hazard control matrix
B. Risk matrix
C. Competency (skills) matrix
D. Inspection checklist

7.1-Q6. Which statement BEST distinguishes a competency from a qualification?

A. They are identical terms
B. Competency is only knowledge; a qualification is only skill
C. Qualifications are never required by OSHA
D. A qualification is a formal credential (certificate, license); competency is the demonstrated ability to perform the task to the required standard

7.1-Q7. Which OSHA regulation contains the general definitions of 'competent person' and 'qualified person' for construction?

A. 29 CFR 1910.120
B. 29 CFR 1926.32
C. 29 CFR 1910.146
D. 29 CFR 1926.21

7.1-Q8. A construction supervisor identifies a cave-in hazard in a trench and immediately stops work until shoring is corrected. Under OSHA, the supervisor is acting as a:

A. Competent person
B. Qualified person
C. Professional engineer
D. Trainer

7.1-Q9. Under 29 CFR 1926.32(m), a qualified person is one who:

A. Has jobsite experience but is not required to solve technical problems
B. Has authority to stop work and take prompt corrective action
C. By a recognized degree, certificate, or professional standing, or by extensive knowledge, training, and experience, has demonstrated the ability to solve problems relating to the subject matter
D. Has completed basic PPE training under supervision

7.1-Q10. An incident trend shows operators skipping a startup checklist step. The plant manager asks the CSP to schedule retraining for all operators. What should the CSP do FIRST?

A. Analyze whether the problem is caused by a lack of knowledge or skill, or by other factors such as time pressure, equipment design, or supervision
B. Schedule retraining for all operators immediately
C. Discipline the operators involved in the incidents
D. Purchase an online refresher course on startup procedures

Answer key and explanations

Q	Ans	Why
1	B	A TNA identifies the gap between current and required performance so training targets real needs. Evaluating past training is effectiveness evaluation, not needs assessment.
2	C	Gap analysis compares actual performance (or KSAs) with the required standard; the difference defines what training must close.
3	D	Training priorities follow risk: tasks whose failure could cause serious injury or loss come first. Frequency alone is a trap; a rare but critical task may matter more.
4	A	Baseline competency is the minimum KSA level required for safe performance; it is the standard against which current ability is compared.
5	C	A competency or skills matrix lists required competencies by role and shows who meets them, making training gaps visible. A risk matrix ranks hazards, not people's skills.
6	D	A credential shows something was earned; competency is proven ability to perform. A worker can hold a certificate yet still lack demonstrated competence on the job.
7	B	29 CFR 1926.32 (definitions) defines competent person (f) and qualified person (m). 1926.21 covers safety training and education; 1910.120 is HAZWOPER; 1910.146 is permit-required confined spaces.
8	A	An OSHA competent person can identify existing and predictable hazards and has authorization to take prompt corrective measures. The key words are 'identify' plus 'authority to correct'.
9	C	The qualified person is defined by demonstrated problem-solving ability (e.g., designing scaffolds or anchorages). Authority to stop work is the competent-person trap.
10	A	A needs assessment first determines whether a performance gap is actually a training problem. If operators know the step but skip it, the fix is a management or design issue, and retraining will not work.

DOMAIN 7 · COMPETENCY 7.2

Training Program Development

CSP11 blueprint: Develop training programs with training materials to address various learning styles (e.g., presentation methods and tools)

1. Definition

Training program development turns needs into structured instruction: writing measurable learning objectives, selecting content, methods and media for different learners, creating lesson plans, instructor guides, trainee materials and job aids, and piloting the course. ADDIE (Analysis, Design, Development, Implementation, Evaluation) is the common instructional design model.

2. Explanation in simple words

Decide exactly what learners must be able to do at the end, then build lessons, activities and materials that get them there - using pictures, talk, reading and hands-on practice so everyone can learn.

3. Key points to remember

- ADDIE: Analyze (needs, audience) → Design (objectives, assessments, sequence, methods) → Develop (materials, lesson plans, media, pilot) → Implement (deliver) → Evaluate (formative during, summative after).
- Learning objectives (ABCD): Audience, Behavior (observable action verb), Condition, Degree (criterion). Avoid 'understand', 'know', 'be aware'.
- Bloom's taxonomy (cognitive, low to high): Remember → Understand → Apply → Analyze → Evaluate → Create. Psychomotor and affective domains also exist.
- Lesson plan contents: objectives, materials, time, content outline, activities, assessment, instructor notes - standardizes delivery and avoids omitting critical material.
- Materials: instructor guide, participant workbook/handouts, job aids (checklists, flowcharts used on the job), visual aids, simulations, case studies.
- Presentation design: 6 × 6 rule (about six lines, six words), large readable fonts, one idea per slide, images over text; lighting, room layout (U-shape for discussion), background noise low.
- Accommodate diversity: literacy levels, language, generations (Gen Z prefers technology-based, interactive, short content), disabilities (captions, large print, audio description, braille), culture.
- Pilot course with a representative sample to test content, timing and methods before full rollout; ethics - copyright/trademark compliance, test security, fairness and confidentiality.

4. Real work scenario

> A plant needs confined-space attendant training. The developer writes the objective: 'Given a gas meter and entry permit, the attendant will verify atmospheric readings and record them with 100% accuracy.' The course combines a short lecture, a video of a real incident, a hands-on gas meter station and a mock rescue drill, and is piloted with one crew before rollout.

5. CSP exam tips and traps

- Good objectives are measurable and observable.
- Lesson plans standardize the presentation.
- Applying a formula in a new situation = Bloom's 'Apply'; designing a new program = 'Create'.
- Pilot with a representative sample of the real audience.

6. Every blueprint point (4 blueprint items + 7 more)

Training materials BLUEPRINT

Training materials are the lesson plans, slides, handouts, manuals, videos, job aids and assessments produced in the Develop phase of ADDIE. They must be accurate, current, job-specific and written at the audience's literacy level.

- Develop phase outputs: lesson plans, slides, study guides, quizzes, case studies, videos, handouts.
- ANSI Z490.1: materials should be clear, relevant and tied to measurable learning objectives.
- Job aids (checklists, pictograms, flowcharts) support performance on the job after training.
- Review and update materials after incidents, regulation changes and MOC.

> **Exam tip:** 'Creating quizzes and handouts' = Develop phase, not Design.

Learning styles BLUEPRINT

Learners differ in how they prefer to take in information; the VARK model groups them as visual, auditory, reading/writing and kinesthetic. Training should use a blend so every learner is reached.

- VARK: Visual, Auditory, Reading/writing, Kinesthetic (detailed under 7.6).
- The guide also lists expanded learner types: stress (learn under pressure), ease (calm settings), scribble (writing notes), trust (authority figures), teach (teaching others), copy (imitating models).
- Kolb styles: diverging, assimilating, converging, accommodating.
- Research gives limited support to matching instruction to a single preferred style; multimodal delivery is the safe design choice.

> **Exam tip:** Best answer for a mixed workforce: use multiple methods/modalities, not one style.

Presentation methods BLUEPRINT

Presentation methods are how content is delivered: lecture, demonstration, discussion, case study, role play, drill and simulation. The method should match the objective and learning domain.

- Knowledge (cognitive) objectives: lecture, reading, discussion, quizzes.
- Skill (psychomotor) objectives: demonstration then supervised practice, drills, simulation.
- Attitude (affective) objectives: case studies, testimonials, discussion, leadership example.
- Adults disengage in theory-heavy lectures - add real examples and hands-on exercises.

> **Exam tip:** Hands-on skill (SCBA donning, extinguisher use) cannot be verified by lecture or written test - choose demonstration and practice.

Training tools BLUEPRINT

Training tools are the aids and technology used to deliver and reinforce learning, from simple job aids to simulators.

- Examples: projectors/slides, videos, props and actual equipment, mock-ups, LMS, VR/AR, simulators, polling/quiz apps, job aids.
- Use real equipment or close mock-ups for skill training.
- An LMS tracks completion, scores and due dates for records.
- Engagement features: polls or quizzes every 10-15 minutes, short case studies, video clips of 2 minutes or less.

ADDIE instructional design model

ADDIE (Analyze, Design, Develop, Implement, Evaluate) is a systematic model for building training. Evaluation feeds back into analysis so the program continuously improves.

- Analyze: needs assessment, audience, gaps.
- Design: write learning objectives, choose methods, outline structure and schedule.
- Develop: create materials, lesson plans, quizzes, videos.
- Implement: deliver sessions, pre-assessments, interaction, post-assessments.
- Evaluate: feedback surveys, pre vs. post test results, revise the program.

> **Exam tip:** Selecting objectives or methods = Design; creating materials = Develop; delivering = Implement; survey after class = Evaluate.

Learning objectives

Objectives state what the learner will be able to do after training, in observable and measurable terms. They drive content selection and assessment design.

- ABCD format: Audience, Behavior (action verb), Condition, Degree (standard).
- Example: 'Given a lockout kit, the maintenance technician will apply LOTO to Press 4 following all 6 steps without error.'
- Use measurable verbs (demonstrate, calculate, identify) - avoid 'understand', 'know', 'appreciate'.
- Assessments must align with objectives (a skill objective needs a performance test).

> **Exam tip:** If an objective cannot be observed or measured, it is poorly written.

Bloom's Taxonomy

A hierarchy of cognitive learning used to write objectives and design assessments, moving from recall up to creating new solutions.

- Revised order (low to high): Remember, Understand, Apply, Analyze, Evaluate, Create.
- Remember: list, define, state (recall exposure limits).
- Understand: explain, summarize, interpret (explain the hierarchy of controls).
- Apply: use in new situations (solve a fall clearance problem).
- Analyze: compare, differentiate, break down (root causes from incident reports).
- Evaluate: judge, assess, recommend (critique a fall protection plan). Create: design, develop (design a new safety program).

> **Exam tip:** Recommending improvements = Evaluate; solving exposure calculations = Apply; designing a program = Create.

Domains of learning (cognitive, affective, psychomotor)

Effective safety training addresses the head (knowledge), heart (attitudes and values) and hands (physical skills).

- Cognitive: knowledge and mental skills - quizzes, reading, written tests (memorizing exposure limits).
- Affective: attitudes, values, motivation - discussions, testimonials, case studies, leadership example (believing PPE is essential).
- Psychomotor: physical skill - hands-on practice, drills, simulation (fire extinguisher drills, safe lifting).
- Bloom's taxonomy describes levels within the cognitive domain.

> **Exam tip:** Changing a worker's belief that safety rules matter = affective domain.

Learner retention (learning pyramid)

The guide teaches that retention rises as learning moves from passive to active: reading 10%, hearing 20%, seeing 30%, hearing + seeing 50%, saying/talking 70%, saying + doing 90%.

- Passive: reading, hearing, seeing. Active: discussing, practicing, teaching others.
- Highest retention = talking + doing (role-plays, hands-on drills, teaching peers).
- Use for high-risk training (confined space, LOTO): practice and explain-back.
- Note: these exact percentages (Dale's Cone / learning pyramid) are not research-validated, but the principle that active learning beats passive learning is well supported.

> **Exam tip:** 'Which method gives the highest retention?' = hands-on practice combined with explaining/teaching (talking + doing).

Disequilibrium in training

A deliberate state of cognitive challenge that disrupts what learners currently believe so they must restructure their thinking. Based on Piaget's constructivist theory.

- Example: showing real incident case studies where skipping PPE caused severe injury disrupts the belief that shortcuts save time.
- Example: VR confined-space emergency simulation forces learners to reconsider procedures.
- Benefits: builds critical thinking, helps unlearn unsafe habits, deepens retention by engaging emotion and problem solving.

> **Exam tip:** Presenting an incident that challenges workers' current practices = disequilibrium.

Presentation slide design

Slides should clarify, not decorate. The guide gives practical rules for count, visuals, fonts, color and density.

- Count: about 25-30 slides for 60 minutes; 8-12 for a 15-20 minute session.
- Visuals: a relevant image or diagram every 2-3 slides; use pictograms (e.g., GHS) rather than text descriptions.
- Fonts: sans-serif; titles 28-32 pt or larger, body text 20-24 pt or larger, footnotes at least 16 pt.
- Contrast: dark on light or light on dark; avoid red-green combinations (color blindness).
- 6 x 6 rule: max 6 lines, max 6 words per line; bullets not sentences; one key idea per slide.
- Accessibility: readable on small screens, alt text for shared slides, consistent templates.

> **Exam tip:** Presentation-effectiveness answers emphasize clarity, engagement and less text/more practice.

Read more: Yates, Safety Professional's Reference and Study Guide (4th ed.): Ch. 19 Training, p. 483

7. Ten CSP-style practice questions

Answer all ten on paper first, then check the key. Score: _____ / 10 Date: _____ Retest: _____ / 10

7.2-Q1. A training designer is choosing between videos, role-plays, and e-learning modules to meet the course objectives. Which ADDIE phase applies?

A. Analyze
B. Design
C. Implement
D. Evaluate

7.2-Q2. While developing a chemical-safety course, you complete the ADDIE Analysis phase by reviewing incidents caused by chemical mishandling. What is the NEXT step?

A. Design the program, writing objectives focused on safe handling and emergency procedures
B. Deliver the training to a small group to test its effectiveness
C. Evaluate feedback from past sessions on similar topics
D. Document the training process for reporting purposes

7.2-Q3. A course team is creating handouts, slides, and quizzes from an approved course outline. Which ADDIE phase is this?

A. Analyze
B. Design
C. Develop
D. Evaluate

7.2-Q4. What is the primary purpose of conducting a pilot class during course development?

A. To give the primary instructors a break
B. To test the course materials and delivery methods before full rollout
C. To finalize the annual training schedule
D. To increase the course length

7.2-Q5. Which document is used to standardize a presentation so that different instructors deliver the same content and no critical material is omitted?

A. Enabling objectives
B. Evaluation forms
C. Slide deck
D. Lesson plan

7.2-Q6. The 6 x 6 rule for presentation slides means:

A. A maximum of about 6 lines per slide and 6 words per line
B. 6 slides per topic and 6 colors per slide
C. 6 points for each question
D. 6 slides every 6 minutes

7.2-Q7. A training objective states: 'Trainees will calculate the required ventilation rate for a solvent storage room.' Which level of Bloom's taxonomy does this objective target?

A. Remember
B. Understand
C. Apply
D. Create

7.2-Q8. When designing training materials for a workforce with mixed literacy levels and learning preferences, the trainer should:

A. Use technical language to convey expertise
B. Assume all participants read at the same level
C. Incorporate varied instructional strategies (visuals, demonstrations, discussion, plain-language text)
D. Remove all text and rely only on pictures

7.2-Q9. What is the role of job aids in a training program?

A. To test trainees' knowledge at the end of the course
B. To introduce new theories unrelated to the course
C. To provide feedback to the instructor
D. To help trainees apply the skills learned to their daily tasks

7.2-Q10. Which of the following is NOT an essential component of a comprehensive lesson plan?

A. Detailed marketing strategy for the course
B. Trainee materials such as handouts and job aids
C. Clear, measurable learning objectives
D. Methods for assessment and evaluation

Answer key and explanations

Q	Ans	Why
1	B	In Design, objectives are written and instructional strategies and delivery methods are selected. Producing the actual materials comes later, in Develop.
2	A	ADDIE runs Analyze, Design, Develop, Implement, Evaluate. After Analysis comes Design (objectives, strategy). A pilot delivery happens only after materials are developed.
3	C	Develop is where the actual training materials (handouts, slides, job aids, tests) are produced to match the design.
4	B	A pilot is a trial run with a representative audience so content, timing, materials, and methods can be corrected before the course is delivered widely.
5	D	A lesson plan lays out objectives, content sequence, methods, timing, and assessments so delivery is consistent across instructors and sessions.
6	A	The 6 x 6 guideline limits text density to prevent cognitive overload; slides should support the speaker, not replace them.
7	C	Using a learned procedure or formula in a new situation is Apply. Remember would be 'list' or 'state'; Create would be 'design a new system'.
8	C	Varied methods reach visual, auditory, reading/writing, and kinesthetic learners and different literacy levels. Removing all text is an overcorrection.

9	**D**	Job aids (checklists, flowcharts, pocket cards) support correct performance on the job and reduce reliance on memory, especially for infrequent or critical tasks.
10	**A**	Lesson plans contain objectives, content, methods, materials, and assessment. Marketing is not part of instructional design. Watch for the NOT.

DOMAIN 7 · COMPETENCY 7.3

Continuous Improvement of Training

CSP11 blueprint: Describe how to implement training programs utilizing the Continuous Improvement model

1. Definition

Continuous improvement applies the PDCA (Plan-Do-Check-Act) cycle to training: plan the program, deliver it, check results through evaluation data, and act on findings to revise content, methods and delivery - so training stays current, effective and aligned with organizational goals.

2. Explanation in simple words

Training is never 'done'. Deliver it, measure whether it worked, fix what didn't, and run it again better. Keep updating it when equipment, rules or incidents change.

3. Key points to remember

- PDCA for training: Plan (needs, objectives, design) → Do (deliver, pilot) → Check (evaluate: reactions, test scores, observations, incident trends) → Act (revise content, retrain, update materials, standardize improvements).
- Formative evaluation: during development and delivery (pilot feedback, module checks) to improve. Summative evaluation: after completion, judges overall effectiveness.
- Triggers for revision: new regulations/standards, MOC (new equipment or processes), incident investigations showing knowledge gaps, audit findings, poor evaluation results, learner feedback.
- Implementation success factors: management support, alignment with organizational goals and culture, clear objectives, scheduling to minimize production impact (opportunity cost), resources (rooms, equipment, trainers), records.
- Feedback loops: post-training reports, interim reports, trainer debriefs, refresher schedules, lessons learned database.
- Remediation: coaching or extra practice for trainees who do not meet objectives (support, not punishment).

4. Real work scenario

> After a new HazCom course, post-tests average 92%, but on-the-job observations three months later show only 60% of workers can find SDS Section 8 for PPE. The trainer adds a hands-on SDS scavenger hunt and quarterly 10-minute refreshers. The next observation cycle shows 90% proficiency.

5. CSP exam tips and traps

- Check = evaluate results; Act = make improvements based on them.
- Updating course materials should be driven first by current regulations and standards, not cost or popularity.
- Formative = during; summative = after.

6. Every blueprint point (1 blueprint items + 6 more)

Continuous improvement of training BLUEPRINT

Training programs are improved through a closed loop: evaluate results, find gaps, revise content and delivery, then re-evaluate. Inputs include incidents, audits, worker feedback, assessment results and regulatory changes.

- PDCA applied to training: Plan (needs/objectives), Do (deliver), Check (evaluate), Act (revise).
- ADDIE Evaluate phase feeds back into Analyze.
- Update triggers: incident or near miss, audit finding, poor test scores, new equipment/MOC, regulation change, worker feedback.
- ANSI Z490.1 and ISO 45001 (clause 10.3) both require ongoing evaluation and improvement.
- Track trends: repeated errors on the same quiz items signal a content or delivery problem.

> **Exam tip:** If post-training observations show workers still not following a procedure, revise the training and look for non-training causes - do not just repeat the same course.

Kaizen (continuous improvement)

Kaizen is the Japanese philosophy of small, incremental, ongoing improvements made by everyone, rather than large one-time projects.

- Guide's Kaizen cycle: make problems visible -> develop countermeasures -> determine root cause (5 Whys, fishbone) -> hypothesize solution -> test (pilot) -> implement -> standardize work.
- Safety uses: employee improvement suggestions, near-miss learning, incremental ergonomic adjustments, continuous refinement of training.
- Links to Domain 2 (culture, SMS), Domain 3 (risk controls) and Domain 7 (training effectiveness).

> **Exam tip:** Employees proposing small weekly safety improvements = Kaizen.

Poka-yoke (error-proofing)

Poka-yoke, from Shigeo Shingo and the Toyota Production System, designs processes so errors cannot happen or are immediately obvious - reducing reliance on human memory and training alone.

- Examples: plug that fits only one way, red/green scaffold tags, interlocked machine guards, keyed LOTO systems where keys cannot be removed until isolation is complete.
- Benefits listed: preventing errors, waste reduction, cost reduction, satisfied customers, competitive advantage.
- Two types: control (prevents the error) and warning (signals the error).
- Aligns with prevention through design and the hierarchy of controls.

> **Exam tip:** 'Prevent errors through design rather than relying on training' = poka-yoke.

Learning curves

A learning curve plots practice or time against performance. CSPs use it to judge training effectiveness, plan refresher training and predict performance.

- Positive accelerating: slow start, then rapid improvement (SCBA use).
- Negative accelerating (diminishing returns): rapid early gains, then levels off (basic PPE).
- S-curve (sigmoid): slow -> fast -> slow (confined space entry skills).
- Plateau: improvement stops (lack of challenge, motivation or poor methods).
- Declining: performance drops from disuse or poor transfer = skill decay - fix with refreshers and drills.

> **Exam tip:** 'Fast initial learning then plateau' = negative accelerating; 'slow, fast, slow' = S-curve.

Refresher training and skill decay

Skills that are rarely used decay over time, so periodic refreshers, drills and practice are scheduled to keep competence current.

- Emergency skills (evacuation, rescue, spill response) decay fastest because they are rarely used - drill them regularly.
- Retraining triggers (forklift example, 1910.178): unsafe operation, incident or near miss, poor evaluation, new truck type, changed workplace conditions.
- Spaced repetition and short refreshers outperform a single long session.

Positive safety culture and training

Training only changes behavior when the culture supports it: open communication, psychological safety and emotionally intelligent leadership.

- Open feedback: toolbox talks, anonymous reporting, leaders walking the floor and listening.
- Psychological safety: workers can report hazards, admit mistakes and suggest ideas without fear of retaliation.
- Emotional intelligence components: self-awareness, self-regulation, empathy, social skills.
- Building culture: lead by example, involve workers (committees), recognize safe behavior, share investigation results, give continuous feedback.

> **Exam tip:** Workers feel safe to report without fear = psychological safety; recognizing and managing emotions = emotional intelligence.

Training records and program review

Documentation supports compliance, competency tracking and improvement decisions.

- ANSI Z490.1 records: trainee name/ID, course title, date, duration, trainer, assessment results, content/outline.
- Use LMS or competency matrix data to spot overdue training and recurring weak topics.
- Periodic program audits verify that training still matches hazards, regulations and job tasks.

Read more: Yates, Safety Professional's Reference and Study Guide (4th ed.): Ch. 19 Training, p. 483; Ch. 31 Safety Management System (PDCA)

7. Ten CSP-style practice questions

Answer all ten on paper first, then check the key. Score: _____ / 10 Date: _____ Retest: _____ / 10

7.3-Q1. During the pilot of a new course, the trainer collects feedback at the end of each module and revises the materials before full rollout. This is an example of:

A. Summative evaluation
B. Formative evaluation
C. Kirkpatrick Level 4 evaluation
D. Needs assessment

7.3-Q2. A company updates its safety courses after every incident review so that lessons learned are built into the next delivery. Which concept is demonstrated?

A. Risk perception
B. Pedagogical review
C. Behavioral control
D. Continuous improvement of training

7.3-Q3. Which principle focuses on small, continuous, incremental improvements rather than major overhauls?

A. Kaizen
B. Poka-yoke
C. Kirkpatrick model
D. Andragogy

7.3-Q4. What is the primary goal of remediation in a training program?

A. To discipline trainees who did not meet the standard
B. To provide additional instruction and support so trainees can meet the learning objectives
C. To evaluate a trainee's future potential in the organization
D. To reduce the training time allocated to each trainee

7.3-Q5. What is the primary purpose of a post-training report?

A. To replace safety incident reports
B. To document only the employees who failed
C. To summarize key outcomes, attendance, and effectiveness of the training
D. To provide legal protection against employee complaints

7.3-Q6. Which of the following is NOT typically included in an interim training report?

A. Participant feedback on training progress
B. Completion status of employees in training
C. Details of an unrelated accident investigation
D. Adjustments made to content based on participant understanding

7.3-Q7. A training coordinator applies the Plan-Do-Check-Act cycle to a fall protection course. Comparing post-training field observation data against the course objectives occurs in which phase?

A. Plan
B. Do
C. Check
D. Act

7.3-Q8. Post-training audits show workers pass the lockout/tagout quiz but often skip the zero-energy verification step in the field. What is the BEST continuous improvement action?

A. Determine why the step is being skipped, revise the training to include hands-on verification practice and supervisor follow-up, then re-measure field performance
B. Make the written quiz harder
C. Repeat the same course for all employees
D. Remove the verification step from the procedure

7.3-Q9. Which event should MOST directly trigger a review and update of an existing training program?

A. The trainer's annual vacation schedule
B. A request for new classroom furniture
C. A change in the color scheme of the company logo
D. Introduction of new equipment or a process change under Management of Change

7.3-Q10. In a continuous improvement model of training, results from the ADDIE Evaluate phase should be used primarily to:

A. Archive the course permanently once evaluation is complete
B. Rank trainers for compensation purposes only
C. Justify removing all refresher training
D. Feed back into analysis and revise the design, materials, and delivery of the program

Answer key and explanations

Q	Ans	Why
1	B	Formative evaluation occurs during development or delivery to improve the program; summative evaluation judges the finished program afterwards.
2	D	Feeding incident findings and evaluation data back into course content is the continuous improvement loop (Plan-Do-Check-Act) applied to training.
3	A	Kaizen means continuous incremental improvement. Poka-yoke is mistake-proofing; Kirkpatrick is an evaluation model; andragogy is adult learning theory.
4	B	Remediation closes individual learning gaps with targeted re-instruction and practice, then reassesses. It is corrective, not punitive.

5	**C**	A post-training report records what was delivered, who attended, and how effective it was; this data drives the next improvement cycle.
6	**C**	Interim reports track progress, feedback, and mid-course adjustments. Unrelated incident details do not belong. Note the NOT.
7	**C**	Check is measuring results against objectives. Act is the next step: standardizing what worked or revising the course based on what Check found.
8	**A**	Passing a knowledge test without transfer to the job signals a gap in practice or reinforcement. Improve the program based on root cause and verify with Level 3 observation. Repeating unchanged training is the trap.
9	**D**	New equipment, processes, hazards, regulations, or incident findings change required competencies, so MOC should include training review. Z490.1 also calls for periodic program review.
10	**D**	ADDIE is cyclic when used for continuous improvement: evaluation findings feed back into analysis and every other phase. Treating evaluation as the end point is the trap.

DOMAIN 7 · COMPETENCY 7.4

Training Effectiveness Evaluation

CSP11 blueprint: Determine the effectiveness of training programs (e.g., surveys, on-the-job compliance, feedback, assessments, demonstrations, quizzes)

1. Definition

Training effectiveness evaluation determines whether training achieved its objectives and produced the intended changes in knowledge, behavior and organizational results. The Kirkpatrick four-level model (with Phillips' fifth level, ROI) is the standard framework.

2. Explanation in simple words

Did they like it, did they learn it, do they use it on the job, and did it make the workplace safer? Measure each level, not just the end-of-class survey.

3. Key points to remember

- Kirkpatrick Level 1 Reaction (smile sheets, surveys); Level 2 Learning (pre-/post-tests, quizzes, demonstrations); Level 3 Behavior (on-the-job observation, compliance audits, supervisor feedback - usually weeks/months later); Level 4 Results (fewer incidents, lower costs, quality); Phillips Level 5 ROI (monetary benefit vs cost).
- Pre-test establishes baseline knowledge; post-test measures learning gained. Delayed post-tests measure retention.
- Demonstrations and return demonstrations are the best way to assess psychomotor skills; written tests assess knowledge.
- Assessment quality: validity (measures what it is supposed to), reliability (consistent results), standardized administration, scoring rubrics, test security.
- Questionnaires: mix of closed-ended (quantifiable) and open-ended (insight) questions; Likert scales measure attitudes.
- Quantitative indicators: employees trained vs scheduled, pass rates, incident rate trends; qualitative: feedback, observations.
- Self-assessment encourages reflection; 360-degree feedback gathers input from multiple perspectives.

4. Real work scenario

> After forklift refresher training, Level 1 surveys show high satisfaction and Level 2 tests average 94%. Three months later, supervisors observe that 97% of operators perform complete pre-use inspections (Level 3), and forklift damage incidents fall by 40% in the next year (Level 4), saving $60,000 against a $15,000 program cost (Level 5 ROI = 300%).

5. CSP exam tips and traps

- Observing behavior on the job after training = Kirkpatrick Level 3.
- Pre- and post-tests measure learning (Level 2).
- For practical skills, the BEST evaluation is observation/demonstration, not a written test.
- The most persuasive argument to management is a cost-benefit comparison of training cost with injury cost.

6. Every blueprint point (6 blueprint items + 2 more)

Surveys BLUEPRINT

Post-training surveys and reaction forms measure participant satisfaction, engagement and perceived relevance - Kirkpatrick Level 1.

- Tools: feedback forms (smile sheets), focus groups, interviews.
- Useful for improving delivery, pacing and materials.
- Does NOT prove learning or behavior change.
- Follow-up surveys of supervisors weeks later can provide Level 3 (behavior) evidence.

> **Exam tip:** A high satisfaction score does not show the training worked - it is the weakest level of evidence.

Compliance BLUEPRINT

Compliance measures whether required training was completed on time and whether workers follow the trained procedures on the job.

- Training compliance: percent of required training completed, overdue refreshers, certifications current.
- Behavioral compliance: observed adherence to procedures (LOTO audits, PPE use) - Kirkpatrick Level 3.
- OSHA citations or audit findings tied to training gaps are lagging evidence.

> **Exam tip:** 100% course completion is an activity measure, not proof of effectiveness.

Feedback BLUEPRINT

Feedback flows in both directions: trainees and supervisors give feedback on the training, and trainers give learners feedback on their performance.

- Sources: participant comments, supervisor observations, incident and near-miss data, audit results.
- Timely, specific feedback during practice improves skill acquisition.
- Feedback is an input to continuous improvement of the program (7.3).

Assessments BLUEPRINT

Assessments measure whether learning objectives were achieved, usually through pre- and post-tests or performance evaluations - Kirkpatrick Level 2.

- Pre-test establishes baseline; post-test shows gain (learning gain = post - pre).
- Must be valid (measures the objective) and reliable (consistent results).
- Knowledge objectives: written tests; skill objectives: performance-based assessment.
- Example: pre-test average 55%, post-test 85% = 30-point gain in knowledge.

Demonstrations BLUEPRINT

Learners physically perform the task while a qualified evaluator observes against a checklist; the best way to verify psychomotor skills.

- Required by many standards: forklift operator practical evaluation (1910.178(l)), respirator donning and seal check, fall protection equipment inspection.
- Teach-back or show-me demonstrations prove competency better than written tests.
- Use an observation checklist with pass/fail criteria.

> **Exam tip:** Verifying a hands-on skill = demonstration/performance evaluation, not a written quiz.

Quizzes BLUEPRINT

Short knowledge checks during or after training that measure recall and understanding and reinforce learning.

- Cognitive domain, Kirkpatrick Level 2.
- Spacing quizzes through a session (every 10-15 minutes) maintains engagement.
- Item analysis: questions many learners miss reveal unclear content or poor items.
- Quizzes alone cannot verify skills or on-the-job behavior.

Kirkpatrick's four levels of training evaluation

A four-level model for evaluating training beyond the classroom: Reaction, Learning, Behavior, Results.

- Level 1 Reaction: satisfaction and engagement - surveys, feedback forms.
- Level 2 Learning: knowledge/skill gained - pre/post tests, quizzes, demonstrations.
- Level 3 Behavior: application on the job - observations, supervisor feedback, performance reviews.
- Level 4 Results: organizational outcomes - incident rate reduction, cost savings, compliance, productivity.
- Phillips adds Level 5 ROI: (monetary benefits - training costs) / costs x 100.
- Higher levels are harder and costlier to measure but give stronger evidence.

> **Exam tip:** On-the-job application = Behavior (Level 3); accident reduction = Results (Level 4); satisfaction = Reaction (Level 1).

Training ROI and outcome measures

Training effectiveness can be expressed in business terms by comparing the monetary benefits of improved safety outcomes with program costs.

- ROI (%) = (net benefits / costs) x 100.
- Example: training costs $20,000; avoided injury costs $50,000; ROI = (50,000 - 20,000) / 20,000 x 100 = 150%.
- Outcome measures: incident/TRIR trend, near-miss reporting rate, observed safe-behavior percentage, audit scores.
- Isolate training effects from other changes (new equipment, staffing) when attributing results.

Read more: Yates, Safety Professional's Reference and Study Guide (4th ed.): Ch. 19 Training, p. 483

7. Ten CSP-style practice questions

Answer all ten on paper first, then check the key. Score: _____ / 10 Date: _____ Retest: _____ / 10

7.4-Q1. Which of the following lists Kirkpatrick's four levels of training evaluation in the correct order?

A. Reaction, Learning, Behavior, Results
B. Learning, Reaction, Results, Behavior
C. Reaction, Behavior, Learning, Results
D. Behavior, Reaction, Results, Learning

7.4-Q2. You will measure the impact of a hazardous materials handling course with tests. What is the MOST effective use of pre-tests and post-tests?

A. Give both tests after training and compare scores with industry standards
B. Give a pre-test to establish prior knowledge and a post-test to measure knowledge gained
C. Use only a post-test to save time
D. Skip the pre-test so low scorers are not demotivated

7.4-Q3. Workers complete post-training surveys rating their satisfaction with the course and trainer. Which Kirkpatrick level is this?

A. Reaction
B. Learning
C. Behavior
D. Results

7.4-Q4. Supervisors check whether workers apply PPE procedures correctly on the job several weeks after training. Which Kirkpatrick level is this?

A. Level 1 - Reaction
B. Level 2 - Learning
C. Level 3 - Behavior
D. Level 4 - Results

7.4-Q5. After a training program, the organization records a 40% reduction in recordable incidents. This outcome corresponds to which Kirkpatrick level?

A. Reaction
B. Learning
C. Behavior
D. Results

7.4-Q6. Six months after a program, the company calculates that reduced injury costs returned $3 for every $1 spent on training. Which evaluation level is this?

A. Kirkpatrick Level 1 - Reaction
B. Kirkpatrick Level 2 - Learning
C. Kirkpatrick Level 3 - Behavior
D. Phillips Level 5 - Return on Investment

7.4-Q7. A written test for forklift operators consists mostly of questions about company history. Operators who score well still drive unsafely. What is the main problem with the test?

A. It has low reliability
B. It has low validity: it does not measure what it is intended to measure
C. It is too short
D. It was not timed

7.4-Q8. What is the MOST effective way to evaluate job performance change following a safety training course?

A. Rely on participant self-assessment
B. Compare incident rates before and after training
C. Observe workers performing their tasks and provide feedback
D. Administer a written test immediately after training

7.4-Q9. When designing a questionnaire to evaluate training, which type of questions gathers the most comprehensive feedback?

A. Open-ended questions only
B. Questions that focus solely on the trainer's performance
C. A mix of open-ended and closed-ended questions
D. Yes/no questions only

7.4-Q10. A CSP must verify that newly trained powered industrial truck operators are competent before they operate independently. Which evaluation method BEST meets this need?

A. A practical performance evaluation of each operator in the workplace by a qualified evaluator
B. A post-course satisfaction survey
C. A written quiz on forklift terminology
D. A signed attendance roster

Answer key and explanations

Q	Ans	Why
1	A	Level 1 Reaction (satisfaction), Level 2 Learning (knowledge/skill gained), Level 3 Behavior (on-the-job transfer), Level 4 Results (organizational outcomes).
2	B	Comparing pre-test and post-test scores isolates what was learned during training (Kirkpatrick Level 2). A post-test alone cannot show improvement.
3	A	Satisfaction surveys (smile sheets) measure Level 1 Reaction. They are easy to collect but do not prove learning or behavior change.

4	**C**	Observing on-the-job application measures Level 3 Behavior (transfer of training). Level 4 would be outcomes such as fewer injuries.
5	**D**	Organizational outcomes such as reduced incidents, costs, or downtime are Level 4 Results.
6	**D**	Converting results into monetary benefit versus cost is Phillips Level 5 (ROI), which extends Kirkpatrick's four levels. ROI = (3 - 1)/1 = 200%.
7	**B**	Validity means the test measures the intended competency. Reliability is consistency of results; a test can be reliable yet invalid.
8	**C**	Direct observation measures actual behavior change. Incident rates are lagging and influenced by many factors; a written test measures learning, not performance.
9	**C**	Closed-ended items (e.g., rating scales) give quantifiable, comparable data; open-ended items explain why. Together they give a complete picture.
10	**A**	For skill-based tasks, competence is verified by demonstration; OSHA 1910.178(l) requires formal instruction, practical training, and evaluation of operator performance in the workplace. Attendance and quizzes do not prove skill.

DOMAIN 7 · COMPETENCY 7.5

Education and Training Methods

CSP11 blueprint: Demonstrate working knowledge of education and training methods and techniques (e.g., classroom, online, simulation, computer-based, Artificial Intelligence, coaching, on-the-job training)

1. Definition

Training methods are the delivery approaches used to transfer knowledge and skills. The best method depends on objectives (knowledge vs skill), audience, risk, frequency of the task, resources and logistics. Blended learning combines several methods.

2. Explanation in simple words

Pick the method that matches what people must be able to do. Facts can be taught online; hands-on skills need practice; rare, high-risk tasks need realistic simulation and drills.

3. Key points to remember

- Classroom/instructor-led: interaction, questions, discussion; costly to schedule; lecture alone is passive - add activities.
- Online/e-learning: synchronous (live webinar, real-time) vs asynchronous (self-paced, any time - good for dispersed workers and schedules); consistent content, tracking in an LMS.
- Computer-based training (CBT) and adaptive/AI-driven learning: personalizes pace and content to learner performance; virtual and augmented reality for immersive hazard recognition; AI tutors and chatbots - require content validation and human oversight.
- Simulation and drills: realistic practice without real consequences - best for high-risk, low-frequency tasks (rescue, emergency response, crane operation simulators).
- On-the-job training (OJT): job instruction training four-step method - prepare the worker, present the operation, try-out performance, follow up; mentoring and apprenticeships.
- Coaching: one-on-one guidance and feedback to improve performance; also used for remediation.
- Other techniques: case studies (real consequences, analysis), role play, brainstorming (generate many ideas without critique), fishbowl, tabletop exercises, toolbox talks (short, specific topic), games, micro-learning.
- Facilitation skills: establish objectives, encourage participation from all, handle disruptive participants privately/politely, correct misinformation tactfully, manage conflict by finding common ground, read body language (kinesics) for disengagement.

4. Real work scenario

> A company must train 600 dispersed field technicians on electrical hazard awareness and 20 rescue-team members on confined-space rescue. It uses an asynchronous e-learning module with knowledge checks for the technicians, and quarterly hands-on simulated rescues in a training vault with debriefs for the rescue team.

5. CSP exam tips and traps

- High-risk, rarely performed tasks → simulation/drills.
- Self-paced LMS module = asynchronous.
- Four-step OJT: prepare, present, perform (try out), follow up.
- Blended learning = online + classroom + hands-on.

6. Every blueprint point (7 blueprint items + 1 more)

Classroom (instructor-led) training BLUEPRINT

A facilitator delivers training in person, allowing high interaction and real-time feedback.

- Best for: small groups, discussions, live demonstrations, drills, complex or high-hazard topics.
- Strengths: immediate questions and feedback, trainer can adjust to audience.
- Limits: limited scalability, travel/time costs, depends on trainer quality.
- Toolbox talks/tailgate meetings are short informal instructor-led sessions focused on one topic.

> **Exam tip:** In-person, hands-on facilitation for small groups = instructor-based learning.

Online (e-learning and virtual) training BLUEPRINT

Web-based training is often self-paced (asynchronous); virtual/blended learning mixes live instructor sessions with online content.

- E-learning: flexible, scalable, cost-effective, consistent; lower engagement if not interactive.
- Virtual/blended: live virtual classroom with Q&A plus online modules - needs reliable technology.
- ANSI/ASSP Z490.2 = accepted practices for e-learning in SH&E training.
- Online alone is insufficient where hands-on evaluation is required (forklift, respirator fit, fire extinguisher).

> **Exam tip:** 'Self-paced, online, recorded' = e-learning; 'live + online together' = virtual/blended.

Simulation BLUEPRINT

Simulation recreates a hazardous or rare situation so learners can practice safely - from tabletop exercises and drills to VR and full-scale simulators.

- Examples: VR confined-space entry, crane or vehicle simulators, mock spill drills, emergency tabletop exercises.
- Allows practice of high-consequence, low-frequency events without real exposure.
- Engages psychomotor and decision-making skills; supports disequilibrium learning.

Computer-based training (CBT) BLUEPRINT

Training delivered through software on a computer or device, often with interactive modules, branching scenarios and built-in quizzes.

- Consistent message, learner-paced, automatic record keeping through an LMS.
- Good for awareness and knowledge topics (HazCom, policies) and refreshers.
- Must include a way to ask a knowledgeable person questions when required (e.g., OSHA expects access to a qualified trainer for questions).
- Weakness: little skill practice, easy to click through without learning.

AI-enabled training BLUEPRINT

Artificial intelligence personalizes training paths and adapts difficulty using learner data.

- AI-personalized learning: adaptive quizzes that adjust difficulty to performance.
- Adaptive learning systems: dynamically change pace and content.
- Other emerging modalities: VR immersive training, social learning platforms (peer forums, group projects).
- Issues: data privacy, accuracy of AI-generated content (must be reviewed by a qualified person), access to technology.

Coaching and mentoring BLUEPRINT

One-on-one guidance from an experienced worker, supervisor or safety professional to develop skills and safe behavior over time.

- Coaching: short-term, task/performance focused, gives specific feedback (e.g., after BBS observations).
- Mentoring: longer-term development relationship.
- Uses adult learning principle of experience; experienced workers sharing lessons learned.
- Effective for transferring tacit knowledge and reinforcing training on the job.

On-the-job training (OJT) BLUEPRINT

Training performed at the actual work location using real equipment, under supervision of a skilled trainer.

- Job Instruction Training four steps: prepare the worker -> present (tell and show) -> try-out performance -> follow up.
- Highly job-relevant with high transfer to work.
- Risks: inconsistent content, bad habits passed on, exposure during learning - use a structured checklist and qualified trainer.
- Often combined with classroom theory (e.g., forklift = formal instruction + practical training + evaluation).

Selecting the training method

Method choice depends on the objective, learning domain, audience, hazard level, number of trainees, cost and available technology.

- Knowledge for large dispersed groups: e-learning/CBT.
- High-risk skills: instructor-led demonstration + hands-on practice + simulation.
- Attitudes/culture: discussions, case studies, leadership involvement.
- Blended approaches usually give the best results.

> **Exam tip:** Match method to objective: you cannot certify a hands-on skill with an online module alone.

Read more: Yates, Safety Professional's Reference and Study Guide (4th ed.): Ch. 19 Training, p. 483

7. Ten CSP-style practice questions

Answer all ten on paper first, then check the key. Score: _____ / 10 Date: _____ Retest: _____ / 10

7.5-Q1. A company introduces an AI-driven adaptive e-learning platform for HazCom refresher training. What is the PRIMARY advantage of adaptive learning?

A. It adjusts content and pace to each learner's demonstrated knowledge, focusing time on weak areas
B. It removes the need for any instructor oversight
C. It guarantees regulatory compliance automatically
D. It eliminates the need for hands-on training

7.5-Q2. You are developing a blended learning program for confined space entry. Which combination BEST exemplifies a blended approach?

A. Printed handouts and a classroom lecture only
B. A live webinar only
C. On-the-job mentoring only
D. An online self-paced module, an in-person session, and a hands-on field exercise

7.5-Q3. A safety manager wants to roll out an electrical hazard awareness course without requiring all employees to attend at the same time. Which delivery method is asynchronous?

A. A live, instructor-led webinar
B. A scheduled group discussion in the break room
C. An in-person demonstration at the jobsite
D. A self-paced e-learning module available on the LMS at any time

7.5-Q4. Which training method is BEST for emergency responders who must perform a high-risk task that rarely occurs in real life, such as confined space rescue?

A. Lecture with slides
B. Self-paced reading
C. Realistic simulation and drills with debriefing
D. A video shown once a year

7.5-Q5. Job Instruction Training (JIT) is a four-step on-the-job instruction method. The four steps are:

A. Provide manufacturer instruction sheets, assign a computer-based module, give an exam, and rank workers by score
B. Schedule training, lecture in person, give a 20-question exam, and assign tasks to those scoring 80% or more
C. Prepare the worker, present the operation, have the worker perform the operation, and follow up
D. Describe the job, list the rules, explain discipline for violations, and state productivity expectations

7.5-Q6. A safety manager notices workers disengage during long lectures but become active during hands-on sessions. Which adjustment is BEST?

A. Add simulations and practical problem-solving exercises
B. Increase lecture duration
C. Provide printed notes only
D. Add more theory quizzes

7.5-Q7. Which collaborative training method encourages participants to freely generate many ideas in a group setting without lectures or slides, deferring judgment until later?

A. Brainstorming
B. Fishbowl
C. Ad lib
D. Icebreaker

7.5-Q8. A new safety technician is paired with an experienced CSP who observes the technician's field audits, asks guiding questions, and gives regular individualized feedback over several months. This method is BEST described as:

A. Computer-based training
B. Coaching
C. Classroom lecture
D. Programmed instruction

7.5-Q9. A plant relies on informal 'follow an experienced worker' on-the-job training. What is the PRIMARY weakness of this approach?

A. It is always more expensive than classroom training
B. It cannot occur at the actual workstation
C. Unsafe shortcuts and inconsistent practices may be passed on unless the training is structured with written procedures, checklists, and competency sign-off
D. It never allows hands-on practice

7.5-Q10. A company plans to use only computer-based training (CBT) to qualify employees on a respirator fit and donning procedure. What is the PRIMARY limitation of this plan?

A. CBT cannot track completion records
B. CBT cannot be updated once published
C. CBT cannot be accessed by remote workers
D. CBT alone cannot verify the psychomotor skill; a hands-on demonstration and fit test are still needed

Answer key and explanations

Q	Ans	Why
1	**A**	Adaptive (AI-based) systems personalize the path based on learner performance. They still require human oversight, content validation, and practical skills verification.
2	**D**	Blended learning combines online and face-to-face methods, and for a high-hazard task it should include hands-on practice. Single-method options are not blended.

3	**D**	Asynchronous learning lets each learner participate at a different time. Live webinars are online but synchronous, a common trap.
4	**C**	Rare, high-consequence tasks need realistic hands-on practice to build and maintain psychomotor skill and decision-making under stress; debriefing reinforces lessons.
5	**C**	JIT: Prepare, Present (tell and show), Try-out performance, Follow-up. It is the classic structured on-the-job training method.
6	**A**	Active, experiential methods increase engagement and retention for adult learners. More lecture or theory addresses the wrong problem.
7	**A**	Brainstorming generates quantity of ideas without criticism. A fishbowl has an inner group discussing while an outer group observes; icebreakers are warm-up activities.
8	**B**	Coaching is ongoing one-on-one guidance and feedback on real work to develop performance. It is individualized and continuous, unlike a one-time class.
9	**C**	Unstructured OJT transfers whatever the mentor does, including bad habits. Structured OJT uses standard procedures, qualified trainers, and documented evaluation.
10	**D**	CBT is efficient for knowledge content, but physical skills must be practiced and demonstrated; respirator use also requires fit testing under 1910.134.

DOMAIN 7 · COMPETENCY 7.6

Adult Learning Principles

CSP11 blueprint: Understand adult learning principles (e.g., visual, auditory, reading and writing, kinesthetic)

1. Definition

Adult learning (andragogy, Malcolm Knowles) recognizes that adults learn differently from children: they are self-directed, bring experience, learn what they need now, and are problem-centered. Learning-style models such as VARK (visual, auditory, reading/writing, kinesthetic) guide the use of varied methods.

2. Explanation in simple words

Adults want to know why they must learn something, how it helps them now, and to use their own experience. Mix pictures, talking, reading and hands-on practice so everyone can learn.

3. Key points to remember

- Knowles' assumptions: need to know (why), self-concept (self-directed), experience (resource for learning), readiness (relevant to life/job roles), orientation (problem/task-centered), motivation (mostly internal).
- VARK: Visual (diagrams, charts, videos, photos), Auditory (lecture, discussion, stories), Reading/Writing (manuals, handouts, notes), Kinesthetic (hands-on practice, demonstrations, simulations). Multimodal approaches reach more learners.
- Retention: people remember more when they practice and teach others than when they only read or listen (active learning); spaced repetition and active recall beat one long session; the forgetting curve (Ebbinghaus) drops quickly without review.
- Make training relevant: real job scenarios, case studies, problem-solving, immediate application, respect for learners' experience.
- Pedagogy (teacher-directed, children) vs andragogy (learner-centered, adults).
- Barriers: low literacy, language, fatigue, fear of failure, irrelevant content; provide a safe environment for questions.
- Feedback: timely, specific, constructive; adults want to see progress.

4. Real work scenario

> Experienced pipefitters resist a new 'basic' hot-work course. The trainer redesigns it around incidents from their own site, asks the crew to identify hazards in photos from their units, and has them practice permit completion in pairs. Engagement improves and post-course audits show complete permits.

5. CSP exam tips and traps

- Adults are motivated when they see the value to their job or life (Knowles).
- Hands-on = kinesthetic; diagrams/flowcharts = visual.
- Andragogy = adult, self-directed learning.
- Spaced, short refreshers improve long-term retention.

6. Every blueprint point (4 blueprint items + 6 more)

Visual learners BLUEPRINT

Visual learners learn best from what they see: pictures, diagrams, charts, maps and videos.

- Strategies: diagrams, flowcharts, process maps, infographics, hazard photos, safety videos.
- Examples: GHS pictograms, evacuation maps, PPE infographics.
- Keywords in questions: charts, slides, visual aids.

Auditory learners BLUEPRINT

Auditory learners learn best by hearing and discussing information.

- Strategies: lectures, storytelling, Q&A, discussions, recorded messages, podcasts.
- Examples: toolbox talks, verbal briefings.
- Keywords: hear, discuss, talk through.

Reading/writing learners BLUEPRINT

Reading/writing (print) learners learn best through text: reading handouts and writing notes or procedures.

- Strategies: handouts, manuals, written assignments, note taking, quizzes.
- Example: writing out the steps of a JHA.
- The guide calls this the print modality; the 'scribble learner' type retains rules by note-taking.

> **Exam tip:** Note-taking or documentation clues = reading/writing (print).

Kinesthetic learners BLUEPRINT

Kinesthetic (tactile/hands-on) learners learn by doing, moving and handling equipment.

- Strategies: role play, drills, field training, practice labs, simulations, handling real equipment.
- Examples: practicing LOTO, extinguisher use, evacuation drills.
- The guide splits tactile (touch/handle/build) from kinesthetic (movement/action) and adds an interactive modality (group discussion, collaboration, peer teaching).

> **Exam tip:** Safety simulations, drills and role-playing = kinesthetic.

Andragogy: Knowles' six principles of adult learning

Andragogy (Malcolm Knowles) describes how adults learn differently from children (pedagogy). Training that ignores these principles produces disengaged learners.

- Need to know: adults must understand why (show real incidents and consequences before LOTO training).
- Self-concept: self-directed, want control (choose module sequence, self-paced options).
- Experience: build on what they know (experienced workers share lessons, peer learning).
- Readiness: learn what is immediately relevant to their job (spill training right after a near miss).
- Orientation: problem-centered rather than subject-centered (case studies, simulations).
- Motivation: internal drivers (growth, mastery, career) outweigh external rewards.

> **Exam tip:** Workers disengage in theory-heavy lectures - add real-world examples and hands-on exercises. Answers with 'explain why', autonomy, job relevance or problem-solving fit andragogy.

Characteristics of adult learners

Adults differ from children in mindset and memory patterns, which shapes how safety training should be designed.

- Rely on prior knowledge and experience - link new content to past practice.
- Value accuracy over speed - slow, step-by-step demonstrations.
- Problem-centered and goal-driven - tie training to personal and organizational goals.
- Self-directed and autonomous - offer flexibility and self-paced options.
- Memory: short-term memory declines with age while long-term memory is strong - reinforce with repetition and scenario practice.

> **Exam tip:** Wrong answers for adults: rote memorization, speed drills, abstract theory.

Adult learning principles applied to safety training

Practical application: involve learners, keep content work-relevant and focus on solving real problems.

- Involve them: have workers demonstrate PPE use instead of just listening.
- Work relevance: emergency response training tailored to actual facility hazards.
- Focus on problems: case study of a past incident caused by poor hazard communication.
- Link to training cycle: needs assessment -> design to learning styles -> implement with continuous improvement -> evaluate with surveys, quizzes and observations.

Kolb's experiential learning cycle

David Kolb's model says learning is a cycle of experience and reflection with four stages.

- Concrete Experience (feeling): do something new - first fire drill.
- Reflective Observation (watching): reflect - debrief what went well.
- Abstract Conceptualization (thinking): connect to theory - compare drill to fire protocols.
- Active Experimentation (doing): apply - modify procedures and try again.
- Kolb learner styles: Diverging (feeling + watching), Assimilating (watching + thinking), Converging (thinking + doing), Accommodating (doing + feeling).

> **Exam tip:** Applying new procedures at work = Active Experimentation; reflecting on what went wrong in a drill = Reflective Observation.

Pedagogy vs. andragogy

Pedagogy is teacher-directed learning designed for children; andragogy is learner-centered learning designed for adults.

- Pedagogy: teacher decides content, learner dependent, subject-centered, external motivation (grades).
- Andragogy: learner self-directed, uses experience, problem-centered, internal motivation.
- Adults learn best with participation, relevance and immediate application.

> **Exam tip:** Exam questions often test the difference between adult and child learners.

Multimodal delivery for mixed audiences

Most workforces include all learning preferences, so effective safety training combines visual, auditory, reading/writing and kinesthetic elements in each session.

- Example LOTO class: short video (visual/auditory), written procedure handout (reading/writing), hands-on lockout on a training panel (kinesthetic), group explain-back (interactive).
- Combining senses and active practice increases retention.
- Consider literacy and language barriers - use pictures and demonstrations.

Read more: Yates, Safety Professional's Reference and Study Guide (4th ed.): Ch. 19 Training, p. 483

7. Ten CSP-style practice questions

Answer all ten on paper first, then check the key. Score: _____ / 10 Date: _____ Retest: _____ / 10

7.6-Q1. A CSP designs training that is self-directed, experience-based, job-relevant, and internally motivated. Which learning theory is being applied?

A. Behaviorism
B. Pedagogy
C. Andragogy
D. Constructivism

7.6-Q2. Which assumption of adult learning (Knowles) is MOST useful when designing safety training for experienced workers?

A. Adults learn best through rote memorization
B. Adults are problem-centered and want to apply learning immediately to real job situations
C. Adults prefer the instructor to make all decisions
D. Adults learn the same way children do

7.6-Q3. A trainee learns best by handling equipment and practicing tasks. Which VARK preference does this describe, and which method fits it best?

A. Visual - diagrams
B. Auditory - lecture
C. Reading/writing - manuals
D. Kinesthetic - hands-on practice and simulation

7.6-Q4. A safety instructor uses videos, diagrams, and flowcharts to explain confined space hazards. Which learning preference benefits MOST?

A. Visual
B. Auditory
C. Kinesthetic
D. Reading/writing

7.6-Q5. A worker retains OSHA rules best by reading the written procedure and taking detailed notes during class. Which VARK learning preference does this describe?

A. Reading/writing
B. Visual
C. Auditory
D. Kinesthetic

7.6-Q6. During a toolbox talk, workers discuss and tell stories about past incidents. This approach PRIMARILY helps which learners?

A. Reading/writing learners
B. Visual learners
C. Auditory learners
D. Kinesthetic learners

7.6-Q7. A trainer allows employees to choose the order in which they complete online safety modules. Which adult learning principle is being applied?

A. Orientation to learning
B. Readiness to learn
C. Motivation
D. Self-concept (self-direction)

7.6-Q8. An experienced worker is invited to share lessons from past incidents during a fall protection class. Which adult learning principle does this demonstrate?

A. Motivation
B. Experience
C. Readiness
D. Orientation

7.6-Q9. Which statement reflects Malcolm Knowles' view of adult learners?

A. Adults are motivated to learn when they see the value of the information to their personal and professional lives
B. Adults dislike learning new things
C. Adults will avoid training whenever possible
D. Adults prefer lecture-style training

7.6-Q10. Which strategy BEST improves long-term retention of safety training content?

A. One long annual session
B. Reading the manual once
C. Spaced practice with periodic short refreshers and active recall
D. Watching a longer video

Answer key and explanations

Q	Ans	Why
1	C	These are Knowles' andragogy assumptions about adult learners. Pedagogy is teacher-directed instruction designed for children.
2	B	Adults are oriented toward problem-solving and immediate application, so job-relevant scenarios and real tasks work best.
3	D	Kinesthetic learners learn by doing; demonstrations followed by hands-on practice, drills, and simulations suit them best.
4	A	Diagrams, charts, and images serve visual learners. Narration or discussion would target auditory learners.
5	A	Reading/writing learners prefer text: manuals, handouts, lists, and note-taking. Visual learners prefer graphics rather than words.
6	C	Listening and speaking (discussion, storytelling, verbal explanation) suit auditory learners.
7	D	Knowles' self-concept principle states that adults are self-directing and want control over their learning. Giving choice respects this.
8	B	Adults bring prior experience that serves as a learning resource. Using it makes training relevant and respected.
9	A	Adults need to know why they are learning something; showing job relevance and benefit increases motivation.
10	C	Spaced repetition and active recall slow the forgetting curve. Single, passive exposures produce rapid decay in retention.

Appendix A · CSP formula sheet

The BCSP exam provides its own reference sheet; check bcsp.org for the current version. Learn when to use each formula, not only the formula.

Incident rates	TRIR or DART = cases × 200,000 ÷ hours worked Fleet rate = crashes × 1,000,000 ÷ miles
Time-weighted average	TWA = (C1T1 + C2T2 + … + CnTn) ÷ 8 h (unsampled time counts as 0 only if there was no exposure)
Mixtures (additive)	Em = C1/T1 + C2/T2 + … ; exceeds when Em > 1 TLVmix = 1 ÷ (f1/TLV1 + f2/TLV2 + …)
ppm ↔ mg/m³	ppm = mg/m³ × 24.45 ÷ MW (25 °C, 1 atm)
Noise dose (OSHA, 5-dB)	T = 8 ÷ 2^((L − 90)/5) h ; D = 100 × Σ(C/T) TWA = 16.61 log10(D/100) + 90
Adding decibels	L = 10 log10(Σ 10^(Li/10)) ; two equal sources = +3 dB
Sound vs distance	L2 = L1 + 20 log10(d1/d2) (−6 dB per doubling of distance)
Inverse square (radiation)	I2 = I1 × (d1/d2)²
Radioactive decay	A = A0 × (½)^(t/T½)
Ventilation	Q = V × A ; V = 4005 √VP ; TP = SP + VP ACH = Q × 60 ÷ room volume Dilution Q = 403×10⁶ × SG × ER × K ÷ (MW × C)
Heat stress	WBGT (sun) = 0.7 NWB + 0.2 GT + 0.1 DB WBGT (indoor) = 0.7 NWB + 0.3 GT
Respirators	MUC = OEL × APF (never above IDLH)
Gas laws	P1V1/T1 = P2V2/T2 (absolute T: K = °C + 273, °R = °F + 460; psia = psig + 14.7) PV = nRT
NIOSH lifting	RWL = 51 × HM × VM × DM × AM × FM × CM (lb) ; HM = 10/H ; LI = load ÷ RWL
Vibration	A(8) = a × √(T/8)
Rigging	Leg tension = (W ÷ legs) ÷ sin θ (θ from horizontal) ; 60° ×1.155, 45° ×1.414, 30° ×2.0 Block & tackle effort = W ÷ supporting parts
Mechanics	F = ma ; W = mg (g = 9.81 m/s², 32.2 ft/s²) ; KE = ½mv² ; PE = mgh ; v = √(2gh) Friction F = μN ; stopping distance = v·t + v²/(2a) Stress σ = F/A ; factor of safety = ultimate ÷ working
Pressure & hydraulics	F = P × A ; P = 0.433 psi per ft of water ; sprinkler Q = K√P
Electrical	V = IR ; P = VI = I²R
Geometry	Circle A = πr² ; cylinder V = πr²h ; c² = a² + b² ; 1 ft³ = 7.48 gal ; 1 m³ = 1,000 L
Life safety	Occupant load = area ÷ load factor ; egress width = occupants × capacity factor
Containment	Required volume = % × largest container ; height = volume (ft³) ÷ floor area
Statistics	Mean = Σx/n ; z = (x − μ)/σ ; 68/95/99.7 rule CI = x̄ ± z(s/√n) ; GM = (x1·x2·…·xn)^(1/n)

Probability	AND (independent) $P = P1 \cdot P2$; OR $P = P1 + P2 - P1P2$ Series reliability $R = R1 \cdot R2$; parallel failure $= F1 \cdot F2$
Epidemiology	Relative risk = Ie ÷ Iu ; odds ratio = ad ÷ bc
Finance	ROI = (benefit − cost) ÷ cost × 100 ; payback = cost ÷ annual savings ; B/C = benefits ÷ costs $F = P(1+i)^n$; $P = F(1+i)^{-n}$; $P = A[(1+i)^n - 1] \div [i(1+i)^n]$ Sales to cover a loss = loss ÷ profit margin ; expected loss = probability × cost
Risk	Risk = likelihood × severity ; FMEA RPN = S × O × D ; risk summation = Σ(frequency × consequence)

Appendix B · Progress tracker

	Competency	Notes read	MCQ score 1	MCQ score 2	Drill %
☐	1.1 Prevention through Design (PtD)		/10	/10	%
☐	1.2 Process Safety		/10	/10	%
☐	1.3 Common Workplace Hazards		/10	/10	%
☐	1.4 Facility Life Safety		/10	/10	%
☐	1.5 Fleet Safety		/10	/10	%
☐	1.6 Materials Handling		/10	/10	%
☐	1.7 Tools, Machines and Equipment		/10	/10	%
☐	2.1 Benchmarking and Gap Analysis		/10	/10	%
☐	2.2 Performance Standards and Action Planning		/10	/10	%
☐	2.3 EHS Culture: Measure, Analyze and Improve		/10	/10	%
☐	2.4 Incident Investigation and Root Cause Analysis		/10	/10	%
☐	2.5 Management of Change (MOC)		/10	/10	%
☐	2.6 System Safety Analysis		/10	/10	%
☐	2.7 Leading and Lagging Indicators		/10	/10	%
☐	2.8 EHS Management and Audit Systems		/10	/10	%
☐	2.9 Plans, Systems and Policies		/10	/10	%
☐	2.10 Document Retention and Management		/10	/10	%
☐	2.11 Budgeting, Finance and Economic Analysis		/10	/10	%
☐	2.12 Management and Leadership		/10	/10	%
☐	2.13 Project Management		/10	/10	%
☐	2.14 Data Analysis		/10	/10	%
☐	3.1 Safety Risk Evaluation Process		/10	/10	%
☐	3.2 Risk Management Strategies and Hierarchy of Controls		/10	/10	%
☐	3.3 Financial Risk Mitigation Strategies		/10	/10	%
☐	3.4 Risk Analysis: Identify, Rank and Monitor		/10	/10	%
☐	4.1 Emergency Response Plans		/10	/10	%
☐	4.2 Disaster Response and Recovery		/10	/10	%
☐	4.3 Fire Prevention, Protection and Suppression		/10	/10	%
☐	4.4 Hazardous Materials Transportation and Security		/10	/10	%
☐	4.5 Workplace Violence Prevention		/10	/10	%
☐	5.1 Environmental Protection and Pollution Prevention		/10	/10	%
☐	5.2 Hazardous Materials Management		/10	/10	%
☐	5.3 Waste Management		/10	/10	%
☐	5.4 Sustainability		/10	/10	%
☐	5.5 Environmental Issues and ESG		/10	/10	%

☐	6.1 Occupational Exposure: Anticipate, Recognize, Evaluate, Control		/10	/10	%
☐	6.2 Public Health and Epidemiology		/10	/10	%
☐	6.3 Toxicology		/10	/10	%
☐	6.4 Ergonomics and Human Factors		/10	/10	%
☐	6.5 Chemistry: Containment Volume and Storage Requirements		/10	/10	%
☐	6.6 Physics: Energy, Weight, Force and Stress		/10	/10	%
☐	7.1 Training Needs Assessment		/10	/10	%
☐	7.2 Training Program Development		/10	/10	%
☐	7.3 Continuous Improvement of Training		/10	/10	%
☐	7.4 Training Effectiveness Evaluation		/10	/10	%
☐	7.5 Education and Training Methods		/10	/10	%
☐	7.6 Adult Learning Principles		/10	/10	%

Error log

Error type: K = knowledge gap, C = calculation, R = misread, D = distractor trap, U = units, F = formula, T = time pressure.

#	Topic / Q	My answer	Correct	Type	Principle I must remember
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